

## FIRE PROTECTION AND LIFE SAFETY SYSTEMS

## General Comments

The provisions required by Chapter 9 are just part of the overall fire protection system of a building or structure. All fire protection requirements contained in the code must be considered as a package or overall system. Noncompliance with any part of the overall system may cause other parts of the system to fail. Failure to install the systems in accordance with code provisions may result in an increased loss of life and property due to a reduction in the level of protection provided.

Every effort must be made to verify the proper design and installation of a given fire protection system, especially those that result in construction alternatives and other code trade-offs.

The title of Chapter 9 has been expanded to include life safety systems. This reflects that the chapter has been expanded over the years to include emergency alarm systems, smoke control systems, explosion control and carbon monoxide detection, which are not fire protection systems. The requirements in Chapter 9 are active fire safety provisions. They are directed at containing and extinguishing a fire once it has erupted. This chapter parallels and duplicates much of Chapter 9 in the *International Building Code®* (IBC®). The code, however, contains additional specific provisions that are applicable only to existing buildings. It also contains periodic testing criteria that are not duplicated in the IBC. Proper testing, inspection and maintenance of the various systems are critical to establishing the reliability of the system. Additionally, Chapter 9 references and adopts numerous National Fire Protection Association (NFPA) standards, including the acceptance testing criteria within the standard. The referenced standards will also contain more specific design and installation criteria than are found in this chapter. As noted in Section 102.7, where differences occur between code requirements and the referenced standard, the code provisions apply.

## Purpose

Fire protection systems may serve one or more purposes in providing adequate protection from fire and hazardous material exposure. The purpose of Chapter 9 is to prescribe the minimum requirements for an active system or systems of fire protection to perform the following functions: to detect a fire, to alert the occupants or fire department of a fire emergency, to control smoke and to control or extinguish the fire. Generally, the requirements are based on the occupancy and height and area of the building, as these are the factors that most affect firefighting capabilities and the relative hazard of a specific space or area.

## SECTION 901—GENERAL

**901.1 Scope.** The provisions of this chapter shall specify where *fire protection* and *life safety systems* are required and shall apply to the design, installation, inspection, operation, testing and maintenance of all *fire protection* and *life safety systems*.

**C** Chapter 9 contains requirements for fire protection and life safety systems that may be installed or located in a building. These include automatic suppression systems, standpipe systems, fire alarm and detection systems, smoke control systems, smoke and heat vents, and portable fire extinguishers. Besides indicating the conditions under which respective systems are required, this chapter contains the design, installation, maintenance, testing and operational criteria for fire protection and life safety systems. While the code requires proper maintenance for the reliability of the systems, the actual maintenance provisions (periodic testing, inspections and maintenance) may be contained in one of the referenced standards.

Chapter 9 is intended to apply to buildings of new construction or where deemed applicable because of a change in occupancy or an addition, unless specifically indicated to be applicable to existing buildings only. The scoping provisions in Chapter 1 should be consulted for determining the proper requirements for additions, alterations and remodelings.

This chapter also addresses the requirements for fire command centers, fire department connections, fire pumps and emergency radio systems. These features all directly relate to the proper function of fire protection and life safety systems.

**901.2 Construction documents.** The *fire code official* shall have the authority to require *construction documents* and calculations for all *fire protection* and *life safety systems* and to require permits be issued for the installation, rehabilitation or modification of any *fire protection* and *life safety systems*. *Construction documents* for *fire protection* and *life safety systems* shall be submitted for review and approval prior to system installation.

**C** The construction documents and related calculations for all fire protection and life safety systems must be reviewed before a permit is issued. The review is performed to determine compliance with the code requirements and the applicable provisions in the referenced standards.

Typical shop drawings for fire protection and life safety systems are usually not prepared during the initial submittal for a construction permit. Many jurisdictions require a separate submittal and issue a separate permit to the contractor

installing the system (see Section 901.3). Factors such as classification of the hazard; amount of agent or water supply available; and the design criteria, including the density or concentration to be achieved by the system, are to be included with the shop drawings. Specific equipment data sheets identifying sprinklers, pipe dimensions, power requirements for smoke detectors, and so on must also be included with the submittal in addition to any required calculations. See also the commentary to Section 907.1.1 regarding fire alarm system construction documents and Section 907.1.2 regarding fire alarm system shop drawings.

**901.2.1 Statement of compliance.** Before requesting final approval of the installation, where required by the *fire code official*, the installing contractor shall furnish a written statement to the *fire code official* that the subject *fire protection or life safety system* has been installed in accordance with *approved* plans and has been tested in accordance with the manufacturer's specifications and the appropriate installation standard. Any deviations from the design standards shall be noted and copies of the approvals for such deviations shall be attached to the written statement.

**C** A certificate or other approved written statement must be submitted to the fire code official with the proper documentation from the installing contractor specifying that the fire protection or life safety system has been installed in accordance with the requirements of the code. The certificate should also indicate that all required inspections and tests of the system have been conducted at the time of application for a certificate of occupancy.

The written statement is to indicate that the system has been installed in accordance with code requirements. As previously stated, contractors may have certificates that specify the criteria of the referenced standards since many of them contain sample certificates. While such certificates may be used, the contractor is required to certify that the system complies with the provisions of the code, which in some instances may vary from the referenced standards. These variations and any other variations involved in the installation, whether by change order, variance or field condition, must also be included with the written statement.

**901.3 Permits.** Permits shall be required as set forth in Sections 105.5 and 105.6.

**C** Section 105 requires permits of two separate but related types. Section 105.5 requires an operational permit. These permits are required to conduct certain types of business or hazardous operations that require a higher level of scrutiny from the fire code official. The second type of permit, required by Section 105.6, is the construction permit, which is required for the installation and modification of all fire protection systems.

**901.4 Fire protection and life safety systems.** *Fire protection and life safety systems* shall be installed, repaired, operated and maintained in accordance with this code and the *International Building Code*.

**C** This section emphasizes that systems are to be installed, repaired, operated and maintained in compliance with the IFC and IBC. It is not the intent of the code to require existing systems that are otherwise not being altered to comply with current code and standard requirements. An existing system, even if it does not meet current standards, is allowed to continue in service as long as it continues to provide no less quality in service than what was originally in place.

**901.4.1 Required fire protection and life safety systems.** *Fire protection and life safety systems* required by this code or the *International Building Code* shall be installed, repaired, operated, tested and maintained in accordance with this code. A *fire protection or life safety system* for which a design option, exception or reduction to the provisions of this code or the *International Building Code* has been granted shall be considered to be a required system.

**C** Fire protection and life safety systems that are required by Chapter 9 or by another section of either the IBC or the code must be considered as required systems. The fire protection and life systems are integral components of the protection and safety features of the building and must be properly installed, repaired, operated, tested and maintained in accordance with the code. Improperly installed or maintained systems can fail to provide the anticipated protection and, in some cases, create a hazard in itself.

Although the code may not require a fire protection or life safety system for a specific building or portion of a building because of its occupancy, the system would still be considered a required system if some other code alternative, exception or reduction was taken based on the installation of that system. For example, a typical small office building may not require an automatic sprinkler system solely because of its Group B occupancy classification. However, if an exit access corridor fire-resistance-rating reduction is taken as allowed by Table 1020.2 for buildings equipped throughout with an NFPA 13 sprinkler system, that sprinkler system is now considered a required system. Code trade-offs, exceptions, reductions or other design options are not unique to the IBC. Such trade-offs occur more than 60 times in the code.

**901.4.2 Nonrequired fire protection and life safety systems.** *Fire protection and life safety systems* or portion thereof not required by this code or the *International Building Code* shall be allowed to be furnished for partial or complete protection provided that such installed systems meet the applicable requirements of this code and the *International Building Code*.

**C** A building owner or designer may elect to install a fire protection or life safety system that is not required in the code. Even though such a system is not required, it must comply with the applicable requirements of Chapter 9. This requirement is based on the concept that any system not installed as required by the code is lacking because it could give a false impression of properly installed protection.

For example, if a building owner chooses to install sprinkler protection in a certain area and that protection is not required by any provisions of the code, the system must be installed in accordance with NFPA 13 or 13R, as applicable, and other applicable requirements of the code, such as water supply and supervision. The extent of the protection provided would not be regulated. Once the system is installed, it is subject to the same code requirements for maintenance and testing with which any other system would be required to comply.

If the optional sprinkler system is intended to be used to provide alternatives found elsewhere in the code, the system would cease to be voluntary. In such a case, it becomes a required system and subject to Section 901.4.1.

**901.4.3 Alterations in buildings and structures.** For any *alteration* within a building or structure, the *fire protection* and *life safety* systems shall be extended, altered or augmented to maintain and continue protection within the building or structure. Persons shall not remove or modify any *fire protection* or life safety system installed or maintained under the provisions of this code or the *International Building Code* without approval from the *fire code official*.

**C** If an alteration or expansion of the system is proposed, the level of protection provided by the system is to be the same or greater than what it was before the alteration or expansion was considered. Any added sections of the system must meet the requirements for new fire protection or life safety according to the system involved. In addition, all work, including the removal or modification of existing systems, is required to be approved by the fire code official before it is started.

**901.4.4 Fire areas.** Where buildings, or portions thereof, are divided into *fire areas* so as not to exceed the limits established for requiring a *fire protection system* in accordance with this chapter, such *fire areas* shall be separated by *fire barriers* constructed in accordance with Section 707 of the *International Building Code* or *horizontal assemblies* constructed in accordance with Section 711 of the *International Building Code*, or both, having a *fire-resistance rating* of not less than that determined in accordance with Section 707.3.10 of the *International Building Code*.

**C** This section provides specific guidance on how a building needs to be divided into fire areas in order to avoid requiring a fire protection system to be installed. A single occupancy group would require fire barriers or horizontal assemblies in order to create multiple fire areas (see IBC Table 707.3.10), each having an area below the threshold for fire protection system installation.

**901.4.5 Additional fire protection systems.** In occupancies of a hazardous nature, where special hazards exist in addition to the normal hazards of the occupancy, or where the *fire code official* determines that access for fire apparatus is unduly difficult, the *fire code official* shall have the authority to require additional safeguards and *fire protection systems*. *Fire protection* and *life safety* systems required under this section shall be installed in accordance with this code and the applicable referenced standards.

**C** This section allows the fire code official to require fire protection safeguards beyond the minimum requirement of Chapter 9 where warranted by potential unsafe conditions. The provisions of the code cannot anticipate every occupancy condition. Hazardous material occupancies or buildings with limited fire department access are potentially a greater hazard to both building occupants and firefighters. Any additional safeguards should be those needed to abate potential hazards. This section does not give the fire code official the right to require additional systems without cause. If the condition is adequately addressed by the code, then additional safeguards are not warranted. Should additional safeguards be deemed necessary, then the fire protection components regulated by this section must be considered required systems.

**901.4.6 Appearance of equipment.** Any device that has the physical appearance of life safety or fire protection equipment but that does not perform that life safety or fire protection function shall be prohibited.

**C** All required or provided life safety or fire protection-related equipment must be continued in use and be maintained to meet the requirements in effect at the time of the original installation. Nonrequired equipment that has been taken out of service or cannot function as intended must be dismantled and removed to prevent creating a false impression of protection.

Simply because a nonrequired system does not meet the current standards is neither cause to require its removal (see Section 901.4), nor a reason to require the system to be upgraded. As long as the system is maintained in the manner in which it was intended when installed, it can be allowed to continue.

**901.4.7 Pump and riser room size.** Where provided, fire pump rooms and *automatic sprinkler system* riser rooms shall be designed with adequate space for all equipment necessary for the installation, as defined by the manufacturer, with sufficient working space around the stationary equipment. Clearances around equipment to elements of permanent construction, including other installed equipment and appliances, shall be sufficient to allow inspection, service, repair or replacement without removing such elements of permanent construction or disabling the function of a required *fire-resistance-rated* assembly. Fire pump and *automatic sprinkler system* riser rooms shall be provided with doors and unobstructed passageways large enough to allow removal of the largest piece of equipment.

**C** Section 901.4.7 establishes that, where a pump or riser room is necessary, those rooms housing fire protection system risers or fire pumps and their components have adequate space to facilitate their maintenance. This section does not require the construction of a room to house fire protection systems; however, if a room is provided, this section requires that it be adequately sized to allow for maintenance.

Instead of prescribing arbitrary dimensions, this provision bases the room area on clearances specified by the equipment manufacturers to ensure adequate space is available for its installation or removal. The design must provide enough area so that walls, finish materials and doors are not required to be removed during maintenance activities. The provision also prescribes that the size of the door serving a riser or pump room is of a size to accommodate the removal of the largest piece of equipment.

Given that the design of fire protection systems generally commences during the period that building construction drawings and specifications are being reviewed by the jurisdiction, it will be especially important for the building's designer to establish dialogue with the fire protection system contractor early in the design process to ensure that the room and passageways can accommodate the largest equipment and provide the space needed for maintenance.

**901.4.7.1 Access.** Automatic sprinkler system risers, fire pumps and controllers shall be provided with *ready access*. Where located in a fire pump room or *automatic sprinkler system* riser room, the door shall be permitted to be locked provided that the key is available at all times.

**C** This section requires that there is available access at all times to the fire pump and system control valves for the purposes of firefighting operations, inspection and testing requirements, and maintenance on the system. Where located in a locked room in order to provide a secure location away from malicious mischief and vandalism, the key must always be available for use by authorized individuals for entry.

**901.4.7.2 Marking on access doors.** Access doors for *automatic sprinkler system* riser rooms and fire pump rooms shall be labeled with an *approved* sign. The lettering shall be in contrasting color to the background. Letters shall have a minimum height of 2 inches (51 mm) with a minimum stroke of  $\frac{3}{8}$  inch (10 mm).

**C** Proper identification of the room is critical in order to allow for quick access for firefighters. The specific requirements of color and lettering size are to ensure that the identification will be uniform at all locations and can be easily read from a distance.

**901.4.7.3 Environment.** *Automatic sprinkler system* riser rooms and fire pump rooms shall be maintained at a temperature of not less than 40°F (4°C). Heating units shall be permanently installed.

**C** The specific requirement for a minimum temperature of the room provides protection from freezing for the sprinkler system and fire pump. Where heating is required to maintain the room temperature, it shall be installed permanently so that it is automatic in operation and dependable when it is necessary.

**901.4.7.4 Lighting.** Permanently installed artificial illumination shall be provided in the *automatic sprinkler system* riser rooms and fire pump rooms.

**C** Sufficient illumination in the fire pump and system control valve room is required to be installed permanently so that it is always available to provide visibility to the equipment within the room for inspection and operation.

**901.5 Administration of installation acceptance testing.** Fire protection and *life safety* systems and appurtenances thereto shall be subject to acceptance tests as contained in the installation standards and as *approved* by the *fire code official*. The *fire code official* shall be notified before any required acceptance testing.

**C** All fire protection and life safety systems must pass an acceptance test to determine that they will operate as required by the code. Acceptance tests are usually part of the final inspection procedures. The referenced standards contain specific acceptance test procedures. In most instances, the acceptance test procedures require 100 percent operation of the testable system components to determine that they are operational and functioning as required. Often, the design professional may require additional testing that may be beyond the code requirements to verify that the system operates as designed. The design professional may also establish the appropriate testing criteria for special systems, such as smoke control systems. These proposed testing protocols, where not addressed elsewhere in the code, must be approved by the fire code official before being performed.

The inclusion of the requirement for acceptance tests in the code is not intended to assign responsibility for witnessing the tests. The responsibility to witness the acceptance test is an administrative issue that each municipality must address. Because the acceptance test is critical during design and construction and is a requirement of occupancy, the requirement is located in the code. The section also clarifies that it is the owner or owner's authorized agent's responsibility to conduct the test and the role of the fire code official to witness—not conduct—the test. Typically, the owner will assign the responsibility of conducting the test to the installing contractor.

**901.5.1 Occupancy.** It shall be unlawful to occupy any portion of a building or structure until the required *fire protection* and *life safety systems* have been tested and *approved*.

**C** Partial occupancy of any structure must not be permitted unless all fire protection and life safety systems for the occupied areas have been tested and approved. Even so, the code assumes that full protection for all areas will be provided as quickly as possible. The installation of many fire protection and life safety systems and the associated code alternatives permitted for a given occupancy assume complete building protection and not just in the occupied areas. All partial occu-

pancy conditions are subject to the final approval of the fire code official. Section 105.3.4 allows the fire code official to issue conditional occupancy permits.

If the building becomes vacant, this section would be applicable again. Here, occupancy is based on a certificate of occupancy. If the vacancy is in a tenant space of an otherwise fully protected building, then the fire protection or life safety system must be continued. The other tenants of the building are subject to the necessary continuation of the fire protection or life safety system. This is similar to partial occupancy. On the other hand, if a building is vacated completely, and the certificate of occupancy is forfeited, then the entire building can be considered as unoccupied, and the fire protection or life safety systems can be discontinued. At such time, utilities can be disconnected and the building be declared nonhabitable. This option carries with it the burden of reuse because if the vacant, nonhabitable building is intended to be used again, it is subject to provisions in the IBC that would require the building to be certified as a new structure and may involve significant modifications in order to receive a new certificate of occupancy.

Often, insurance companies will require that automatic sprinkler systems be kept in operation during a vacancy. If such is the case, utilities will likely need to be kept in place in order to keep the system within operating temperature requirements. Buildings in which occupancy is discontinued for an extended period of time may be considered hazardous and require action to remediate the problem, possibly by demolition.

If the system being installed is a nonrequired system, then this section is not applicable and partial occupancy can be granted.

**901.6 Inspection, testing and maintenance.** *Fire protection and life safety systems* shall be maintained in an operative condition at all times, and shall be replaced or repaired where defective. Nonrequired *fire protection and life safety systems* and equipment shall be inspected, tested and maintained or removed in accordance with Section 901.8.

**C** Adequate maintenance, inspection and periodic testing of all systems, equipment and devices, including equipment and systems related to mechanical smoke exhaust and smoke and heat vents, is necessary so that they are ready to perform their intended functions should fire occur.

An inspection consists of a visual check of a system or device to verify that it is in operating condition and free from defects or damage. Indicating valves, gauges and indicator lamps are a few of the features required by the codes to facilitate this activity. Obvious damage and the general condition of the system, particularly the presence of corrosion, both external and internal, must always be noted and recorded. Partially because they are less detailed, inspections are conducted more frequently than tests and maintenance. Because special knowledge and tools are not required, inspections may be performed by any reasonably competent person.

Periodic tests following standardized methods are intended to confirm the results of inspections and determine that all components function properly and that systems meet their original design specifications. Tools, equipment or devices are usually required for these tests.

Because tests are more detailed than inspections, they are usually conducted only once or twice per year in most cases. Some tests, however, may be required as frequently as bimonthly or quarterly (for example, some fire alarm system equipment) or as infrequently as 5-, 6- or 12-year intervals (for example, portable fire-extinguisher hydrostatic tests). Since specialized knowledge and equipment are required, testing is usually done by technicians or specialists trained in the proper conduct of the test methods involved.

Periodic maintenance keeps systems in good working order and may be used to repair damage or defects discovered during inspections or testing. Specialized tools and training are required to perform maintenance. Only properly trained technicians or specialists should perform required periodic maintenance. Most maintenance is required only as needed, but many manufacturers suggest or require regular periodic replacement of parts subject to wear or abuse.

Nonrequired fire protection or life safety systems, where installed, require the same level of maintenance as required systems. If required maintenance is not being done, there is no way to determine if the system will function as intended. Therefore, inadequately maintained, nonrequired systems must be removed to avoid creating a false impression of adequate protection.

**901.6.1 Standards.** *Fire protection systems* shall be inspected, tested and maintained in accordance with the referenced standards listed in Table 901.6.1.

**C** Specific requirements related to inspection practices, testing schedules and maintenance procedures are dependent on the type of fire protection system and its corresponding referenced NFPA standard as indicated in Table 901.6.1.

**TABLE 901.6.1—FIRE PROTECTION SYSTEM INSPECTION, TESTING AND MAINTENANCE STANDARDS**

| SYSTEM                                    | STANDARD  |
|-------------------------------------------|-----------|
| Portable fire extinguishers               | NFPA 10   |
| Carbon dioxide fire-extinguishing systems | NFPA 12   |
| Halon 1301 fire-extinguishing systems     | NFPA 12A  |
| Dry-chemical extinguishing systems        | NFPA 17   |
| Wet-chemical extinguishing systems        | NFPA 17A  |
| Water-based fire protection systems       | NFPA 25   |
| Fire alarm systems                        | NFPA 72   |
| Fire dampers                              | NFPA 80   |
| Smoke dampers                             | NFPA 105  |
| Smoke and heat vents                      | NFPA 204  |
| Water-mist systems                        | NFPA 750  |
| Clean-agent extinguishing systems         | NFPA 2001 |
| Aerosol fire-extinguishing systems        | NFPA 2010 |

**C** This table lists the NFPA referenced standards to be used for the inspection, testing and maintenance criteria for various fire protection systems. Many of the testing and maintenance requirements are included elsewhere in Chapter 9 of the code.

**901.6.2 Integrated testing.** Where two or more *fire protection* or *life safety systems* are interconnected, the intended response of subordinate *fire protection* and *life safety systems* shall be verified when required testing of the initiating system is conducted. In addition, integrated testing shall be performed in accordance with Sections 901.6.2.1 and 901.6.2.2.

**C** The code intends to require proper operation of integrated features of fire protection and life safety systems. In some cases, such as a fire alarm system initiating a complex response of doors, dampers, elevators and fans in a high-rise building, the integration is highly complex, involving cooperation of many different trades, controls and systems. In other cases, such as notifying a monitoring service when a fire sprinkler operates, the integration is relatively simple. The intent of this section is that the required testing of integrated features is scaled in a manner that is reasonable for a wide range of applications.

**901.6.2.1 High-rise buildings.** For high-rise buildings, integrated testing shall comply with NFPA 4, with an integrated test performed prior to issuance of the certificate of occupancy and at intervals not exceeding 10 years, unless otherwise specified by an integrated system test plan prepared in accordance with NFPA 4. If an equipment failure is detected during integrated testing, a repeat of the integrated test shall not be required, except as necessary to verify operation of fire protection or life safety functions that are initiated by equipment that was repaired or replaced.

**C** NFPA 4 is referenced for compliance in this section for the integrated testing of fire protection systems in high-rise buildings. The integrated test is to be completed before the building is allowed to be occupied and additionally every 10 years or less to ensure that the required fire protection systems continue to function as intended. In addition, it is stated that a repeat of the entire integrated test is not required when failed initiating equipment is repaired or replaced.

**901.6.2.2 Smoke control systems.** Where a *fire alarm system* is integrated with a smoke control system as outlined in Section 909, integrated testing shall comply with NFPA 4, with an integrated test performed prior to issuance of the certificate of occupancy and at intervals not exceeding 10 years, unless otherwise specified by an integrated system test plan prepared in accordance with NFPA 4. If an equipment failure is detected during integrated testing, a repeat of the integrated test shall not be required, except as necessary to verify operation of fire protection or life safety functions that are initiated by equipment that was repaired or replaced.

**C** NFPA 4 is referenced for compliance in this section for the integrated testing of fire alarm systems that are required to operate in combination with smoke control systems. The integrated test is to be completed before the building is allowed to be occupied and additionally every 10 years or less to ensure that the required fire protection systems continue to function as intended. In addition, it is stated that a repeat of the entire integrated test is not required when failed initiating equipment is repaired or replaced.

**901.6.3 Records.** Records of all system inspections, tests and maintenance shall be maintained in accordance with Section 110.3.

**C** Accurate, up-to-date records are required to document the history of system inspection, testing and maintenance. Record keeping is not intended simply to prove to the fire code official that required inspection, testing and maintenance are being performed, but to assist the owner or his or her agent in performing these functions. A well-kept log helps an owner or technician determine how the system is performing over time and how changes inside and outside of the protected premises are affecting system performance. For example, automatic sprinkler system main drain test results may indicate whether the public water supply is being degraded by development, thereby impairing sprinkler system capabilities. Similarly, a history of accidental alarms at a specific smoke detector may indicate that the device requires cleaning or maintenance.

**901.6.3.1 Records information.** Initial records shall include the name of the installation contractor, type of components installed, manufacturer of the components, location and number of components installed per floor. Records shall include the manufacturers' operation and maintenance instruction manuals. Such records shall be maintained for the life of the installation.

**C** When the fire protection systems are first installed, an accurate inventory must be compiled so that future owners and officials can refer back to the documents for maintenance and operational requirements. Additionally, if a recall is required, the installation inventory will be able to identify if any of the components subject to the recall are installed. By including the installation contractor's information in the list, a resolution of any legal aspects of defective components can be more readily obtained. Also, if operations or maintenance change during the life of the installation and a public notice is provided, the building owner will have the information necessary to know and apply the new requirements. The requirement to collect and maintain this information is already within many of the standards referenced in the code. This requirement gives the fire code official enforcement language and assists the building owner in understanding the responsibilities associated with having these fire protection systems installed in the building.

**901.7 Systems out of service.** Where a required *fire protection system* is out of service, the fire department and the *fire code official* shall be notified immediately and, where required by the *fire code official*, the building shall be either evacuated or an *approved* fire watch shall be provided for all occupants left unprotected by the shutdown until the *fire protection system* has been returned to service.

Where utilized, fire watches shall be provided with not less than one *approved* means for notification of the fire department and their only duty shall be to perform constant patrols of the protected premises and keep watch for fires.

**Exception:** Facilities with an *approved* notification and impairment management program. The notification and impairment program for water-based *fire protection systems* shall comply with NFPA 25.

**C** The protection afforded by a required fire protection system must not be diminished in any existing building except for the purpose of conducting tests, maintenance or repairs. The length of service interruptions must be kept to a minimum. The fire department and the fire code official must be notified of any service interruptions. They must carefully evaluate the continued operation or occupancy of buildings and structures where protection is interrupted. Whenever possible, all unaffected portions of the system should be kept in service. Until protection is restored, hazardous processes or operations should be suspended and alternative special protection should be considered in addition to an approved fire watch.

The code text only addresses when a required system is placed out of service. However, if a system is in place, even though it is not required by the code or the IBC, it would be an appropriate courtesy to inform the fire department of any fire protection system being discontinued or temporarily taken out of service. If the nonrequired system is to be placed out of service for an extended period of time, the fire code official has the authority to address the condition under Section 901.4.6 and require that the system be either placed back into operation or removed so as not to create a false impression of protection.

The exception provides an alternative to the immediate notification of the fire department and fire code official when systems are taken out of service. In some occupancies, such as existing Group I-2 buildings, in addition to periodic maintenance and testing, the possible extent of construction activity would result in the fire department and fire code official being constantly notified of fire protection systems being placed out of service and then needing to make a determination as to whether a fire watch or building evacuation are required. In many instances, the duration of the system being out of service is relatively short. For example, a dry pipe sprinkler system is impaired at the conclusion of every trip test for the time period necessary to reset the system. NFPA 25 addresses these situations and allows 10 hours in a 24-hour period of time before the building is required to be evacuated or a fire watch is implemented. NFPA 25 still requires that the fire department, code official and other parties (such as the insurance carrier) be notified.

**901.7.1 Impairment coordinator.** The building *owner* shall assign an impairment coordinator to comply with the requirements of this section. In the absence of a specific designee, the *owner* shall be considered to be the impairment coordinator.

**C** The impairment coordinator is the person responsible for maintaining the building fire protection systems. The impairment coordinator may be the building owner or other designee, such as the plant manager or building engineer, if he or she is trained to comply with the provisions of Section 901.7.

**901.7.2 Tag required.** A tag shall be used to indicate that a system, or portion thereof, has been removed from service.

**C** When any fire protection system is taken out of service, it must be clearly identified with a visible tag that indicates the conditions of the impairment and who to notify. The tag is intended to alert building occupants and fire department personnel that the system in question is impaired. It must remain visibly in place until full protection is restored.

**901.7.3 Placement of tag.** The tag shall be posted at each fire department connection, system control valve, fire alarm control unit, fire alarm annunciator and *fire command center*, indicating which system, or part thereof, has been removed from service. The *fire code official* shall specify where the tag is to be placed.

**C** This section specifies some of the impaired locations where a tag must be used. Tagging a fire department connection, for example, is intended to alert the responding fire department that a normal operating condition does not exist for the portion of the system beyond the connection. While it is also important to tag system control valves, an impairment tag in the

sprinkler riser room may not get noticed until accessed by fire department personnel. The final location of all impairment tags is subject to the approval of the fire code official.

**901.7.4 Preplanned impairment programs.** Preplanned impairments shall be authorized by the impairment coordinator. Before authorization is given, a designated individual shall be responsible for verifying that all of the following procedures have been implemented:

1. The extent and expected duration of the impairment have been determined.
2. The areas or buildings involved have been inspected and the increased risks determined.
3. Recommendations have been submitted to management or the building *owner*/manager.
4. The fire department has been notified.
5. The insurance carrier, the alarm company, the building *owner*/manager and other authorities having jurisdiction have been notified.
6. The supervisors in the areas to be affected have been notified.
7. A tag impairment system has been implemented.
8. Necessary tools and materials have been assembled on the impairment site.

**C** This section specifies the procedures that must be followed in a thorough preplanned impairment program. These procedures must be followed whenever systems are purposely impaired, such as for routine sprinkler system alarm testing. Proper notification of responsible parties eliminates the chance of false alarms, reduces disruption of normal business activities and encourages quick resumption of normal operations.

**901.7.5 Emergency impairments.** Where unplanned impairments occur, appropriate emergency action shall be taken to minimize potential injury and damage. The impairment coordinator shall implement the steps outlined in Section 901.7.4.

**C** Unplanned impairments, of course, go beyond typical testing and maintenance procedures but are also not necessarily indicative of a fire event. For example, an unplanned emergency impairment might occur if a sprinkler head or pipe was found leaking or was accidentally impacted by a forklift. To reduce water damage and to repair the sprinkler system, the valve controlling the water supply to the affected area would need to be closed, thereby impairing protection to the area protected by that portion of the sprinkler system. The impairment coordinator must follow the procedures in Section 901.7.4 to restore protection in minimum time.

**901.7.6 Restoring systems to service.** Where impaired equipment is restored to normal working order, the impairment coordinator shall verify that all of the following procedures have been implemented:

1. Necessary inspections and tests have been conducted to verify that affected systems are operational.
2. Supervisors have been advised that protection is restored.
3. The fire department has been advised that protection is restored.
4. The building *owner*/manager, insurance carrier, alarm company and other involved parties have been advised that protection is restored.
5. The impairment tag has been removed.

**C** Regardless of whether a system is taken out of service for either a planned impairment or for an emergency, this section specifies the procedures to follow when restoring a system to service. By following these procedures, all responsible parties who were informed of the initial impairment will also be made aware that the system is now fully operational. Restoring the system to service assumes the affected part of the system has been corrected and is in proper working condition.

**901.8 Removal of or tampering with equipment.** It shall be unlawful for any person to remove, tamper with or otherwise disturb any *fire protection or life safety system* required by this code except for the purposes of extinguishing fire, training, recharging or making necessary repairs or where *approved* by the *fire code official*.

**C** Tampering or otherwise unauthorized altering of any fire protection or life safety system or component is illegal. A person who unlawfully tampers with equipment could face potential criminal charges. Tampering could include intentionally pulling a manual fire alarm box when no emergency exists, playing with matches to set off a smoke detector or flowing a city fire hydrant. The use of fire protection or life safety systems, equipment and other fire appliances is limited to those people authorized to conduct repairs and maintenance unless approved by the fire code official.

**901.8.1 Removal of or tampering with appurtenances.** Locks, gates, doors, barricades, chains, enclosures, signs, tags or seals that have been installed by or at the direction of the *fire code official* shall not be removed, unlocked, destroyed, tampered with or otherwise vandalized in any manner.

**C** Tampering with or vandalizing appurtenances that are in place to prevent tampering with the system components is also prohibited. For example, sprinkler system control valves are routinely chained and locked in the open position in addition to being equipped with electronically monitored tamper switches. Gates at fire apparatus roads, authorized by Section 503.6 of the code, must not have the locks changed or operation altered unless approved by the fire code official. Any unauthorized removal or tampering with these types of devices is strictly prohibited.

**901.8.2 Removal of existing occupant-use hose lines.** The *fire code official* is authorized to permit the removal of existing occupant-use hose lines where both of the following conditions exist:

1. The hose line would not be utilized by trained personnel or the fire department.
2. The remaining outlets are compatible with local fire department fittings.

**C** The current code and the IBC do not require occupant-use hoses in as many locations as were required in previous editions of the codes and referenced standards. There also has been a shift in the philosophy of fire service leaders as to whether occupants should attempt to extinguish the fire via occupant-use hose stations or evacuate the structure. Most of the population is now being taught to evacuate the building, not to fight the fire. This shift is mainly due to the safety risk of having an untrained person attempting to fight a fire with more than a fire extinguisher. Another consideration was the ongoing cost to building owners to comply with maintenance and testing requirements for the hoses and their racks. Many jurisdictions have enacted local legislation or code changes to address this issue, and the addition of this new section updates the code to reflect conventional wisdom on this topic.

This section provides safeguards in the form of several conditions for the fire code official to consider in formulating an approval for removal of occupant-use hoses. The first condition assumes that there is not a trained fire brigade on the premises that might be qualified to use the hose stations as first responders in, for example, an industrial setting. Condition 1 also recognizes that fire departments typically will not use private hose stations for fire attack due to the questionable quality of their maintenance. Condition 2 was included to ensure that any outlets that remain are useable by the fire department by requiring that their threads be compatible with local fire department fire hose fittings, such as American National Fire Hose Connection Screw Threads (NH).

**901.9 Termination of monitoring service.** For fire alarm systems required to be monitored by this code, notice shall be made to the *fire code official* whenever alarm monitoring services are terminated. Notice shall be made in writing by the provider of the monitoring service being terminated.

**C** This section requires the monitoring service itself to notify the fire code official of service being terminated. Although the ultimate responsibility rests with the building owner, he or she is not cited in this section since if they discontinued the service, they would likely not understand the implications, and if they did would have no incentive to contact the fire code official.

**901.10 Recall of fire protection components.** Any *fire protection system* component regulated by this code that is the subject of a voluntary or mandatory recall under federal law shall be replaced with *approved, listed* components in compliance with the referenced standards of this code. The *fire code official* shall be notified in writing by the building owner when the recalled component parts have been replaced.

**C** This section provides the fire code official with a valuable tool for monitoring recalls. Under this provision, a code section can be cited that will allow the fire code official to enforce the provisions of the recall on systems within the jurisdiction. Product listings and compliance with referenced standards are paramount in the effectiveness of fire protection systems. Companies may be under an agreement with federal agencies to “voluntarily” recall certain components or face legal action. While this is technically voluntary, it in no way reduces the need for the recall. To the extent that the company’s livelihood depends on the recall being performed properly, the action is not voluntary. This section allows the fire code official to compel the building owner to comply with the recall and replace the component with one that is not subject to suspicion.

## SECTION 902—DEFINITIONS

**902.1 Definitions.** The following terms are defined in Chapter 2:

ALARM NOTIFICATION APPLIANCE.

ALARM SIGNAL.

ALARM VERIFICATION FEATURE.

ANNUNCIATOR.

AUDIBLE ALARM NOTIFICATION APPLIANCE.

AUTOMATIC.

AUTOMATIC FIRE-EXTINGUISHING SYSTEM.

AUTOMATIC SMOKE DETECTION SYSTEM.

AUTOMATIC SPRINKLER SYSTEM.

AUTOMATIC WATER MIST SYSTEM.

AVERAGE AMBIENT SOUND LEVEL.

CARBON DIOXIDE EXTINGUISHING SYSTEM.

CLEAN AGENT.

**FIRE PROTECTION AND LIFE SAFETY SYSTEMS**

*COMMERCIAL MOTOR VEHICLE.*  
*CONSTANTLY ATTENDED LOCATION.*  
*DELUGE SYSTEM.*  
*DETECTOR, HEAT.*  
*DRY-CHEMICAL EXTINGUISHING AGENT.*  
*ELEVATOR GROUP.*  
*EMERGENCY ALARM SYSTEM.*  
*EMERGENCY VOICE/ALARM COMMUNICATIONS.*  
*FIRE ALARM BOX, MANUAL.*  
*FIRE ALARM CONTROL UNIT.*  
*FIRE ALARM SIGNAL.*  
*FIRE ALARM SYSTEM.*  
*FIRE AREA.*  
*FIRE DETECTOR, AUTOMATIC.*  
*FIRE PROTECTION SYSTEM.*  
*FIRE SAFETY FUNCTIONS.*  
*FIXED BASE OPERATOR (FBO).*  
*FOAM-EXTINGUISHING SYSTEM.*  
*GAS DETECTION SYSTEM.*  
*HALOGENATED EXTINGUISHING SYSTEM.*  
*IMPAIRMENT COORDINATOR.*  
*INITIATING DEVICE.*  
*LIFE SAFETY SYSTEMS.*  
*MANUAL FIRE ALARM BOX.*  
*MULTIPLE-STATION ALARM DEVICE.*  
*MULTIPLE-STATION SMOKE ALARM.*  
*NOTIFICATION ZONE.*  
*NUISANCE ALARM.*  
*PRIVATE GARAGE.*  
*RECORD DRAWINGS.*  
*SINGLE-STATION SMOKE ALARM.*  
*SLEEPING UNIT.*  
*SMOKE ALARM.*  
*SMOKE DETECTOR.*  
*STANDPIPE, TYPES OF.*  
     Automatic dry.  
     Automatic wet.  
     Manual dry.  
     Manual           wet.  
     Semiautomatic dry.  
*STANDPIPE SYSTEM, CLASSES OF.*  
     Class I system.  
     Class II system.  
     Class III system.  
*SUPERVISING STATION.*  
*SUPERVISORY SERVICE.*  
*SUPERVISORY SIGNAL.*  
*SUPERVISORY SIGNAL-INITIATING DEVICE.*  
*TIRES, BULK STORAGE OF.*  
*TRANSIENT AIRCRAFT.*

TROUBLE SIGNAL.

VISIBLE ALARM NOTIFICATION APPLIANCE.

WET-CHEMICAL EXTINGUISHING AGENT.

WIRELESS PROTECTION SYSTEM.

ZONE.

ZONE, NOTIFICATION.

**C** Definitions of terms can help in the understanding and application of the code requirements. This section directs the code user to Chapter 2 for the proper application of the indicated terms used in this chapter. Terms may be defined in Chapter 2, in another International Code® (I-Code®) as indicated in Section 201.3 or the dictionary meaning may be all that is needed (see also commentary, Sections 201.1 through 201.4).

Certain requirements in the code are based on code provisions in the IBC, *International Mechanical Code*® (IMC®) and *International Plumbing Code*® (IPC®). A review of definitions included in those codes will aid in the understanding of many of the requirements contained in the code.

## SECTION 903—AUTOMATIC SPRINKLER SYSTEMS

**903.1 General.** *Automatic sprinkler systems* shall comply with this section.

**C** This section identifies the conditions requiring an automatic sprinkler system for all occupancies. The need for an automatic sprinkler system may depend on not only the occupancy but also the occupant load, fuel load, height and area of the building as well as firefighting capabilities. Section 903.2 addresses all occupancy conditions requiring an automatic sprinkler system. Section 903.3 contains the installation requirements for all sprinkler systems in addition to the requirements of NFPA 13, NFPA 13R and NFPA 13D. The supervision and alarm requirements for sprinkler systems are contained in Section 903.4, whereas Section 903.5 refers to testing and maintenance requirements for sprinkler systems found in Section 901 and NFPA 25. Section 903.6 addresses existing buildings and references Chapter 11.

Unless specifically allowed by the code or the IBC, residential sprinkler systems installed in accordance with NFPA 13R or NFPA 13D are not recognized for reductions or exceptions permitted by other sections of this code or the IBC. NFPA 13 systems provide the level of protection associated with adequate fire suppression for all occupancies. NFPA 13R and NFPA 13D systems are intended more to provide adequate time for egress but not necessarily for complete suppression of the fire. Commentary Figure 903.2 lists examples of where the various sprinkler thresholds differ in application.

The area values contained in this section are intended to apply to fire areas, which are comprised of all floor areas bounded by fire barriers, fire walls or exterior walls. The minimum required fire-resistance rating of fire barrier assemblies that define a fire area is specified in Table 707.3.10 of the IBC. Because the areas are defined as fire areas, fire barriers, horizontal assemblies, fire walls or exterior walls are the only acceptable means of subdividing a building into smaller areas instead of installing an automatic sprinkler system. Whereas fire barriers and exterior walls define multiple fire areas within a single building, a fire wall defines separate buildings within one structure. Also note that some of the threshold limitations result in a requirement to install an automatic sprinkler system throughout the building while others may require only specific fire areas to be sprinklered.

Another important point is that one fire area may include floor areas in more than one story of a building (see commentary to “Fire area,” Section 202).

The application of mixed occupancies and fire areas must be carefully researched. Often the required separation between occupancies for the purposes of applying the separated mixed-use option in Section 508.4 of the IBC will result in a separation that is less than what is required to define the boundaries of a fire area. It is possible to have two different occupancies within a given fire area, treated as separated uses but with code requirements applicable to both occupancies since they are not separated by the rating required for fire areas.

**903.1.1 Alternative protection.** Alternative automatic fire-extinguishing systems complying with Section 904 shall be permitted instead of *automatic sprinkler system* protection where recognized by the applicable standard and *approved* by the *fire code official*.

**C** This section permits the use of an alternative automatic fire-extinguishing system when approved by the fire code official as a means of compliance with the occupancy requirements of Section 903. Although the use of an alternative extinguishing system allowed by Section 904, such as a carbon dioxide system or clean-agent system, would satisfy the requirements of Section 903.2, it would not be considered an acceptable alternative for the purposes of exceptions, reductions or other code alternatives that would be applicable if an automatic sprinkler system were installed.

**903.2 Where required.** *Approved automatic sprinkler systems* in new buildings and structures shall be provided in the locations described in Sections 903.2.1 through 903.2.12.

**Exception:** Spaces or areas in telecommunications buildings used exclusively for telecommunications equipment, associated electrical power distribution equipment, batteries not required to have an *automatic sprinkler system* by Section 1207 for energy storage systems and standby engines, provided that those spaces or areas are equipped throughout with an *automatic smoke detection system* in accordance with Section 907.2 and are separated from the remainder of the building by not less than 1-hour *fire barriers* constructed in accordance with Section 707 of the *International Building Code* or not less than 2-hour *horizontal assemblies* constructed in accordance with Section 711 of the *International Building Code*, or both.

**C** Sections 903.2.1 through 903.2.12 identify the conditions requiring an automatic sprinkler system (see Commentary Figure 903.2). The type of sprinkler system must be one that is permitted for the specific occupancy condition. An NFPA 13R sprinkler system, for example, may not be installed to satisfy the sprinkler threshold requirements for a mercantile occupancy (see Section 903.2.7). As indicated in Section 903.3.1.2, the use of an NFPA 13R sprinkler system is limited to Group R occupancies not exceeding four stories in height.

There is one exception for those spaces or areas used exclusively for telecommunications equipment. The clarification for batteries provides a reference to the exceptions in Section 1207.5.5. The telecommunications industry has continually stressed the need for the continuity of telephone service, and the ability to maintain this service is of prime importance. This service is a vital link between the community and the various life safety services, including fire, police and emergency medical services. The integrity of this communications service can be jeopardized not only by fire, but also by water, from whatever the source.

It must be emphasized that the exception applies only to those spaces or areas that are used exclusively for telecommunications equipment. Historically, those spaces have a low incidence of fire events. Fires in telecommunications equipment are difficult to start and, if started, grow slowly, thus permitting early detection. Such fires are typically of the smoldering type, do not spread beyond the immediate area and generally self-extinguish.

Note, however, that this exception requires fire resistive separation from other portions of the building and, most importantly, that the building cannot qualify for any code trade-offs for fully sprinklered buildings.

Commentary Figure 903.2—Summary of Occupancy-Related Automatic Sprinkler Thresholds<sup>a</sup>

| Occupancy                             | Threshold                                                                                                                                                                                                        | Exception                                                                                                                                                                                        |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| All occupancies                       | Buildings with floor level $\geq$ 55 feet above or below fire department vehicle access and occupant load $\geq$ 30.                                                                                             | Open parking structures (F-2)                                                                                                                                                                    |
| Assembly (A-1, A-3, A-4)              | Fire area > 12,000 sq. ft. or fire area occupant load $\geq$ 300 or fire area above/below level of exit discharge. Multitheater complex (A-1 only).                                                              | None                                                                                                                                                                                             |
| Assembly (A-2)                        | Fire area > 5,000 sq. ft. or fire area occupant load $\geq$ 100 or fire area above/below level of exit discharge.                                                                                                | None                                                                                                                                                                                             |
| Assembly (A-5)                        | Accessory areas > 1,000 sq. ft.<br>Enclosed spaces under grandstands or bleachers $\leq$ 1,000 sq. ft. and not constructed in accordance with Section 1029.1.1.1 or > 1,000 sq. ft.                              | None                                                                                                                                                                                             |
| Ambulatory care facility (B)          | $\geq$ 4 care recipients incapable of self preservation or any care recipients incapable of self preservation above or below level of exit discharge.                                                            | Floors classified as an open parking garage                                                                                                                                                      |
| Educational (E)                       | Fire area > 12,000 sq. ft. or on a floor other than a level of exit discharge serving such occupancies. Fire area occupant load $\geq$ 300.                                                                      | Every classroom has not fewer than one exterior exit door at ground level, an automatic sprinkler system is not required in any area below the lowest level of exit discharge serving that area. |
| Factory (F-1)                         | Fire area > 12,000 sq. ft. or fire area located > three stories above grade, or combined fire area > 24,000 sq. ft.                                                                                              | None                                                                                                                                                                                             |
| Mercantile (M)                        | Woodworking > 2,500 sq. ft. (F-1 only)<br>Manufacture > 2,500 sq. ft. (F1),<br>display and sale > 5,000 sq. ft. (M),<br>storage > 2,500 sq. ft. (S-1) of upholstered furniture or mattresses.                    |                                                                                                                                                                                                  |
| Storage (S-1)                         | Bulk storage of tires > 20,000 cu. ft. (S-1 only)                                                                                                                                                                |                                                                                                                                                                                                  |
| High-hazard (H-1, H-2, H-3, H-4, H-5) | Sprinklers required.                                                                                                                                                                                             |                                                                                                                                                                                                  |
| Institutional (I-1, I-2, I-3, I-4)    | Sprinklers required.                                                                                                                                                                                             | Day care at level of exit discharge and each classroom has exterior exit door.                                                                                                                   |
| Residential (R)                       | Sprinklers required.                                                                                                                                                                                             | None                                                                                                                                                                                             |
| Repair garage (S-1)                   | Fire area > 12,000 sq. ft. or $\geq$ two stories above grade with fire area > 10,000 sq. ft. or repair garage servicing vehicles in basement or servicing commercial motor vehicles in fire area > 5,000 sq. ft. | None                                                                                                                                                                                             |
| Parking garage (S-1)                  | Commercial motor vehicles parking area > 5,000 sq. ft.                                                                                                                                                           |                                                                                                                                                                                                  |
| Parking garage (S-2)                  | Fire area > 12,000 sq. ft. or fire area > 5,000 sq. ft. for storage of commercial motor vehicles; or beneath other groups (enclosed parking).                                                                    | Not if beneath Group R-3.                                                                                                                                                                        |
| Covered and open malls (914.2.1)      | Sprinklers required.                                                                                                                                                                                             | Attached open parking structures.                                                                                                                                                                |
| High-rises (914.3.1)                  | Sprinklers required.                                                                                                                                                                                             | Open garages; certain telecommunications equipment buildings.                                                                                                                                    |
| Unlimited area buildings (IBC 507)    | A-3, A-4, B, F, M, S: one story. B, F, M, S: two story.                                                                                                                                                          | One story F-2 or S-2.                                                                                                                                                                            |

For SI: 1 foot = 304.8 mm, 1 square foot = 0.0929 m<sup>2</sup>.

a. Thresholds located in Section 903.2 unless noted. See also Table 903.2.11.6 for additional required suppression systems.

**903.2.1 Group A.** An *automatic sprinkler system* shall be provided throughout buildings and portions thereof used as Group A occupancies as provided in this section.

**C** Group A occupancies are characterized by a significant number of people who are not familiar with their surroundings. The requirement for a suppression system reflects the additional time needed for egress. The extent of protection is also intended to extend to the occupants of the assembly group from unobserved fires in other building areas located between the story containing the assembly occupancy and all levels of exit discharge serving such occupancies. The only exception to the coverage is for Group A-5 occupancies that are open to the atmosphere. Such occupancies require only certain aspects to be sprinklered, such as concession stands (see commentary, Section 903.2.1.5).

The requirement for sprinklers is based on the location and function of the space. It is not dependent on whether the area is provided with exterior walls. IFC Committee Interpretation No. 25-05 to this section discusses this issue and states, in part, that “where no surrounding exterior walls are provided along the perimeter of the building, the building area is used to identify and determine applicable fire area.” Outdoor areas, such as pavilions and patios, may have no walls but will have an occupant load and other factors that identify the assembly occupancy as such. If any of the thresholds are reached requiring sprinkler protection, then sprinkler protection must be provided whether there are exterior walls or not.

**903.2.1.1 Group A-1.** An *automatic sprinkler system* shall be provided throughout stories containing Group A-1 occupancies and throughout all stories from the Group A-1 occupancy to and including the *levels of exit discharge* serving that occupancy where one of the following conditions exists:

1. The *fire area* exceeds 12,000 square feet (1115 m<sup>2</sup>).
2. The *fire area* has an *occupant load* of 300 or more.
3. The *fire area* is located on a floor other than a *level of exit discharge* serving such occupancies.
4. The *fire area* contains a multiple-theater complex.

**C** Group A-1 occupancies are identified as assembly occupancies with fixed seating, such as theaters. In addition to the high occupant load associated with these types of facilities, egress is further complicated by the possibility of low lighting levels customary during performances. The fuel load in these buildings is usually of a type and quantity that would support fairly rapid fire development and sustained duration.

Theaters with stages pose a greater hazard. IBC Section 410.6 requires stages to be equipped with an automatic sprinkler system. The proscenium opening must also be protected. This compensates for the additional hazards associated with stages in Group A-1 occupancies.

This section lists four conditions that require installing a suppression system in a Group A-1 occupancy including the entire story where the A-1 occupancy is located and all intervening floors. Condition 1 requires that, if any one fire area of Group A-1 exceeds 12,000 square feet (1115 m<sup>2</sup>), the automatic sprinkler system is to be installed throughout the entire story where a Group A-1 occupancy is located, regardless of whether the building is divided into more than one fire area. However, if all fire areas are less than 12,000 square feet (1115 m<sup>2</sup>) (and less than the other thresholds), then sprinklers would not be required. Compartmentalization into multiple fire areas in compliance with IBC Chapter 7 is deemed an adequate alternative to sprinkler protection.

Condition 2 establishes the minimum number of occupants for which an automatic sprinkler system is considered necessary. The determination of the actual occupant load must be based on Section 1004.

Condition 3 accounts for occupant egress delay when traversing a stairway requiring a sprinkler system, regardless of the size of occupant load. In such cases, alternative emergency escape elements such as windows may not be available; making the suppression needs all the greater. It is not necessary for the occupant load to exceed 300 on a level other than the level of exit discharge serving such occupancy. Any number of Group A-1 occupants on the alternative level would be cause to apply the requirement for sprinklers. The text does not make reference to “story” but uses the term “floor,” which could include mezzanines and basements.

Condition 4 states that a sprinkler system is required for multiplex theater complexes to account for the delay associated with the notification of adjacent compartmentalized spaces where the occupants may not be immediately aware of an emergency.

**903.2.1.2 Group A-2.** An *automatic sprinkler system* shall be provided throughout stories containing Group A-2 occupancies and throughout all stories from the Group A-2 occupancy to and including the *levels of exit discharge* serving that occupancy where one of the following conditions exists:

1. The *fire area* exceeds 5,000 square feet (464 m<sup>2</sup>).
2. The *fire area* has an *occupant load* of 100 or more.
3. The *fire area* is located on a floor other than a *level of exit discharge* serving such occupancies.

**C** Group A-2 assembly occupancies are intended for food or drink consumption, such as banquet halls, nightclubs and restaurants. Occupancies in Group A-2 involve life safety factors such as a high occupant density, flexible fuel loading, movable

furnishings and limited lighting; therefore, they must be protected with an automatic sprinkler system under any of the listed conditions.

In the case of an assembly use, the purpose of the automatic sprinkler system is to provide life safety from fire as well as preserving property. By requiring fire suppression in areas through which the occupants may egress, including the level of exit discharge serving such occupancies, the possibility of unobserved fire development affecting occupant egress is minimized.

The 5,000-square-foot (464 m<sup>2</sup>) threshold for the automatic sprinkler system reflects the higher degree of life safety hazard associated with Group A-2 occupancies. As alluded to earlier, Group A-2 occupancies could have low lighting levels, loud music, late hours of operation, dense seating with ill-defined aisles and alcoholic beverage service. These factors in combination could delay fire recognition, confuse occupant response and increase egress time.

Although the calculated occupant load for a 5,000-square-foot (465 m<sup>2</sup>) space at 15 square feet (1.4 m<sup>2</sup>) per occupant would be over 100, the occupant load threshold in Condition 2 is meant to reflect the concern for safety in these higher density occupancies. Although the major reason for establishing the occupant threshold at 100 was because of several nightclub incidents, the requirement is not limited to nightclubs or banquet facilities but to all Group A-2 occupancies. Any restaurant with an occupant load greater than 100 would require sprinkler protection as well. This includes fast food facilities with no low lighting or alcohol sales. The similar intent of Condition 3 is addressed in the commentary to Section 903.2.1.1.

Note that as with Group A-1 occupancies, when sprinklers are required, they are required on the story where the Group A-2 occupancy is located and on all intervening floors leading to the levels of exit discharge.

**903.2.1.3 Group A-3.** An *automatic sprinkler system* shall be provided throughout stories containing Group A-3 occupancies and throughout all stories from the Group A-3 occupancy to and including the *levels of exit discharge* serving that occupancy where one of the following conditions exists:

1. The *fire area* exceeds 12,000 square feet (1115 m<sup>2</sup>).
2. The *fire area* has an *occupant load* of 300 or more.
3. The *fire area* is located on a floor other than a *level of exit discharge* serving such occupancies.

**C** Group A-3 occupancies are assembly occupancies intended for worship, recreation or amusement and other assembly uses not classified elsewhere in Group A, such as churches, museums and libraries. While Group A-3 occupancies could potentially have a high occupant load, they normally do not have the same potential combination of life safety hazards associated with Group A-2 occupancies. As with most assembly occupancies, however, most of the occupants are typically not completely familiar with their surroundings. Where any of the three listed conditions are applicable, an automatic sprinkler system is required throughout the fire area containing the Group A-3 occupancy, including the entire story where the Group A-3 occupancy is located and throughout all floors between the Group A occupancy and exit discharge that serves that occupancy (see commentary, Sections 903.2.1 and 903.2.1.1).

**903.2.1.4 Group A-4.** An *automatic sprinkler system* shall be provided throughout stories containing Group A-4 occupancies and throughout all stories from the Group A-4 occupancy to and including the *levels of exit discharge* serving that occupancy where one of the following conditions exists:

1. The *fire area* exceeds 12,000 square feet (1115 m<sup>2</sup>).
2. The *fire area* has an *occupant load* of 300 or more.
3. The *fire area* is located on a floor other than a *level of exit discharge* serving such occupancies.

**C** Group A-4 occupancies are assembly uses intended for viewing of indoor sporting events and activities such as arenas, skating rinks and swimming pools. The occupant load density may be high, depending on the extent and style of seating, such as bleachers or fixed seats, and the potential for standing-room viewing.

Where any of the three listed conditions are applicable, an automatic sprinkler system is required throughout the fire area containing the Group A-4 occupancy, including the entire story where the Group A-4 occupancy is located, and in all floors between the Group A occupancy and exit discharge (see commentary, Sections 903.2.1 and 903.2.1.1).

**903.2.1.5 Group A-5.** An *automatic sprinkler system* shall be provided for all enclosed Group A-5 accessory use areas in excess of 1,000 square feet (93 m<sup>2</sup>).

**C** Group A-5 occupancies are assembly uses intended for viewing of outdoor activities. This occupancy classification could include amusement park structures, grandstands and open stadiums. A sprinkler system is not required in the open area of Group A-5 occupancies because the buildings would not accumulate smoke and hot gases. A fire in open areas would also be obvious to all spectators.

Enclosed areas such as retail areas, press boxes and concession stands require sprinklers if they are in excess of 1,000 square feet (93 m<sup>2</sup>). This section does not address, provide requirements or specifically allow a larger area to be subdivided into multiple smaller areas that are 1,000 square feet or less using rated separations in order to avoid the requirement to provide an automatic sprinkler system.

The provision is meant to mirror that in Section 1030.6.2.3, which exempts press boxes and storage facilities less than 1,000 square feet (93 m<sup>2</sup>) in area from sprinkler requirements in smoke-protected assembly seating areas.

**903.2.1.5.1 Spaces under grandstands or bleachers.** Enclosed spaces under *grandstands* or *bleachers* shall be equipped with an *automatic sprinkler system* in accordance with Section 903.3.1.1 where either of the following exist:

1. The enclosed area is 1,000 square feet (93 m<sup>2</sup>) or less and is not constructed in accordance with Section 1030.1.1.1.
2. The enclosed area exceeds 1,000 square feet (93 m<sup>2</sup>).

**C** This section provides the area threshold for automatic sprinklers to be required in an enclosed space under grandstands or bleachers. Areas that are 1,000 square feet or less are required to be constructed in accordance with Section 1030.1.1.1 in order to be exempt from requiring automatic sprinkler protection. Enclosed areas that are greater than 1,000 square feet are to be protected with automatic sprinklers regardless of the construction type and level of separation.

**903.2.1.6 Assembly occupancies on roofs.** Where an occupied roof has an assembly occupancy with an *occupant load* exceeding 100 for Group A-2 and 300 for other Group A occupancies, all floors between the occupied roof and the *level of exit discharge* shall be equipped with an *automatic sprinkler system* in accordance with Section 903.3.1.1 or 903.3.1.2.

**Exception:** Open parking garages of Type I or Type II construction.

**C** Frequently, rooftops are being used and occupied as assembly occupancies. Building owners will provide an open-air rooftop bar or lounge, or other use similar to a Group A-2 occupancy on the roof of a building. A roof does not meet the definition of a fire area. As such, protection of the occupants can be less than what would otherwise be required were the occupancy located on a floor rather than on the roof. In addition, even if a fire occurs within the building itself, it puts these occupants at risk. The provisions requiring sprinklers are based on the type of assembly occupancy located on the roof. The roof itself is not required to be sprinklered. The reference to Section 903.3.1.2 is added, since this use can occur on the roof of multifamily housing facilities.

The exception for open parking garages is consistent with the existing code requirement exception for open parking garages under Section 903.2.11.3 for “Buildings 55 feet or more in height.” It is becoming more common in the urban renewal areas throughout the United States that jurisdictions are asking developers to provide additional recreational and green spaces for their citizens to enjoy within their own communities. Because of the limited space available, it is not uncommon for such recreational and green spaces to be provided on the roofs of open parking garages. Based on the existing wording of this section, these recreational and green spaces greater than 700 square feet (based on 7 square feet net per occupant) or 1,500 square feet (based on 15 square feet net per occupant) would now require the open parking garage to be sprinklered. In other words, an open recreational or green space on the roof of an open parking structure that is more than approximately 39 by 39 feet square would require the garage to be sprinklered with a dry pipe sprinkler system that is initially a major cost to the project as well as a major monthly and yearly maintenance expense. Such an expense would most likely have an adverse effect on developers doing major city urban renewal projects, deterring them from providing such amenities for the local jurisdiction.

Additionally, there are considerable data supporting the exception’s elimination of automatic sprinkler systems in open parking garages. Two sample reports that evaluated fire behavior in parking garages are:

1. 2006 NFPA Fire Data Report, “Structure and Vehicle Fires in General Vehicle Parking Garages.”
2. 2008 Parking Consultants Council Fire Safety Committee Report, “Parking Structure Fire Facts.”

These reports provided the following conclusions:

- There was an average of only 660 fires per year in all types of parking garages in the United States. This represented a mere 0.006 percent of all fires annually.
- There were no fire fatalities in open parking garages of Type I or II construction. On average, there were only 2 injuries per year.
- There was no structural damage in 98.7 percent of the fires in parking garages.
- Vehicle fires in parking garages typically do not spread from vehicle to vehicle. Fire spread from vehicle to vehicle occurred in only 7 percent of the incidents.

Automatic sprinkler systems are required in occupancies other than open parking garages to protect the assembly occupancy above the fire and to protect the means of egress. Based on the inherent fire safety provided by open parking garages of Type I or II construction, an automatic sprinkler system is not required when an assembly use is located on the roof.

**903.2.1.7 Multiple fire areas.** An *automatic sprinkler system* shall be provided where multiple *fire areas* of Group A-1, A-2, A-3 or A-4 occupancies share *exit* or *exit access* components and the combined *occupant load* of these *fire areas* is 300 or more.

**C** There are two conditions required to trigger the requirements of this section. The first is that exit or exit access components of Group A-1, A-2, A-3 or A-4 fire areas are shared. The second is that the combined occupant load of the fire areas that share these components exceeds 300 persons.

This section addresses the issue of multiple small assembly occupancies placed in a single-story building and not triggering a sprinkler system requirement because of the installation of a rated corridor and separation walls. The code now requires that sprinkler systems be added when the convergence of more than 300 persons share an exit. This is consistent with the intent of automatic sprinkler systems being required for life safety and to maintain tenable exiting in a fire event. A fire event that is near an exit is the same whether there are 300 occupants in one room or three rooms with 100 occupants each sharing an exit. This is also consistent with the requirement in the “multiple-theater complex” for Group A-1, which is a requirement for anytime two or more theaters are in the same tenancy and does not consider occupant load as a trigger.

This will still allow single-story buildings with multiple tenancies that have separate exits and that utilize the fire area separation concept, such as buildings with multiple restaurants with separate entrances and strip-style mall buildings.

**903.2.2 Group B.** An *automatic sprinkler system* shall be provided for Group B occupancies as required in Sections 903.2.2.1 and 903.2.2.2.

**C** Sections 903.2.2.1 and 903.2.2.2 list the types and conditions of specific Group B occupancies that are required to be provided with an automatic sprinkler system.

**903.2.2.1 Ambulatory care facilities.** An *automatic sprinkler system* shall be installed throughout the entire floor containing an ambulatory care facility where either of the following conditions exist at any time:

1. Four or more care recipients are incapable of self-preservation.
2. One or more care recipients that are incapable of self-preservation are located at other than the *level of exit discharge* serving such a facility.

In buildings where ambulatory care is provided on levels other than the *level of exit discharge*, an *automatic sprinkler system* shall be installed throughout the entire floor as well as all floors below where such care is provided, and all floors between the level of ambulatory care and the nearest *level of exit discharge*, the *level of exit discharge*, and all floors below the *level of exit discharge*.

**Exception:** Floors classified as an open parking garage are not required to be sprinklered.

**C** Ambulatory care facilities are Group B occupancies, which have an enhanced set of requirements that account for the fact that patients may be incapable of self-preservation and require rescue by other occupants or fire personnel. There are several aspects to the enhanced features, including smoke compartments, sprinklers and fire alarms. More specifically, the requirements for sprinklers are based on the presence of four or more care recipients at any given time that are incapable of self-preservation or any number of care recipients that are incapable of self-preservation located on a floor other than the level of exit discharge that serves the ambulatory care facility. The sprinkler requirement is limited to the floor area that contains the Group B ambulatory care facility and any floors between the ambulatory care facility and level of exit discharge (see commentary, IBC Section 422).

**903.2.2.2 Laboratories involving research and development or testing.** An *automatic sprinkler system* shall be installed throughout the fire areas utilized for the research and development or testing of lithium-ion or lithium metal batteries.

**C** The section requires that all laboratories, regardless of the size, that are used for the research, development or testing of lithium-ion or lithium metal batteries have an automatic sprinkler system installed throughout the fire area. These uses can have an increased risk of thermal runaway, which can be intentional in some cases.

**903.2.3 Group E.** An *automatic sprinkler system* shall be provided for Group E occupancies as follows:

1. Throughout all Group E *fire areas* greater than 12,000 square feet (1115 m<sup>2</sup>) in area.
2. The Group E *fire area* is located on a floor other than a *level of exit discharge* serving such occupancies.

**Exception:** In buildings where every classroom has not fewer than one exterior exit door at ground level, an *automatic sprinkler system* is not required in any area below the lowest *level of exit discharge* serving that area.

3. The Group E *fire area* has an *occupant load* of 300 or more.

**C** Group E occupancies are limited to educational purposes through the 12th grade and day care centers serving children older than 2 1/2 years of age. The 12,000-square-foot (1115 m<sup>2</sup>) fire area threshold for the sprinkler system was established to allow smaller schools and day care centers to be nonsprinklered to minimize the economic impact on these facilities. The 12,000-square-foot (1115 m<sup>2</sup>) threshold is similar to that used for several other occupancies, such as Group M occupancies.

Sprinklers would also be required in portions of the building located below the level of exit discharge serving that occupancy. However, there is an exception that allows the omission of the automatic sprinkler system for the Group E fire area if there is a direct exit to the exterior from each classroom at ground level. The occupants must be able to go from the classroom directly to the outside without passing through intervening corridors, passageways or interior exit stairways.

**903.2.4 Group F-1.** An *automatic sprinkler system* shall be provided throughout all buildings containing a Group F-1 occupancy where one of the following conditions exists:

1. A Group F-1 *fire area* exceeds 12,000 square feet (1115 m<sup>2</sup>).

2. A Group F-1 *fire area* is located more than three stories above *grade plane*.
3. The combined area of all Group F-1 *fire areas* on all floors, including any mezzanines, exceeds 24,000 square feet (2230 m<sup>2</sup>).
4. A Group F-1 occupancy is used to manufacture lithium-ion or lithium metal batteries.
5. A Group F-1 occupancy is used to manufacture vehicles, energy storage systems or equipment containing lithium-ion or lithium metal batteries where the batteries are installed as part of the manufacturing process.

**C** Group F-1 occupancies must meet several different conditions as to when the fire area or occupancy must be sprinklered. The three conditions are related to the difficulty of manually suppressing a fire involving a large area. Therefore, occupancies of Group F-1 must be protected throughout with an automatic sprinkler system if the fire area is in excess of 12,000 square feet (1115 m<sup>2</sup>); if the total of all fire areas is in excess of 24,000 square feet (2230 m<sup>2</sup>); or if the fire area is located more than three stories above grade plane. This is one of the few locations in the code where the total floor area of the building is aggregated for application of a code requirement. The stipulated conditions for when an automatic sprinkler system is required also apply to Group M (see Section 903.2.7) and S-1 (see Section 903.2.9) occupancies.

Condition 4 includes the manufacturing of the batteries, and Condition 5 includes the manufacturing of vehicles, ESS and equipment where the battery is installed as part of the manufacturing process. These specific Group F-1 occupancy uses are required to have an automatic sprinkler system installed throughout the entire building regardless of the size of the area. This is due to the greater potential for a fire event and the increased risk of thermal runaway as a result of the manufacturing process.

The following examples illustrate how the criteria of Conditions 1 through 3 of this section are intended to be applied if Conditions 4 and 5 do not apply:

- If a building contains a single fire area of Group F-1 and the fire area is 13,000 square feet (1208 m<sup>2</sup>), an automatic sprinkler system is required throughout the entire building; however, if this fire area is separated into two fire areas and neither is in excess of 12,000 square feet (1115 m<sup>2</sup>), an automatic fire sprinkler system is not required. To be considered separate fire areas, the areas must be separated by fire barriers or horizontal assemblies having a fire-resistance rating as required in IBC Table 707.3.10.
- If a 30,000-square-foot (2787 m<sup>2</sup>) Group F-1 building is equally divided into separate fire areas of 10,000 square feet (929 m<sup>2</sup>) each, an automatic sprinkler system would still be required throughout the entire building. Because the aggregate area of all fire areas exceeds 24,000 square feet (2230 m<sup>2</sup>), additional compartmentation will not eliminate the need for an automatic sprinkler system. However, the use of a fire wall to separate the structure into two buildings would reduce the aggregate area of each building to less than 24,000 square feet (2230 m<sup>2</sup>) and each fire area to less than 12,000 square feet (1115 m<sup>2</sup>), which would offset the need for an automatic sprinkler system.

**903.2.4.1 Woodworking operations.** An *automatic sprinkler system* shall be provided throughout all Group F-1 occupancy *fire areas* that contain woodworking operations in excess of 2,500 square feet (232 m<sup>2</sup>) in area that generate finely divided combustible waste or use finely divided combustible materials.

**C** Because of the potential amount of combustible dust that could be generated during woodworking operations, an automatic sprinkler system is required throughout a fire area when it contains a woodworking operation that exceeds 2,500 square feet (232 m<sup>2</sup>) in area. Facilities where woodworking operations take place, such as cabinet making, are considered Group F-1 occupancies. The intent of the phrase “finely divided combustible waste” is to describe particle concentrations that are in the explosive range (see Chapter 22 for discussion of dust-producing operations).

The extent of sprinkler coverage is only intended to be for the Group F-1 occupancy involved in the woodworking activity. If the fire area is larger than 2,500 square feet (232 m<sup>2</sup>) but the woodworking area is 2,500 square feet (232 m<sup>2</sup>) or less, sprinklers are not required. It is not the intent to require the installation of sprinklers throughout the building but rather in the fire area where the hazard may be present.

**903.2.4.2 Group F-1 distilled spirits.** An *automatic sprinkler system* shall be provided throughout a Group F-1 *fire area* used for the manufacture of distilled spirits.

**C** Group F-1 occupancies that manufacture distilled spirits, defined in Section 203.5.1 as beverages that contain over 20 percent alcohol content, are required to have an automatic sprinkler system throughout the fire area that contains the occupancy. This includes the aggregate floor area enclosed and bounded by fire walls, fire barriers, exterior walls or horizontal assemblies of the building. Areas of the building that are not provided with surrounding walls shall be included in the fire area if such areas are included within the horizontal projection of the roof or floor above.

**903.2.4.3 Group F-1 upholstered furniture or mattresses.** An *automatic sprinkler system* shall be provided throughout a Group F-1 *fire area* that exceeds 2,500 square feet (232 m<sup>2</sup>) used for the manufacture of upholstered furniture or mattresses.

**C** Group F-1 occupancies that manufacture upholstered furniture or mattresses in a fire area that exceeds 2,500 square feet are required to be provided with an automatic sprinkler system throughout the fire area. Upholstered furniture and mattresses have the potential for rapid-growing and high-heat-release fires and this hazard is increased substantially where there are numerous upholstered furniture items or mattresses being manufactured. Such fires put the occupants and

emergency responders at risk. This requirement exists regardless of whether the upholstered furniture has passed any fire-retardant tests.

For more information on upholstered furniture and mattresses in a Group M occupancy, see the commentary to Section 903.2.7.2 and Section 903.2.9.4 for a Group S-1 occupancy.

**903.2.5 Group H.** *Automatic sprinkler systems* shall be provided in high-hazard occupancies as required in Sections 903.2.5.1 through 903.2.5.3.

**C** Group H occupancies are those intended for the manufacturing, processing or storage of hazardous materials that constitute a physical or health hazard. To be considered a Group H occupancy, the amount of hazardous materials is assumed to be in excess of the maximum allowable quantities permitted by Tables 5003.1.1(1) and 5003.1.1(2).

**903.2.5.1 General.** An *automatic sprinkler system* shall be installed in Group H occupancies.

**C** This section requires an automatic sprinkler system in all Group H occupancies. Even though in some instances the hazard associated with the occupancy may be one that is not a fire hazard, an automatic sprinkler system is still required to minimize the potential for fire spreading to the high-hazard use; that is, the sprinklers protect the high-hazard area from fire outside the area. This section does not prohibit the use of an alternative automatic fire-extinguishing system in accordance with Section 904. When a water-based system is not compatible with the hazardous materials involved and thus creates a dangerous condition, an alternative fire-extinguishing system should be used. For example, combustible metals, such as magnesium and titanium, have a serious record of involvement with fire and are typically not compatible with water (see commentary, Chapter 59).

Where control areas are used to regulate the quantity of hazardous material within a building, the building is not considered a Group H occupancy. Unless a building would be required by some other code provision to be protected with sprinklers, control areas can be used to control the allowable quantities of hazardous materials in a building so as to not warrant a Group H classification and its mandatory sprinkler requirements.

**903.2.5.2 Group H-5 occupancies.** An *automatic sprinkler system* shall be installed throughout buildings containing Group H-5 occupancies. The design of the *automatic sprinkler system* shall be not less than that required under the *International Building Code* for the occupancy hazard classifications in accordance with Table 903.2.5.2.

Where the design area of the *automatic sprinkler system* consists of a *corridor* protected by one row of sprinklers, the maximum number of sprinklers required to be calculated is 13.

**C** Group H-5 occupancies are structures that are typically used as semiconductor fabrication facilities and comparable research laboratory facilities that use hazardous production materials (HPM). Many of the materials used in semiconductor fabrication present unique hazards. Several of the materials are toxic, while some are corrosive, water reactive or pyrophoric. Fire protection for these facilities is aimed at preventing incidents from escalating and producing secondary threats beyond a fire, such as the release of corrosive or toxic materials. Because of the nature of Group H-5 facilities, the overall amount of hazardous materials can far exceed the maximum allowable quantities given in Tables 5003.1.1(1) and 5003.1.1(2). Although the amount of HPM is restricted in fabrication areas, the quantities of HPM in storage rooms normally will be in excess of those allowed by the tables. Additional requirements for Group H-5 facilities are located in Chapter 27 and Section 415.11 of the IBC.

This section also specifies the automatic sprinkler design criteria, based on NFPA 13, for various areas in a Group H-5 occupancy (see commentary, Table 903.2.5.2). When the corridor design area sprinkler option is used, a maximum of 13 sprinklers must be calculated. This exceeds the requirements of NFPA 13 for typical egress corridors, which requires a maximum of either five or seven calculated sprinklers, depending on the extent of protected openings in the corridor. The increased number of calculated corridor sprinklers is based on the additional hazard associated with the movement of hazardous materials in corridors of Group H-5 facilities.

**TABLE 903.2.5.2—GROUP H-5 AUTOMATIC SPRINKLER SYSTEM DESIGN CRITERIA**

| LOCATION                         | OCCUPANCY HAZARD CLASSIFICATION |
|----------------------------------|---------------------------------|
| Fabrication areas                | Ordinary Hazard Group 2         |
| Service corridors                | Ordinary Hazard Group 2         |
| Storage rooms without dispensing | Ordinary Hazard Group 2         |
| Storage rooms with dispensing    | Extra Hazard Group 2            |
| Corridors                        | Ordinary Hazard Group 2         |

**C** Table 903.2.5.2 designates the appropriate occupancy hazard classification for the various areas within a Group H-5 facility. The listed occupancy hazard classifications correspond to specific automatic sprinkler system design criteria in NFPA 13.

Ordinary Hazard Group 2 occupancies, for example, require a minimum design density of 0.20 gpm/ft<sup>2</sup> (8.1 L/min/m<sup>2</sup>) with a minimum design area of 1,500 square feet (139 m<sup>2</sup>). An Extra Hazard Group 2 occupancy, in turn, requires a minimum design density of 0.40 gpm/ft<sup>2</sup> (16.3 L/min/m<sup>2</sup>) with a minimum operating area of 2,500 square feet (232 m<sup>2</sup>). The increased overall automatic sprinkler demand for Extra Hazard Group 2 occupancies is based on the potential use and handling of substantial amounts of hazardous materials, such as flammable or combustible liquids.

**903.2.5.3 Pyroxylin plastics.** An *automatic sprinkler system* shall be provided in buildings, or portions thereof, where cellulose nitrate film or pyroxylin plastics are manufactured, stored or handled in quantities exceeding 100 pounds (45 kg).

**C** Cellulose nitrate (pyroxylin) plastics pose unusual and substantial fire risks. Pyroxylin plastics are the most dangerous and unstable of all plastic compounds. The chemically bound oxygen in their structure causes them to burn vigorously in the absence of atmospheric oxygen. Although these compounds produce approximately the same amount of energy as paper when they burn, pyroxylin plastics burn at a rate as much as 15 times greater than comparable common combustibles. When burning, these materials release highly flammable and toxic combustion byproducts. Consequently, cellulose nitrate fires are very difficult to control. Although this section specifies a sprinkler threshold quantity of 100 pounds, the need for additional fire protection should be considered for pyroxylin plastics in any amount.

Although the code includes cellulose nitrate “film” in its requirements, cellulose nitrate motion picture film has not been used in the United States since the 1950s. All motion picture film produced since that time is what is typically called “safety film.” Consequently, the only application for this section relative to motion picture film is where it may be used in laboratories or storage vaults that are dedicated to film restoration and archives. The protection of these facilities is addressed in Sections 306.2 and 6504.2.

**903.2.6 Group I.** An *automatic sprinkler system* shall be provided throughout buildings with a Group I *fire area*.

**Exceptions:**

1. An *automatic sprinkler system* installed in accordance with Section 903.3.1.2 shall be permitted in Group I-1, Condition 1 facilities.
2. An *automatic sprinkler system* is not required where Group I-4 day care facilities are at the *level of exit discharge* and where every room where care is provided has not fewer than one exterior *exit door*.
3. In buildings where Group I-4 day care is provided on levels other than the *level of exit discharge*, an *automatic sprinkler system* in accordance with Section 903.3.1.1 shall be installed on the entire floor where care is provided, all floors between the level of care and the *level of exit discharge* and all floors below the *level of exit discharge* other than areas classified as an open parking garage.

**C** The Group I occupancy is divided into four individual occupancy classifications based on the degree of detention, supervision and physical mobility of the occupants. The evacuation difficulties associated with the building occupants creates the need to incorporate a defend-in-place philosophy of fire protection in occupancies of Group I. For this reason, all such occupancies are to be protected with an automatic sprinkler system. Note that this section is applicable to the entire building that contains a Group I occupancy.

Of particular note, this section encompasses all Group I-3 occupancies where more than five persons are detained (see IBC Section 308.5). There has been considerable controversy concerning the use of automatic sprinklers in detention and correctional occupancies. Special design considerations can be taken into account to alleviate the perceived problems with sprinklers in sleeping units. Sprinklers that reduce the likelihood of vandalism as well as the potential to hang oneself are commercially available. Knowledgeable designers can incorporate certain design features to increase reliability and decrease the likelihood of damage to the system.

Group I-4 occupancies would include either adult-only care facilities or occupancies that provide personal care for more than five children 2 1/2 years of age or less on a less-than-24-hour basis. Because the degree of assistance and the time needed for egress cannot be gauged, an automatic sprinkler system is required.

There are three exceptions to this section. Exception 1 permits Group I-1 Condition 1 occupancies to be protected throughout with an NFPA 13R system instead of an NFPA 13 system. This is the lower risk condition for Group I-1 occupancies. Group I-1 Condition 2 occupancies would be required to use an NFPA 13 system.

Exception 2 exempts sprinkler systems completely if the day care center is at the level of exit discharge and every room has at least one exterior exit door. Note that day cares to which this section applies are considered by IBC Section 308.5.1 to be Group E occupancies. An automatic sprinkler system would not be required unless dictated by the requirements in Section 903.2.2 (see commentary, IBC Section 308.5.1).

Exception 3 is also related to day cares that are still classified as Group I-4 by nature of the location in the building. In that case, an NFPA 13 system would be required on the floor where the center is located and all floors between and including the level of exit discharge. This is less stringent than the main requirement in Section 903.2.6 that requires the entire building to be sprinklered. As defined in Section 202, a Group I-4 child care facility located at the level of exit discharge and accommodating no more than 100 children, with each child care room having an exit directly to the exterior, would be classified as a Group E occupancy.

**903.2.7 Group M.** An *automatic sprinkler system* shall be provided throughout buildings containing a Group M occupancy where one of the following conditions exists:

1. A Group M *fire area* exceeds 12,000 square feet (1115 m<sup>2</sup>).
2. A Group M *fire area* is located more than three stories above *grade plane*.
3. The combined area of all Group M *fire areas* on all floors, including any mezzanines, exceeds 24,000 square feet (2230 m<sup>2</sup>).

**C** The sprinkler threshold requirements for Group M occupancies are identical to those of Group F-1 and S-1 occupancies (see commentary, Section 903.2.4). Automatic sprinkler systems for mercantile occupancies are typically designed for an Ordinary Hazard Group 2 classification in accordance with NFPA 13. If high-piled storage (see Section 903.2.7.1) is anticipated, additional levels of fire protection may be required. Also, some merchandise in mercantile occupancies, such as aerosols, rubber tires, paints and certain plastic commodities, even at limited storage heights, are considered beyond the standard Class I through IV commodity classification assumed for mercantile occupancies in NFPA 13 and may warrant additional fire protection.

**903.2.7.1 High-piled storage.** An *automatic sprinkler system* shall be provided as required in Chapter 32 in all buildings of Group M where storage of merchandise is in high-piled or rack storage arrays.

**C** Regardless of the size of the Group M fire area, an automatic sprinkler system may be required in a high-piled storage area. High-piled storage includes piled, palletized, bin box, shelf or rack storage of Class I through IV commodities to a height greater than 12 feet (3658 mm) and certain high-hazard commodities greater than 6 feet (1829 mm). Chapter 23 provides a package of requirements that may include sprinkler protection, depending on the size of the high-piled storage area. The design standard for the sprinkler protection of high-piled storage is NFPA 13, which addresses the many different types and configurations of high-piled storage.

**903.2.7.2 Group M upholstered furniture or mattresses.** An *automatic sprinkler system* shall be provided throughout a Group M *fire area* where the area used for the display and sale of upholstered furniture or mattresses exceeds 5,000 square feet (464 m<sup>2</sup>).

**C** Group M occupancies that display and sell upholstered furniture or mattresses in a fire area that exceeds 5,000 square feet are required to be provided with an automatic sprinkler system throughout the fire area. These occupancies are provided with an increased area for display of upholstered furniture and mattresses of 5,000 square feet (464 m<sup>2</sup>) versus 2,500 square feet (232 m<sup>2</sup>) required for Group F-1 and S-1 occupancies. As noted in the commentary for Group F-1 occupancies, upholstered furniture and mattresses have the potential for rapidly growing and high-heat-release fires. This hazard is increased substantially where there are numerous upholstered furniture items or mattresses on display. Such fires put the occupants and emergency responders at risk. This requirement exists regardless of whether the upholstered furniture has passed any fire-retardant tests.

The code does not specifically address what constitutes upholstered furniture, but by simple dictionary definition, upholstered furniture has seats covered with padding, springs, webbing and fabric or leather covers. The code does not make any distinction between levels of padding and upholstery provided on furniture, which was intentional. The proponent's reason statement for code change F135-07/08 stated, in part, "the American Home Furnishings Alliance (AHFA) and the National Home Furnishings Association (NHFA) have examined proposals for exempting vendors of certain constructions of furniture and concluded that such exemptions would be impractical for local code officials to enforce. This is the case because the internal construction of furniture cannot be established reliably without deconstructing it."

**903.2.7.3 Lithium-ion or lithium metal battery storage.** An *automatic sprinkler system* shall be provided in a room or space within a Group M occupancy where required for the storage of lithium-ion or lithium metal batteries by Section 320 or Chapter 32.

**C** This section provides the specific references for Group M occupancies that are used for the storage of lithium-ion or lithium metal batteries. These occupancies must have an automatic sprinkler system installed throughout the room or space where required.

**903.2.8 Group R.** An *automatic sprinkler system* installed in accordance with Section 903.3 shall be provided throughout all buildings with a Group R *fire area*.

**C** This section requires sprinklers in any building that contains a Group R fire area. This includes uses such as hotels, apartment buildings, group homes and dormitories. There are no minimum criteria and no exceptions.

It should be noted that buildings constructed under the *International Residential Code*® (IRC®) are not included in Group R and would not, therefore, be subject to these particular requirements. The 2009 IRC requires sprinklers in all new townhouses and in all new one- and two-family dwellings. The IRC is a stand-alone code for the construction of detached one- and two-family dwellings and multiple single-family dwellings (townhouses) not more than three stories in height with a separate means of egress and addresses the requirements for sprinklers in a different way. That is, all of the provisions for new construction that affect those buildings are to be covered exclusively by the IRC and are not to be covered by another I-Code. Buildings that do not fall within the scope of the IRC would be classified in Group R and be subject to these provisions. This is stated clearly in IFC Committee Interpretation No. 29-03.

With respect to life safety, the need for a sprinkler system is dependent on the occupants' proximity to the fire and the ability to respond to a fire emergency. Group R occupancies could contain occupants who may require assistance to evacuate, such as infants, those with a disability or who may simply be asleep. While the presence of a sprinkler system cannot always protect occupants in residential buildings who are aware of the ignition and either do not respond or respond inappropriately, it can prevent fatalities outside of the area of fire origin regardless of the occupants' response. Section 903.3.2 requires quick-response or residential sprinklers in all Group R occupancies. Full-scale fire tests have demonstrated the ability of quick-response and residential sprinklers to maintain tenability from flaming fires in the room of fire origin.

Where a different occupancy is located in a building with a residential occupancy, the provisions of this section still apply and the entire building is required to be provided with an automatic sprinkler system regardless of the type of mixed-use condition considered. This is consistent with the mixed-use provisions in IBC Chapter 5. The type of sprinkler system permitted in the different types of Group R occupancies is further clarified in Sections 903.2.8.1 through 903.2.8.3.

**903.2.8.1 Group R-3.** An *automatic sprinkler system* installed in accordance with Section 903.3.1.3 shall be permitted in Group R-3 occupancies.

**C** Group R-3 occupancies are essentially one- and two-family dwellings that fall outside the scope of the IRC; thus, an NFPA 13D system is appropriate. It should be noted there is no restriction on the use of NFPA 13 or NFPA 13R systems.

**903.2.8.2 Group R-4, Condition 1.** An *automatic sprinkler system* installed in accordance with Section 903.3.1.3 shall be permitted in Group R-4, Condition 1 occupancies.

**C** Group R-4, Condition 1 is the lesser of the risk categories for Group R-4 occupancies. The occupants are more capable of evacuating without assistance. Therefore, they are treated no differently than a Group R-3 occupancy.

**903.2.8.3 Care facilities.** An *automatic sprinkler system* installed in accordance with Section 903.3.1.3 shall be permitted in care facilities with five or fewer individuals in a single-family *dwelling*.

**C** This section is similar to Sections 903.2.8.1 and 903.2.8.2 and allows the use of an NFPA 13D system in place of an NFPA 13 or 13R system. In this case, it is specific to smaller care facilities with five or fewer residents. Again, while not technically a single-family dwelling, they are very similar in nature based on the type and actual use of the building.

**903.2.9 Group S-1.** An *automatic sprinkler system* shall be provided throughout all buildings containing a Group S-1 occupancy where one of the following conditions exists:

1. A Group S-1 *fire area* exceeds 12,000 square feet (1115 m<sup>2</sup>).
2. A Group S-1 *fire area* is located more than three stories above *grade plane*.
3. The combined area of all Group S-1 *fire areas* on all floors, including any mezzanines, exceeds 24,000 square feet (2230 m<sup>2</sup>).
4. A Group S-1 *fire area* used for the storage of commercial motor vehicles where the *fire area* exceeds 5,000 square feet (464 m<sup>2</sup>).
5. A Group S-1 *fire area* used for the storage of lithium-ion or lithium metal powered vehicles where the *fire area* exceeds 500 square feet (46.4 m<sup>2</sup>).

**C** An automatic sprinkler system must be provided throughout all buildings containing a Group S-1 occupancy fire area where the fire area exceeds 12,000 square feet (1115 m<sup>2</sup>); is more than three stories above grade plane; combined, on all floors including mezzanines, exceeds 24,000 square feet (2230 m<sup>2</sup>); or is used for the storage of commercial motor vehicles and exceeds 5,000 square feet (464 m<sup>2</sup>). See the commentary to "Commercial motor vehicle" in Chapter 2.

The first three sprinkler threshold requirements for Group S-1 occupancies are identical to those of Groups F-1 and M (see commentary, Sections 903.2.4 and 903.2.7). Group S-1 occupancies, such as warehouses and self-storage buildings, are assumed to be used for the storage of combustible materials. While high-piled storage does not change the Group S-1 occupancy classification, automatic sprinkler protection, if required, may have to comply with the additional requirements of Chapter 32. High-piled stock or rack storage in any occupancy must comply with the code and the IBC.

Condition 5 includes the storage of lithium-ion or lithium metal battery powered vehicles due to the increased risk. The 500-square-foot fire area correlates with the threshold at which Chapter 32 would require suppression for just lithium-ion battery storage.

**903.2.9.1 Repair garages.** An *automatic sprinkler system* shall be provided throughout all buildings used as repair garages in accordance with Section 406.8 of the *International Building Code*, as shown:

1. Buildings having two or more stories above *grade plane*, including *basements*, with a *fire area* containing a repair garage exceeding 10,000 square feet (929 m<sup>2</sup>).
2. Buildings not more than one story above *grade plane*, with a *fire area* containing a repair garage exceeding 12,000 square feet (1115 m<sup>2</sup>).
3. Buildings with repair garages servicing vehicles parked in *basements*.
4. A Group S-1 *fire area* used for the repair of commercial motor vehicles where the *fire area* exceeds 5,000 square feet (464 m<sup>2</sup>).

5. A Group S-1 *fire area* used for the storage of lithium-ion or lithium metal powered vehicles where the *fire area* exceeds 500 square feet (46.4 m<sup>2</sup>).

**C** Automatic sprinklers may be required in repair garages, depending on the quantity of combustibles present, their location and floor area. In addition, any Group S-1 fire area intended for the repair of commercial motor vehicles that exceeds 5,000 square feet (464 m<sup>2</sup>) would require sprinklers. This is the same criteria as Group S-1 occupancies and Group S-2 enclosed parking garages storing commercial motor vehicles. Repair garages may contain significant quantities of flammable liquids and other combustible materials. These occupancies are typically considered Ordinary Hazard Group 2 occupancies as defined in NFPA 13. Portions of repair garages used for parts cleaning using flammable or combustible liquids may require automatic sprinkler protection. If quantities of hazardous materials exceed the limitations in Section 307 for maximum allowable quantities per control area, the repair garage would be reclassified as a Group H occupancy. Note that the term “Commercial motor vehicles” is specifically defined in Chapter 2.

Item 5 includes garages that are used for the repair of lithium-ion or lithium metal battery powered vehicles due to the increased risk. The 500-square-foot fire area correlates with the threshold at which Chapter 32 would require suppression for just lithium-ion battery storage.

**903.2.9.2 Bulk storage of tires.** Buildings and structures where the area for the storage of tires exceeds 20,000 cubic feet (566 m<sup>3</sup>) shall be equipped throughout with an *automatic sprinkler system* in accordance with Section 903.3.1.1.

**C** This section specifies when an automatic sprinkler system is required for the bulk storage of tires based on the volume of the storage area as opposed to a specific number of tires. Even in fully sprinklered buildings, tire fires pose significant problems to fire departments. Tire fires produce thick smoke and are difficult to extinguish by sprinklers alone. NFPA 13 contains specific fire protection requirements for the storage of rubber tires.

Whether the volume of tires is divided into different fire areas or not is irrelevant to the application of this section. If the total for all areas where tires are stored is great enough that the resultant storage volume exceeds 20,000 cubic feet (566 m<sup>3</sup>), the building must be sprinklered throughout. See the commentary to “Tires, bulk storage of” in Section 202 for further information.

**903.2.9.3 Group S-1 distilled spirits or wine.** An *automatic sprinkler system* shall be provided throughout a Group S-1 *fire area* used for the bulk storage of distilled spirits or wine.

**C** Group S-1 occupancies that are used for the bulk storage of distilled spirits, defined in Section 203.10.3 as beverages that contain over 20 percent alcohol content, or wine are required to have an automatic sprinkler system throughout the fire area that contains the occupancy. This includes the aggregate floor area enclosed and bounded by fire walls, fire barriers, exterior walls or horizontal assemblies of the building. Areas of the building that are not provided with surrounding walls shall be included in the fire area if such areas are included within the horizontal projection of the roof or floor above.

**903.2.9.4 Group S-1 upholstered furniture and mattresses.** An *automatic sprinkler system* shall be provided throughout a Group S-1 *fire area* where the area used for the storage of upholstered furniture or mattresses exceeds 2,500 square feet (232 m<sup>2</sup>).

**Exception:** Self-service storage facilities not greater than one story above *grade plane* where all storage spaces can be accessed directly from the exterior.

**C** The area threshold to provide an automatic sprinkler system is the same as for Group F-1 except that, in this case, upholstered furniture and mattresses are being stored and not manufactured. Group M has a similar threshold but is required for larger occupancies containing such items with an area of 5,000 square feet (464 m<sup>2</sup>) versus what is required for Group S-1 and F-1 occupancies of 2,500 square feet (232 m<sup>2</sup>). See the commentary to Section 903.2.7.2 for Group M and Section 903.2.4.3 for Group F-1. Again, it is important to note that the threshold is based on the square footage of the fire area.

This section specifically includes self-storage warehouses since it is highly likely that upholstered furniture and mattresses will be stored in such units. The automatic sprinkler system is required to be provided based on the square footage of the fire area and not on the amount of upholstered furniture or mattresses present.

The exception acknowledges that single-story self-storage buildings, regardless of size, without interior corridors and in which all the individual spaces have direct access to the outside, do not have to be provided with an automatic sprinkler system. This is based on the reduced life safety risk in these facilities and the direct and independent means of egress that each storage space has.

**903.2.10 Group S-2 parking garages.** An *automatic sprinkler system* shall be provided throughout buildings classified as parking garages where any of the following conditions exist:

1. Where the *fire area* of the enclosed parking garage, in accordance with Section 406.6 of the *International Building Code*, exceeds 12,000 square feet (1115 m<sup>2</sup>).
2. Where the enclosed parking garage, in accordance with Section 406.6 of the *International Building Code*, is located beneath other groups.

**Exception:** Enclosed parking garages located beneath Group R-3 occupancies.

3. Where the *fire area* of the open parking garage, in accordance with Section 406.5 of the *International Building Code*, exceeds 48,000 square feet (4460 m<sup>2</sup>).

**C** This section provides requirements for both enclosed and open parking garages. An enclosed parking garage does not allow the dissipation of smoke and hot gases as readily as an open parking structure, which is also considered a Group S-2 occupancy. For this reason, the fire area threshold for an enclosed parking garage is much lower than for an open parking garage. If the enclosed parking garage has a fire area greater than 12,000 square feet (1115 m<sup>2</sup>) or is located beneath another occupancy group, the enclosed parking garage must be protected with an automatic sprinkler system. This requirement that the enclosed parking garage located beneath other occupancy groups is required to be sprinklered is based on the potential for a fire to develop undetected, which would endanger the occupants of the other occupancy. The 12,000-square-foot (1115 m<sup>2</sup>) threshold is similar to other occupancies such as Groups M and S-1.

The exception exempts enclosed garages in buildings where the garages are located below a Group R-3 occupancy. The exception is essentially moot since the code requires all buildings with a Group R occupancy to be sprinklered throughout. Because the entire building with the residential occupancy is required to be sprinklered according to Section 903.2.8, the garage would be sprinklered as well. It should be noted that if the Group R-3 occupancy was protected with an NFPA 13D system, the enclosed parking garage would not require sprinklers.

Historically, open parking garages have been considered to have a very low fire risk, which led to dozens of special allowances for reduced code requirements in these occupancies. Fire tests run decades ago offered some support for this perspective. However, much of what makes up a vehicle today is combustible, and bodies and interior components that may previously have been of steel are now primarily plastic, rubber, fiberglass and lightweight metals. This has increased the potential for vehicle-to-vehicle fire spread and the production of dense combustible smoke layers. Stored energy systems in electric vehicles have also increased the fuel load.

An open parking garage fire in Liverpool, UK, on January 1, 2018 demonstrated the fire risk associated with the new type of vehicle construction. The concrete building and 1,400 cars were destroyed by a fire that reportedly started in a single vehicle's engine compartment. The approach of allowing smoke to escape from an open parking garage perimeter to entirely mitigate the fire risk was shown to be incorrect by this incident, and this section requirement recognizes that these structures and their contents can present significant challenges to the fire service and result in catastrophic fire losses. It provides an area-based threshold, which is much higher compared to other occupancies that might be argued as having similar, or even lesser, fire loads. The value is four times as large as what is applicable to enclosed garages, recognizing that, while there may be some benefit to perimeter openings, the fire service will ultimately be relied on to control these fires if automatic sprinklers are not provided.

**903.2.10.1 Commercial parking garages.** An *automatic sprinkler system* shall be provided throughout buildings used for storage of commercial motor vehicles where the *fire area* exceeds 5,000 square feet (464 m<sup>2</sup>).

**C** Because of the larger-sized vehicles involved in commercial parking structures, such as those housing commercial motor vehicles as defined in Section 202, a more stringent sprinkler threshold is required. Bus garages may also be located adjacent to passenger terminals (Group A-3) that have a substantial occupant load. Commercial parking requires only a single vehicle in order to be classified as commercial parking.

The criterion for sprinkler protection is based on the size of the fire area and not the size of the commercial parking. If the commercial parking involves only 1,000 square feet (93 m<sup>2</sup>) but the fire area exceeds 5,000 square feet (464 m<sup>2</sup>), sprinkler protection is required.

**903.2.10.2 Mechanical-access enclosed parking garages.** An *approved automatic sprinkler system* shall be provided throughout buildings used for the storage of motor vehicles in a mechanical-access enclosed parking garage. The portion of the building that contains the mechanical-access enclosed parking garage shall be protected with a specially engineered *automatic sprinkler system*.

**C** These occupancies are different from the traditional open mechanical-access parking garage in that there are no openings and the entire structure is enclosed. They are more similar to automated high-rack storage systems with no floors, no stairwells and no above-ground level access, with the exception of maintenance walkways and ladders.

An enclosed mechanical-access parking garage has many firefighting challenges; most are constructed in a building shell, without a floor system. The vehicles are parked in a cage/rack system, with no safe elevated access to the interior of the structure. With firefighter safety in mind and to have the ability to use fixed fire suppression to extinguish or control these fires, this section prescriptively requires a performance-based designed sprinkler system specifically within the portion of the building that contains the mechanical-access parking garage and an approved automatic sprinkler system in the other areas of the building.

**903.2.11 Specific buildings areas and hazards.** In all occupancies other than Group U, an *automatic sprinkler system* shall be installed for building design or hazards in the locations set forth in Sections 903.2.11.1 through 903.2.11.6.

**C** Sections 903.2.11.1 through 903.2.11.6 specify certain conditions under which an automatic sprinkler system is required, even in otherwise nonsprinklered buildings. As indicated, the listed conditions in the noted sections are applicable to all occupancies except Group U. Most structures that qualify as Group U do not typically have the type of conditions stipulated in Sections 903.2.11.1 through 903.2.11.1.6.

**903.2.11.1 Stories without openings.** An *automatic sprinkler system* shall be installed throughout all *stories*, including *basements*, of all buildings where the floor area exceeds 1,500 square feet (139 m<sup>2</sup>) and where the story does not comply with the following criteria for *exterior wall* openings:

1. Openings below grade that lead directly to ground level by an exterior *stairway* complying with Section 1011 or an outside *ramp* complying with Section 1012. Openings shall be located in each 50 linear feet (15 240 mm), or fraction thereof, of *exterior wall* in the story on not fewer than one side. The required openings shall be distributed such that the lineal distance between adjacent openings does not exceed 50 feet (15 240 mm).
2. Openings entirely above the adjoining ground level totaling not less than 20 square feet (1.86 m<sup>2</sup>) in each 50 linear feet (15 240 mm), or fraction thereof, of *exterior wall* in the story on not fewer than one side. The required openings shall be distributed such that the lineal distance between adjacent openings does not exceed 50 feet (15 240 mm). The height of the bottom of the clear opening shall not exceed 44 inches (1118 mm) measured from the floor.

**C** Because of both the lack of openings in exterior walls for access by the fire department for firefighting and rescue and the problems associated with venting the products of combustion during fire suppression operations, all stories, including any basements of buildings that do not have adequate openings as defined in this section, must be equipped with an automatic sprinkler system. This section applies to stories without sufficient exterior openings where the floor area exceeds 1,500 square feet (139 m<sup>2</sup>) and where the building is not otherwise required to be fully sprinklered. The requirement for an automatic sprinkler system in this section applies only to the affected area and does not mandate sprinkler protection throughout the entire building.

Stories without openings, as defined in this section, are stories that do not have at least 20 square feet (1.9 m<sup>2</sup>) of opening leading directly to ground level in each 50 linear feet (15 240 mm), or fraction thereof, on at least one side. Since exterior doors will provide openings of 20 square feet (1.9 m<sup>2</sup>), or slightly less in some occupancies, exterior stairways and ramps in each 50 linear feet (15 240 mm) are considered acceptable.

This section specifically states that the required openings be distributed such that the lineal distance between adjacent openings does not exceed 50 feet (15 240 mm). If the openings in the exterior wall are located without regard to the location of the adjacent openings, it is possible that segments of the exterior wall will not have the required access to the interior of the building for firefighting purposes. Any arrangement of required stairways, ramps or openings that results in a portion of the wall 50 feet (15 240 mm) or more in length with no openings to the exterior does not meet the intent of the code that access be provided in each 50 linear feet (15 240 mm) (see Commentary Figure 903.2.11.1).

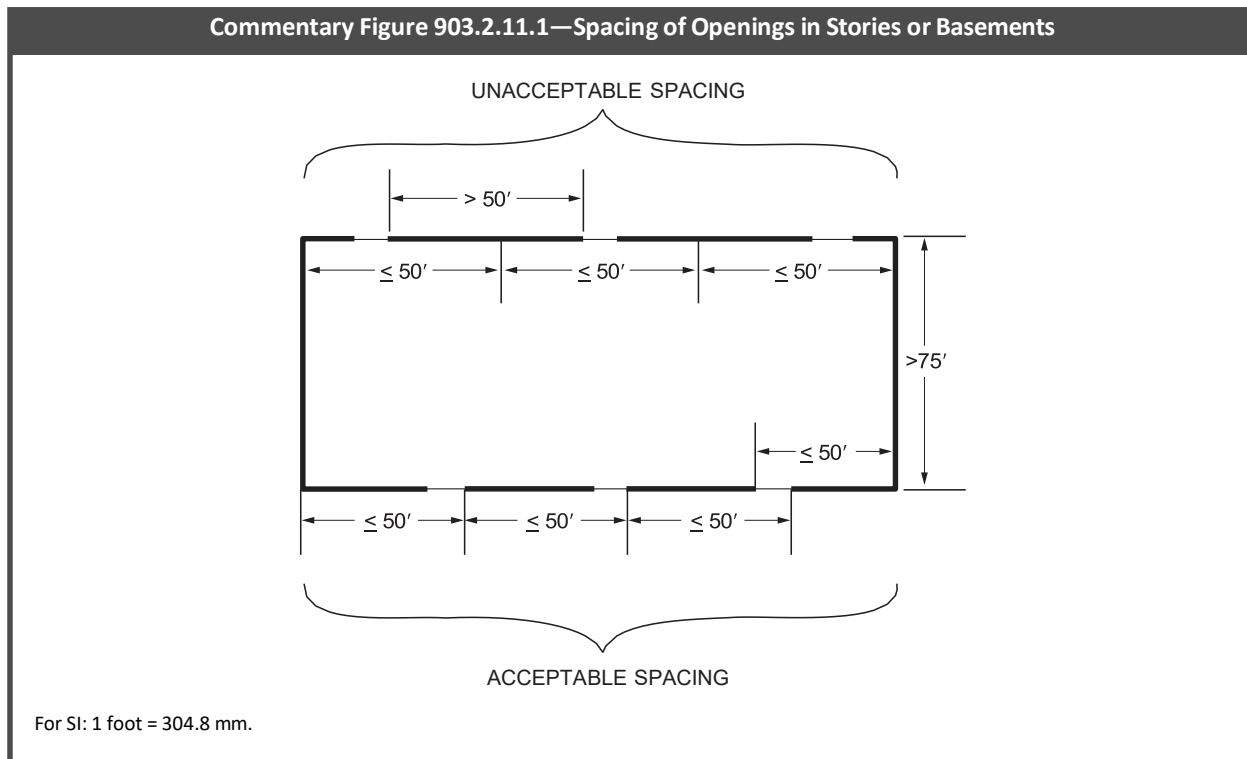
There is a further restriction on openings that are entirely above grade. More specifically, to support firefighting operations, the openings need to be accessible and usable. Therefore, Item 2 specifies that the maximum sill height be 44 inches (1118 mm) above the floor. This height is consistent with the height provided for emergency escape and rescue openings in Section 1031.3.3.

One application of this section has been addressed and is in the 2015 edition of *International Code Interpretations*. It addresses automotive service shops that have below-grade service areas where employees perform oil changes and other minor maintenance services. The below-grade areas are typically open to the grade-level service bays via openings providing access to the underside of the vehicles without requiring the vehicle to be lifted into the air. Inasmuch as the below-grade space has no openings directly to the exterior, the question was asked if it would be regulated as a windowless story and thus be required to be equipped with an automatic fire suppression system in accordance with Section 903.2.11.1.

The answer to that question is no. Because of the openness between the adjacent service levels, the below-grade area would be more appropriately regulated similar to a mezzanine rather than a story. A mezzanine is not regulated as a separate story but rather as part of the same story that it serves. Therefore, if the below-grade service level is in compliance with the applicable provisions of IBC Section 505, the windowless story provisions of Section 903.2.11.1 would be evaluated based on the exterior wall openings of the main level and not the service mezzanine below. The direct interconnections between the two adjacent floor levels by multiple service openings provide access to the lower service area for firefighting and rescue operations. As such, it would not be regulated as a windowless story.

The requirement to sprinkler the basement is independent of mixed-use conditions. Whether the basement is separated or nonseparated is irrelevant to the need for sprinkler protection, nor does the requirement to provide sprinklers in the basement imply that sprinklers must be provided elsewhere. This requirement is applicable to the basement or any story without openings, irrespective of other code provisions.

Also, these provisions are not based on the size of a fire area but rather on the size of the basement. Thus, subdividing the basement into multiple fire areas would have no effect on the requirement. However, one benefit of the multiple fire areas could be that each fire area could have a separate limited area sprinkler system with less than 20 sprinklers.



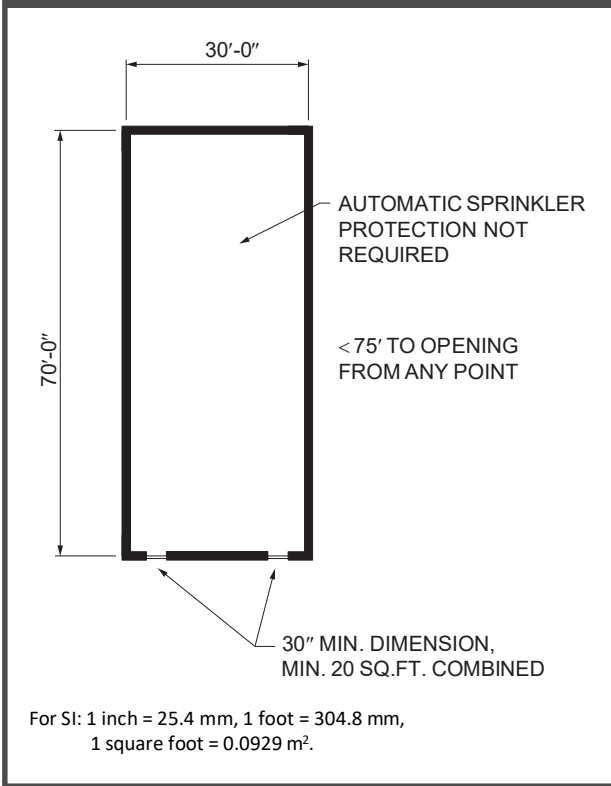
**903.2.11.1.1 Opening dimensions and access.** Openings shall have a minimum dimension of not less than 30 inches (762 mm). Access to such openings shall be provided for the fire department from the exterior and shall not be obstructed in a manner such that firefighting or rescue cannot be accomplished from the exterior.

**C** To qualify, an opening must not be less than 30 inches (762 mm) in least dimension and must be accessible to the fire department from the exterior. The minimum opening dimension gives fire department personnel access to the interior of the story or basement for firefighting and rescue operations and provides openings that are large enough to vent the products of combustion.

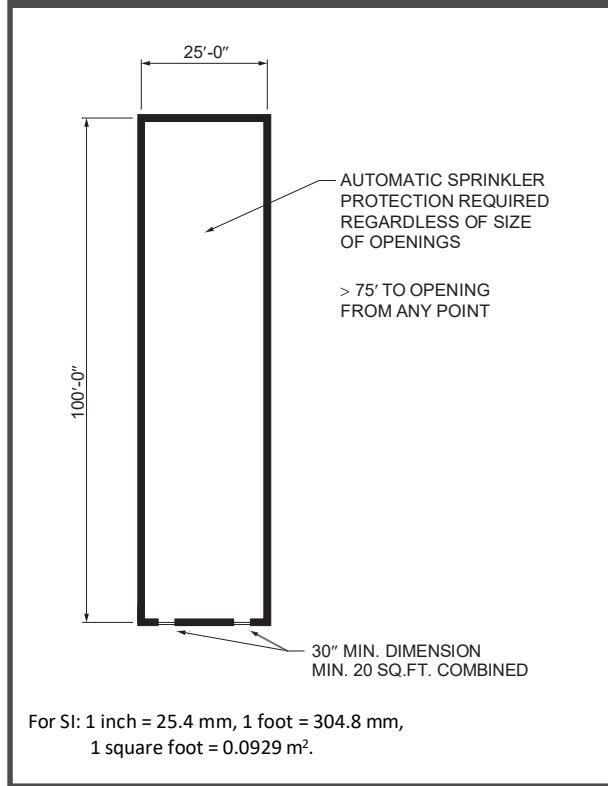
**903.2.11.1.2 Openings on one side only.** Where openings in a story are provided on only one side and the opposite wall of such story is more than 75 feet (22 860 mm) from such openings, the story shall be equipped throughout with an *approved automatic sprinkler system*, or openings shall be provided on not fewer than two sides of the story.

**C** If openings are provided on only one side, an automatic sprinkler system would still be required if the opposite wall of the story is more than 75 feet (22 860 mm) from existing openings. An alternative to providing the automatic sprinkler system would be to design openings on at least two sides of the exterior of the building. As long as the story being considered is not a basement, the openings on two sides can be greater than 75 feet (22 860 mm) from any portion of the floor. In basements, if any portion is more than 75 feet (22 860 mm) from the openings, the entire basement must be equipped with an automatic sprinkler system, as indicated in Section 903.2.11.1.3. Providing openings on more than one wall allows cross ventilation to vent the products of combustion [see Commentary Figures 903.2.11.1.2(1–4)].

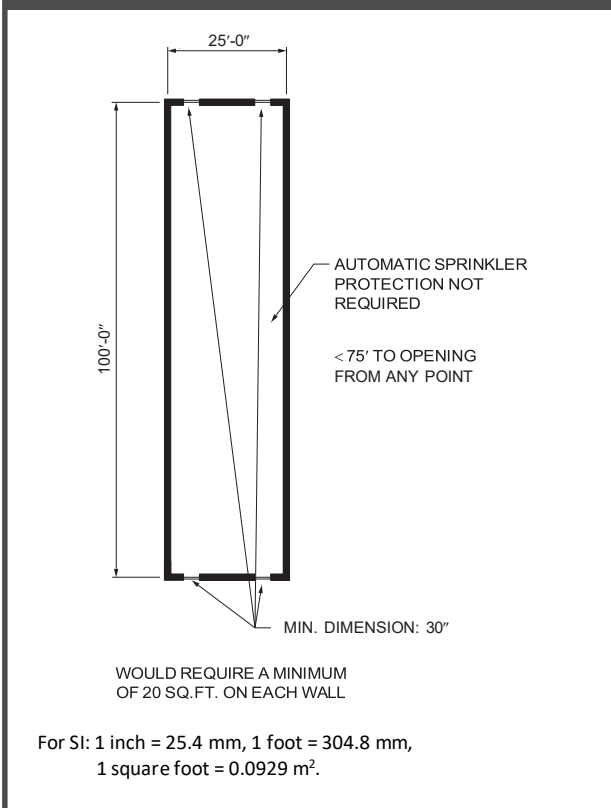
**Commentary Figure 903.2.11.1.2(1)—Openings in Stories or Basements Less than 75 Feet from Any Point to an Opening**



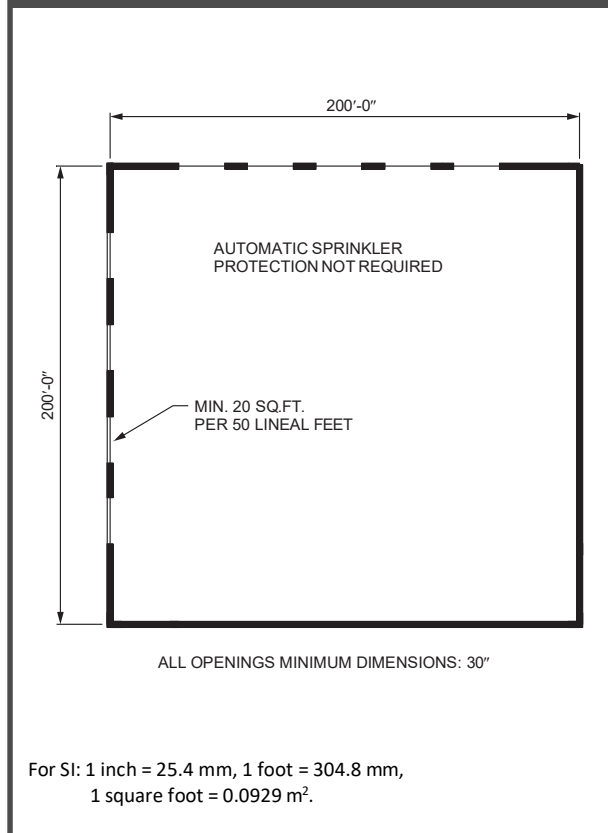
**Commentary Figure 903.2.11.1.2(2)—Openings in Stories or Basements Greater than 75 Feet from Any Point to an Opening**



**Commentary Figure 903.2.11.1.2(3)—Openings in Stories or Basements Less than 75 Feet from Any Point to an Opening**



**Commentary Figure 903.2.11.1.2(4)—Openings in Stories or Basements**



**903.2.11.1.3 Basements.** Where any portion of a *basement* is located more than 75 feet (22 860 mm) from openings required by Section 903.2.11.1, or where walls, partitions or other obstructions are installed that restrict the application of water from hose streams, the *basement* shall be equipped throughout with an *approved automatic sprinkler system*.

**C** The 75-foot (22 860 mm) distance is intended to be measured in the line of travel—not in a straight line perpendicular to the wall. Where obstructions, such as walls or other partitions, are present in a basement, the walls and partitions enclosing any room or space must have openings that provide an equivalent degree of fire department access to that provided by the openings prescribed in Section 903.2.11.1 for exterior walls. When obstructions such as walls or partitions are installed in the basement, the ability to apply hose streams through these openings and reach the basement area is reduced or eliminated. The configuration and clear-opening requirements become useless when an interior wall or other obstruction is placed inside the basement. In that case, it is reasonable to require automatic fire sprinklers to provide adequate protection in the basement. If an equivalent degree of fire department access to all portions of the floor area is not provided, the basement would require an automatic sprinkler system.

**903.2.11.2 Rubbish and linen chutes.** An *automatic sprinkler system* shall be installed at the top of rubbish and linen chutes and in their terminal rooms. Chutes shall have additional sprinkler heads installed at alternate floors and at the lowest intake. Where a rubbish chute extends through a building more than one floor below the lowest intake, the extension shall have sprinklers installed that are recessed from the drop area of the chute and protected from freezing in accordance with Section 903.3.1.1. Such sprinklers shall be installed at alternate floors, beginning with the second level below the last intake and ending with the floor above the discharge. Access to sprinklers in chutes shall be provided for servicing.

**C** Gravity rubbish (waste) and linen chutes can pose a significant hazard to building occupants if they are not properly installed and protected. Generally, these systems are installed in high-occupancy buildings where the occupants will be sleeping or are incapable of self-rescue, such as in Group I, R-1 and R-2 occupancies. For occupant convenience, openings to the chutes are commonly provided in areas accessible to the public and, in older buildings, the chute opening may be located in an exit access corridor. In comparison to other building shafts, gravity rubbish and linen chutes always contain fuel. As bags of waste debris or linen fall through the chute, they can deposit fluids such as waste cooking oil, which adheres to the shaft surface. This waste material and other debris provide fuel that can support and accelerate vertical fire spread. The greatest accumulation of fuel will be in the termination room; however, a significant amount of fuel that covers the interior surface area of the chute will be found in the sections of chutes closest to the collection or termination room. Therefore, it is important that the automatic sprinklers be properly placed and protected so they are available in the event of a fire in the termination room and to protect waste compaction equipment where such equipment is installed.

Installation of gravity chutes for rubbish or linen requires compliance with IBC Section 713.13. Under the code, permanent rubbish and linen chutes are constructed inside of a fire-resistance-rated shaft assembly with a minimum 1-hour fire-resistance rating in buildings less than four stories in height; in buildings four or more stories in height, the fire-resistance rating is increased to 2 hours by IBC Section 713.4. The design of the shaft system and its openings must also comply with the requirements in IBC Sections 713.11 and 713.13, which require the termination room receiving the discharged material to be separated from the building by a fire-resistance rating equivalent to that of the shaft that it serves.

IBC Section 713.13.6 requires the installation of an automatic sprinkler system in rubbish and linen chutes to comply with the requirements of Section 903.2.11.2. Section 903.2.11.2 correlates with the requirements in Chapter 19 of NFPA 13. Chapter 19 of NFPA 13 contains the special occupancy requirements for all buildings, including gravity waste and linen chutes. The provisions align the code and IBC requirements with those in NFPA 82 and NFPA 13.

A critical term in this section is “extension.” The word was selected to address chutes installed in buildings of pedestal construction or other designs in which the fire-resistant construction shaft and chute pass through a less hazardous occupancy, such as a Group S-2 parking garage, or other floors that do not have access to the shaft. In these areas, chute openings are generally not provided. As a result, this section now contains a specific provision that may impose a requirement for sprinklers in the portion of the chute that serves as an extension beyond the last intake and the termination room or discharge area.

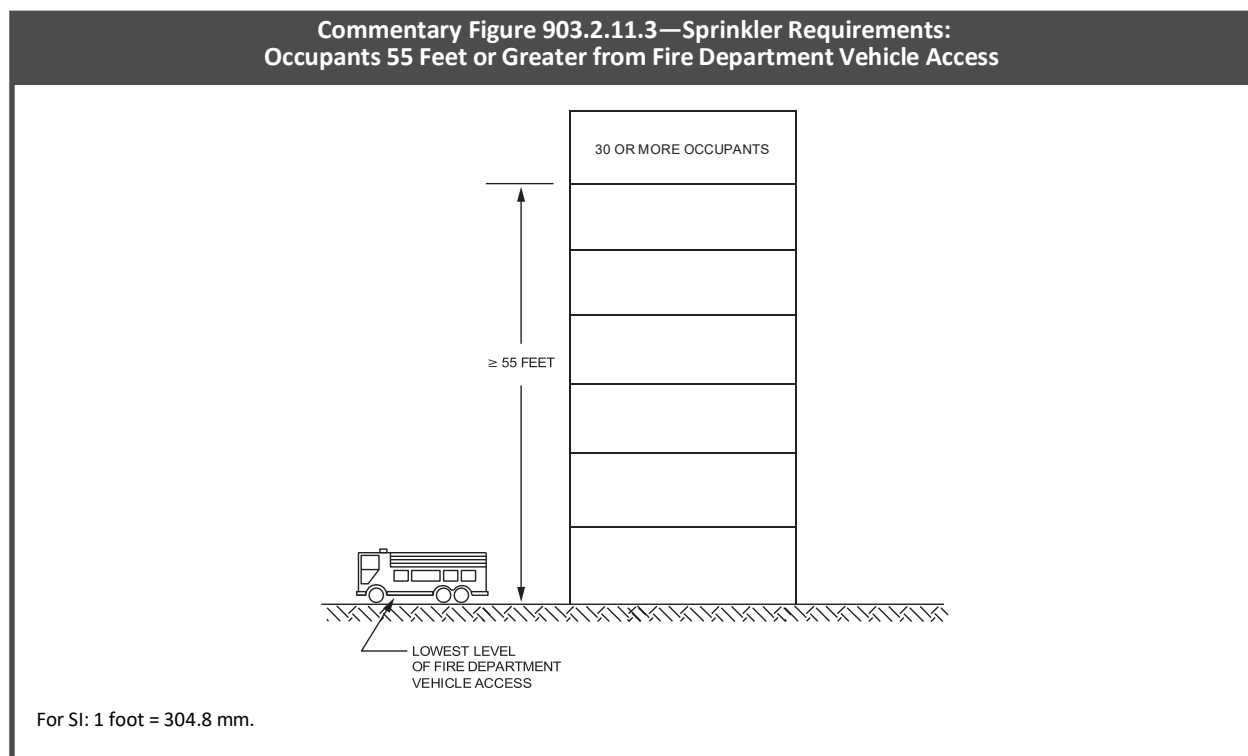
Because objects will be falling through the chute, the code requires the chute sprinklers to be recessed and protected from impact. Sprinklers are not required at every story housing a chute. The code requires automatic sprinklers at the top of the chute and at its termination. In addition, sprinkler heads are required at alternate floors within the chute, with a head being installed at the floor level with the lowest intake point into the chute. Previously, these additional sprinkler heads were only required where the shaft extended through three or more floors. These revisions, plus the previously discussed requirements for extensions, may result in additional sprinkler heads within some shafts as compared to the previous requirements.

Sprinklers in chutes that are in locations subject to freezing require freeze protection in accordance with the requirements of Section 903.3.1.1 and, therefore, the NFPA 13 standard. This can be accomplished using a dry-pendant sprinkler or constructing a dry-pipe sprinkler system.

**903.2.11.3 Buildings 55 feet or more in height.** An *automatic sprinkler system* shall be installed throughout buildings that have one or more stories with an *occupant load* of 30 or more located 55 feet (16 764 mm) or more above the lowest level of fire department vehicle access, measured to the finished floor.

**Exception:** Occupancies in Group F-2.

**C** Because of the difficulties associated with manual suppression of a fire in mid-rise buildings in excess of 55 feet (16 764 mm) above the lowest level of fire department vehicle access, an automatic sprinkler system is required throughout the building for all occupancies, with the exception of Group F-2. Buildings that qualify for a sprinkler system under this section are not necessarily high-rise buildings as defined in Section 202 and are focused also on those with occupants located on the upper floors. These provisions apply only to buildings with occupied floors having an occupant load of 30 or more located on stories 55 feet or greater from fire department vehicle access. The 55 feet is measured to the finished floor (see Commentary Figure 903.2.11.3).



**903.2.11.4 Ducts conveying hazardous exhausts.** Where required by the *International Mechanical Code*, automatic sprinklers shall be provided in ducts conveying hazardous exhaust or flammable or combustible materials.

**Exception:** Ducts where the largest cross-sectional diameter of the duct is less than 10 inches (254 mm).

**C** Section 509 of the IMC addresses the requirements for hazardous exhaust systems. To protect against the spread of fire within a hazardous exhaust system and to prevent a duct fire from involving the building, an automatic sprinkler system must be installed to protect the exhaust duct system. Where materials conveyed in the ducts are not compatible with water, alternative extinguishing agents should be used. The fire suppression requirement is intended to apply to exhaust systems having an actual fire hazard. An automatic sprinkler system in the duct would be of little value for an exhaust system that conveys only nonflammable or noncombustible materials, fumes, vapors or gases.

The exception recognizes the reduced hazard associated with smaller ducts and the impracticality of installing sprinkler protection. Another exception in the IMC indicates that laboratory hoods that meet specific provisions of the IMC are not required to be suppressed. Because the IMC is more specific in this regard, it should be consulted for the proper application of the exception.

**903.2.11.5 Commercial cooking operations.** An *automatic sprinkler system* shall be installed in commercial kitchen exhaust hood and duct systems where an *automatic sprinkler system* is used to comply with Section 904.

**C** An automatic suppression system is required for commercial kitchen exhaust hood and duct systems where required by Section 606 or by the IMC to have a Type I hood. Type I hoods are required for commercial cooking equipment that produces grease-laden vapors or smoke. Section 904.14 recognizes that alternative extinguishing systems other than an automatic sprinkler system may be used. Where an automatic sprinkler system is used for commercial cooking operations, it must comply with the requirements identified in Section 904.14.4.

**903.2.11.6 Other required fire protection systems.** In addition to the requirements of Section 903.2, the provisions indicated in Table 903.2.11.6 require the installation of a fire protection system for certain buildings and areas.

**C** In addition to Section 903.2, requirements for automatic fire protection systems are also found elsewhere in the code as indicated in Table 903.2.11.6.

| TABLE 903.2.11.6—ADDITIONAL REQUIRED FIRE PROTECTION SYSTEMS |                                                 |
|--------------------------------------------------------------|-------------------------------------------------|
| SECTION                                                      | SUBJECT                                         |
| 320.2                                                        | Lithium-ion and lithium metal battery storage   |
| 903.2.10.2                                                   | Mechanical-access enclosed parking garages      |
| 914.2.1                                                      | Covered and open mall buildings                 |
| 914.3.1                                                      | High-rise buildings                             |
| 914.4.1                                                      | Atriums                                         |
| 914.5.1                                                      | Underground structures                          |
| 914.6.1                                                      | Stages                                          |
| 914.7.1                                                      | Special amusement area                          |
| 914.8.2                                                      | Airport traffic control towers                  |
| 914.8.3, 914.8.6                                             | Aircraft hangars                                |
| 914.9                                                        | Flammable finishes                              |
| 914.10                                                       | Drying rooms                                    |
| 914.11.1                                                     | Ambulatory care facilities                      |
| 1030.6.2.3                                                   | Smoke-protected assembly seating                |
| 1103.5.1                                                     | Existing Group A occupancies                    |
| 1103.5.5                                                     | Pyroxylin plastic storage in existing buildings |
| 1103.5.2                                                     | Existing Group I-2 occupancies                  |
| 1103.5.3                                                     | Existing Group I-2, Condition 2 occupancies     |
| 1103.5.5                                                     | Pyroxylin plastics                              |
| Table 1206.9, Table 1206.10, Table 1207.7, Table 1207.8      | Stationary and mobile energy storage systems    |
| 2108.2                                                       | Dry cleaning plants                             |
| 2108.3                                                       | Dry cleaning machines                           |
| 2309.3.1.5.2                                                 | Hydrogen motor fuel-dispensing area canopies    |
| 2404.4                                                       | Spray finishing in Group A, E, I or R           |
| 2404.6                                                       | Spray booths and spray rooms                    |
| 2405.2                                                       | Dip-tank rooms in Group A, I or R               |
| 2405.4.1                                                     | Dip tanks                                       |
| 2405.9.4                                                     | Hardening and tempering tanks                   |
| 2703.10                                                      | HPM facilities                                  |
| 2703.10.1.1                                                  | HPM workstation exhaust                         |
| 2703.10.2                                                    | HPM gas cabinets and exhausted enclosures       |
| 2703.10.3                                                    | HPM exit access corridor                        |
| 2703.10.4                                                    | HPM exhaust ducts                               |
| 2703.10.4.1                                                  | HPM noncombustible ducts                        |
| 2703.10.4.2                                                  | HPM combustible ducts                           |
| 2807.3                                                       | Lumber production conveyor enclosures           |
| 2808.7                                                       | Recycling facility conveyor enclosures          |
| 3006.1                                                       | Class A and B ovens                             |
| 3006.2                                                       | Class C and D ovens                             |
| Table 3206.2                                                 | Storage fire protection                         |
| 3206.4                                                       | Storage                                         |

| TABLE 903.2.11.6—ADDITIONAL REQUIRED FIRE PROTECTION SYSTEMS—continued |                                                                   |
|------------------------------------------------------------------------|-------------------------------------------------------------------|
| SECTION                                                                | SUBJECT                                                           |
| 3210.1.1                                                               | Record storage over 12 feet                                       |
| 3704.5                                                                 | Storage of more than 1,000 cubic feet of loose combustible fibers |
| 5003.8.4.1                                                             | Gas rooms                                                         |
| 5003.8.5.3                                                             | Exhausted enclosures                                              |
| 5004.5                                                                 | Indoor storage of hazardous materials                             |
| 5005.1.8                                                               | Indoor dispensing of hazardous materials                          |
| 5104.4.1                                                               | Aerosol product warehouses                                        |
| 5106.3.2                                                               | Aerosol display and merchandising areas                           |
| 5306.2.1                                                               | Exterior medical gas storage room                                 |
| 5306.2.2                                                               | Interior medical gas storage room                                 |
| 5306.2.3                                                               | Medical gas storage cabinet                                       |
| 5606.5.2.1                                                             | Storage of smokeless propellant                                   |
| 5606.5.2.3                                                             | Storage of small arms primers                                     |
| 5704.3.7.5.1                                                           | Flammable and combustible liquid storage rooms                    |
| 5704.3.8.4                                                             | Flammable and combustible liquid storage warehouses               |
| 5705.3.7.3                                                             | Flammable and combustible liquid Group H-2 or H-3 areas           |
| 6004.1.2                                                               | Gas cabinets for highly toxic and toxic gas                       |
| 6004.1.3                                                               | Exhausted enclosures for highly toxic and toxic gas               |
| 6004.2.2.6                                                             | Gas rooms for highly toxic and toxic gas                          |
| 6004.3.3                                                               | Outdoor storage for highly toxic and toxic gas                    |
| 6504.1.1                                                               | Pyroxylin plastic storage cabinets                                |
| 6504.1.3                                                               | Pyroxylin plastic storage vaults                                  |
| 6504.2                                                                 | Pyroxylin plastic storage and manufacturing                       |
| For SI: 1 cubic foot = 0.023 m <sup>3</sup> .                          |                                                                   |

**C** Table 903.2.11.6 identifies other sections of the code that require an automatic fire protection system based on the specific occupancy or use because of the unique hazards of such use or occupancy. The table does not identify the various sections of the code that contain design alternatives based on the use of an automatic fire protection system, typically an automatic sprinkler system.

**903.2.12 During construction.** *Automatic sprinkler systems* required during construction, *alteration* and demolition operations shall be provided in accordance with Section 3307.

**C** Chapter 33 of the code and Chapter 33 of the IBC address fire safety requirements during construction, alteration or demolition work. Working sprinkler systems should remain operative at all times unless it is absolutely necessary to shut down the system because of the proposed work. All sprinkler system impairments should be rectified as quickly as possible unless specific prior approval has been obtained from the fire code official. Buildings with a required sprinkler system should not be occupied unless the sprinkler system has been installed and tested consistent with Section 901.5. If the system must be placed out of service, the requirements of Section 901.7 are necessary to address the temporary impairment to the fire protection system.

**903.3 Installation requirements.** *Automatic sprinkler systems* shall be designed and installed in accordance with Sections 903.3.1 through 903.3.8.

**C** Specific design, installation and testing criteria are given for automatic sprinkler systems in the sections and subsections that follow, as well as an indication of the applicability of a nationally recognized standard in the area. The information required to complete a thorough review of an automatic sprinkler system is listed in Commentary Figure 903.3.

### Commentary Figure 903.3—Sample Sprinkler System Drawing and Data Submittals

#### 1. Information required on shop drawings includes:

- Name of owner and occupant
- Location, including street address
- Point of compass
- Graphic indication of scale
- Ceiling construction
- Full-height cross section
- Location of fire walls
- Location of partitions
- Occupancy of each area or room
- Location and size of blind spaces and closets
- Any questionable small enclosures in which no sprinklers are to be installed
- Size of city main in street, pressure and whether dead end or circulation and, if dead end, direction and distance to nearest circulating main, city main test results
- Other source of water supply, with pressure or elevation
- Make, type and orifice size of sprinkler
- Temperature rating and location of high-temperature sprinklers
- Limitations on extended coverage sprinklers or other special sprinkler types
- Number of sprinklers on each riser and on each system by floors and total area by each system on each floor
- Make, type, model and size of alarm or dry-pipe valve
- Make, type, model and size of preaction or deluge valve
- Type and location of alarm bells
- Backflow prevention method and details
- Total number of sprinklers on each dry-pipe system or preaction deluge system
- Approximate capacity in gallons or each dry-pipe system
- Setting for pressure-reducing valves
- Pipe size, type, and schedule of wall thickness
- Cutting lengths of pipe (or center-to-center dimensions)
- Type of fittings, riser nipples and size, and all welds and bends
- Type and location of hangers, inserts and sleeves
- Calculations of loads and details for sway bracing
- All control valves, checks, drain pipes, flushing, and test pipes
- Size and location of standpipe risers and hose outlets
- Small hand hose equipment
- Underground pipe size, length, location, weight, material, point of connection to city main; the type of valves, meters and valve pits; and the depth that top of the pipe is laid below grade
- Size and location of hydrants along with hose-houses
- Size and location of fire department connections
- When the equipment is to be installed as an addition to an old group of sprinklers without additional feed from the yard system, enough of the old system shall be indicated on the plans to show the total number of sprinklers to be supplied and to make all connections clear
- Information to be provided on the hydraulic nameplate
- Name, address and phone number of contractor and sprinkler designer
- Hydraulic reference points shall be shown by a number and/or letter designation and shall correspond with comparable reference points shown on the hydraulic calculation sheets
- System design criteria showing the minimum rate of water application (density), the design area of water application and the water required for hose streams both inside and outside
- Actual calculated requirements showing the total quantity of water and the pressure required at a common reference point for each system
- Elevation data showing elevations of sprinklers, junction points and supply or reference points
- Protected wall openings if room design method is used

#### 2. Information required on calculations includes:

- Location
- Name of owner and occupant
- Building identification
- Description of hazard
- Name and address of contractor and designer
- Name of approving agency

#### 3. System design requirements include:

- Design area of water application
- Minimum rate of water application (density)
- Area of sprinkler coverage
- Hazard or commodity classification
- Building height
- Storage height
- Storage method
- Total water requirements, as calculated, including allowance for hose demand water supply information and allowance for in-rack sprinklers
- Location and elevation static and residual test gauge with relation to the riser reference point
- Size and location of hydrants used for flow test data
- Flow location
- Static pressure, psi
- Residual pressure, psi
- Flow, gpm
- Date
- Time
- Test conducted by whom
- Sketch to accompany gridded system calculations to indicate flow quantities and directions for lines with sprinklers operated in the remote area

#### 4. Additional information necessary for complete review includes:

- Sprinkler description and discharge constant (K-value)
- Hydraulic reference points
- Flow, gpm
- Pipe diameter (actual internal diameter)
- Pipe length
- Equivalent pipe length for fittings and components
- Friction loss in psi per foot of pipe
- Total friction loss between reference points
- Elevation difference between reference points
- Required pressure in psi at each reference point
- Velocity pressures and normal pressure if included in calculations
- Notes to indicate starting points, reference to other sheets or clarification of data
- Information on antifreeze solution (type and quantity)
- Water treatment system information including reason for treatment and program details

#### 5. Included with the submittal must be a graph sheet showing water supply curves and system requirements including:

- Hose demand plotted on semilogarithmic graph paper so as to present a graphic summary of the complete hydraulic calculations
- Sprinkler system demand including in-rack sprinklers (if applicable)

**903.3.1 Standards.** Automatic sprinkler systems shall be designed and installed in accordance with Section 903.3.1.1, unless otherwise permitted by Sections 903.3.1.2 and 903.3.1.3 and other chapters of this code, as applicable.

**C** Automatic sprinkler systems are to be installed to comply with the code and NFPA 13, 13R or 13D. As provided for in Section 102.7.1, where differences occur between the code and NFPA 13, 13R or 13D, the code applies. The fire code official also has the authority to approve the type of sprinkler system to be installed. See Commentary Figure 903.3.1 for typical design parameters for each type of sprinkler system.

This section also provides a pointer to other sections of the code that might provide more specific or detailed sprinkler requirements such as those found in IBC Chapter 4.

**Commentary Figure 903.3.1—NFPA 13, NFPA 13R, NFPA 13D Systems**

|                      | NFPA 13                                 | NFPA 13R                               | NFPA 13D                               |
|----------------------|-----------------------------------------|----------------------------------------|----------------------------------------|
| Extent of protection | Equip throughout<br>(Section 903.3.1.1) | Occupied spaces<br>(Section 903.3.1.2) | Occupied spaces<br>(Section 903.3.1.3) |
| Scope                | All occupancies                         | Low-rise residential                   | One- and two-family dwellings          |
| Sprinkler design     | Density/area concept                    | 4-head design                          | 2-head design                          |
| Sprinklers           | All types                               | Residential only                       | Residential only                       |
| Duration             | 30 minutes (minimum)                    | 30 minutes                             | 10 minutes                             |
| Advantages           | Property and life protection            | Life safety/tenability                 | Life safety/tenability                 |

**903.3.1.1 NFPA 13 sprinkler systems.** Where the provisions of this code require that a building or portion thereof be equipped throughout with an *automatic sprinkler system* in accordance with this section, sprinklers shall be installed throughout in accordance with NFPA 13 except as provided in Sections 903.3.1.1.1 through 903.3.1.1.3.

**C** NFPA 13 contains the minimum requirements for the design and installation of automatic water sprinkler systems and exposure protection sprinkler systems. The requirements contained in the standard include the character and adequacy of the water supply and the selection of sprinklers, piping, valves and all of the materials and accessories. The standard does not include requirements for installation of private fire service mains and their appurtenances, installation of fire pumps, or construction and installation of gravity and pressure tanks and towers.

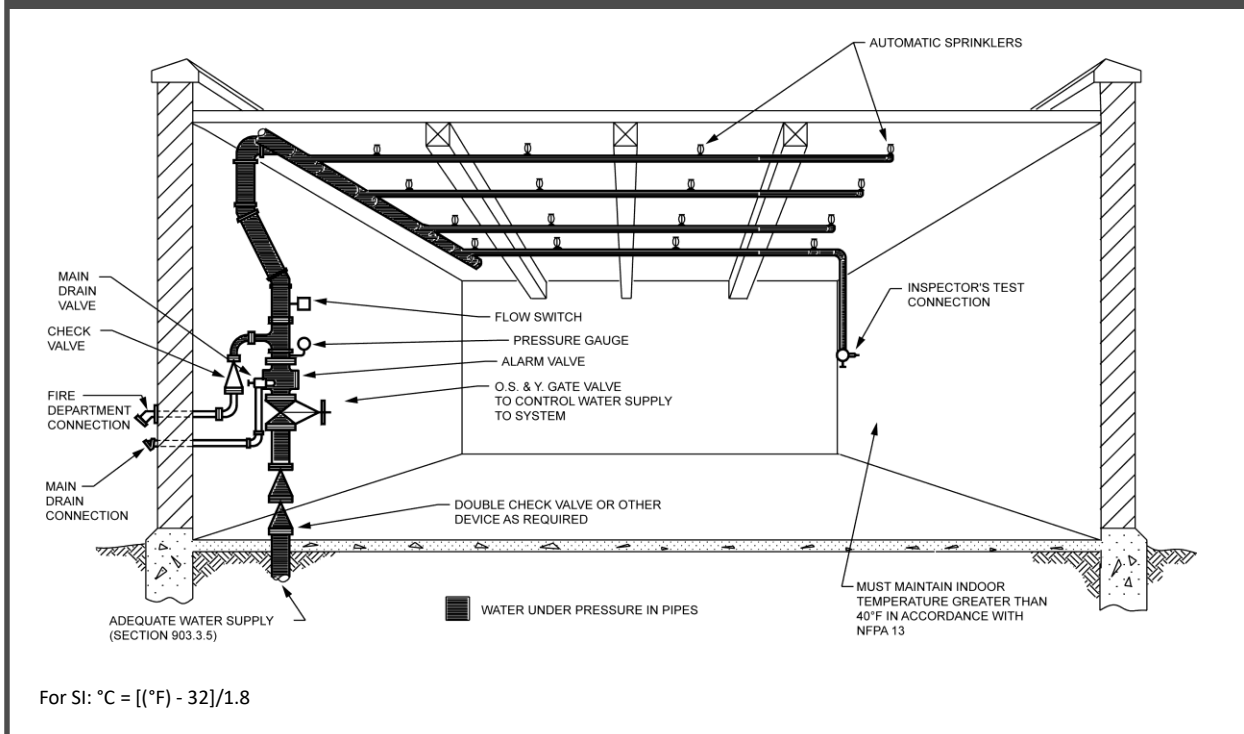
NFPA 13 defines seven classifications or types of water sprinkler systems: wet-pipe (see Commentary Figure 903.3.1.1), dry pipe, preaction or deluge, combined dry pipe and preaction, antifreeze systems, sprinkler systems that are designed for a special purpose and outside sprinklers for exposure protection. While numerous variables must be considered in selecting the proper type of sprinkler system the wet-pipe sprinkler system is recognized as the most effective and efficient. The wet-pipe system is also the most reliable type of sprinkler system because water under pressure is available at the sprinkler. Therefore, wet-pipe sprinkler systems are recommended wherever possible.

The extent of coverage and distribution of sprinklers is based on the NFPA 13 standard. Numerous conditions exist in the standard where sprinklers are specifically required and also where they may or may not be located. Once it is determined that the sprinkler system is to be in accordance with NFPA 13, that standard must be reviewed for installation details. For example, exterior spaces such as combustible canopies are required to be equipped with sprinklers according to Section 8.15.7 of NFPA 13 where the canopy extends for a distance of 4 feet (1219 mm) or more. A 3-foot (914 mm) combustible canopy would not require sprinklers nor would a 6-foot (1829 mm) canopy constructed of noncombustible materials, provided there is no combustible storage under the canopy.

Because installation is required to be in accordance with NFPA 13, if the standard allows for the omission of sprinklers in any location, then the building is still considered as sprinklered throughout. For example, Section 8.15.8.1.1 of NFPA 13 allows sprinklers to be omitted from bathrooms in dwelling units in motels and hotels. If sprinklers are not provided in the bathrooms because of the conditions stipulated in NFPA 13, the building would still be considered as sprinklered throughout in accordance with the code, NFPA 13 and the IBC.

Exceptions for the use of NFPA 13R and 13D systems are addressed throughout the code when exceptions based on the use of sprinklers are provided. More specifically, if the use of these other standards is appropriate, it will be noted within the exception. For a building to be considered “equipped throughout” with an NFPA 13 sprinkler system, complete protection must be provided in accordance with the referenced standard, subject to the exempt locations indicated in Section 903.3.1.1.1. See Commentary Figure 904.2.1 for examples of requirements modified through the use of sprinkler systems.

Commentary Figure 903.3.1.1—Typical Wet-Pipe System



**903.3.1.1.1 Exempt locations.** Automatic sprinklers shall not be required in the following rooms or areas where such rooms or areas are protected with an *approved* automatic fire detection system in accordance with Section 907.2 that will respond to visible or invisible particles of combustion. Sprinklers shall not be omitted from a room merely because it is damp, of *fire-resistance-rated* construction or contains electrical equipment.

1. A room or space where sprinklers constitute a serious life or fire hazard because of the nature of the contents, where *approved* by the fire code official.
2. Generator and transformer rooms separated from the remainder of the building by walls and floor/ceiling or roof/ceiling assemblies having a *fire-resistance rating* of not less than 2 hours.
3. Rooms or areas that are of noncombustible construction with wholly noncombustible contents.
4. Fire service access elevator machine rooms and machinery spaces.
5. Machine rooms, machinery spaces, control rooms and control spaces associated with occupant evacuation elevators designed in accordance with Section 3008 of the *International Building Code*.

**C** This section allows the omission of sprinkler protection in certain locations if an approved automatic fire detection system is installed. Buildings in compliance with one of the five listed conditions would still be considered fully sprinklered throughout in compliance with the code and NFPA 13 and thus are eligible for all applicable code alternatives, exceptions or reductions. Elimination of the sprinkler system in a sensitive area is subject to the approval of the fire code official.

Condition 1 addresses restrictions where the application of water could create a hazardous condition. For example, sprinkler protection is to be avoided where it is not compatible with certain stored materials (i.e., some water-reactive hazardous materials such as calcium carbide). Combustible metals, such as magnesium and aluminum, may burn so intensely that the use of water to attempt fire control will only intensify the reaction.

It is not the intent of Condition 1 to omit sprinklers solely because of a potential for water damage. A desire to not sprinkle a certain area (such as a computer room or operating room) does not fall within the limitations of the exception unless there is something unique about the space that would result in water being incompatible. A computer room can be adequately protected using an automatic sprinkler system or an alternative gaseous suppression agent system or a combination of these systems. The intent of Condition 2 is to consider whether the contents would react adversely to the application of water. It is important to note that the fire code official must approve the use of this item. Note also that with respect to computer rooms, NFPA 75, *Standard for the Fire Protection of Information Technology Equipment*, recognizes automatic sprinklers as the primary fire protection system for computer rooms.

Condition 2 recognizes the low fuel load and low occupancy hazards associated with generator and transformer rooms and, therefore, allows the omission of sprinkler protection if the rooms are separated from adjacent areas by 2-hour fire-resistance-rated construction. This condition assumes the room is not used for any combustible storage. This condition is similar to Section 8.15.11.3 of NFPA 13, which exempts electrical equipment rooms from sprinkler protection, provided the

room is dedicated to the use of dry-type electrical equipment, is constructed as a 2-hour fire-resistance-rated enclosure, and is not used for combustibile storage.

Condition 3 requires the construction of the room or area, as well as the contents, to be noncombustible. An example would be an area in an unprotected steel-frame building (Type IIB construction) used for steel or concrete block storage. Neither involves any significant combustibile packaging or sources of ignition, and few combustibles are present (see Commentary Figure 903.3.1).

Condition 4 addresses the concern for elevator machine rooms and machinery spaces associated with fire service access elevators, as required by IBC Sections 403.6.1 and 3007, for buildings with occupied floors greater than 120 feet (36.58 m) from the lowest level of fire department access. These elevators need to work during fire situations and their operation cannot be threatened by the activation of a sprinkler in a machine room or space that may affect the operation of the elevators. Fire service access elevators are required to be continuously monitored at the fire command center in accordance with IBC Section 3007.7.

Condition 5, similar to Condition 4, exempts sprinklers from the machine rooms, machinery spaces, control rooms and control spaces for occupant evacuation elevators. Like fire service access elevators, these elevators need to work during fire situations and their operation cannot be threatened by the activation of a sprinkler in machine rooms, machinery spaces, control rooms and control spaces. Such elevators are required to be monitored at the fire command center in accordance with IBC Section 3008.7.

**903.3.1.1.2 Bathrooms.** In Group R occupancies, sprinklers shall not be required in bathrooms that do not exceed 55 square feet (5 m<sup>2</sup>) in area and are located within individual *dwelling units* or *sleeping units*, provided that walls and ceilings, including the walls and ceilings behind a shower enclosure or tub, are of noncombustible or limited-combustible materials with a 15-minute thermal barrier rating.

**C** This provision was added to the code to reinstate an exception of NFPA 13 that had been in existence since 1976 but was deleted for all but dwelling units in motels and hotels (NFPA 8.15.8.1.1) from the 2013 edition of NFPA 13. Although reinstating the small bathroom exception will have a limited impact on new construction because many modern bathrooms exceed the 55-square-foot area limit to accommodate wheelchair access, the more important consequence will be removing an unnecessary cost increase for building owners who choose to retrofit existing properties with small bathrooms that were built before it was common to provide wheelchair access. Codes and standards should not erect any unnecessary barriers to retrofitting sprinklers into existing properties, such as existing high-rise buildings.

In the 1976 edition of the Life Safety Code (NFPA 101), to encourage cost-effective fire protection systems for apartment buildings, Section 11-3.8.3.4.1 provided an exception to permit bathrooms that did not exceed 55 square feet within individual dwelling units to omit sprinklers when the apartment building was sprinklered in accordance with NFPA 13. The basis of the 55-square-foot area is that this area accommodates a “typical” small bathroom that contains a standard tub, a toilet, a sink and nothing more. This exception was later duplicated from NFPA 101 into the 1991 edition of NFPA 13 with the understanding that the next edition of NFPA 101 (1994) could delete the exception since NFPA 13 would have it covered. NFPA 101-1994 then, as planned, deleted the exception.

The history of apartment unit bathroom fires is statistically minimal. According to the NFPA Home Structure Fire Report, January 2009, Table 9B, “Reported Apartment Structure Fires by Area of Origin 2003–2006 Annual Averages,” out of 113,000 fires/year, only 1,600 (1 percent) are in bathrooms. Given that there is more than 35 years of experience with the bathroom sprinkler exception being in place (since it was put into NFPA 101 in 1976), it would certainly be reasonable to expect anecdotal or statistical experience to indicate the existence of a problem if there were one.

**903.3.1.1.3 Lithium-Ion or lithium metal batteries.** Where *automatic sprinkler systems* are required by this code for areas containing lithium-ion or lithium metal batteries, the design of the system shall be based on a series of fire tests. Such tests shall be conducted or witnessed and reported by an approved testing laboratory involving test scenarios that address the range of variables associated with the intended arrangement of the hazards to be protected.

**C** Automatic sprinkler systems are relied upon for fire protection. In the case of areas containing lithium-ion or lithium metal batteries, the design of the automatic sprinkler system is to be based upon a large-scale fire testing at an approved laboratory because there is no guidance in the 2022 edition of NFPA 13. In the commodity classification portion of NFPA 13—2019 this issue is highlighted by Table A.20.4(a), “Examples of Commodities Not Addressed by Classifications in Section 20.4,” which specifically lists lithium-ion and lithium metal batteries.

Fire code officials, as well as designers and installers, may miss the need for the submitted design to be based upon witnessed fire tests. The purpose of this section is to ensure that the submitted design is documented and able to address the potential for a high heat release event.

**903.3.1.2 NFPA 13R sprinkler systems.** *Automatic sprinkler systems* in Group R occupancies shall be permitted to be installed throughout in accordance with NFPA 13R where the Group R occupancy meets all of the following conditions:

1. Four stories or less above *grade plane*.

2. For other than Group R-2 occupancies, the floor level of the highest story is 30 feet (9144 mm) or less above the lowest level of fire department vehicle access.

For Group R-2 occupancies, the roof assembly is less than 45 feet (13 716 mm) above the lowest level of fire department vehicle access. The height of the roof assembly shall be determined by measuring the distance from the lowest required fire vehicle access road surface adjacent to the building to the eave of the highest pitched roof, the intersection of the highest roof to the *exterior wall*, or the top of the highest parapet, whichever yields the greatest distance.

3. The floor level of the lowest story is 30 feet (9144 mm) or less below the lowest level of fire department vehicle access.

The number of stories of Group R occupancies constructed in accordance with Sections 510.2 and 510.4 of the *International Building Code* shall be measured from *grade plane*.

**C** NFPA 13R contains design and installation requirements for a sprinkler system to aid in the detection and control of fires in low-rise (four stories or less) residential occupancies.

Sprinkler systems designed in accordance with NFPA 13R are intended to prevent flashover (total involvement) in the room of fire origin and to improve the chance for occupants to escape or be evacuated. The design criteria in NFPA 13R are similar to those in NFPA 13 except that sprinklers may be omitted from areas in which fatal fires in residential occupancies do not typically originate (bathrooms, closets, attics, porches, garages and concealed spaces).

A common question is whether a mixed-occupancy building that contains a Group R occupancy could still use NFPA 13R for the design. If one of the mixed-use occupancies would require a sprinkler system throughout the building in accordance with NFPA 13, then a 13R system would not be allowed. If, however, the only reason a sprinkler system is being installed is because there is a Group R fire area within the building, then an NFPA 13R system would be an appropriate design choice. The areas that are not classified as Group R would require protection in accordance with NFPA 13.

It must be noted that although the building would be considered sprinklered throughout in accordance with NFPA 13R, not all of the code sprinkler alternatives could be applied. Any alternative that requires the installation of an NFPA 13 system would not be applicable if a portion of the building utilizes an NFPA 13R system.

The code provisions that allow for an increase in building height according to Section 504.3 do not compound this section. If the design of a residential building intends to take advantage of the sprinkler height increase so that the building is five stories or more, the sprinkler system must be an NFPA 13 system. Because this section limits the height to four stories and, for other than Group R-2 occupancies, limits the highest or lowest floor level to 30 feet or less above or below, respectively, the lowest level of fire department vehicle access, that is the maximum height for a building that can utilize an NFPA 13R system in accordance with the code.

The limitation of four stories in height is to be measured with respect to the established grade plane, which is consistent with IFC Interpretation No. 43-03. As such, a basement would not be considered a story above grade for purposes of determining the applicability of this section.

Condition 2 recognizes the common use of mezzanines in upper levels of Group R-2 occupancies and provides a roof assembly limit. From the exterior, a mezzanine level in the fourth story would appear to be a fifth story, and such mezzanines often include a sleeping area. This roof assembly limit only applies to Group R-2 occupancies since these have different operational, occupant and architectural attributes in comparison to other occupancies.

The second paragraph recognizes the application of the requirements of IBC Sections 510.2 and 510.4, which are essentially exceptions to the height and area requirements. They are based on providing a horizontal fire separation similar to the concept of a fire wall to create separate buildings. This establishes that the height in stories is to be measured from grade plane. Such buildings are often referred to as “pedestal buildings.”

**903.3.1.2.1 Balconies and decks.** Sprinkler protection shall be provided for exterior balconies, decks and ground floor patios of *dwelling units* and *sleeping units* where either of the following conditions exists:

1. The building is of Type V construction, provided that there is a roof or deck above.
2. Exterior balconies, decks and ground floor patios of *dwelling units* and *sleeping units* are constructed in accordance with Section 705.2.3.1, Exception 3 of the *International Building Code*.

Sidewall sprinklers that are used to protect such areas shall be permitted to be located such that their deflectors are within 1 inch (25 mm) to 6 inches (152 mm) below the structural members and a maximum distance of 14 inches (356 mm) below the deck of the exterior balconies and decks that are constructed of open wood joist construction.

**C** Balconies, decks and patios in buildings of Type V construction and used for Group R occupancies are required to have sprinkler protection when there is a roof or deck above. This is in addition to the requirements of NFPA 13R, which primarily addresses the life safety of occupants and not property protection. The intent is to address hazards such as grilling and similar activities. Since NFPA 13R does not require such coverage, there is potential that a fire on a balcony could grow much too large for the system within the building to handle. The concern is that a potential exterior balcony fire could spread to unprotected floor/ceiling assemblies and attic spaces and result in major property damage. Section 4104.2 specifically addresses restrictions on open flame cooking devices used on combustible balconies. Note that sprinklers are not intended to be provided in closets found on such balconies.

Regardless of whether the exterior walking surface is attached to the building and called a balcony or is a freestanding structure such as a deck or patio, the concern for fire ignition in the area adjacent to the exterior wall is the same. Sidewall sprinklers should be selected based on the area of coverage and climate. If the potential for freezing exists, a dry sidewall sprinkler should be used. Where the overhanging deck or balcony is extensive, an extended coverage sprinkler should be selected.

**903.3.1.2.2 Corridors and balconies in the means of egress.** Sprinkler protection shall be provided in *corridors* and for balconies in the *means of egress* where any of the following conditions apply:

1. *Corridors* with combustible floor or walls.
2. *Corridors* with an interior change of direction exceeding 45 degrees (0.79 rad).
3. *Corridors* that are less than 50 percent open to the outside atmosphere at the ends.
4. Open-ended *corridors* and associated exterior *stairways* and *ramps* as specified in Section 1027.6, Exception 3.
5. Egress balconies not complying with Sections 1021.2 and 1021.3.

**C** This section requires automatic sprinkler protection in any of the five conditions listed when using an NFPA 13R system. It establishes the criteria for corridor and balcony protection in the means of egress within buildings having NFPA 13R systems. Specifically, in conditions 1 to 3, corridors that have combustible floors or walls, have an interior change of direction exceeding 45 degrees, or are less than 50 percent open to the outside atmosphere at the ends are to be provided with automatic sprinklers to allow for a protected and safe means of egress.

Condition 4 references the requirement in Section 1027.6, Exception 3. It allows the separation between the open-ended corridor and exterior exit stairway to be omitted, but only where several conditions are met. The primary condition is that the corridor be sprinklered. A definition of “Open-ended corridor” is provided in Chapter 2. See Commentary Figure 903.3.1.2.2 for an example of an open-ended corridor.

Condition 5 references two sections in Chapter 10 and requires automatic sprinklers to be provided for egress balconies when the requirements in those sections are not met. Section 1021.2 requires exterior egress balconies to be separated from the interior of the building by walls and opening protectives the same as required for corridors in Section 1020. If this separation is not provided, the requirement to provide automatic sprinklers is necessary to protect the exterior egress balconies. Section 1021.3 requires that the long side of an egress balcony is to be not less than 50 percent open, and the open area above the guards shall be so distributed as to minimize the accumulation of smoke or toxic gases. Similar to condition 3 for corridors, if egress balconies are not at least 50 percent open, the requirement to provide automatic sprinklers is necessary to protect the exterior egress balconies.

Commentary Figure 903.3.1.2.2—Open-Ended Corridor



**903.3.1.2.3 Attics.** Attic protection shall be provided as follows:

1. Attics that are used or intended for living purposes or storage shall be protected by an *automatic sprinkler system*.
2. Where fuel-fired equipment is installed in an unsprinklered attic, not fewer than one quick-response intermediate temperature sprinkler shall be installed above the equipment.
3. Where located in a building of Type III, Type IV or Type V construction designed in accordance with Section 510.2 or 510.4 of the *International Building Code*, attics not required by Item 1 to have sprinklers shall comply with one of the following if the roof assembly is located more than 55 feet (16 764 mm) above the lowest level of fire department vehicle access needed to meet the provisions in Section 503:
  - 3.1. Provide *automatic sprinkler system* protection.
  - 3.2. Construct the attic using noncombustible materials.
  - 3.3. Construct the attic using fire-retardant-treated wood complying with Section 2303.2 of the *International Building Code*.
  - 3.4. Fill the attic with noncombustible insulation.

The height of the roof assembly shall be determined by measuring the distance from the lowest required fire vehicle access road surface adjacent to the building to the eave of the highest pitched roof, the intersection of the highest roof to the *exterior wall*, or the top of the highest parapet, whichever yields the greatest distance. For the purpose of this measurement, required fire vehicle access roads shall include only those roads that are necessary for compliance with Section 503.
4. Group R-4, Condition 2 occupancy attics not required by Item 1 to have sprinklers shall comply with one of the following:
  - 4.1. Provide *automatic sprinkler system* protection.
  - 4.2. Provide a heat detection system throughout the attic that is arranged to activate the building *fire alarm system*.
  - 4.3. Construct the attic using noncombustible materials.
  - 4.4. Construct the attic using fire-retardant-treated wood complying with Section 2303.2 of the *International Building Code*.
  - 4.5. Fill the attic with noncombustible insulation.

**C** This section clarifies that if the attic is used for living purposes or if fuel-fired equipment or storage is in these areas, full coverage in accordance with NFPA 13R is required. In attics where people are not expected and where storage or fuel-fired equipment is not located, some protection is required in addition to what NFPA 13R would require. Four different options of protection are provided. The first is simply to provide more warning time to the occupants if a fire should occur in the attic via a heat detector that activates the fire alarm system. The second is simply to reduce the risk of fire by requiring noncombustible construction materials. The third, similar to the second, is reducing the fire hazard by using fire-retardant-treated wood. This will slow the growth of a fire should one occur or prevent the start of a fire. The final option is simply to provide sprinkler protection to the attic. However, if a sprinkler system is provided in the attic, issues such as freezing temperatures need to be addressed.

**903.3.1.3 NFPA 13D sprinkler systems.** *Automatic sprinkler systems* installed in one- and two-family *dwelling*s; Group R-3; Group R-4, Condition 1; and *townhouses* shall be permitted to be installed throughout in accordance with NFPA 13D.

**C** NFPA 13D contains design and installation requirements for a sprinkler system to aid in the detection and control of fires in one- and two-family dwellings, mobile homes and townhouses. This section also specifically allows the use of an NFPA 13D system for occupancies classified as Group R-3 and Group R-4, Condition 1. Group R-3 occupancies are one- and two-family dwellings that fall outside the scope of the IRC and also those buildings housing small congregate living facilities and boarding houses. These facilities operate very similarly to a single-family home, and the level of the ability of the occupants is such that little assistance is needed for self-evacuation. This is consistent with the NFPA 13D requirements and is also consistent with FHA court cases based on nondiscrimination for group homes.

Similar to NFPA 13R, sprinkler systems designed in accordance with NFPA 13D are intended to prevent flashover (total involvement) in the room of fire origin and to improve the chance for occupants to escape or be evacuated. Although the allowable omission of sprinklers in certain areas of the dwelling unit in NFPA 13D is similar to that in NFPA 13R, the water supply requirements are less restrictive. NFPA 13D uses a two-head sprinkler design with a 10-minute duration requirement, while NFPA 13R uses a four-head sprinkler design with a 30-minute duration requirement. The decreased water supply requirement emphasizes the main intent of NFPA 13D to control the fire and maintain tenability during evacuation of the residence.

Since the fire code official has the authority to approve the type of sprinkler system, this section may be used to prevent the use of a specific type of sprinkler system that may be inappropriate for a particular type of occupancy.

**903.3.2 Quick-response and residential sprinklers.** Where *automatic sprinkler systems* are required by this code, quick-response or residential automatic sprinklers shall be installed in all of the following areas in accordance with Section 903.3.1 and their listings:

1. Throughout all spaces within a *smoke compartment* containing care recipient *sleeping units* in Group I-2 in accordance with the *International Building Code*.

2. Throughout all spaces within a *smoke compartment* containing gas fireplace appliances and decorative gas appliances in Group I-2 in accordance with the *International Building Code*.
3. Throughout all spaces within a *smoke compartment* containing treatment rooms in ambulatory care facilities.
4. *Dwelling units* and *sleeping units* in Group I-1 and R occupancies.
5. Light-hazard occupancies as defined in NFPA 13.

**C** This section requires the use of either listed quick-response or residential automatic sprinklers, depending on the type of sprinkler system required to achieve faster and more effective suppression in certain areas. Residential sprinklers are required in all types of residential buildings that would permit the use of an NFPA 13R or 13D sprinkler system.

Quick-response and residential sprinklers are similar in nature. They use a lighter material for the operating mechanism, thus reducing the heat lag in the element. The faster the heat can be absorbed, the sooner the sprinkler will begin to discharge water. Quick-response sprinklers have shown that they operate up to 25 percent faster than traditional sprinklers and create conditions in the room of origin that significantly increase the tenability of the environment. In tests performed by Factory Mutual (FM) for the Federal Emergency Management Agency (FEMA), the gas temperature in the room of origin was 550°F (288°C) with quick-response sprinklers, while it was 1,470°F (799°C) for conventional sprinklers at the time of sprinkler activation. More importantly, while the carbon monoxide (CO) level was 1,860 ppm for conventional sprinklers, the CO level when tested with quick-response sprinklers was only around 350 ppm. Comparatively, the National Institute of Occupational Safety and Health (NIOSH) considers the IDLH (immediately dangerous to life and health) level of CO to be 1,200 ppm. Thus, quick-response sprinklers have been shown to add significantly to the life safety effects of standard sprinkler systems.

Condition 1 requires the use of approved quick-response or residential sprinklers in smoke compartments containing care recipient sleeping units in Group I-2 occupancies. Even though properly operating standard sprinklers are effective, the extent of fire growth and smoke production that can occur before sprinkler activation creates the need for early warning to enable faster response by care providers and initiation of egress that is critical in occupancies containing persons incapable of self-preservation. The faster response time associated with quick-response or residential sprinklers increases the probability that the sprinklers will actuate before the care recipient's life would be threatened by a fire in his or her room.

Condition 2 requires the use of approved quick-response or residential sprinklers throughout all spaces within a smoke compartment containing gas fireplace appliances and decorative gas appliances in Group I-2. In addition to the life safety protection provided by carbon monoxide alarms, as already stated in this section commentary, quick-response and residential sprinklers will initiate operation at a much lower level of carbon monoxide, which provides enhanced protection where these fuel-fired appliances are located.

Condition 3 requires the use of approved quick-response or residential sprinklers in smoke compartments containing treatment rooms in ambulatory care facilities. The justification is the same as that for Condition 1. When there is a potential for care recipients to be incapable of self-preservation, the use of residential sprinklers or quick-response sprinklers is critical.

Because of the kind of occupants sleeping in Group R and I-1 occupancies, as indicated in Condition 4, a faster-responding type of sprinkler is desirable. Similar to the first condition, because occupants will be sleeping, the use of quick-response sprinklers creates additional safety by reducing sprinkler response time, thereby increasing the time available for egress and allowing for the time necessary for occupants to wake up and recognize the emergency event.

Condition 5 recognizes light-hazard occupancies in accordance with NFPA 13. These could include restaurants, schools, office buildings, places of religious worship and similar occupancies where the fire load and potential heat release of combustible contents are low.

**903.3.3 Obstructed locations.** Automatic sprinklers shall be installed with regard to obstructions that will delay activation or obstruct the water distribution pattern and shall be in accordance with the applicable *automatic sprinkler system* standard that is being used. Automatic sprinklers shall be installed in or under covered kiosks, displays, booths, concession stands or equipment that exceeds 4 feet (1219 mm) in width. Not less than a 3-foot (914 mm) clearance shall be maintained between automatic sprinklers and the top of piles of *combustible fibers*.

**Exception:** Kitchen equipment under exhaust hoods protected with a fire-extinguishing system in accordance with Section 904.

**C** To provide adequate sprinkler coverage, sprinkler protection must be extended under any obstruction that exceeds 4 feet (1219 mm) in width. Large air ducts are another common obstruction where sprinklers are routinely extended beneath the duct. The 3-foot (914 mm) storage clearance requirement for combustible fibers is caused by their potential high heat release. Most storage conditions require only a minimum 18-inch (457 mm) storage clearance to combustibles, depending on the type of sprinklers used and their actual storage conditions.

The exception recognizes that an alternative extinguishing system is permitted for commercial cooking systems in place of sprinkler protection for exhaust hoods that may be more than 4 feet (1219 mm) wide.

The application of this section is more critical to the ongoing use of the space. The obstruction conditions, therefore, should have already been addressed during plan review and installation inspection. This section gives the fire code official and building owner adequate information to avoid the most typical obstruction-related issues in terms of proper sprinkler coverage.

**903.3.4 Actuation.** *Automatic sprinkler systems* shall be automatically actuated unless specifically provided for in this code.

**C** The intent of this section is to eliminate the need for occupant intervention during a fire. As such, it is assumed that it will not be necessary for a person to manually open a valve or perform some other physical activity in order to allow the sprinkler system to activate.

Wet-pipe and dry-pipe sprinkler systems, for example, are essentially fail-safe systems in the sense that, if the system is in proper operating condition, it will operate once a sprinkler fuses. Dry systems have an inherent time lag for water to reach the sprinkler; therefore, the response is not as fast as for a wet-pipe system. Other types of sprinkler systems, such as preaction and deluge, rely on the actuation of a detection system to operate the sprinkler valve.

**903.3.5 Water supplies.** Water supplies for *automatic sprinkler systems* shall comply with this section and the standards referenced in Section 903.3.1. The potable water supply shall be protected against backflow in accordance with the requirements of this section and the *International Plumbing Code*. For connections to public waterworks systems, the water supply test used for design of *fire protection systems* shall be adjusted to account for seasonal and daily pressure fluctuations based on information from the water supply authority and as *approved by the fire code official*.

**C** To be effective, all sprinkler systems must have an adequate supply of water. The criteria for an acceptable water supply are contained in the standards referenced in Section 903.3.1. For example, NFPA 13 contains criteria for different types of water supplies as well as the methods to determine the pressure, flow capabilities and capacity necessary to get the intended performance from a sprinkler system. An acceptable water supply could consist of a reliable municipal supply, a gravity tank or a fire pump with a pressure tank or a combination of these.

This section also establishes the requirements for protecting the potable water system against a nonpotable source, such as stagnant water retained within the sprinkler piping. As stated in Section 608.17.4 of the IPC, an approved double check valve device or reduced pressure principle backflow preventer is required.

The other issue addressed by this section is fluctuations in water pressure. Information on pressure fluctuation is necessary to not only ensure that the minimum required pressure will be available for the automatic sprinkler system, but also to ensure that high pressures do not exceed the pressure limitations of the sprinkler system. If the water pressure on a sprinkler system exceeds 100 psi, changes in the hanging methods are required. Also, if a fire pump is provided, it might be possible to exceed 175 psi, which is typically considered the maximum working pressure for a sprinkler system. These are just additional reasons why it is critical to account for pressure fluctuations in the water supply. Obviously, fire flows can be affected by this, as well as other water-based fire protection systems, such as standpipes, which require a minimum 100 psi at the roof of high-rise buildings, and such may not be available due to pressure fluctuations. With regards to gathering of data, this requirement is simply intended to make sure that pressure fluctuations are addressed in accordance with the water supply authority to the extent that such information is available. Further authority is provided to the fire code official to accept other documentation.

**903.3.5.1 Domestic services.** Where the domestic service provides the water supply for the *automatic sprinkler system*, the supply shall be in accordance with this section.

**C** This section establishes the scope of domestic services for residential combination services. Essentially, compliance with Section 903.3.5 is required.

**903.3.5.2 Residential combination services.** A single combination water supply shall be allowed provided that the domestic demand is added to the sprinkler demand as required by NFPA 13R.

**C** NFPA 13R permits a common supply main to a building to serve both the sprinkler system and domestic services if the domestic demand is added to the sprinkler demand. NFPA 13R systems do not provide the same level of property protection as NFPA 13 systems.

**903.3.6 Hose threads.** Fire hose threads and fittings used in connection with *automatic sprinkler systems* shall be as prescribed by the *fire code official*.

**C** The threads on connections and fittings that the fire department will use to connect a hose must be compatible with the fire department threads.

Design documents must specify the type of thread to be used in order to be compatible with the local fire department equipment after consultation and coordination with the fire code official. The criteria typically apply to fire department connections for sprinkler and standpipe systems, standpipe hose connections, yard hydrants and wall hydrants.

The majority of fire departments in the United States use the American National Fire Hose Connection Screw Threads, also commonly known as NST and NS. NFPA 1963 gives the screw thread dimensions and the thread size of threaded

connections, with nominal sizes ranging from 3/4 inch (19 mm) to 6 inches (152 mm) for the NS thread. Although efforts to standardize fire hose threads began after the Boston conflagration in 1872, there are still many different screw threads, some of which give the appearance of compatibility with the NH thread. While NFPA 1963 may be used as a guide, the code does not require that any particular standard be used. Rather, it is important that the fire code official be consulted for the appropriate thread selection. The intent is that the threads match those of the local department identically so that adapters are not required within the fire department's own district.

**903.3.7 Fire department connections.** Fire department connections for *automatic sprinkler systems* shall be installed in accordance with Section 912.

**C** Section 912, to which this section points, provides a comprehensive set of requirements for fire department connections (FDCs), reducing the opportunity for any of its requirements to be overlooked. See the commentary to Section 912.

**903.3.8 Limited area sprinkler systems.** Limited area sprinkler systems shall be in accordance with the standards listed in Section 903.3.1 except as provided in Sections 903.3.8.1 through 903.3.8.5.

**C** The use of limited area sprinkler systems is restricted to cases in which the code requires a limited number of sprinklers to protect a specific hazard or area and not a complete automatic sprinkler system. For example, limited area sprinkler systems may be used to protect areas including, but not limited to, stages, storage and workshop areas, painting rooms, trash rooms and chutes, furnace rooms, kitchens and hazardous exhaust systems, and incidental uses as regulated in IBC Section 509.

**903.3.8.1 Number of sprinklers.** Limited area sprinkler systems shall not exceed six sprinklers in any single *fire area*.

**C** In the 2015 edition of the code, the number of sprinklers allowed on a limited area sprinkler system in a fire area was reduced to only six. In previous editions, up to 19 sprinklers were allowed. This reduced number limits the type of applications for such systems. In the past, 19 sprinklers may have been able to address an entire building, depending on the system design, more easily allowing the use of the domestic water supply.

**903.3.8.2 Occupancy hazard classification.** Only areas classified by NFPA 13 as Light Hazard or Ordinary Hazard Group 1 shall be permitted to be protected by limited area sprinkler systems.

**C** The use of limited area systems is restricted to only the two lowest hazard occupancy classifications in accordance with NFPA 13. Such systems are fairly limited and can only contain six sprinklers per fire area. Because of this, the types of hazards protected should be limited.

**903.3.8.3 Piping arrangement.** Where a limited area sprinkler system is installed in a building with an automatic wet standpipe system, sprinklers shall be supplied by the standpipe system. Where a limited area sprinkler system is installed in a building without an automatic wet standpipe system, water shall be permitted to be supplied by the plumbing system provided that the plumbing system is capable of simultaneously supplying domestic and sprinkler demands.

**C** Two options are provided for how water to the limited area sprinkler system is to be supplied. If the building contains automatic wet standpipes, then the system must be supplied by the standpipe system. If there is no standpipe system available, connection to the domestic water supply is permitted. The water supply must be analyzed to determine if it is sufficient to supply simultaneously both domestic usage and the sprinkler system demand. This will mean looking at peak water use throughout the day. See Section 903.3.5 and the IPC.

**903.3.8.4 Supervision.** Control valves shall not be installed between the water supply and sprinklers unless the valves are of an *approved* indicating type that are supervised or secured in the open position.

**C** No shutoff valves are permitted in the sprinkler system piping unless the valves are specifically approved and are either supervised or secured in the open position. These restrictions increase the likelihood that the sprinkler system will be operational should a fire occur. Valve supervision or securing a valve in the open position are considered equally reliable by this section.

**903.3.8.5 Calculations.** Hydraulic calculations in accordance with NFPA 13 shall be provided to demonstrate that the available water flow and pressure are adequate to supply all sprinklers installed in any single *fire area* with discharge densities corresponding to the hazard classification.

**C** Hydraulic calculations are required to be in accordance with NFPA 13 to demonstrate that the water system is adequate to supply the sprinkler demand in any particular fire area.

**903.3.9 High-rise building floor control valves.** *Approved* supervised indicating control valves shall be provided at the point of connection to the riser on each floor in high-rise buildings.

**C** In high-rise buildings, sprinkler control valves with supervisory initiating devices must be installed at the point of connection to the riser on each floor. Sprinkler control valves on each floor are intended to permit servicing activated systems without impairing the water supply to large portions of the building.

**903.4 Sprinkler system supervision and alarms.** *Automatic sprinkler system* supervision and alarms shall comply with Sections 903.4.1 through 903.4.3.

**C** This section provides the supervision and alarm requirements for automatic sprinkler systems.

**903.4.1 Electronic supervision.** Valves controlling the water supply for *automatic sprinkler systems*, pumps, tanks, water levels and temperatures, critical air pressures and waterflow switches on all *automatic sprinkler systems* shall be electrically supervised by a *listed* fire alarm control unit.

**Exceptions:**

1. *Automatic sprinkler systems* protecting one- and two-family dwellings.
2. Limited area sprinkler systems in accordance with Section 903.3.8, provided that backflow prevention device test valves located in limited area sprinkler system supply piping shall be locked in the open position unless supplying an occupancy required to be equipped with a *fire alarm system*, in which case the backflow preventer valves shall be electrically supervised by a tamper switch installed in accordance with NFPA 72 and separately annunciated.
3. *Automatic sprinkler systems* installed in accordance with NFPA 13R where a common supply main is used to supply both domestic water and the *automatic sprinkler system*, and a separate shutoff valve for the *automatic sprinkler system* is not provided.
4. Jockey pump control valves that are sealed or locked in the open position.
5. Control valves to commercial kitchen hoods, paint spray booths or dip tanks that are sealed or locked in the open position.
6. Valves controlling the fuel supply to fire pump engines that are sealed or locked in the open position.
7. Trim valves to pressure switches in dry, preaction and *deluge sprinkler systems* that are sealed or locked in the open position.
8. Underground key or hub gate valves in roadway boxes.

**C** The reliability data on automatic sprinkler systems clearly indicate that a closed valve is the leading cause of sprinkler system failure. There are also a number of other critical elements that contribute to successful sprinkler system operation, including, but not limited to, pumps, water tanks and air pressure maintenance devices; therefore, this section requires that the various critical elements that contribute to an available water supply and to the function of the sprinkler system be electrically supervised.

Automatic sprinkler systems in one- and two-family dwellings are typically designed to comply with NFPA 13D, which does not require electrical supervision (see Exception 1).

Limited area sprinkler systems are generally supervised by their connection to the domestic water service, although the use of a supervised indicating valve is permitted. Compliance with Section 903.3.8 means that the alarm provisions of this section are not applicable to limited area systems. Consequently, limited area sprinkler systems do not require local alarms or supervision. Again, electrical supervision is required only if a control valve is installed between the riser control valve and the sprinkler system piping (see Exception 2). Additionally, Exception 2 acknowledges that local water utilities and environmental authorities in many instances require, by local ordinances, that backflow prevention devices be installed in limited-area sprinkler system piping. To make the testing and maintenance of backflow prevention devices easier, test valves are installed on each side of the device. These valves are typically indicating-type valves and can function as shutoff valves for the sprinkler system, thus requiring some level of supervision.

Because these infrequently used valves may be the only feature of protection requiring supervision in occupancies not otherwise required to be equipped with a fire alarm system, the exception permits these valves to be locked in the open position; however, if the occupancy is protected by a fire alarm system, these valves must be equipped with approved valve supervisory devices connected to the fire alarm control panel on a separate (supervisory) zone so that the supervisory signal is transmitted to the designated receiving station. Installation and testing of backflow preventers in sprinkler systems are regulated in Sections 312.11 and 608.17.4 of the IPC.

Similar to limited area sprinkler systems, electrical supervision is not required for NFPA 13R residential combination services when a shutoff valve is not installed (see Exception 3). NFPA 13R sprinkler systems are supervised in that the only way to shut off the sprinkler system is to also shut off the domestic water supply.

The valves discussed in Exceptions 4 through 7 can be sealed or locked in the open position because they do not control the sprinkler system water supply. Exception 8 recognizes that since underground key and hub gate valves are equipped with special keys, they are not required to be supervised or monitored.

**903.4.2 Monitoring.** Alarm, supervisory and trouble signals shall be distinctly different and shall be automatically transmitted to an *approved* supervising station or, where *approved* by the *fire code official*, shall sound an audible signal at a constantly attended location.

**C** Automatic sprinkler systems must be supervised as a means of determining that the system is operational. A valve supervisory switch operating as a normally open or normally closed switch is usually used. NFPA 72 does not permit valve supervisory switches to be connected to the same zone circuit as the waterflow switch unless it is specifically arranged to actuate a signal that is distinctive from the circuit trouble condition signal.

Required sprinkler systems are to be monitored by an approved supervising service to comply with NFPA 72. Types of supervising stations recognized in NFPA 72 include central stations, remote supervising stations or proprietary supervising stations.

A central station is an independent, off-site facility operated and maintained by personnel whose primary business is to furnish, maintain, record and supervise a signaling system. A proprietary system is similar to a central station system; however, a proprietary system is typically an on-site facility monitoring a number of buildings on the same site for the same owner. A remote station system has an alarm signal that is transmitted to a remote location acceptable to the authority having jurisdiction and that is attended 24 hours a day. The receiving equipment is usually located at a fire station, police station, regional emergency communications center or telephone answering service. An alternative use to the three previous supervising methods is an audible signal that can be transmitted to a constantly attended location approved by the fire code official.

**903.4.3 Alarms.** An *approved* audible and visual sprinkler waterflow alarm device, located on the exterior of the building in an *approved* location, shall be connected to each *automatic sprinkler system*. Such sprinkler waterflow alarm devices shall be activated by water flow equivalent to the flow of a single sprinkler of the smallest orifice size installed in the system. Where a waterflow switch is required by Section 903.4.1 to be electrically supervised, such sprinkler waterflow alarm devices shall be powered by a fire alarm control unit or, where provided, a *fire alarm system*. Where a *fire alarm system* is provided, actuation of the *automatic sprinkler system* shall actuate the building *fire alarm system*.

**Exception:** *Automatic sprinkler systems* protecting one- and two-family dwellings.

**C** The audible and visual sprinkler waterflow alarm, sometimes referred to as the “outside ringer” or “water-motor gong,” sounds when the sprinkler system has activated. The alarm device may be electrically operated or it may be a true water-motor gong operated by a paddle-wheel-type attachment to the sprinkler system riser that responds to the flow of water in the piping. Though no longer the alarm device of choice, water-motor gongs do have the advantage of not being subject to power failures within or outside the protected building (see Sections 6.9 and 8.17 of NFPA 13 for further information on these devices). The alarm must be installed on the exterior of the building in a location approved by the fire code official. This location is often in close proximity to the fire department connection (FDC), serving a collateral function of helping the responding fire apparatus engineer more promptly locate the FDC.

The alarm is not intended to be an evacuation alarm. The requirement is also not intended to be an indirect requirement for a fire alarm system. Unless a fire alarm system is required by some other code provision, only the exterior alarm device is required. However, when a fire alarm system is installed, the sprinkler system must also be interconnected with the fire alarm system so that when the sprinkler system actuates, it sounds the evacuation alarms required for the fire alarm system.

The primary purpose of the exterior alarm is to notify people outside the building that the sprinkler system is in operation. Originally, it was to act as a supplemental alert so that passersby could notify the fire department of the condition. However, because the code now requires electronic supervision of sprinkler systems, that function is mostly moot. The exterior notification now primarily serves the function of alerting the arriving fire department of which building or sprinkler system is in operation before staging firefighting activities for the building.

The exception includes automatic sprinkler systems in one- and two-family dwellings that are typically designed to comply with NFPA 13D, which does not require electrical supervision or alarms.

**903.5 Inspection, testing and maintenance.** *Automatic sprinkler systems* shall be inspected, tested and maintained in accordance with Section 901.

**C** Section 901 contains requirements for the inspection, testing and maintenance of automatic sprinkler systems. Acceptance tests are necessary to verify that the system performs as intended by design and by the code. Periodic inspection, testing and maintenance are essential to verify that the level of protection designed into the building will be operational whenever a fire occurs. Water-based extinguishing systems must be tested and maintained as required by NFPA 25.

**903.6 Where required in existing buildings and structures.** An *automatic sprinkler system* shall be provided in existing buildings and structures where required in Chapter 11.

**C** Chapter 11 of this code, specifically Section 1103.5, requires sprinklers for existing buildings used for the manufacture and storage of pyroxylin plastics in certain Group A-2 occupancies and in existing Group I-2 occupancies. See the commentary to Sections 1103.5.1 through 1103.5.4 for more detail on the retroactive sprinkler requirements.

## SECTION 904—ALTERNATIVE AUTOMATIC FIRE-EXTINGUISHING SYSTEMS

**904.1 General.** *Automatic fire-extinguishing systems*, other than *automatic sprinkler systems*, shall be designed, installed, inspected, tested and maintained in accordance with the provisions of this section and the applicable referenced standards.

**C** Section 904 covers alternative fire-extinguishing systems that use extinguishing agents other than water. Alternative automatic fire-extinguishing systems include wet-chemical, dry-chemical, foam, carbon dioxide, halon and clean-agent sup-

pression systems. In addition to the provisions of Section 904, the indicated referenced standards include specific installation, maintenance and testing requirements for all systems.

**904.1.1 Certification of service personnel for fire-extinguishing equipment.** Service personnel providing or conducting maintenance on *automatic fire-extinguishing systems*, other than *automatic sprinkler systems*, shall possess a valid certificate issued by an *approved governmental agency*, or other *approved organization* for the type of system and work performed.

**C** Maintenance of fire protection systems and devices are minimum Chapter 9 requirements. Fire protection systems, like other technologies, have advanced new designs that require a clear understanding of their construction and maintenance. To ensure that systems and devices are properly maintained, the code now requires individuals performing these activities be certified. Certification must be issued by an approved organization or governmental agency.

These provisions align the code with NFPA standards governing the design, construction, inspection and maintenance of alternative automatic fire-extinguishing systems and portable fire extinguishers. The personnel qualification requirements specified by the applicable NFPA standards for the various alternative automatic fire-extinguishing systems are summarized in Commentary Figure 904.1.1.

Several organizations offer certifications that will confirm to code officials that individuals are qualified to perform maintenance on alternative automatic fire-extinguishing systems and portable fire extinguishers. The International Code Council® (ICC®), in conjunction with the National Association of Fire Equipment Distributors (NAFED), offers certifications to individuals who service and inspect portable fire extinguishers and preengineered alternative fire-extinguishing systems protecting commercial kitchen hoods or industrial equipment such as spray booths. The National Institute for Certification in Engineering Technologies (NICET) offers four progressive levels of certification for individuals who design, erect and inspect special hazard suppression systems. Special hazard suppression systems include all of the various systems recognized in Section 904 and Commentary Figure 904.1.1. Finally, most manufacturers of these systems offer corporate certifications to individuals who will design, install and inspect them. Such certifications are beneficial to code officials because they confirm the individuals are qualified to install or maintain fire protection equipment produced by a specific manufacturer.

**Commentary Figure 904.1.1—Personnel Qualification Requirements by Type of Fire-Extinguishing Agent**

| NFPA STANDARD<br>NUMBER & EDITION | SUBJECT        | SECTION | PERSONNEL QUALIFICATION REQUIREMENT                                                                                                                                                                                                                              |
|-----------------------------------|----------------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12, 2011                          | Carbon dioxide | 4.4.1.1 | Specifications for carbon dioxide fire-extinguishing systems shall be prepared under the supervision of a person fully experienced and qualified in the design of carbon dioxide extinguishing systems and with the advice of the authority having jurisdiction. |
| 12A, 2009                         | Halon 1301     | 5.1.2.1 | Plans and calculations shall be submitted for approval to the authority having jurisdiction before installation begins. Their preparation shall be entrusted only to persons fully experienced and qualified in the design of Halon 1301 extinguishing systems.  |
| 17, 2013                          | Dry chemical   | 1.6     | Only trained persons shall be considered competent to design or lay out, install and service dry-chemical systems.                                                                                                                                               |
| 17A, 2013                         | Wet chemical   | 1.7     | Only trained persons shall be considered competent to design or lay out, install, and service wet chemical systems.                                                                                                                                              |
| 2001, 2012                        | Clean agent    | 7.6.1   | All persons who could be expected to inspect, test, maintain, or operate fire-extinguishing systems shall be thoroughly trained and kept thoroughly trained in the functions they are expected to perform.                                                       |

**904.2 Where permitted.** *Automatic fire-extinguishing systems* installed as an alternative to the required *automatic sprinkler systems* of Section 903 shall be *approved* by the *fire code official*.

**C** One of the main considerations in selecting an extinguishing agent should be the compatibility of the agent with the hazard. The fire code official is responsible for approving an alternative extinguishing agent. The approval should be based on the compatibility of the agent with the hazard and the potential effectiveness of the agent to suppress a fire involving the hazards present.

**904.2.1 Restriction on using automatic sprinkler system exceptions or reductions.** *Automatic fire-extinguishing systems* shall not be considered alternatives for the purposes of exceptions or reductions allowed for *automatic sprinkler systems* or by other requirements of this code.

**C** Although Section 904.2 allows the use of alternative fire-extinguishing systems with specific approval, this section prohibits the use of such systems as alternatives to reductions or exceptions allowed for automatic sprinkler systems throughout the code. Therefore, the building will not be considered as equipped throughout with an automatic sprinkler system when using systems such as automatic water mist or other alternative systems. See Commentary Figure 904.2.1.

**Commentary Figure 904.2.1—Selected Examples of Requirements Modified Through Use of Automatic Sprinkler Systems**

| CODE SECTION <sup>a</sup>       | MODIFICATION                                     | NFPA 13                | NPFA 13R            | NFPA 13D         |
|---------------------------------|--------------------------------------------------|------------------------|---------------------|------------------|
| <b>Increases</b>                |                                                  |                        |                     |                  |
| 504.3 and Table 504.3           | Height increase in feet                          | yes                    | yes                 | no               |
| 504.4 and Table 504.4           | Height increase in number of stories             | yes                    | yes                 | no               |
| 506.2 and Table 506.2           | Area increase                                    | yes                    | yes                 | no               |
| 1005.3.1, 1005.3.2 <sup>b</sup> | Egress width                                     | yes                    | yes                 | no               |
| Table 1017.2 <sup>b</sup>       | Travel distance                                  | yes                    | yes                 | no               |
| <b>Rating Reductions</b>        |                                                  |                        |                     |                  |
| Table 508.4                     | Separated occupancies                            | yes                    | no                  | no               |
| 708.3                           | Fire partitions (dwelling units, sleeping units) | yes                    | no                  | no               |
| Table 1020.1 <sup>b</sup>       | Corridor walls                                   | yes                    | yes                 | no               |
| <b>Miscellaneous</b>            |                                                  |                        |                     |                  |
| Tables 307.1(1), 307.1(2)       | Hazardous material increase                      | yes                    | no                  | no               |
| 404.2                           | Atriums                                          | yes                    | no                  | no               |
| 507.4, 507.5                    | Unlimited area buildings                         | yes                    | no                  | no               |
| Table 705.8                     | Allowable area of openings                       | yes                    | no                  | no               |
| 705.8.5                         | Vertical separation of openings                  | yes                    | yes                 | no               |
| 718.3.2                         | Residential floor/ceiling draftstopping          | yes                    | yes <sup>c</sup>    | no               |
| 718.3.3                         | Nonresidential draftstopping                     | yes                    | no                  | no               |
| 718.4.2                         | Groups R-1, R-2 attic draftstopping              | yes                    | yes <sup>c</sup>    | no               |
| 718.4.3                         | Other group draftstopping                        | yes                    | no                  | no               |
| Table 803.9                     | Interior finish                                  | yes                    | yes                 | no               |
| 804.4.2                         | Floor finish                                     | yes                    | yes                 | no               |
| 907.2.1–907.2.10 <sup>b</sup>   | Manual fire alarm system                         | yes<br>(A, B, E, F, M) | yes<br>(R-1, R-2)   | no               |
| 1009.2.1 <sup>b</sup>           | Accessible egress                                | yes                    | yes                 | no               |
| 1028.1 <sup>b</sup>             | Exit discharge                                   | yes                    | yes                 | no               |
| 1406.3                          | Balconies                                        | yes                    | yes <sup>c, d</sup> | yes <sup>c</sup> |

a. Section numbers refer to sections in the code.

b. Section numbers in Chapters 9 and 10 apply to both the IBC and the code.

c. Sprinkler protection must be extended to the affected areas.

d. For additional balcony requirements, see Section 903.3.1.2.1.

**904.2.2 Commercial hood and duct systems.** Each required commercial kitchen exhaust hood and duct system required by Section 606 to have a Type I hood shall be protected with an *approved automatic fire-extinguishing system* installed in accordance with this code.

**C** This section requires an effective suppression system to combat fire on the cooking surfaces of grease-producing appliances and within the hood and exhaust system of a commercial kitchen installation. Type I hoods, including the duct system, must be protected with an approved automatic fire-extinguishing system because they are used for handling grease-laden vapors or smoke, whereas Type II hoods handle fumes, steam, heat and odors. Type I hoods are typically required for commercial food heat-processing equipment, such as deep fryers, griddles, charbroilers, broilers and open burner stoves and ranges. For additional guidance on the requirements for Type I and II hoods, see the commentary to Section 507 of the IMC.

**904.3 Installation.** *Automatic fire-extinguishing systems* shall be installed in accordance with this section.

**C** The installation of automatic fire-extinguishing systems must comply with the requirements of Sections 904.3.1 through 904.3.5 in addition to the installation criteria contained in the referenced standard for the proposed type of alternative extinguishing system.

**904.3.1 Electrical wiring.** Electrical wiring shall be in accordance with NFPA 70.

**C** NFPA 70 regulates the design and installation of electrical systems and equipment. All electrical work must also be in compliance with any specific electrical classifications and conditions contained in the referenced standards for each type of system.

Chapter 27 of the IBC and Section 605 contain provisions that also reference NFPA 70. Those sections also contain additional information that must be applied when addressing electrical issues.

**904.3.2 Actuation.** *Automatic fire-extinguishing systems* shall be automatically actuated and provided with a manual means of actuation in accordance with Section 904.14.1. Where more than one hazard could be simultaneously involved in fire due to their proximity, all hazards shall be protected by a single system designed to protect all hazards that could become involved.

**Exception:** Multiple systems shall be permitted to be installed if they are designed to operate simultaneously.

**C** Section 904.3.2 requires alternative fire-extinguishing systems to be designed for automatic activation. Activation commonly occurs when a heat, fire or smoke detection system activates. In Type I commercial kitchen hoods, Section 904.14 requires a manual and automatic means of activating the fire-extinguishing system. Designing a fire-extinguishing system to operate only upon manual actuation is prohibited by the code and many of the NFPA fire protection system standards.

The requirements for fire-extinguishing system actuation correlate with the requirements of NFPA 17 and NFPA 17A. The requirement prescribes that when a hazard is protected by two or more fire-extinguishing systems, all of the systems must be designed to operate simultaneously. The reason for the revision is that a typical alternative automatic fire-extinguishing system has a limited amount of fire-extinguishing agent. The amount of agent that is available is based on the area or volume of the hazard and the fire behavior of the fuel. Because the amount of agent is limited, the simultaneous operation of all the fire-extinguishing systems ensures that enough agent is applied to extinguish the fire and prevent its spread from the area of origin.

It is fairly common for a single hazard to be protected by two or more alternative automatic fire-extinguishing systems. For example, protection of a spray booth used for the application of flammable finishes using dry chemical commonly requires two or three alternative automatic fire-extinguishing systems. The reason is that many dry-chemical and all wet-chemical systems are preengineered systems. Utilizing listed nozzles, preengineered systems are designed and constructed based on the manufacturer's installation requirements. Because these systems are assembled using listed nozzles and extinguishing agents, one system may not be able to protect the spraying space and exhaust plenum. As a result, two or more systems may be required as a provision of an extinguishing system's listing to protect certain hazards.

Another example is commercial kitchen cooking operations. Consider a flat grill broiler and a deep fat fryer located beneath the same Type I hood. It is quite common for each of these commercial cooking appliances to be protected by separate automatic fire-extinguishing systems. Based on the revision to Section 904.3.2, both extinguishing systems must simultaneously operate in the event a fire involves either of the example appliances.

**904.3.3 System interlocking.** Automatic equipment interlocks with fuel shutoffs, ventilation controls, door closers, window shutters, conveyor openings, smoke and heat vents and other features necessary for proper operation of the fire-extinguishing system shall be provided as required by the design and installation standard utilized for the hazard.

**C** Shutting off fuel supplies will eliminate potential ignition sources in the protected area. Automatic door and window closers and dampers for forced-air ventilation systems are intended to maintain the desired concentration level of the extinguishing agent in the protected area. See the commentary to Section 904.14.2 for information on system interconnections in commercial cooking fire-extinguishing systems.

**904.3.4 Alarms and warning signs.** Where alarms are required to indicate the operation of *automatic fire-extinguishing systems*, distinctive audible, visible alarms and warning signs shall be provided to warn of pending agent discharge. Where exposure to automatic-extinguishing agents poses a hazard to persons and a delay is required to ensure the evacuation of occupants before agent discharge, a separate warning signal shall be provided to alert occupants once agent discharge has begun. Audible signals shall be in accordance with Section 907.5.2.

**C** Safeguards are necessary to prevent injury or death to personnel in areas where the atmosphere will be made hazardous by oxygen depletion due to agent discharge in a confined space. The “where alarms are required” phrase is referring to requirements that will be found in the referenced installation standards indicated in Sections 904.5 through 904.14, as applicable. Predischage alarms that will operate on fire detection system activation must be installed within and at entrances to the affected areas.

Where required by the appropriate installation standard, an extinguishing agent discharge delay feature shall also be provided to allow evacuation of personnel prior to agent discharge. Warning and instructional signs are also to be posted, preferably at the entrances to and within the protected area. See Section 4.5.6.1 of NFPA 12 for additional information on carbon dioxide system alarms, Section 4.3.5 of NFPA 12A for additional information on Halon system alarms and Section 4.3.5 of NFPA 2001 for additional information on clean agent system alarms.

**904.3.5 Monitoring.** Where a building *fire alarm system* is installed, *automatic fire-extinguishing systems* shall be monitored by the building *fire alarm system* in accordance with NFPA 72.

**C** Automatic fire-extinguishing systems need not be electrically supervised unless the building is equipped with a fire alarm system. This section recognizes the fact that a fire alarm system is not required in all buildings. However, because most alternative fire-extinguishing systems require the space to be evacuated before the system is discharged, they are equipped with evacuation alarms. Interconnection of the fire-extinguishing system evacuation alarm with the building evacuation alarm results in an increased level of hazard notification for the occupants in addition to the electrical supervision of the fire-extinguishing system.

**904.4 Inspection and testing.** *Automatic fire-extinguishing systems* shall be inspected and tested in accordance with the provisions of this section prior to acceptance.

**C** The completed installation must be tested and inspected to determine that the system has been installed in compliance with the code and will function as required. Full-scale acceptance tests must be conducted as required by the applicable referenced standard.

**904.4.1 Inspection.** Prior to conducting final acceptance tests, all of the following items shall be inspected:

1. Hazard specification for consistency with design hazard.
2. Type, location and spacing of automatic- and manual-initiating devices.
3. Size, placement and position of nozzles or discharge orifices.
4. Location and identification of audible and visible alarm devices.
5. Identification of devices with proper designations.
6. Operating instructions.

**C** This section identifies those items that need to be verified or visually inspected prior to the final acceptance tests. All equipment should be listed, approved and installed in accordance with the manufacturer’s recommendations.

**904.4.2 Alarm testing.** Notification appliances, connections to fire alarm systems and connections to *approved* supervising stations shall be tested in accordance with this section and Section 907 to verify proper operation.

**C** Components of fire-extinguishing systems related to alarm devices and their supervision must be tested before the system is approved. Alarm devices must be tested to satisfy the requirements of NFPA 72.

**904.4.2.1 Audible and visible signals.** The audibility and visibility of notification appliances signaling agent discharge or system operation, where required, shall be verified.

**C** This section requires verification, upon completion of the system installation, of the audibility and visibility of notification appliances in the area affected by the extinguishing agent discharge of the alternative automatic fire-extinguishing system.

**904.4.3 Monitor testing.** Connections to protected premises and supervising station fire alarm systems shall be tested to verify proper identification and retransmission of alarms from *automatic fire-extinguishing systems*.

**C** Where monitoring of fire-extinguishing systems is required, such as by Section 904.3.5, all connections related to the supervision of the system must be tested to verify they are in proper working order.

**904.5 Wet-chemical systems.** Wet-chemical extinguishing systems shall be installed, maintained, periodically inspected and tested in accordance with NFPA 17A and their listing. Records of inspections and testing shall be maintained.

**C** NFPA 17A contains minimum requirements for the design, installation, operation, testing and maintenance of wet-chemical preengineered extinguishing systems. Equipment that is typically protected with wet-chemical extinguishing systems

includes restaurant, commercial and institutional hoods, plenums, ducts, and associated cooking equipment. Strict compliance with the manufacturer's installation instructions is vital for a viable installation.

Wet-chemical solutions used in extinguishing systems are relatively harmless and there is usually no lasting significant effect on a person's skin, respiratory system or clothing. These solutions may produce a mild, temporary irritation, but the symptoms will usually disappear when contact is eliminated.

This section also specifically requires that records be maintained of inspections and testing to increase the effectiveness of such systems. Without records of such inspections and testing, it is more difficult to determine if the systems will be effective when activated and when future inspection and testing is necessary.

**904.5.1 System test.** Systems shall be inspected and tested for proper operation at 6-month intervals. Tests shall include a check of the detection system, alarms and releasing devices, including manual stations and other associated equipment. Extinguishing system units shall be weighed and the required amount of agent verified. Stored pressure-type units shall be checked for the required pressure. The cartridge of cartridge-operated units shall be weighed and replaced at intervals indicated by the manufacturer.

**C** This section specifies the frequency for inspection and testing of wet-chemical extinguishing systems. The system and its essential components must be inspected and checked every 6 months to determine that the system is in full operating condition.

**904.5.2 Fusible link maintenance.** Fixed temperature-sensing elements shall be maintained to ensure proper operation of the system.

**C** Wet-chemical extinguishing systems are commonly used to protect commercial cooking equipment. The fusible metal alloy sensing elements are subject to the accumulation of grease or other contaminants that could affect the operation of the fusible link. The sensing elements must be inspected routinely and replaced as needed.

**904.6 Dry-chemical systems.** Dry-chemical extinguishing systems shall be installed, maintained, periodically inspected and tested in accordance with NFPA 17 and their listing. Records of inspections and testing shall be maintained.

**C** NFPA 17 contains the minimum requirements for the design, installation, testing, inspection, approval, operation and maintenance of dry-chemical extinguishing systems. The fire code official has the authority to approve the type of dry-chemical extinguishing system to be used. NFPA 17 identifies three types of dry-chemical extinguishing systems: total flooding, local application and hand hose-line systems. Only total flooding and local application systems are considered automatic extinguishing systems.

The types of hazards and equipment that can be protected with dry-chemical extinguishing systems include: flammable and combustible liquids and combustible gases; combustible solids, which melt when involved in a fire; electrical hazards, such as transformers or oil circuit breakers; textile operations subject to flash surface fires; ordinary combustibles such as wood, paper or cloth; and restaurant and commercial hoods, ducts and associated cooking appliance hazards, such as deep fat fryers and some plastics, depending on the type of material and configuration.

Total flooding dry-chemical extinguishing systems are used only where there is a permanent enclosure surrounding the hazard that is adequate to enable the required concentration to be built up. The total area of unclosable openings must not exceed 15 percent of the total area of the sides, top and bottom of the enclosure. Consideration must be given to eliminating the probable sources of reignition within the enclosure because the extinguishing action of dry-chemical systems is transient.

Local application of dry-chemical extinguishing systems is to be used for extinguishing fires where the hazard is not enclosed or where the enclosure does not conform to the requirements for total-flooding systems. Local application systems have successfully protected hazards involving flammable or combustible liquids, gases and shallow solids, such as paint deposits.

NFPA 17 also discusses preengineered dry-chemical systems consisting of components designed to be installed in accordance with pretested limitations as tested and labeled by a testing agency. Preengineered systems must be installed within the limitations that have been established by the testing agency and may include total flooding, local application or a combination of both types of systems.

The type of dry chemical used in the extinguishing system is a function of the hazard to be protected. The type of dry chemical used in a system should not be changed unless it has been proven changeable by a testing laboratory, is recommended by the manufacturer of the equipment and is acceptable to the fire code official for the hazard being protected. Additional guidance on the use of various dry-chemical agents can be found in NFPA 17.

This section also specifically requires that records be maintained of inspections and testing to increase the effectiveness of such systems. Without records of such inspections and testing, it is more difficult to determine if the systems will be effective when activated and when future inspection and testing is necessary.

**904.6.1 System test.** Systems shall be inspected and tested for proper operation at 6-month intervals. Tests shall include a check of the detection system, alarms and releasing devices, including manual stations and other associated equipment. Extinguishing

system units shall be weighed, and the required amount of agent verified. Stored pressure-type units shall be checked for the required pressure. The cartridge of cartridge-operated units shall be weighed and replaced at intervals indicated by the manufacturer.

**C** This section specifies the frequency for inspection and testing of dry-chemical extinguishing systems. The system and its essential components must be inspected and checked every 6 months to determine that the system is in full operating condition.

**904.6.2 Fusible link maintenance.** Fixed temperature-sensing elements shall be maintained to ensure proper operation of the system.

**C** Dry-chemical extinguishing systems are commonly used to protect commercial cooking systems and other hazardous use conditions. In these applications, the fusible metal alloy sensing elements are subject to the accumulation of grease or other contaminants that could affect the operation of the fusible link. The sensing elements must be inspected routinely and replaced as needed.

**904.7 Foam systems.** Foam-extinguishing systems shall be installed, maintained, periodically inspected and tested in accordance with NFPA 11 and their listing. Records of inspections and testing shall be maintained.

**C** NFPA 11 covers the characteristics of foam-producing materials used for fire protection and the requirements for design, installation, operation, testing and maintenance of equipment and systems, including those used in combination with other fire-extinguishing agents. The minimum requirements are covered for flammable and combustible liquid hazards in local areas within buildings, storage tanks and indoor and outdoor processing areas.

Low-expansion foam is defined as an aggregation of air-filled bubbles resulting from the mechanical expansion of a foam solution by air with a foam-to-solution volume ratio of less than 20:1. It is most often used to protect flammable and combustible liquid hazards. Also, low-expansion foam may be used for heat radiation protection. Combined-agent systems involve the application of low-expansion foam to a hazard simultaneously or sequentially with dry-chemical powder.

NFPA 11 gives minimum requirements for the installation, design, operation, testing and maintenance of medium- and high-expansion foam systems. Medium-expansion foam is defined as an aggregation of air-filled bubbles resulting from the mechanical expansion of a foam solution by air or other gases with a foam-to-solution volume ratio of 20:1 to 200:1. High-expansion foam has a foam-to-solution volume ratio of 200:1 to approximately 1,000:1.

Medium-expansion foam may be used on solid fuel and liquid fuel fires where some degree of in-depth coverage is necessary (for example, for the total flooding of small, enclosed or partially enclosed volumes, such as engine test cells, transformer rooms, etc.). High-expansion foam is most suitable for filling volumes in which fires exit at various levels. For example, high-expansion foam can be used effectively against high-rack storage fires in enclosures such as in underground passages, where it may be dangerous to send personnel to control fires involving liquefied natural gas (LNG) and liquefied petroleum gas (LP-gas), and to provide vapor dispersion control for LNG and ammonia spills. High-expansion foam is particularly suited for indoor fires in confined spaces, since it is highly susceptible to wind and lack-of-confinement effects.

NFPA 11 also contains the minimum requirements for open-head deluge-type foam-water sprinkler systems and foam-water spray systems. The systems are especially applicable to the protection of most flammable liquid hazards and have been used successfully to protect aircraft hangars and truck loading racks.

This section also specifically requires that records be maintained of inspections and testing to increase the effectiveness of such systems. Without records of such inspections and testing, it is more difficult to determine if the systems will be effective when activated and when future inspection and testing is necessary.

**904.7.1 System test.** Foam-extinguishing systems shall be inspected and tested at intervals in accordance with NFPA 25.

**C** Although Section 904.7.1 references NFPA 25 as the standard for the inspection and testing of foam-extinguishing systems, NFPA 25 is limited to water-based extinguishing systems. NFPA 25 technically addresses only foam-water sprinkler systems as specified in NFPA 11. It should be consulted for inspection and testing intervals for low-, medium- and high-expansion foam systems.

As with other alternative fire-extinguishing systems, the inspection and testing of foam systems is necessary to determine that the system is fully operational. In addition to general maintenance of equipment, the condition of the foam concentrate, and its storage tanks or containers should be inspected at least once a year to verify adequate quality. The desired concentration of the foam concentrate in a stagnant storage situation may deteriorate over time.

**904.8 Carbon dioxide systems.** Carbon dioxide extinguishing systems shall be installed, maintained, periodically inspected and tested in accordance with NFPA 12 and their listing. Records of inspections and testing shall be maintained.

**C** NFPA 12 provides minimum requirements for the design, installation, testing, inspection, approval, operation and maintenance of carbon dioxide extinguishing systems.

Carbon dioxide extinguishing systems are useful in extinguishing fires in specific hazards or equipment in occupancies where an inert electrically nonconductive medium is essential or desirable and where cleanup of other extinguishing

agents, such as dry-chemical residue, presents a problem. Carbon dioxide systems have satisfactorily protected the following: flammable liquids; electrical hazards, such as transformers, oil switches, rotating equipment and electronic equipment; engines using gasoline and other flammable liquid fuels; ordinary combustibles, such as paper, wood and textiles; and hazardous solids.

The fire code official has the authority to approve the type of carbon dioxide system to be installed. NFPA 12 defines four types of carbon dioxide systems: total flooding, local application, hand hose lines and standpipe and mobile supply systems. Only total flooding and local application systems are automatic suppression systems.

Total-flooding systems may be used where there is a permanent enclosure around the hazard that is adequate to allow the required concentration to be built up and maintained for the required period of time, which varies for different hazards. Examples of hazards that have been successfully protected by total-flooding systems include rooms, vaults, enclosed machines, ducts, ovens, and containers and their contents.

Local application systems may be used for extinguishing surface fires in flammable liquids, gases and shallow solids where the hazard is not enclosed or the enclosure does not conform to the requirements for a total-flooding system. Examples of hazards that have been successfully protected by local application systems include dip tanks, quench tanks, spray booths, oil-filled electric transformers and vapor vents.

This section also specifically requires that records be maintained of inspections and testing to increase the effectiveness of such systems. Without records of such inspections and testing, it is more difficult to determine if the systems will be effective when activated and when future inspection and testing is necessary.

**904.8.1 System test.** Systems shall be inspected and tested for proper operation at 12-month intervals.

**C** To determine adequate operation, carbon dioxide systems must be inspected and tested at least once a year.

**904.8.2 High-pressure cylinders.** High-pressure cylinders shall be weighed and the date of the last hydrostatic test shall be verified at six-month intervals. Where a container shows a loss in original content of more than 10 percent, the cylinder shall be refilled or replaced.

**C** Because of the potential for unobserved leakage of high-pressure cylinders, they need to be weighed semiannually to verify the concentration level is always within at least 10 percent of the original content.

**904.8.3 Low-pressure containers.** The liquid-level gauges of low-pressure containers shall be observed at 1-week intervals. Where a container shows a content loss of more than 10 percent, the container shall be refilled to maintain the minimum gas requirements.

**C** A weekly visual observation of the liquid-level gauges of low-pressure containers is required to verify that there has been no significant leakage.

**904.8.4 System hoses.** System hoses shall be examined at 12-month intervals for damage. Damaged hoses shall be replaced or tested. At 5-year intervals, all hoses shall be tested.

**C** The maintenance of system hoses is essential to ensuring the reliability of their use in an emergency. Although system hoses need to be visually checked on an annual basis only, a complete pressure test as indicated in Section 904.8.4.1 must be performed every 5 years.

**904.8.4.1 Test procedure.** Hoses shall be tested at not less than 2,500 pounds per square inch (psi) (17 238 kPa) for high-pressure systems and at not less than 900 psi (6206 kPa) for low-pressure systems.

**C** Every 5 years, system hoses for both high-pressure/low-pressure systems must be pressure tested to verify they are still in proper operating condition. The test typically involves filling the hose with water. The hose is then pressurized at the desired test pressure for at least 1 minute to observe any potential distortions or leakage in the hose. All hose assemblies that do not pass the test should be marked, destroyed and replaced with new hose assemblies. Hose assemblies that pass the test should be marked, dated and returned to service.

**904.8.5 Auxiliary equipment.** Auxiliary and supplementary components, such as switches, door and window releases, interconnected valves, damper releases and supplementary alarms, shall be manually operated at 12-month intervals to ensure that such components are in proper operating condition.

**C** The effectiveness of the carbon dioxide extinguishing system is also dependent on the operation of its auxiliary components. These components must also be manually operated at least once a year.

**904.9 Halon systems.** Halogenated extinguishing systems shall be installed, maintained, periodically inspected and tested in accordance with NFPA 12A and their listing. Records of inspections and testing shall be maintained.

**C** NFPA 12A contains minimum requirements for the design, installation, testing, inspection, approval, operation and maintenance of Halon 1301 extinguishing systems. Halon 1301 fire-extinguishing systems are useful in specific hazards, equipment or occupancies where an electrically nonconductive medium is essential or desirable and where cleanup of other extinguishing agents presents a problem.

Halon 1301 systems have satisfactorily protected gaseous and liquid flammable materials; electrical hazards, such as transformers, oil switches and rotating equipment; engines using gasoline and other flammable fuels; ordinary combustibles, such as paper, wood and textiles; and hazardous solids. Halon 1301 systems have also satisfactorily protected electronic computers, data processing equipment and control rooms.

The fire code official has the authority to approve the type of halogenated extinguishing system to be installed. NFPA 12A defines two types of halogenated extinguishing systems: total flooding and local application. Total-flooding systems may be used where there is a fixed enclosure around the hazard that is adequate to enable the required halon concentration to be built up and maintained for the required period of time to enable the effective extinguishing of the fire. Total-flooding systems may provide fire protection for rooms, vaults, enclosed machines, ovens, containers, storage tanks and bins.

Local application systems are used where there is not a fixed enclosure around the hazard or where the fixed enclosure around the hazard is not adequate to enable an extinguishing concentration to be built up and maintained in the space. Hazards that may be successfully protected by local application systems include dip tanks, quench tanks, spray booths, oil-filled electric transformers and vapor vents.

Two other considerations in selecting the proper extinguishing system are ambient temperature and the personnel hazards associated with the agent. The ambient temperature of the enclosure for a total-flooding system must be above 70°F (21°C) for Halon 1301 systems. Special consideration must also be given to the use of Halon systems when the temperatures are in excess of 900°F (482°C) because halon will readily decompose at such temperatures and the products of decomposition can be extremely irritating if inhaled, even in small amounts.

Halon 1301 total-flooding systems must not be used in concentrations greater than 10 percent in normally occupied areas. Where personnel cannot vacate the area within 1 minute, Halon 1301 total-flooding systems must not be used in normally occupied areas with concentrations greater than 7 percent. Halon 1301 total-flooding systems may be used with concentrations of up to 15 percent if the area is not normally occupied and the area can be evacuated within 30 seconds.

The use of halogenated extinguishing systems has become a concern with respect to the potential environmental effects of halon. These fire-extinguishing agents have been identified as a source of emissions, resulting in the depletion of the stratospheric ozone layer and, in accordance with the Montreal Protocol, the ceasing of its production in January 1994. Therefore, the supply of halon is limited and new supplies of halogenated extinguishing agents will not be available in the future. Existing supplies of halon can, however, continue to be used in existing, undischarged systems or to recharge discharged systems. This newfound need for halon supplies has given rise to new industries geared to the ranking, recycling and reclamation of existing halon supplies. Alternative “clean agent” extinguishing agents have been developed to replace halogenated agents (see Section 904.10).

This section also specifically requires that records be maintained of inspections and testing to increase the effectiveness of such systems. Without records of such inspections and testing, it is more difficult to determine if the systems will be effective when activated and when future inspection and testing is necessary.

**904.9.1 System test.** Systems shall be inspected and tested for proper operation at 12-month intervals.

**C** To determine proper operation, Halon systems must be inspected and tested at least once a year.

**904.9.2 Containers.** The extinguishing agent quantity and pressure of containers shall be checked at 6-month intervals. Where a container shows a loss in original weight of more than 5 percent or a loss in original pressure (adjusted for temperature) of more than 10 percent, the container shall be refilled or replaced. The weight and pressure of the container shall be recorded on a tag attached to the container.

**C** Because of the potential for unobserved leakage of the Halon containers, they must be checked at least semiannually to verify the original weight and pressure are within the designated tolerances. When necessary, containers must be refilled or replaced when the desired levels are not maintained. The containers should also be checked for evidence of corrosion or mechanical damage.

**904.9.3 System hoses.** System hoses shall be examined at 12-month intervals for damage. Damaged hoses shall be replaced or tested. At 5-year intervals, all hoses shall be tested.

**C** Maintenance of system hoses is essential to ensuring the reliability of their use in an emergency. System hoses need to be visually checked annually. A complete pressure test as indicated in Section 904.9.3.1 must be performed every 5 years.

**904.9.3.1 Test procedure.** For Halon 1301 systems, hoses shall be tested at not less than 1,500 psi (10 343 kPa) for 600 psi (4137 kPa) charging pressure systems and not less than 900 psi (6206 kPa) for 360 psi (2482 kPa) charging pressure systems. For Halon 1211 hand-hose line systems, hoses shall be tested at 2,500 psi (17 238 kPa) for high-pressure systems and 900 psi (6206 kPa) for low-pressure systems.

**C** Every 5 years, system hoses must be pressure tested to verify they are still in proper operating condition. This section specifies the test pressure for the various types of Halon systems. The pressure test is intended to check for any potential distortion or leaking in the hose (see commentary, Section 904.8.4.1).

**904.9.4 Auxiliary equipment.** Auxiliary and supplementary components, such as switches, door and window releases, interconnected valves, damper releases and supplementary alarms, shall be manually operated at 12-month intervals to ensure such components are in proper operating condition.

**C** The effectiveness of halogenated extinguishing systems is also dependent on the operation of its auxiliary components. These components must be manually operated at least once a year.

**904.10 Clean-agent systems.** Clean-agent fire-extinguishing systems shall be installed, maintained, periodically inspected and tested in accordance with NFPA 2001 and their listing. Records of inspections and testing shall be maintained.

**C** NFPA 2001 contains minimum requirements for the design, installation, testing, inspection and operation of clean-agent fire-extinguishing systems. A clean agent is an electrically nonconducting suppression agent that is volatile or gaseous at discharge and does not leave a residue on evaporation. Clean-agent fire-extinguishing systems are installed in locations that are enclosed and have openings in the protected area that can be sealed on activation of the alarm to provide effective clean-agent concentrations. A clean-agent fire-extinguishing system should not be installed in locations that cannot be sealed unless testing has shown that adequate concentrations can be developed and maintained.

The two categories of clean agents are halocarbon compounds and inert gas agents. Halocarbon compounds include bromine, carbon, chlorine, fluorine, hydrogen and iodine. Halocarbon compounds suppress fire by a combination of breaking the chemical chain reaction of the fire, reducing the oxygen supporting the fire and reducing the ambient temperature of the fire origin to reduce the propagation of the fire. Inert gas agents contain primary components consisting of helium, neon, argon or a combination of these. Inert gases work by reducing the oxygen concentration around the fire origin to a level that does not support combustion.

Clean-agent fire-extinguishing systems were developed in response to the demise of halon as an acceptable fire-extinguishing agent because of its harmful effect on the environment. Although the original hope for a halon substitute was that these new clean agents could be directly and proportionally substituted for halon agents in existing systems (drop-in replacements), research has shown that clean agents are less efficient in extinguishing fires than are the halons they were intended to replace and require approximately 60 percent more agent by weight and volume in storage to do the same job. Additionally, the physical and chemical characteristics of clean agents differ sufficiently from halon to require different nozzles in addition to the need for larger storage vessels. Existing piping systems should be salvaged for use with clean agents only if they are carefully evaluated and determined to be hydraulically compatible with the flow characteristics of the new agent.

This section also relies on strict adherence to the system manufacturer's design and installation instructions for code compliance. As with many of the alternative fire suppression systems covered in this chapter, clean-agent systems are, for the most part, subjected by their manufacturers to a testing and listing program conducted by an approved testing agency. In such testing and listing programs, the clean agent is listed for use with specific equipment and equipment is listed for use with specific clean agents. The resultant listings include reference to the manufacturer's installation manuals, thereby giving the fire code official another valuable resource for reviewing and approving clean-agent systems.

Although clean agents have found a limited market for local application uses, such as a replacement for Halon 1211 in portable fire extinguishers, their primary application is in total-flooding systems, and they are available in both engineered and preengineered configurations.

Engineered clean-agent systems are specifically designed for protection of a particular hazard, whereas preengineered systems are designed to operate within predetermined limitations up to the noted maximums, thus allowing broader applicability to a variety of hazard applications.

Total-flooding systems are used where there is a fixed enclosure around the hazard that is adequate to enable the required clean-agent concentration to build up and be maintained within the space long enough to extinguish the fire. Such applications can include vaults, ovens, containers, tanks, computer rooms, paint lockers or enclosed machinery. In selecting the clean agent to be used in a given application, careful consideration must be given to whether the protected area is a normally occupied space, because different agents have different levels of concentration at which they may be a health hazard to occupants of the area.

The fire code official has the authority to approve the type of clean-agent system to be installed and should become familiar with the unique characteristics and hazards of clean-agent extinguishing systems using all available resources on the subject.

This section also specifically requires that records be maintained of inspections and testing to increase the effectiveness of such systems. Without records of such inspections and testing, it is more difficult to determine if the systems will be effective when activated and when future inspection and testing is necessary.

**904.10.1 System test.** Systems shall be inspected and tested for proper operation at 12-month intervals.

**C** To determine proper operation, all clean-agent systems must be inspected and tested at least once a year.

**904.10.2 Containers.** The extinguishing agent quantity and pressure of the containers shall be checked at 6-month intervals. Where a container shows a loss in original weight of more than 5 percent or a loss in original pressure, adjusted for temperature, of more than 10 percent, the container shall be refilled or replaced. The weight and pressure of the container shall be recorded on a tag attached to the container.

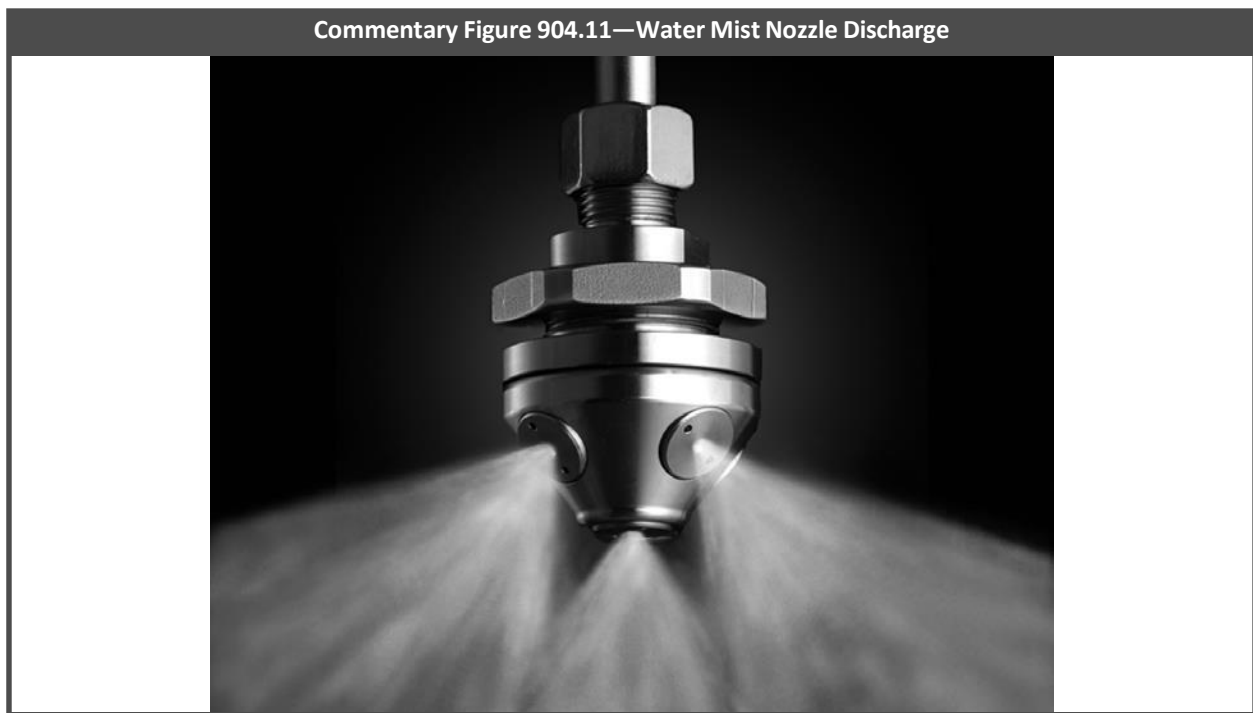
**C** Because of the potential for unobserved leakage of the clean-agent containers, they must be checked at least semi-annually to verify the original weight and pressure are within the designated tolerances. When necessary, containers must be refilled or replaced when the desired levels are not maintained.

**904.10.3 System hoses.** System hoses shall be examined at 12-month intervals for damage. Damaged hoses shall be replaced or tested. All hoses shall be tested at 5-year intervals.

**C** The maintenance of system hoses is essential to ensuring their reliability in an emergency. System hoses must be visually checked annually. A complete pressure test must be performed every 5 years.

**904.11 Automatic water mist systems.** *Automatic water mist systems* shall be permitted in applications that are consistent with the applicable listing or approvals and shall comply with Sections 904.11.1 through 904.11.3.

**C** This section provides the ability to use automatic water mist systems in specific applications. These installations are required to be consistent with the listings or approvals to which such systems have been tested. See Commentary Figure 904.11 for a picture showing the discharge of a water mist nozzle.



**904.11.1 Design and installation requirements.** *Automatic water mist systems* shall be designed and installed in accordance with Sections 904.11.1.1 through 904.11.1.4.

**C** The subsections that follow provide the various installation and design requirements related to water mist systems. This relates to the systems themselves and providing a reliable water supply.

**904.11.1.1 General.** *Automatic water mist systems* shall be designed and installed in accordance with NFPA 750 and the manufacturer's instructions.

**C** This section simply provides reference to the subsections with the detailed requirements related to the design and installation of water mist systems.

**904.11.1.2 Actuation.** *Automatic water mist systems* shall be automatically actuated.

**C** If a water mist system is used, it is required to be automatically activated.

**904.11.1.3 Water supply protection.** Connections to a potable water supply shall be protected against backflow in accordance with the *International Plumbing Code*.

**C** The water supply quality is more critical with automatic water mist systems than automatic sprinkler systems. Backflow prevention is required as it would be for water supply for automatic sprinkler systems.

**904.11.1.4 Secondary water supply.** Where a secondary water supply is required for an *automatic sprinkler system*, an *automatic water mist system* shall be provided with an *approved* secondary water supply.

**C** Since these systems work to protect buildings and spaces in a fashion similar to a sprinkler system and possibly as an alternative to automatic sprinkler systems, the water supply must be appropriate. If an automatic sprinkler system is required to have a secondary water supply in accordance with IBC Section 403.3.3, then the water mist system must also provide an approved secondary water supply.

**904.11.2 Water mist system supervision and alarms.** Supervision and alarms shall be provided as required for *automatic sprinkler systems* in accordance with Section 903.4.1.

**C** This section plays a similar role to an automatic sprinkler system with regard to supervision and alarms. It is critical to make sure valves are open. In addition, an alarm to notify of an activation must be provided as it is for a sprinkler system. Note that this is not intended to be an occupant notification system. See the commentary to Section 903.4.1.

**904.11.2.1 Monitoring.** Monitoring shall be provided as required for *automatic sprinkler systems* in accordance with Section 903.4.2.

**C** See the commentary to Sections 903.4.2 and 904.11.2.

**904.11.2.2 Alarms.** Alarms shall be provided as required for *automatic sprinkler systems* in accordance with Section 903.4.3.

**C** Again, this is not intended to be occupant notification but to simply alert someone that the system has activated. See the commentary to Sections 903.4.3 and 904.11.2.

**904.11.2.3 Floor control valves.** Floor control valves shall be provided as required for *automatic sprinkler systems* in accordance with Section 903.3.9.

**C** See the commentary to Section 903.3.9.

**904.11.3 Testing and maintenance.** *Automatic water mist systems* shall be tested and maintained in accordance with Section 901.6.

**C** Testing and maintenance are critical to the long-term viability of automatic water mist systems. The code focuses not simply on initial installation but also on maintenance.

**904.12 Hybrid fire-extinguishing systems.** Hybrid fire-extinguishing systems shall be designed, installed, maintained, periodically inspected and tested in accordance with NFPA 770. Records of inspection and testing shall be maintained.

**C** This section provides the appropriate reference standard for hybrid fire-extinguishing systems. In addition, it requires that the inspection and testing records be retained to provide for future evidence of compliance.

**904.13 Aerosol fire-extinguishing systems.** Aerosol fire-extinguishing systems shall be installed, maintained, periodically inspected and tested in accordance with NFPA 2010 and their listing.

Such devices and appurtenances shall be *listed* and installed in compliance with manufacturer's instructions.

**C** This section provides the requirements for the use of aerosol fire-extinguishing systems intended for total-flooding applications. As stated, the installation, maintenance, inspection and testing are to conform to NFPA 2010 and the specific system listing. The appropriate standard is UL 2775, *Standard for Fixed Condensed Aerosol Extinguishing System Units*.

Condensed aerosol fire suppression systems can be used as an alternative to other fire suppression systems for the protection of Class A (surface), Class B and Class C hazards. These systems do not use compressed gas cylinders or pressure-rated piping. Generally, they are electrically operated when integrated with IFC-approved fire alarm and releasing control systems and approved/listed releasing panels or are deployed as automatic stand-alone fire-extinguishing units. Typically, these extinguishing units are designed as disposable devices with a minimum 10-year shelf life.

**904.13.1 Maintenance.** Not less than semiannually, an inspection shall be conducted by a trained person to assess whether the system is in working order. Not less than annually, a certified fire suppression contractor having knowledge of and training in the installation, operation and maintenance of the specific fire-extinguishing system shall inspect, test, service and maintain such system in accordance with this section and the manufacturer's specifications and servicing manuals. Records of inspections and testing shall be maintained.

**C** This section provides the maintenance frequency for aerosol fire-extinguishing systems. The systems are required to be inspected by a trained person at least twice a year. Note that the person performing the inspection should be trained by the installing certified fire suppression contractor or system manufacturer's representative. The annual maintenance requirements are to be as prescribed by the specific manufacturer's specifications and servicing manual and must be completed by a fire suppression contractor that is certified by the system manufacturer. In order to provide evidence to the fire code official that the required inspection and testing have been completed, the records are to be retained.

**904.14 Commercial cooking systems.** The *automatic fire-extinguishing system* for commercial cooking systems shall be of a type recognized for protection of commercial cooking equipment and exhaust systems of the type and arrangement protected. Preengineered automatic dry- and wet-chemical extinguishing systems shall be tested in accordance with UL 300 and *listed* and *labeled* for the intended application. Other types of *automatic fire-extinguishing systems* shall be *listed* and *labeled* for specific use as protection for commercial cooking operations. The system shall be installed in accordance with this code, NFPA 96, its listing and the manufac-

turer's installation instructions. *Automatic fire-extinguishing systems* of the following types shall be installed in accordance with the referenced standard indicated, as follows:

1. Carbon dioxide extinguishing systems, NFPA 12.
2. *Automatic sprinkler systems*, NFPA 13.
3. Automatic water mist systems, NFPA 750.
4. Foam-water sprinkler system or foam-water spray systems, NFPA 11.
5. Dry-chemical extinguishing systems, NFPA 17.
6. Wet-chemical extinguishing systems, NFPA 17A.

**Exception:** Factory-built commercial cooking recirculating systems that are tested in accordance with UL 710B and *listed, labeled* and installed in accordance with Section 304.1 of the *International Mechanical Code*.

**C** The history of commercial kitchen exhaust systems shows that the mixture of flammable grease and effluents carried by such systems and the potential for the cooking equipment to act as an ignition source contribute to a higher level of hazard for kitchen exhaust systems than is normally found in many other exhaust systems. Furthermore, fire in a grease exhaust duct can produce temperatures of 2,000°F (1093°C) or more and heat radiating from the duct can ignite nearby combustibles. As a result, the code requires exhaust systems serving grease-producing equipment to include fire suppression to protect the cooking surfaces, hood, filters and exhaust duct to confine a fire to the hood and duct system, thus reducing the likelihood of it spreading to the structure.

In addition to the general requirements of this section, five industry standards are referenced for the installation of fire-extinguishing systems protecting commercial food heat-processing equipment and kitchen exhaust systems. Design professionals should specify and design fire-extinguishing systems to comply with these referenced standards. Only the installation of fire-extinguishing systems is regulated by these references. Where preengineered automatic dry- and wet-chemical extinguishing systems are installed, they must be listed and labeled for the specific cooking operation and tested in accordance with UL 300. Design and construction requirements for the specific types of fire-extinguishing systems are found in the respective sections of the referenced standards.

Regulatory requirements for the approval and installation of fire-extinguishing systems are the same as the approval required for all mechanical equipment and appliances. This section, therefore, requires extinguishing systems to be listed and labeled by an approved agency and installed in accordance with their listing and the manufacturer's installation instructions.

The exception allows factory-built commercial cooking recirculating systems to be installed if they have been tested and listed in accordance with UL 710B. It is important that they be installed in accordance with the manufacturer's installation instructions so that the listing requirements are met. An improper installation could result in hazardous vapors being discharged back into the kitchen.

Commercial cooking recirculating systems consist of an electric cooking appliance and an integral or matched packaged hood assembly. The hood assembly consists of a fan, collection hood, grease filter, fire damper, fire-extinguishing system and air filter, such as an electrostatic precipitator. These systems are tested for fire safety and emissions. The grease vapor (condensable particulate matter) in the effluent at the system discharge is not allowed to exceed a concentration of 5.0 mg/m<sup>3</sup>. Recirculating systems are not used with fuel-fired appliances because the filtering systems do not remove combustion products. Kitchens require ventilation in accordance with IMC Chapter 4.

Although the provisions in Section 904.14 address many of the specifics for commercial kitchens, additional information regarding commercial cooking suppression systems is located in Sections 904.3.2 and 904.3.3. This information is supplemental to that and should be considered together in developing the design for commercial cooking suppression systems.

**904.14.1 Manual system operation.** A manual actuation device shall be located at or near a *means of egress* from the cooking area not less than 10 feet (3048 mm) and not more than 20 feet (6096 mm) from the kitchen exhaust system. The manual actuation device shall be installed not more than 48 inches (1200 mm) nor less than 42 inches (1067 mm) above the floor and shall clearly identify the hazard protected. The manual actuation shall require a maximum force of 40 pounds (178 N) and a maximum movement of 14 inches (356 mm) to actuate the fire suppression system.

**Exceptions:**

1. *Automatic sprinkler systems* shall not be required to be equipped with manual actuation means.
2. Where locating the manual actuation device between 10 feet (3048 mm) and 20 feet (6096 mm) from the cooking area is not feasible, the *fire code official* is permitted to accept a location at or near a *means of egress* from the cooking area, where the manual actuation device is unobstructed and in view from the *means of egress*.

**C** The manual device, usually a pull station, mechanically activates the suppression system. The typical system uses a mechanical circuit of cables under tension to hold the system in the armed (cocked) mode. Melting of a fusible link or actuation of a manual pull station causes the cable to lose tension, which, in turn, starts the discharge of the suppression agent.

The manual actuation device must be readily and easily usable by the building occupants; therefore, the device must not require excessive force or range of movement to cause actuation.

In order to allow the actuation device to be used most effectively, the specified mounting height is intended to be consistent with the NFPA 17A standards and be handicapped accessible. This includes the requirement to identify the actuation device with the hazard protected. Where multiple kitchen appliances are provided, properly identifying which device relates to which appliance is very important. Required signage should be readily visible in the hazard area and capable of conveying information quickly and concisely.

Exception 1 clarifies that manual actuation is not required for automatic sprinkler systems because the typical system design will employ closed heads and wet system piping. A manual actuation valve would serve no purpose because sprinkler heads are already supplied with pressurized water and will discharge water only when the individual fusible elements open the heads.

Exception 2 allows the fire code official to accept a location for the manual actuation device that may be closer or further than the 10- to 20-foot range prescribed. This can be applied when strict conformance of these distances cannot be readily achieved. In some instances, the cooking appliance is a self-contained device that is mobile and can be rolled around to various locations. These appliances have built-in suppression systems, and the manual actuation device is mounted on the appliance. In other instances, the kitchen is so large that there is no available wall space within 20 feet of the cooking area for the mounting of the actuation device. It is useful to look at other code provisions that address protection for cooking appliances. First, there is an allowance for Class K extinguishers to be placed up to 30 feet from commercial cooking equipment in Section 906.1. It is reasonable to assume that the location of the Class K extinguisher with the manual actuation device of automatic suppression may be advantageous. Second, the 2017 edition of NFPA 96 was changed to entirely remove the distance range of 10 to 20 feet. Instead, the 2021 edition of NFPA 96, Section 10.5.1.1, reads as follows: "At least one manual actuation device shall be located in a means of egress or at a location acceptable to the authority having jurisdiction."

**904.14.2 System interconnection.** The actuation of the fire-extinguishing system shall automatically shut down the fuel or electrical power supply to the cooking equipment. The fuel and electrical supply reset shall be manual.

**C** The actuation of any fire suppression system must automatically shut off all sources of fuel or power to all cooking equipment located beneath the exhaust hood protected by the suppression system. This requirement is intended to shut off all heat sources that could reignite or intensify a fire. Shutting off a fuel and power supply to cooking appliances will eliminate an ignition source and allow the cooking surfaces to cool down. This shutdown is accomplished with mechanical or electrical interconnections between the suppression system and a shutoff valve or switch located on the fuel or electrical supply.

Common fuel shutoff valves include mechanical-type gas valves and electrical solenoid-type gas valves. Contactor-type switches or shunt-trip circuit breakers can be used for electrically heated appliances. The fuel or electric source must not be automatically restored after the suppression system has been actuated.

Chemical-type fire-extinguishing systems discharge for only a limited time and can discharge only once before recharge and reset; therefore, precautions must be taken to prevent a fire from reigniting. After a fire is detected and the initial suppressant discharge begins, the fuel and power supply will be locked out, thereby preventing the operation of the appliances until all systems are again ready for operation. Fuel and power supply shutoff must be manually restored by resetting a mechanical linkage or holding/latching-type circuit.

**904.14.3 Carbon dioxide systems.** Where carbon dioxide systems are used, there shall be a nozzle at the top of the ventilating duct. Additional nozzles that are symmetrically arranged to give uniform distribution shall be installed within vertical ducts exceeding 20 feet (6096 mm) and horizontal ducts exceeding 50 feet (15 240 mm). Dampers shall be installed at either the top or the bottom of the duct and shall be arranged to operate automatically upon activation of the fire-extinguishing system. Where the damper is installed at the top of the duct, the top nozzle shall be immediately below the damper. Automatic carbon dioxide fire-extinguishing systems shall be sufficiently sized to protect all hazards venting through a common duct simultaneously.

**C** This section states specific design requirements for nozzle locations, dampers and ducts for carbon dioxide extinguishing systems that may be used to protect commercial cooking systems. These requirements are intended to supersede similar, more general provisions in NFPA 12. Because carbon dioxide (CO<sub>2</sub>) is a gaseous suppressant, dampers are required in the ductwork to define the atmosphere where the fire event would be. A specific concentration of CO<sub>2</sub> is necessary and dampers are required to define and contain the suppressant. The discharge cools exposed surfaces in addition to depriving the fire of oxygen. Although not mentioned specifically in this section, the applicable provisions of NFPA 12 should also be applied because the system is a CO<sub>2</sub> system as referenced in Section 904.8.

**904.14.3.1 Ventilation system.** Commercial-type cooking equipment protected by an automatic carbon dioxide extinguishing system shall be arranged to shut off the ventilation system upon activation.

**C** Shutting down the ventilation system upon activation of the CO<sub>2</sub> extinguishing system maintains the desired concentration of carbon dioxide to suppress the fire. Leakage of gas from the protected area should be kept to a minimum. Where leakage is anticipated, additional quantities of carbon dioxide must be provided to compensate for any losses.

**904.14.4 Special provisions for automatic sprinkler systems.** *Automatic sprinkler systems* protecting commercial-type cooking equipment shall be supplied from a separate, indicating-type control valve that is identified. Access to the control valve shall be provided.

**C** This section requires a separate control valve in the water line to the sprinklers protecting the cooking and ventilating system. The additional valve allows the flexibility to shut off the system for repairs or for cleanups after sprinkler discharge without taking the entire system out of service.

**904.14.4.1 Listed sprinklers.** Sprinklers used for the protection of fryers shall be tested in accordance with UL 199E, *listed* for that application and installed in accordance with their listing.

**C** Sprinklers specifically listed for such use must be used when protecting deep fat fryers. These specially listed sprinklers use finer water droplets than standard spray sprinklers. The water spray lowers the temperature below a point where the fire can sustain itself and reduces the possibility of the fire expanding. UL 199E addresses these special sprinklers and includes performance tests for deep fat fryer extinguishment and also deep fat fryer cooking temperature splash. The selection of inappropriate sprinklers for deep fat fryer protection can increase the hazards during water application rather than suppressing the fire.

**904.14.5 Operations and maintenance.** *Automatic fire-extinguishing systems* protecting commercial cooking systems shall be maintained in accordance with Sections 904.14.5.1 through 904.14.5.3.

**C** Most fires in commercial kitchens involve the cooking appliance and exhaust system in some way. Proper operation of the system in accordance with the IMC, as well as routine maintenance, can reduce the hazards related to the collection and removal of smoke and grease-laden vapors.

**904.14.5.1 Existing automatic fire-extinguishing systems.** Where changes in the cooking media, positioning of cooking equipment or replacement of cooking equipment occur in existing commercial cooking systems, the *automatic fire-extinguishing system* shall be required to comply with the applicable provisions of Sections 904.14 through 904.14.4.

**C** The provisions of Section 904.14 have long required new commercial cooking systems with preengineered fire-extinguishing systems to be tested in accordance with UL 300 and listed and labeled for that use. The question has often arisen, however, as to whether or how those provisions should be applied to existing systems. This section provides guidance and adds clarity as to when existing automatic fire-extinguishing systems protecting commercial cooking operations need to be modified, upgraded or replaced to meet UL 300 requirements. These provisions are thought to be consistent with provisions adopted by some states and local jurisdictions on this topic.

**904.14.5.2 Extinguishing system service.** *Automatic fire-extinguishing systems* shall be serviced not less frequently than every six months and after activation of the system. Inspection shall be by qualified individuals, and a certificate of inspection shall be forwarded to the *fire code official* upon completion.

**C** Range hood fire-extinguishing systems must be inspected and serviced at regular intervals to determine that they are ready to perform the intended function. Obviously, service is required after the suppression system discharges. The extent of service and maintenance depends on the type of fire-extinguishing system installed. The NFPA standard corresponding to the installed extinguishing agent should be consulted for additional service requirements (see commentary, Section 904.14).

**904.14.5.3 Fusible link and sprinkler head replacement.** Fusible links and automatic sprinkler heads shall be replaced annually, and other protection devices shall be serviced or replaced in accordance with the manufacturer's instructions.

**Exception:** Frangible bulbs are not required to be replaced annually.

**C** Because of the potential accumulation of grease or other contaminants that could adversely affect proper operation, fusible links and automatic sprinkler heads must be replaced at least annually. The sensing elements of the fusible link devices as well as the sprinkler heads must be routinely visually inspected and replaced as needed, at least annually.

The exception allows frangible bulb-type sprinklers to not be replaced as long as the annual examination shows no accumulation of grease or other contaminants.

**904.15 Domestic cooking facilities.** Cooktops and ranges installed in the following occupancies shall be protected in accordance with Section 904.15.1:

1. In Group I-1 occupancies where domestic cooking facilities are installed in accordance with Section 420.9 of the *International Building Code*.
2. In Group I-2 occupancies where domestic cooking facilities are installed in accordance with Section 407.2.7 of the *International Building Code*.
3. In Group R-2 college dormitories where domestic cooking facilities are installed in accordance with Section 420.11 of the *International Building Code*.

**C** As nursing homes move away from institutional models, it is critical to have a functioning kitchen that can serve as the "hearth of the home." Instead of a large, centralized, institutional kitchen where all meals are prepared and delivered to a central dining room or the resident's room, the new "household model" nursing home uses decentralized kitchens and

small dining areas to create the feeling and focus of home. For persons with dementia, it is particularly important to have spaces that look familiar, like the kitchen in their former home, to increase their understanding and ability to function at their highest level. This section addresses the fire protection system needs for these kitchens. Note that these occupancies already contain quick-response sprinklers. This section provides the requirements for the hood fire-extinguishing system needed for the cooking hood located over the cooktop or range. That system is required to be in compliance with UL 300A.

**904.15.1 Protection from fire.** Cooktops and ranges shall be protected in accordance with Section 904.15.1.1 or 904.15.1.2.

**C** There are two options available for the protection of cooktops and ranges. The first is to provide an automatic fire-extinguishing system in accordance with Section 904.15.1.1 and the second is to provide burners that have been tested and listed to prevent ignition of cooking oil in accordance with Section 904.15.1.2.

**904.15.1.1 Automatic fire-extinguishing system.** The domestic recirculating or exterior vented cooking hood provided over the cooktop or range shall be equipped with an *approved automatic fire-extinguishing system* complying with the following:

1. The *automatic fire-extinguishing system* shall be of a type recognized for protection of domestic cooking equipment. Preengineered *automatic fire-extinguishing systems* shall be *listed* and *labeled* in accordance with UL 300A and installed in accordance with the manufacturer's instructions.
2. Manual actuation of the fire-extinguishing system shall be provided in accordance with Section 904.14.1.
3. Interconnection of the fuel and electric power supply shall be in accordance with Section 904.14.2.

**C** The first option for the protection of cooktops and ranges is to provide an approved automatic fire-extinguishing system. The system is to conform to the three conditions listed. It is to be listed in accordance with UL 300A and installed in conformance with the manufacturer's written instructions. A means of manual operation is to be provided in conformance with Section 904.15.1. In addition, the fuel and electric power supply are to be interconnected as stated in Section 904.15.2.

**904.15.1.2 Ignition prevention.** Cooktops and ranges shall include burners that have been tested and *listed* to prevent ignition of cooking oil with burners turned on to their maximum heat settings and allowed to operate for 30 minutes.

**C** The second option for the protection of cooktops and ranges is to use specifically listed ignition resistant burners that do not allow cooking oils to ignite during testing. If these burners are used, an automatic cooktop fire-extinguishing system is not required. Work by the Fire Protection Research Foundation has confirmed that burners meeting these specifications are highly unlikely to ignite cooking materials. UL 858, *Household Electric Ranges*, includes Section 60A, "Abnormal Operation—Coil Surface Unit Cooking Oil Ignition Test," which evaluates the ability of coiled surface units to not ignite cooking oil.

## SECTION 905—STANDPIPE SYSTEMS

**905.1 General.** Standpipe systems shall be provided in new buildings and structures in accordance with Sections 905.2 through 905.11. In buildings used for *high-piled combustible storage*, fire protection shall be in accordance with Chapter 32.

**C** Standpipe systems are required in buildings to provide a quick, convenient water source for fire department use where hose lines would otherwise be impractical, such as in high-rise buildings. Standpipe systems can also be used prior to deployment of hose lines from fire department apparatus. The requirements for standpipes are based on practical requirements of typical firefighting operations and the nationally recognized standard NFPA 14.

The threads on connections to which the fire department may connect a hose must be compatible with the fire department hose threads (see commentary, Section 903.3.6). Chapter 32 requires a Class I standpipe system in exit passageways of buildings used for high-piled storage. Note that if a building containing high-piled storage does not contain an exit passageway, then standpipes would not be required. High-piled storage involves the solid-piled, bin box, palletized or rack storage of Class I through IV commodities over 12 feet (3658 mm) high. High-hazard commodities stored higher than 6 feet (1829 mm) are also considered high piled.

**905.2 Installation standard.** Standpipe systems shall be installed in accordance with this section and NFPA 14. Fire department connections for standpipe systems shall be in accordance with Section 912.

**C** This section requires the installation of standpipe systems to comply with the applicable provisions of NFPA 14 in addition to Section 905. NFPA 14 contains the minimum requirements for the installation of standpipe and hose systems for buildings and structures. The standard addresses additional requirements not addressed in the code, such as pressure limitations, minimum flow rates, piping specifications, hose connection details, valves, fittings, hangers and the testing and inspection of standpipes. The periodic inspection, testing and maintenance of standpipe systems must comply with NFPA 25.

Section 905 and NFPA 14 recognize three classes of standpipe systems: Class I, II or III. The type of system required depends on building height, building area, type of occupancy and the extent of automatic sprinkler protection. Section 905 also recognizes five types of standpipe systems: automatic dry, automatic wet, manual dry, manual wet and semiautomatic dry. The use of each type of system is limited to the building conditions and locations identified in Section 905.3. The classes and types of standpipe systems are defined in Section 202.

Section 912, to which this section refers, provides a comprehensive set of requirements for FDCs, reducing the opportunity for any of its requirements to be overlooked. See the commentary to Section 912.

**905.3 Required installations.** Standpipe systems shall be installed where required by Sections 905.3.1 through 905.3.7. Standpipe systems are allowed to be combined with *automatic sprinkler systems*.

**Exceptions:**

1. Standpipe systems are not required in Group R-2 *townhouses*.
2. Standpipe systems are not required in Group R-3 occupancies.

**C** Standpipe systems are installed in buildings based on the occupancy, fire department accessibility and special conditions that may require manual fire suppression exceeding the capacity of a fire extinguisher. Standpipe systems are most commonly required for buildings that exceed the height threshold requirement in Section 905.3.1; for buildings with certain features of a specific occupancy; or for buildings such as covered and open mall buildings, stages and underground buildings.

This section also states that a standpipe system does not have to be separate from an installed sprinkler system. It is common practice in multiple-story buildings for the standpipe system risers to also serve as risers for the automatic sprinkler systems.

In these instances, precautions need to be taken so that the operation of one system will not interfere with the operation of the other system. Therefore, control valves for the sprinkler system must be installed where the sprinklers are connected to the standpipe riser at each floor level. This allows the standpipe system to remain operational, even if the sprinkler system is shut off at the floor control valve. Exception 1 recognizes that there are no locations in a Group R-2 townhouse that would require hose connections. Regardless of whether a standpipe is technically required by this section, such a system would not be installed if hose connections are never required. It is not the intent of the code to require standpipes in individual townhouse units, even in cases where townhouses might exceed the story or height thresholds in Section 905.3.1.

Exception 2 recognizes that standpipe systems in Group R-3 occupancies would be of minimal value to the fire department and would send the wrong message to the occupants of a dwelling unit. In the case of multiple single-family dwellings, each dwelling unit has a separate entrance and is separated from the other units by 1-hour fire partitions. These conditions permit ready access to fires and also provide for a degree of fire containment through compartmentation, which is not always present in other occupancies.

**905.3.1 Height.** Class III standpipe systems shall be installed throughout buildings where any of the following conditions exist:

1. Four or more stories are above or below *grade plane*.
2. The floor level of the highest story is located more than 30 feet (9144 mm) above the lowest level of the fire department vehicle access.
3. The floor level of the lowest story is located more than 30 feet (9144 mm) below the highest level of fire department vehicle access.

**Exceptions:**

1. Class I standpipes are allowed in buildings equipped throughout with an *automatic sprinkler system* in accordance with Section 903.3.1.1 or 903.3.1.2.
2. Class I standpipes are allowed in Group B and E occupancies.
3. Class I standpipes are allowed in parking garages.
4. Class I standpipes are allowed in *basements* equipped throughout with an *automatic sprinkler system*.
5. Class I standpipes are allowed in buildings where occupant-use hose lines will not be utilized by trained personnel or the fire department.
6. In determining the lowest level of fire department vehicle access, it shall not be required to consider either of the following:
  - 6.1. Recessed loading docks for four vehicles or less.
  - 6.2. Conditions where topography makes access from the fire department vehicle to the building impractical or impossible.

**C** Given the available manpower on the fire department vehicle, standard firefighting operations and standard hose sizes, a 30-foot (9144 mm) vertical distance is generally considered the maximum height to which a typical fire department engine company can practically and readily extend its hose lines. Thus, the maximum vertical travel (height) threshold is based on the time it would take a typical fire department engine (pumper) company to manually suppress a fire. The standpipe connection reduces the time needed for the fire department to extend hose lines up or down stairways to advance and apply water to the fire. For this use, a minimum Class III standpipe system is required.

With respect to the height of the building, the threshold is measured from the level at which the fire department can gain access to the building directly from its vehicle and begin vertical movement. Floor levels above grade are measured from the lowest level of fire department vehicle access to the highest floor level above [see Commentary Figure 905.3.1(1)]. If a

building contains floor levels below the level of fire department vehicle access, the measurement is made from the highest level of fire department vehicle access to the lowest floor level. In cases where a building has more than one level of fire department vehicle access, the most restrictive measurement is used because it is not known at which level the fire department will access the building. In other words, the vertical distance is to be measured from the more restrictive level of fire department vehicle access to the level of the highest (or lowest, if below) floor [see Commentary Figure 905.3.1(2)].

The threshold based on the height of the building is independent of the occupancy of the building, the area of the building or the presence of an automatic sprinkler system. This is based on the universal need to be able to provide a water supply for fire suppression in any building and on the limitations of the physical effort necessary to extend hose lines vertically.

Before discussing the exceptions, it is important to understand the differences between the different classes and operational characteristics of standpipes. More detailed information is included in Section 202 for the definitions of the different classes and types of standpipes.

Standpipes can be dry or wet, manual, automatic or semiautomatic. Automatic systems can be either wet or dry. Manual systems can be either wet or dry. A semiautomatic system is always in association with a dry system.

The code is written such that it could be assumed the default is an automatic wet system. This is, however, not the case. The requirement is left to the design standard, NFPA 14. Section 5.4.1.1 of NFPA 14 indicates that Class I standpipes can be manual if the building is not a high-rise. Section 5.4.1.4 of the standard indicates that a Class I standpipe must be wet except where the pipe is subject to freezing. Thus, where a Class I standpipe is installed, possibly as a part of Exception 1, the system can be manual wet if the building is not a high-rise. This is consistent with IFC Committee Interpretation No. 33-03. As long as the building is not a high-rise, it can be provided with a Class I standpipe system that is manual wet.

Class II and III standpipes are required to be automatic wet or semiautomatic wet except where the piping is subject to freezing, according to Section 5.4.2 of NFPA 14. They cannot be manual. Only Class I standpipes can be manual and can only be used under the conditions noted in this code. Note that other sections of the code may specify whether the system must be automatic. If the requirement is not noted elsewhere in the code, then the decision to use an automatic or manual system is left to the designer.

Exception 1 recognizes that with a fully operational automatic sprinkler system, the time that the fire department has to extend hoses within the building is substantially increased and that the amount of effort required is greatly reduced. Consequently, a single Class I connection can be provided. The second, 1 1/2-inch (38 mm) connection is allowed to be omitted. NFPA 14 also has a similar provision but is more restrictive as it only eliminates the hose station and additionally requires a 2 1/2-inch by 1 1/2-inch (65 mm by 40 mm) reducer and a cap attached with a chain (see NFPA 14, Section 7.3.4.1). In accordance with Section 102.7.1, the code would take precedence and the reducer and cap would not be required.

Exception 2 allows for the use of Class I standpipes in Group B and E occupancies. Occupant-use hose stations are considered to be a legacy method of fire protection, existing for decades in model code requirements. In recent years, many fire safety and evacuation plans have all but abandoned occupant-use hoses in their training of building occupants, relying instead on fire extinguisher training (which is required in all new and existing Group B and E occupancies) and the primary focus of evacuation. Also, fire behavior can be different in these occupancies due to compartment fire loading. This creates fires that develop faster, create more heat in most situations, and produce greater amounts of toxic smoke. Collectively, the ability for occupants to safely and effectively utilize occupant-use hoses without firefighting gear and respiratory protection is considered to be greatly minimized. Occupant-use hose is permitted to not be installed in sprinkler-protected buildings that otherwise require standpipes. Even though this may seem to be trade-off, the determination of whether to replace the manual method (occupant-use hose) with an automatic sprinkler system should be based on the occupants' ability to suppress a fire manually.

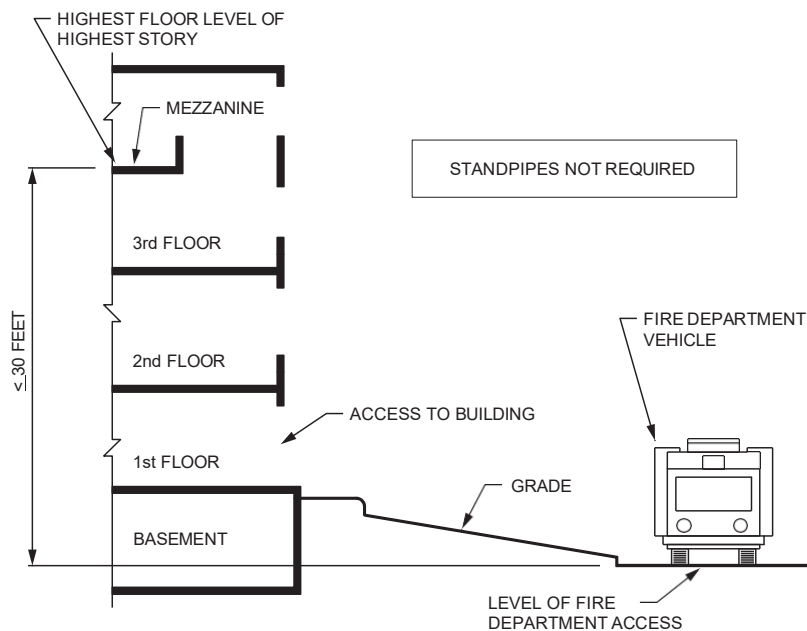
Exception 3 allows for Class I standpipes to be installed in any parking garage. The standpipes are there for firefighter use and in the rare occurrence a parking garage has trained personnel, they can add the necessary outlet reducer and hose line.

Exception 4 is similar to Exception 1 but only addresses sprinklers in the basement. Thus, it is possible to use this exception if only the basement is protected by automatic sprinklers. However, Class I connections can only be provided in the basements—not on the upper floors. The exception cannot be used for stories above grade unless the entire building is sprinklered and, therefore, compliant with Exception 1.

Exception 5 allows for the use of Class I standpipes in place of Class III standpipes in buildings where occupant-use hose lines will not be used by trained personnel or the fire department. If occupants are not trained in their use and the fire department will not utilize them, then the occupant-use hose lines serve no purpose and could potentially slow down the fire department's attempt to connect its own hose to the 2 1/2-inch (64 mm) hose valve outlet. Thus, a Class I standpipe can be used in this situation.

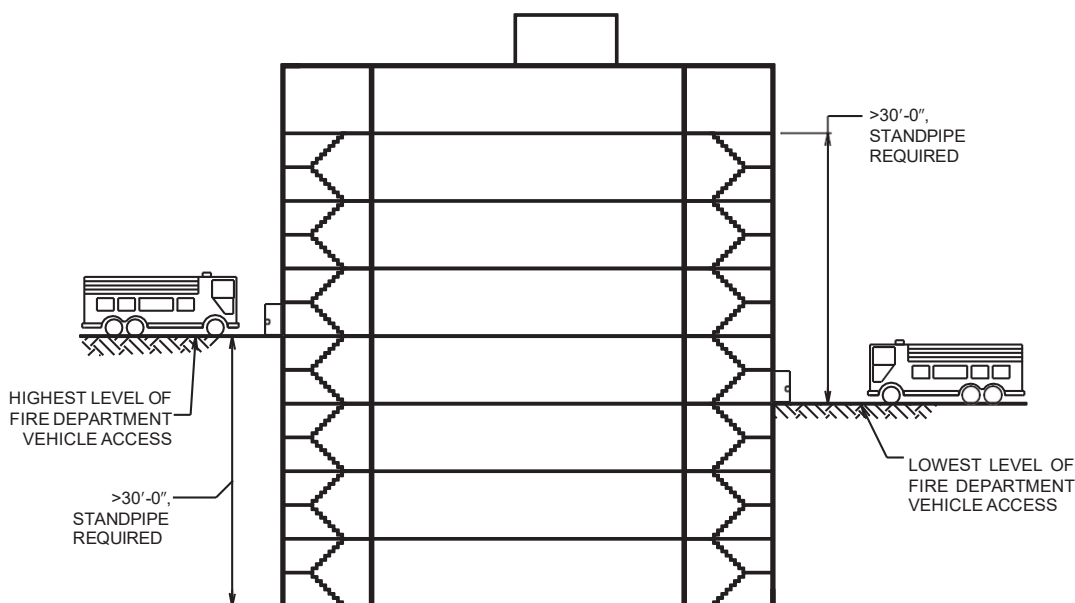
Exception 6 provides additional information about what must be considered when determining building height with respect to the level of fire department vehicle access. The first item is a practical one that excludes loading docks of a limited size. The second item notes that although it may be possible to have a fire department vehicle arrive adjacent to the building at a low level, it may not be possible for the fire department to access the building from that level. An example of this condition would be where a road surface is located below a building built on a bluff. Although the fire department vehicles can approach from the lower road, fire department personnel cannot access the building from that lower level. Thus, the standpipe requirement would not be based on the road below the bluff.

Commentary Figure 905.3.1(1)—Height Threshold for Standpipes



For SI: 1 inch = 25.4 mm, 1 foot = 304.8 mm.

Commentary Figure 905.3.1(2)—Height Requirements for Standpipes with Multiple Levels of Fire Department Vehicle Access



For SI: 1 inch = 25.4 mm, 1 foot = 304.8 mm.

**905.3.2 Group A.** Class I automatic wet standpipes shall be provided in nonsprinklered Group A buildings having an *occupant load* exceeding 1,000 persons.

**Exceptions:**

1. Open-air-seating spaces without enclosed spaces.
2. Class I automatic dry and semiautomatic dry standpipes or manual wet standpipes are allowed in buildings that are not high-rise buildings.

**C** The main concern in assembly occupancies with a high occupant load is evacuation. Many occupants may not be familiar with either their surroundings or the egress arrangement in the building. This section also assumes the building is not sprinklered; therefore, control and suppression of the fire is left to the fire department.

Exception 1 exempts open-air seating without enclosed spaces, such as grandstands and bleachers. In such occupancies, a buildup of smoke and hot gases is not possible because these structures are open to the atmosphere.

Exception 2 states that in lieu of a Class I automatic wet standpipe, automatic dry and semiautomatic dry Class I standpipes are permitted in buildings that are not considered to be a high-rise.

**905.3.3 Covered and open mall buildings.** Covered mall and open mall buildings shall be equipped throughout with a standpipe system where required by Section 905.3.1. Mall buildings not required to be equipped with a standpipe system by Section 905.3.1 shall be equipped with Class I hose connections connected to the *automatic sprinkler system* sized to deliver water at 250 gallons per minute (946.4 L/min) at the hydraulically most remote hose connection while concurrently supplying the *automatic sprinkler system* demand. The standpipe system shall be designed not to exceed a 50 pounds per square inch (psi) (345 kPa) residual pressure loss with a flow of 250 gallons per minute (946.4 L/min) from the fire department connection to the hydraulically most remote hose connection. Hose connections shall be provided at each of the following locations:

1. Within the mall at the entrance to each *exit passageway* or *corridor*.
2. At each floor-level landing within *interior exit stairways* opening directly on the mall.
3. At exterior public entrances to the mall of a covered mall building.
4. At public entrances at the perimeter line of an open mall building.
5. At other locations as necessary so that the distance to reach all portions of a tenant space does not exceed 200 feet (60 960 mm) from a hose connection.

**C** Covered and open mall buildings are only required to have a standpipe system if Section 905.3.1 requires such features. If standpipes are not required because of building height, Class I hose connections that are connected to the automatic sprinkler system are still required. Also, to ensure that both the sprinkler system and hose connections will function at an acceptable level, the system must be sized for both the sprinkler demand and the hose connection demand. This section specifies a minimum flow rate and a maximum pressure loss to the most remote hose connection so that the fire department can gain full use of the hose connection during a fire. Hose connections are required when a standpipe system is not at key locations, such as entrances to exit passageways and at entrances to the covered or open mall. Note that these locations are essentially the same locations required for Class I hose connections in Section 905.4, except that this section also requires that the distance to reach any portion of a tenant space from a hose connection must not exceed 200 feet (60 960 mm).

**905.3.4 Underground buildings.** Underground buildings shall be equipped throughout with a Class I automatic wet or manual wet standpipe system.

**C** Underground buildings pose unique hazards to life safety because of their isolation and inaccessibility. Additional fire protection and firefighting measures for the fire department are required to compensate for the lack of exterior access for fire suppression and rescue operations (see IBC Section 405).

**905.3.5 Helistops and heliports.** Buildings with a rooftop *helistop* or *heliport* shall be equipped with a Class I or III standpipe system extended to the roof level on which the *helistop* or *heliport* is located in accordance with Section 2007.5.

**C** Buildings containing rooftop helistops or heliports are required to be equipped with a Class I or III standpipe. A heliport is a distinct hazard that will involve flammable fuels. In the event of an emergency, rapid deployment of hand hose lines will be necessary to attack a resulting fire, effectuate rescue and protect exposures and the remainder of the building.

The requirement results in a standpipe system throughout the building, not just a connection at the roof level. This is critical in firefighting operations because many times the connection below the rooftop level may be needed just to gain access to the roof. If the only connection is on the roof, it is of no use if the firefighters cannot get to it.

Additionally, a heliport includes fueling operations. It is entirely possible for a spill to not only affect the rooftop, but also floors below as the liquid fuel spreads. The standpipe system will again be utilized in these situations.

Section 2007.5 requires a 2 1/2-inch (64 mm) standpipe outlet to be within 150 feet (45 675 mm) of all portions of the heliport or helistop area and be either Class I or III.

**905.3.6 Marinas and boatyards.** Standpipes in marinas and boatyards shall comply with Chapter 36.

**C** Section 3604.2 contains the specifics as to when standpipes are required at marinas. Marinas and boatyards have unique challenges for firefighting. Although there is water readily available, it is not easily or effectively capable of being applied to a fire at such a facility. A fire in such facilities can spread from structure to structure and from vessel to vessel with no effective way to attack and control it. Section 3604.2 references NFPA 303 for the standpipe requirements and additionally requires that no point on the marina pier or float system exceeds 150 feet (45 720 mm) from a standpipe hose connection (see commentary, Section 3604.2).

**905.3.7 Vegetative roofs and landscaped roofs.** Buildings or structures that have landscaped roofs or *vegetative roofs* and that are equipped with a standpipe system shall have the standpipe system extended to the roof level on which the *landscaped roof* or *vegetative roof* is located.

**C** This section requires that if the building is equipped with a standpipe system, whether or not such systems are required, it must be extended to a roof that is landscaped or vegetative. These requirements relate to the requirements in Section 317 that address the increased fuel load being added to roofs.

**905.4 Location of Class I standpipe hose connections.** Class I standpipe hose connections shall be provided in all of the following locations:

1. In every required *interior exit stairway* or *exterior exit stairway*, a hose connection shall be provided for each story above and below *grade plane*. Hose connections shall be located at the main floor landing unless otherwise *approved* by the *fire code official*.  
**Exception:** A single hose connection shall be permitted to be installed in the open *corridor* or open breezeway between open stairs that are not greater than 75 feet (22 860 mm) apart.
2. On each side of the wall adjacent to the exit opening of a horizontal exit.  
**Exception:** Where floor areas adjacent to a horizontal *exit* are reachable from an *interior exit stairway* or *exterior exit stairway* hose connection by a 30-foot (9144 mm) hose stream from a nozzle attached to 100 feet (30 480 mm) of hose, a hose connection shall not be required at the horizontal *exit*.
3. In every exit *passageway*, at the entrance from the *exit passageway* to other areas of a building.  
**Exception:** Where floor areas adjacent to an *exit passageway* are reachable from an *interior exit stairway* or *exterior exit stairway* hose connection by a 30-foot (9144 mm) hose stream from a nozzle attached to 100 feet (30 480 mm) of hose, a hose connection shall not be required at the entrance from the *exit passageway* to other areas of the building.
4. In covered mall buildings, adjacent to each exterior public entrance to the mall and adjacent to each entrance from an *exit passageway* or *exit corridor* to the mall. In open mall buildings, adjacent to each public entrance to the mall at the perimeter line and adjacent to each entrance from an *exit passageway* or *exit corridor* to the mall.
5. Where the roof has a slope less than 4 units vertical in 12 units horizontal (33.3 percent slope), a hose connection shall be located to serve the roof or at the highest landing of an *interior exit stairway* with access to the roof provided in accordance with Section 1011.12.
6. Where the most remote portion of a nonsprinklered floor or story is more than 150 feet (45 720 mm) from a hose connection or the most remote portion of a sprinklered floor or story is more than 200 feet (60 960 mm) from a hose connection, the *fire code official* is authorized to require that additional hose connections be provided in *approved* locations.

**C** Hose connections are required for the fire department to make use of the standpipe system. Since the fire department will typically access the building using the stairways, and most fire departments do not permit entry to the fire floor without an operating hose line, a hose connection must be installed for each floor level of each required enclosed stairway.

Item 1 also specifies that the hose connections are to be located at the main floor landing. The hose connections, however, are permitted to be at the intermediate landing instead of at the main floor landing if this arrangement is approved by the fire code official. In the specific case of an open corridor or open breezeway that is located in between open stairs, a single hose connection is permitted to be installed within the open corridor or open breezeway.

Because horizontal exits are also primary entrances to the fire floor, Item 2 states that hose connections must also be provided at each horizontal exit. The construction of the fire separation assembly used as the horizontal exit will protect the firefighters while they are connecting to the standpipe system. The hose connections are to be located on each side of the horizontal exit to enable firefighters to be in a protected area, regardless of the location of the fire. The exception acknowledges that there may already be a hose connection in close proximity to the horizontal exit if there is a stairway adjacent to the horizontal exit. The intent is to allow fewer standpipe outlets if the area can be adequately covered by the standpipes in stairways since those are the standpipes typically used by the fire department.

Item 3 states that an exit passageway in a building required to have a standpipe system is typically used as an extension of a required exit stairway. This allows use of the exit passageway for firefighting staging operations in the same way as an exit stair. The exception acknowledges that there may already be a hose connection in close proximity to the exit passageway. If there is a stairway containing a hose connection in close proximity to the exit passageway that can meet the 30-foot (9144 mm) hose stream from a nozzle attached to 100 feet (30 480 mm) of hose, then an additional standpipe is not

required. The intent is to allow fewer standpipe outlets if the area can be adequately covered by the standpipes in stairways since those are the standpipes typically used by the fire department.

In covered and open mall buildings, Item 4 requires hose connections at each entrance to an exit passageway or exit corridor. In addition, covered mall buildings would be required to have connections at each exterior public entrance. Open malls would require connections at the public entrance perimeter line. These locations allow fire personnel to have a support line as soon as they enter the building.

Item 5 is consistent with NFPA 14 regarding the installation of Class I standpipe hose connections on the roofs of buildings. This requirement requires only one standpipe to extend to the roof level or highest landing of the stairway serving the roof. This coordinates with Section 1011.12, which only requires one stairway to extend to the roof.

Hose connections in each interior exit stairway result in hose connections being located based on the travel distances permitted in Table 1017.2, which recognizes that most fire departments carry standpipe hose packs with 150 feet (45 720 mm) of hose or possibly with 100 feet (30 480 mm) of hose and an additional 50-foot (15 240 mm) section that could be easily connected.

With the typical travel distance permitted in nonsprinklered buildings of 200 feet (60 960 mm), reasonable coverage is provided when the effective reach of a fire stream is considered. Depending on the arrangement of the floor, however, all areas may not be effectively protected. Although this situation could easily be corrected by locating additional hose connections on the floor, such connections may rarely be used because of the difficulty in identifying their location during a fire and the fact that most fire departments require an operational hose line before they enter the fire floor. Because longer travel distances are allowed in sprinklered buildings, the problem is increased, but the need for prompt manual suppression is reduced by the presence of the sprinkler system. Item 6 gives the fire code official the authority to require additional hose connections if needed.

**905.4.1 Protection.** Risers and laterals of Class I standpipe systems not located within an *interior exit stairway* shall be protected by a degree of *fire resistance* equal to that required for vertical enclosures in the building in which they are located.

**Exception:** In buildings equipped throughout with an *approved automatic sprinkler system*, laterals that are not located within an *interior exit stairway* are not required to be enclosed within fire-resistance-rated construction.

**C** To minimize the potential for damage to the standpipe systems from a fire, the risers and laterals (i.e., the horizontal segments of standpipe system piping) must be located in an enclosure having the same fire-resistance rating as required for a vertical or shaft enclosure within the building. The required fire-resistance rating for the enclosure can be determined as detailed in IBC Section 713.4.

The exception states that the enclosure for laterals is not required if the building is equipped throughout with an approved automatic sprinkler system. The potential for damage to the standpipe system is minimized by the protection provided by the sprinkler system. The automatic sprinkler system may be either an NFPA 13 or 13R system, depending on what was permitted for the building occupancy.

If the interior exit stairway is not required to have a rated enclosure, such as in an open parking garage, the laterals are similarly not required to be in an enclosure. The protection afforded the vertical riser in the stairway must be the same as that afforded the laterals. If the stairway is not required by other sections of the code to be located in a rated enclosure, then the laterals are not required to be in rated protection either.

**905.4.2 Interconnection.** In buildings where more than one standpipe is provided, the standpipes shall be interconnected in accordance with NFPA 14.

**C** In cases where there are multiple Class I standpipe risers, the risers must be supplied from and interconnected to a common supply line. The required fire department connection must serve all of the sprinklers or standpipes in the building.

**905.5 Location of Class II standpipe hose connections.** Class II standpipe hose connections shall be located so that all portions of the building are within 30 feet (9144 mm) of a nozzle attached to 100 feet (30 480 mm) of hose. Class II standpipe hose connections shall be located where they will have *ready access*.

**C** Sections 905.5.1 through 905.5.3 specify the requirements for Class II standpipe hose connections. Class II standpipe systems are primarily intended for use by the building occupants.

This section for Class II standpipes does not specifically require the hose station and uses the term “hose connection” with a location based on 100 feet (30 480 mm) of hose. However, the definition of Class II and III standpipes and Section 7.3.3.1 of NFPA 14 specifically require hose stations. Section 905.2 specifically references NFPA 14.

Although NFPA 14 requires a hose station, the decision as to whether a hose station is required may be one that is affected by the policies and procedures of the local fire department. It should be remembered that Class II hose connections and hose stations are intended for occupant use and not necessarily for fire department use. The fire department typically uses the Class I connection that is compatible with 2 1/2-inch (64 mm) hose.

**905.5.1 Groups A-1 and A-2.** In Group A-1 and A-2 occupancies with *occupant loads* of more than 1,000, hose connections shall be located on each side of the rear of the auditorium and on each side of the balcony.

**C** Because of the high occupant load density in Group A-1 and A-2 occupancies, providing additional means for controlling fires in their initial stage is important to enable prompt evacuation of the building.

**905.5.2 Protection.** *Fire-resistance-rated* protection of risers and laterals of Class II standpipe systems is not required.

**C** Class II standpipe systems are normally not located in exit stairways; standpipe hose connections are located near the protected area to allow quick access. Therefore, it is likely that neither the risers nor the laterals would be located in any enclosure.

**905.5.3 Class II system 1-inch hose.** A minimum 1-inch (25 mm) hose shall be allowed to be used for hose stations in light-hazard occupancies where investigated and *listed* for this service and where *approved* by the *fire code official*.

**C** This section permits the use of 1-inch (25 mm) listed noncollapsible hose as an alternative to 1 1/2-inch (38 mm) hose, subject to the approval of the fire code official. This alternative is limited to light-hazard occupancies, such as office buildings and certain assembly occupancies that tend to have lower fuel loads, since a smaller hose generally discharges less water.

**905.6 Location of Class III standpipe hose connections.** Class III standpipe systems shall have hose connections located as required for Class I standpipes in Section 905.4 and shall have Class II hose connections as required in Section 905.5.

**C** Class III standpipe systems that have both a 2 1/2-inch (64 mm) hose connection and a 1 1/2-inch (38 mm) hose connection must comply with the applicable requirements of Sections 905.4, 905.5 and 905.6. Thus, it is necessary to review and comply with all applicable provisions.

**905.6.1 Protection.** Risers and laterals of Class III standpipe systems shall be protected as required for Class I systems in accordance with Section 905.4.1.

**C** Because Class III standpipe systems are intended for use by fire-suppression personnel, they must be located in construction that has a fire-resistance rating equivalent to that of the vertical or shaft enclosure requirements of the building (see commentary, Section 905.4.1).

**905.6.2 Interconnection.** In buildings where more than one Class III standpipe is provided, the standpipes shall be interconnected in accordance with NFPA 14.

**C** As indicated in Section 905.4.2 for Class I standpipe systems, multiple standpipe risers must be interconnected with a common supply line. An indicating valve is typically installed at the base of each riser so that individual risers can be taken out of service without affecting the water supply or the operation of other standpipe risers.

**905.7 Cabinets.** Cabinets containing firefighting equipment, such as standpipes, fire hose, fire extinguishers or fire department valves, shall not be blocked from use or obscured from view.

**C** This section does not require that cabinets be provided to contain fire protection equipment. However, if they are provided, cabinets must be readily visible and accessible at all times. Sections 905.7.1 and 905.7.2 contain additional criteria for the construction and identification of the cabinets. Where cabinets are located in fire-resistance-rated assemblies, the integrity of the assembly must be maintained. Cabinet design for hose connections, control valves or other devices that require manual operation should be such that there is sufficient clearance between the cabinet body and the device to allow grasping of the device (quite likely with a gloved hand) and prompt operation of it.

**905.7.1 Cabinet equipment identification.** Cabinets shall be identified in an *approved* manner by a permanently attached sign with letters not less than 2 inches (51 mm) high in a color that contrasts with the background color, indicating the equipment contained therein.

**Exceptions:**

1. Doors not large enough to accommodate a written sign shall be marked with a permanently attached pictogram of the equipment contained therein.
2. Doors that have either an *approved* visual identification clear glass panel or a complete glass door panel are not required to be marked.

**C** This section specifies the minimum criteria to make the signs readily visible. Different color combinations may be approved by the fire code official if the color contrast between the letters and the background is vivid enough to make the sign visible at an approved distance. The exceptions address alternatives to letter signage if the cabinet is still conspicuously identified or the contents are readily visible.

**905.7.2 Locking cabinet doors.** Cabinets shall be unlocked.

**Exceptions:**

1. Visual identification panels of glass or other *approved* transparent frangible material that is easily broken and allows access.

2. *Approved* locking arrangements.
3. Group I-3 occupancies.

**C** Ready access to all firefighting equipment in the cabinet is essential. The exceptions, however, recognize the need to lock cabinets for security reasons and to prevent theft or vandalism (see commentary, Section 906.8).

**905.8 Dry standpipes.** Dry standpipes shall not be installed.

**Exception:** Where subject to freezing and in accordance with NFPA 14.

**C** Wet standpipe systems are preferred because they tend to be the most reliable type of standpipe system; therefore, dry standpipes are prohibited unless subject to freezing. For example, Class I manual standpipe systems, which do not have a permanent water supply, are permitted in open parking structures. This recognizes that open parking structures are not heated and that most fires are limited to the vehicle of origin. The use of any dry standpipe system instead of a wet standpipe should take into consideration the added response time and its effect on the occupancy characteristics of the building.

**905.9 Valve supervision.** Valves controlling water supplies shall be supervised in the open position so that a change in the normal position of the valve will generate a supervisory signal at the supervising station required by Section 903.4.1. Where a *fire alarm system* is provided, a signal shall be transmitted to the control unit.

**Exceptions:**

1. Valves to underground key or hub valves in roadway boxes do not require supervision.
2. Valves locked in the normal position and inspected as provided in this code in buildings not equipped with a *fire alarm system*.

**C** As with sprinkler systems, water control valves for standpipe systems must be electrically supervised as a means of determining that the system is operational (see commentary, Section 903.4.1).

Exception 1 recognizes that underground key or hub valves in roadway boxes are not normally supervised or need to be supervised, whether the building contains a standpipe system or an automatic sprinkler system.

Exception 2 does not require the control valves for the standpipes to be electrically monitored if they are locked in the normal position and a fire alarm system is not installed in the building. When a fire alarm system is installed, the control valves for the standpipes must be electrically monitored and tied into the supervision required for the fire alarm system.

**905.10 During construction.** Standpipe systems required during construction and demolition operations shall be provided in accordance with Section 3307.

**C** As stated in Section IBC Section 3311, at least one standpipe is required during construction of buildings four stories or more in height or during demolition of standpipe-equipped buildings. Standpipe systems must be accessible and operable during construction and demolition operations to assist in any potential fire (see commentary, Sections 3311.1 and 3311.2 of the IBC, and Sections 3307.1 and 3307.2 of the code).

**905.11 Locking standpipe outlet caps.** The *fire code official* is authorized to require locking caps on the outlets on standpipes where the responding fire department carries key wrenches for the removal that are compatible with locking FDC connection caps.

**C** Standpipe connection caps are vulnerable to theft. Vandalism is a concern when trash and debris are introduced into the outlet. This provision, when applied, will require the protected FDC and protected standpipe caps to have a compatible and standard opening mechanism.

**905.12 Existing buildings.** Where required in Chapter 11, existing structures shall be equipped with standpipes installed in accordance with Section 905.

**C** This section simply refers to Chapter 11, which addresses all requirements that apply to existing buildings (see commentary, Section 1103.6).

## SECTION 906—PORTABLE FIRE EXTINGUISHERS

**906.1 Where required.** Portable fire extinguishers shall be installed in all of the following locations:

1. In new and existing Group A, B, E, F, H, I, M, R-1, R-2, R-4 and S occupancies.

**Exceptions:**

1. In Group R-2 occupancies, portable fire extinguishers shall be required only in locations specified in Items 2 through 6 where each *dwelling unit* is provided with a portable fire extinguisher having a minimum rating of 1-A:10-B:C.
2. In Group E occupancies, portable fire extinguishers shall be required only in locations specified in Items 2 through 6 where each classroom is provided with a portable fire extinguisher having a minimum rating of 2-A:20-B:C.

3. In storage areas of Group S occupancies where forklift, powered industrial truck or powered cart operators are the primary occupants, fixed extinguishers, as specified in NFPA 10, shall not be required where in accordance with all of the following:
  - 3.1. Use of vehicle-mounted extinguishers shall be *approved* by the *fire code official*.
  - 3.2. Each vehicle shall be equipped with a 10-pound, 40A:80B:C extinguisher affixed to the vehicle using a mounting bracket *approved* by the extinguisher manufacturer or the *fire code official* for vehicular use.
  - 3.3. Not less than two spare extinguishers of equal or greater rating shall be available on-site to replace a discharged extinguisher.
  - 3.4. Vehicle operators shall be trained in the proper operation, use and inspection of extinguishers.
  - 3.5. Inspections of vehicle-mounted extinguishers shall be performed daily.
2. Within 30 feet (9144 mm) distance of travel from commercial cooking equipment and from domestic cooking equipment in Group I-1; I-2, Condition 1; and R-2 college dormitory occupancies.
3. In areas where *flammable* or *combustible liquids* are stored, used or dispensed.
4. On each floor of structures under construction, except Group R-3 occupancies, in accordance with Section 3306.6.
5. Where required by the sections indicated in Table 906.1.
6. Special-hazard areas, including but not limited to laboratories, computer rooms and generator rooms, where required by the *fire code official*.

**Exception:** Portable fire extinguishers are not required at normally unmanned Group U occupancy buildings or structures where a portable fire extinguisher suitable to the hazard of the location is provided on the vehicle of visiting personnel.

**C** Portable fire extinguishers are required in certain instances to give the occupants the means to suppress a fire in its incipient stage. The capability for manual fire suppression can contribute to the protection of the occupants, especially if there are evacuation difficulties associated with the occupancy or the specific hazard in the area. To be effective, personnel must be properly trained in the use of portable fire extinguishers.

Because of the high-hazard nature of building contents, portable fire extinguishers are required in occupancies in Group H.

Portable fire extinguishers are required in occupancies in Groups A, B, E, F, H, I, M, R-1, R-2, R-4 and S because of the need to control the fire in its early stages and because evacuation can be slowed by the density of the occupant load, the capability of the occupants to evacuate or the overall fuel load in the building. Because the code typically focuses on new buildings, this section is applicable to new buildings.

Portable fire extinguishers are required in areas containing special hazards, such as commercial cooking equipment, and specific hazardous operations, as indicated in Table 906.1. Because of the potentially extreme fire hazard associated with such areas or occupancy conditions, prompt extinguishment of the fire is critical.

Portable fire extinguishers are required in all buildings under construction, except in occupancies in Group R-3. The extinguishers are intended for use by construction personnel to suppress a fire in its incipient stages.

Portable fire extinguishers are also required in laboratories, computer rooms and other workspaces in which fire hazards may exist based on the use of the space. Many of these will be addressed by the required occupancy group criteria or by the specific hazard provisions of Table 906.1. Laboratories, for example, may not be considered Group H, but still use limited amounts of hazardous materials that would make manual means of fire extinguishment desirable.

The first exception to Item 1 permits smaller portable fire extinguishers (PFEs) in dwelling units of Group R-2 occupancies instead of larger PFEs in the common areas. Under the exception, the installation of 1-A:10-B:C PFEs within individual dwelling units allows apartment owners to eliminate their installation in common areas such as corridors, laundry rooms and swimming pool areas. PFEs in these areas are susceptible to vandalism or theft. Another issue is that larger PFEs are more difficult for the infirm and elderly to safely deploy and operate.

The second exception to Item 1 permits PFEs in Group E occupancies to be installed in only the locations listed in Items 2 through 6 if each classroom is provided with an extinguisher having a rating of 2-A:20-B:C. The reason for this option is that where schools are required to develop lockdown plans to protect students and faculty from intruders, these plans effectively prevent access to portable fire extinguishers normally located in hallways during lockdown situations. Locating extinguishers in classrooms provides accessibility during normal conditions as well as when a school is forced to go into lockdown. This exception provides an option for schools implementing lockdown plans to relocate extinguishers from hallways to classrooms.

The third exception to Item 1 allows large storage warehouses where the occupants are primarily riding on powered industrial trucks to provide a suitable extinguisher on the truck with the operator versus distributing them around the facility. The intent is that these will be more readily accessible to the operator and will improve the speed of deployment. The use of this exception requires daily inspections, operator training and the availability of spare extinguishers in order to improve the reliability of fire extinguisher use.

For the period of 2003 through 2007, NFPA reported that approximately 38,000 fires occurred annually in apartment buildings. Sixty percent of these fires occurred inside of dwelling units versus 14 percent that occurred in common areas

covered by Items 3 and 6 of Section 906.1. It is more logical to place PFEs inside dwelling units versus common areas because it locates the extinguisher in an area where statistically most fires occur. If the occupant cannot control the fire using the PFE, he or she can escape and allow the automatic sprinkler system to operate and control the fire. This exception improves the safety of Group R-2 residents because it does not require them to leave a dwelling involved in a fire, find a PFE and then return to the fire-involved dwelling unit to attempt incipient fire attack.

Including this requirement in the code alerts designers and building officials that the extinguishers are required. This will allow designers to plan for thicker walls where recessed cabinets may be used or to design locations where the extinguishers will not project into or obstruct the egress or circulation path.

The exception at the end of the section applies only to Group U occupancies. It recognizes that while the use of PFEs for incipient fire control allows for increased time for evacuation, this is not applicable to unmanned locations. In these occupancies, it is not practical to install and maintain PFEs in accordance with NFPA 10 at normally unmanned locations. Compliance with NFPA 10 entails a monthly visit for the sole purpose of inspecting the extinguisher, which is not reasonable for rarely visited remote locations. Certain remote and isolated locations are impractical to visit or not accessible during winter months as access is blocked by snow. Further, if there were a fire, there is no one on-site the vast majority of the time to use the extinguisher. Having a PFE on the technician's vehicle when servicing the site is a more effective alternative to a fixed site-mounted extinguisher. The use of this exception requires that the PFE carried on the vehicle be inspected and maintained in accordance with NFPA 10 and be ready for use if necessary.

**TABLE 906.1—ADDITIONAL REQUIRED PORTABLE FIRE EXTINGUISHERS**

| SECTION   | SUBJECT                                    |
|-----------|--------------------------------------------|
| 303.5     | Asphalt kettles                            |
| 307.5     | Open burning                               |
| 308.1.3   | Open flames—torches                        |
| 309.4     | Powered industrial trucks                  |
| 1204.10   | Portable generators                        |
| 2005.2    | Aircraft towing vehicles                   |
| 2005.3    | Aircraft welding apparatus                 |
| 2005.4    | Aircraft fuel-servicing tank vehicles      |
| 2005.5    | Aircraft hydrant fuel-servicing vehicles   |
| 2005.6    | Aircraft fuel-dispensing stations          |
| 2007.7    | Heliports and helistops                    |
| 2108.4    | Dry cleaning plants                        |
| 2305.5    | Motor fuel-dispensing facilities           |
| 2310.6.4  | Marine motor fuel-dispensing facilities    |
| 2311.6    | Repair garages                             |
| 2404.6.1  | Spray-finishing operations                 |
| 2405.4.2  | Dip-tank operations                        |
| 2406.4.2  | Powder-coating areas                       |
| 2804.3    | Lumberyards/woodworking facilities         |
| 2808.8    | Recycling facilities                       |
| 2809.5    | Exterior lumber storage                    |
| 2903.5    | Organic-coating areas                      |
| 3006.3    | Industrial ovens                           |
| 3108.9    | Tents and membrane structures              |
| 3206.10   | High-piled storage                         |
| 3306.6    | Buildings under construction or demolition |
| 3305.10.2 | Roofing operations                         |
| 3408.2    | Tire rebuilding/storage                    |
| 3504.2.6  | Welding and other hot work                 |
| 3604.4    | Marinas                                    |

**TABLE 906.1—ADDITIONAL REQUIRED PORTABLE FIRE EXTINGUISHERS—continued**

| SECTION      | SUBJECT                                                                              |
|--------------|--------------------------------------------------------------------------------------|
| 3703.6       | Combustible fibers                                                                   |
| 5703.2.1     | Flammable and combustible liquids, general                                           |
| 5704.3.3.1   | Indoor storage of flammable and combustible liquids                                  |
| 5704.3.7.5.2 | Liquid storage rooms for flammable and combustible liquids                           |
| 5705.4.9     | Solvent distillation units                                                           |
| 5706.2.7     | Farms and construction sites—flammable and combustible liquids storage               |
| 5706.4.10.1  | Bulk plants and terminals for flammable and combustible liquids                      |
| 5706.5.4.5   | Commercial, industrial, governmental or manufacturing establishments—fuel dispensing |
| 5706.6.4     | Tank vehicles for flammable and combustible liquids                                  |
| 5707.5.4     | On-demand mobile fueling                                                             |
| 5906.5.7     | Flammable solids                                                                     |
| 6108.2       | LP-gas                                                                               |

**C** Table 906.1 lists those sections that represent specific occupancy conditions requiring portable fire extinguishers for incipient fire control. Wherever the code requires a fire extinguisher because of one of the listed occupancy conditions, it may identify the required rating of the extinguisher that is compatible with the hazard involved in addition to referencing Section 906.

**906.2 General requirements.** Portable fire extinguishers shall be selected, installed and maintained in accordance with this section and NFPA 10.

**Exceptions:**

1. The distance of travel to reach an extinguisher shall not apply to the spectator seating portions of Group A-5 occupancies.
2. Thirty-day inspections shall not be required and maintenance shall be allowed to be once every 3 years for dry-chemical or halogenated agent portable fire extinguishers that are supervised by a *listed* and *approved* electronic monitoring device, provided that all of the following conditions are met:
  - 2.1. Electronic monitoring shall confirm that extinguishers are properly positioned, properly charged and unobstructed.
  - 2.2. Loss of power or circuit continuity to the electronic monitoring device shall initiate a trouble signal.
  - 2.3. The extinguishers shall be installed inside of a building or cabinet in a noncorrosive environment.
  - 2.4. Electronic monitoring devices and supervisory circuits shall be tested every 3 years when extinguisher maintenance is performed.
  - 2.5. A written log of required hydrostatic test dates for extinguishers shall be maintained by the *owner* to verify that hydrostatic tests are conducted at the frequency required by NFPA 10.
3. In Group I-3, portable fire extinguishers shall be permitted to be located at staff locations.

**C** NFPA 10 contains minimum requirements for the selection, installation and maintenance of portable fire extinguishers. Portable fire extinguishers are investigated and rated in conformance to NFPA 10 and listed under a variety of standards. Portable fire extinguishers must be labeled and rated for use on fires of the type, severity and hazard class protected.

NFPA 10 notes that more frequent inspections may be necessary where conditions warrant. For existing installations, a history of recent fires, vandalism, physical abuse and theft should be considered in determining if more frequent inspections are needed. For both existing and new facilities, determining the frequency of inspections should take into account the environmental conditions in which the extinguisher will be located, including corrosiveness and temperature variations, and the possibility of obstructions that may place the extinguisher out of reach in case of an emergency.

Exception 1 recognizes the openness to the atmosphere associated with Group A-5 occupancies. A fire in open areas is more obvious to all spectators. Group A-5 occupancies also do not accumulate smoke and hot gases because they are not enclosed spaces. These reasons, in addition to the large and expansive layout within seating areas, make it reasonable and practical not to apply the distance of travel to a PFE criteria in Group A-5. Revised distance of travel allowances would need to be approved by the fire code official. Group A-5 occupancies also tend to be more subject to the corrosive conditions of an outdoor environment and may include freeze/thaw cycles that can be detrimental to fire extinguishers.

Exception 2 acknowledges a 30-day inspection interval similar to NFPA 10. An electronic monitoring device can determine whether the fire extinguisher is still present and whether its contents are still at the proper charge. The use of such devices, being relatively new, is allowed if it is limited to dry-chemical and halogenated agents with the additional

safeguards noted in the list. Where inspections may be at more frequent intervals, as discussed earlier, the use of electronic monitoring may have even greater benefit and is acknowledged as such in NFPA 10. The log, noted in Exception 2.5, can be a written log or a printout of the electronic log maintained by the electronic monitoring device. This exception provides the building owner with an alternative to the contract inspections popularly used.

Exception 3 recognizes that portable fire extinguishers located throughout the facility are at times tampered with, removed or used for weapons by inmates in a detention or correctional setting. This exception would protect the extinguishers from damage or removal by inmates while still making them available to staff and employees for use in an emergency situation.

**906.2.1 Certification of service personnel for portable fire extinguishers.** Service personnel providing or conducting maintenance on portable fire extinguishers shall possess a valid certificate issued by an *approved* governmental agency, or other *approved* organization for the type of work performed.

**C** Maintenance of fire protection systems and devices are minimum Chapter 9 requirements. Fire protection systems, like other technologies, have advanced, new designs that require a clear understanding of their construction and maintenance. To ensure that systems and devices are properly maintained, the code now requires individuals performing these activities be certified. Certification must be issued by an approved organization or governmental agency.

These provisions align the code with NFPA standards governing the inspection and maintenance of portable fire extinguishers.

Qualifications for individuals who service portable fire extinguishers are established in the 2018 edition of NFPA 10. Section 7.1.2.1 of NFPA 10 requires individuals inspecting and servicing portable fire extinguishers be trained and certified to reliably perform these activities.

**906.3 Size and distribution.** The size and distribution of portable fire extinguishers shall be in accordance with Sections 906.3.1 through 906.3.4.

**C** Proper selection and distribution of portable fire extinguishers are essential to having adequate protection for the building structure and the occupancy conditions within. This section introduces the sections that provide those requirements. Determination of the desired type of portable fire extinguisher depends on the character of the fire anticipated, building occupancy, specific hazards and ambient temperature conditions [see commentary, Tables 906.3(1) and 906.3(2)].

**TABLE 906.3(1)—FIRE EXTINGUISHERS FOR CLASS A FIRE HAZARDS**

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>LIGHT (Low)<br/>HAZARD OCCUPANCY</b> | <b>ORDINARY (Moderate)<br/>HAZARD OCCUPANCY</b> | <b>EXTRA (High)<br/>HAZARD OCCUPANCY</b> |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|-------------------------------------------------|------------------------------------------|
| Minimum-rated single extinguisher                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2-A <sup>c</sup>                        | 2-A                                             | 4-A <sup>a</sup>                         |
| Maximum floor area per unit of A                                                                                                                                                                                                                                                                                                                                                                                                                                             | 3,000 square feet                       | 1,500 square feet                               | 1,000 square feet                        |
| Maximum floor area for extinguisher <sup>b</sup>                                                                                                                                                                                                                                                                                                                                                                                                                             | 11,250 square feet                      | 11,250 square feet                              | 11,250 square feet                       |
| Maximum distance of travel to extinguisher                                                                                                                                                                                                                                                                                                                                                                                                                                   | 75 feet                                 | 75 feet                                         | 75 feet                                  |
| For SI: 1 foot = 304.8 mm, 1 square foot = 0.0929 m <sup>2</sup> , 1 gallon = 3.785 L.<br>a. Two 2½-gallon water-type extinguishers shall be deemed the equivalent of one 4-A rated extinguisher.<br>b. Annex E.3.3 of NFPA 10 provides more details concerning application of the maximum floor area criteria.<br>c. Two water-type extinguishers, each with a 1-A rating, shall be deemed the equivalent of one 2-A rated extinguisher for Light (Low) Hazard Occupancies. |                                         |                                                 |                                          |

**C** Table 906.3(1), which parallels Table 6.2.1.1 of NFPA 10, establishes the minimum number and rating of fire extinguishers for Class A fires in any particular occupancy. The occupancy classifications are further defined in NFPA 10. The maximum area that a single fire extinguisher can protect is determined based on the rating of the fire extinguisher. The distance of travel limitation of 75 feet (22 860 mm) is intended to be the actual walking distance along a normal path of travel to the extinguisher. For this reason, it is necessary to select fire extinguishers that comply with both the distribution criteria and distance of travel limitation for a specific occupancy classification.

**TABLE 906.3(2)—FIRE EXTINGUISHERS FOR FLAMMABLE OR COMBUSTIBLE LIQUIDS WITH DEPTHS OF LESS THAN OR EQUAL TO 0.25 INCH<sup>a</sup>**

| <b>TYPE OF HAZARD</b> | <b>BASIC MINIMUM EXTINGUISHER RATING</b> | <b>MAXIMUM DISTANCE OF TRAVEL TO EXTINGUISHERS<br/>(feet)</b> |
|-----------------------|------------------------------------------|---------------------------------------------------------------|
| Light (Low)           | 5-B                                      | 30                                                            |
|                       | 10-B                                     | 50                                                            |

**TABLE 906.3(2)—FIRE EXTINGUISHERS FOR FLAMMABLE OR COMBUSTIBLE LIQUIDS WITH DEPTHS OF LESS THAN OR EQUAL TO 0.25 INCH<sup>a</sup>—continued**

| TYPE OF HAZARD      | BASIC MINIMUM EXTINGUISHER RATING | MAXIMUM DISTANCE OF TRAVEL TO EXTINGUISHERS (feet) |
|---------------------|-----------------------------------|----------------------------------------------------|
| Ordinary (Moderate) | 10-B                              | 30                                                 |
|                     | 20-B                              | 50                                                 |
| Extra (High)        | 40-B                              | 30                                                 |
|                     | 80-B                              | 50                                                 |

For SI: 1 inch = 25.4 mm, 1 foot = 304.8 mm.  
a. For requirements on water-soluble flammable liquids and alternative sizing criteria, see Section 5.5 of NFPA 10.

**C** Fires involving flammable or combustible liquids present a severe hazard challenge regardless of occupancy. Table 906.3(2), which parallels Table 6.3.1.1 of NFPA 10, prescribes the minimum portable fire extinguisher requirements where flammable or combustible liquids are limited in depth [0.25 inch (6 mm) or less]. As can be seen in the table, the size of the extinguisher is directly related to the distance of travel to the extinguisher for each given occupancy classification. These fire extinguisher provisions are independent of whether other fixed automatic fire-extinguishing systems are installed. For occupancy conditions involving flammable or combustible liquids in potential depths greater than 0.25 inch (6 mm), the selection and spacing criteria of NFPA 10 must be used in addition to any applicable requirements in Chapter 57 and NFPA 30.

**906.3.1 Class A fire hazards.** The minimum sizes and distribution of portable fire extinguishers for occupancies that involve primarily Class A fire hazards shall comply with Table 906.3(1).

**C** Class A fires generally involve materials considered to be “ordinary combustibles,” such as wood, cloth, paper, rubber and most plastics [see commentary, Table 906.3(1)].

**906.3.2 Class B fire hazards.** Portable fire extinguishers for occupancies involving *flammable or combustible liquids* with depths less than or equal to 0.25 inch (6.4 mm) shall be selected and placed in accordance with Table 906.3(2).

Portable fire extinguishers for occupancies involving *flammable or combustible liquids* with a depth of greater than 0.25 inch (6.4 mm) shall be selected and placed in accordance with NFPA 10.

**C** Class B fires involve flammable and combustible liquids, oil-based paints, alcohols, solvents, flammable gases and similar materials. Selection of these extinguishers is made based on the depth of the liquid that could become involved in a fire. If the depth is 0.25 inch (6 mm) or less, selection is made using Table 906.3(2). Class B extinguishers for greater liquid depth, characterized in NFPA 10 as “appreciable depth,” must be selected and installed in accordance with Section 6.3.2 of NFPA 10 [see commentary, Table 906.3(2)].

**906.3.3 Class C fire hazards.** Portable fire extinguishers for Class C fire hazards shall be selected and placed on the basis of the anticipated Class A or B hazard.

**C** Class C fires involve energized electrical equipment where the electrical nonconductivity of the extinguishing agent is critical. The need for this class of extinguisher is simply based on the presence of the hazard in an occupancy and no numerical rating is required.

**906.3.4 Class D fire hazards.** Portable fire extinguishers for occupancies involving combustible metals shall be selected and placed in accordance with NFPA 10.

**C** Class D fires involve flammable solids, the bulk of which are combustible metals including, but not limited to, magnesium, potassium, sodium and titanium. Most Class D extinguishers will have a special low-velocity nozzle or discharge wand to gently apply the agent in large volumes to avoid disrupting any finely divided burning materials. Extinguishing agents are also available in bulk and can be applied with a scoop or shovel. While Class D extinguishers are often referred to as “dry chemical” fire extinguishers, they are more properly called “dry powder” fire extinguishers because their mechanism of extinguishment is by a smothering action rather than by chemical reaction with the combustion process.

There are several Class D fire-extinguisher agents available—some will handle multiple types of metal fires, others will not. Sodium carbonate-based extinguishers are used to control sodium, potassium, and sodium-potassium alloy fires but have limited use on other metals. This material smothers and forms a crust. Sodium chloride-based extinguishers contain sodium chloride salt and a thermoplastic additive. The plastic melts to form an oxygen-excluding crust over the metal, and the salt dissipates heat. This powder is useful on most alkali metals, including magnesium, titanium, aluminum, sodium, potassium and zirconium. Graphite-based extinguishers contain dry graphite powder that smothers burning metals. Unlike sodium chloride powder extinguishers, the graphite powder fire extinguishers can be used on very hot burning metal fires such as lithium, but the powder will not stick to and extinguish flowing or vertical lithium fires. The graphite powder acts as a heat sink as well as a means of smothering the metal fire. See the commentary to Section 5906.5.7 for a discussion of extinguishing flammable solid fires.

**906.4 Cooking equipment fires.** Fire extinguishers provided for the protection of cooking equipment shall be of an *approved* type compatible with the *automatic fire-extinguishing system* agent. Cooking equipment involving solid fuels or vegetable or animal oils and fats shall be protected by a Class K-rated portable extinguisher in accordance with Sections 906.1, Item 2, 906.4.1 and 906.4.2 as applicable.

**C** The combination of high-efficiency cooking appliances and hotter burning cooking media creates a potentially severe fire hazard. Although commercial cooking systems must have an approved exhaust hood and be protected by an approved automatic fire-extinguishing system, a manual means of extinguishment is desirable to attack a fire in its incipient stage.

As indicated in Section 906.1, Item 2, a Class K-rated portable fire extinguisher must be located within 30 feet (9144 mm) of travel distance of commercial-type cooking equipment. Class K-rated extinguishers have been specifically tested on commercial cooking appliances using vegetable or animal oils or fats. These portable fire extinguishers are usually of sodium bicarbonate or potassium bicarbonate dry-chemical type.

**906.4.1 Portable fire extinguishers for solid fuel cooking appliances.** Solid fuel cooking appliances, whether or not under a hood, with fireboxes 5 cubic feet (0.14 m<sup>3</sup>) or less in volume shall have a minimum 2.5-gallon (9 L) or two 1.5-gallon (6 L) Class K wet-chemical portable fire extinguishers located in accordance with Section 906.1.

**C** The fuels used in solid fuel-fired cooking appliances present significantly more potential burning surface area than the flat surface of a grill or deep fat fryer. This surface area is also often shielded by other solid fuel elements. As a result, a large extinguisher or two moderate-sized extinguishers are required. The 2 1/2-gallon (9 L) extinguisher roughly equates to a 2A rating. The K-rating is necessary rather than using a water-based agent because the discharge from water-based extinguishers is usually in the form of a straight stream rather than a less concentrated, flooding type of coverage. A straight stream can dislodge the burning solid fuel material and possibly spread the burning coals to other areas where they could pose both a secondary fire risk, as well as a life safety hazard. The same travel distance to an extinguisher is required for solid fuel extinguishers as for deep fat fryer extinguishers so that manual suppression can be provided, if necessary, in a reasonable time. See the commentary to Section 906.4.2 for further discussion of Class K extinguishers.

**906.4.2 Class K portable fire extinguishers for deep fat fryers.** Where hazard areas include deep fat fryers, *listed* Class K portable fire extinguishers shall be provided as follows:

1. For up to four fryers having a maximum cooking medium capacity of 80 pounds (36.3 kg) each: one Class K portable fire extinguisher of a minimum 1.5-gallon (6 L) capacity.
2. For every additional group of four fryers having a maximum cooking medium capacity of 80 pounds (36.3 kg) each: one additional Class K portable fire extinguisher of a minimum 1.5-gallon (6 L) capacity shall be provided.
3. For individual fryers exceeding 6 square feet (0.55 m<sup>2</sup>) in surface area: Class K portable fire extinguishers shall be installed in accordance with the extinguisher manufacturer's recommendations.

**C** Newer commercial cooking operations use improved, more efficient deep fat fryer-type cooking appliances and more healthful, unsaturated cooking oils that require a much higher cooking temperature than the former saturated oils. The Class K extinguishing agent and extinguishers were developed to deal with this new hazard. Class K extinguishers use a wet-chemical, potassium acetate-based agent that has proven to be more effective in fighting these fires and provides a cooling effect for the deep fat fryer hazard. Though primarily intended for cooking fires, many Class K extinguishers can also be effectively used on Class A, B and C hazards.

Class K fire extinguishers do not have letter ratings similar to other types of extinguishers. The capacity of the Class K extinguisher becomes the effective rating. Based on the extinguishing capability of a moderate-sized Class K extinguisher, the maximum quantity of typical fat frying medium can be determined. This quantity is determined by weight based on the typical deep fat fryer. A fryer capacity of 80 pounds (36.3 kg) can provide a surface area between 4 1/2 to 6 square feet (0.42 to 0.55 m<sup>2</sup>), depending on the manufacturer. Where the surface area exceeds 6 square feet (0.55 m<sup>2</sup>), guidelines for Class K extinguishers are no longer applicable. Consequently, for the larger surface area fryers, the size of Class K extinguisher should be based on the manufacturer's recommendations. Although not specifically indicated in the code text, the understanding is that if the weight capacity of the fryer exceeds 80 pounds (36.3 kg) but the surface area is less than 6 square feet (0.55 m<sup>2</sup>), the manufacturer's recommendations should be applied for those conditions as well.

**906.5 Conspicuous location.** Portable fire extinguishers shall be located in conspicuous locations where they will have *ready access* and be immediately available for use. These locations shall be along normal paths of travel, unless the *fire code official* determines that the hazard posed indicates the need for placement away from normal paths of travel.

**C** Fire extinguishers must be located in readily accessible locations along normal egress paths. This increases the occupants' familiarity with the location of the fire extinguishers. When considering location, the most frequent occupants should be considered. These are the occupants who would become most familiar with the fire-extinguisher placement. For most buildings, it is the employees who are most familiar with their surroundings; therefore, a good understanding of employee operations is important for proper extinguisher placement.

**906.6 Unobstructed and unobscured.** Portable fire extinguishers shall not be obstructed or obscured from view. In rooms or areas in which visual obstruction cannot be completely avoided, means shall be provided to indicate the locations of extinguishers.

**C** Portable fire extinguishers must be located where they are readily visible at all times. If visual obstruction cannot be avoided, the location of the extinguishers must be marked by an approved means of identification. This could include additional signage, lights, arrows or other means approved by the fire code official. Unobstructed does not necessarily mean visible from all angles within the space. Often, columns or furnishings may obscure the extinguisher from one direction or another. These are not by themselves obstructions. The intent is that the extinguisher is not hidden but rather can be readily found. If the extinguisher is placed in the wall behind a door, it is clearly obstructed since it cannot be easily viewed. An extinguisher on a wall that is visible from most of the space would be considered unobstructed.

**906.7 Hangers and brackets.** Hand-held portable fire extinguishers, not housed in cabinets, shall be installed on the hangers or brackets supplied. Hangers or brackets shall be securely anchored to the mounting surface in accordance with the manufacturer's installation instructions.

**C** Portable fire extinguishers not housed in cabinets are usually mounted on walls or columns using securely fastened hangers. Brackets must be used where the fire extinguishers need to be protected from impact or other potential physical damage.

**906.8 Cabinets.** Cabinets used to house portable fire extinguishers shall not be locked.

**Exceptions:**

1. Where portable fire extinguishers subject to malicious use or damage are provided with a means of *ready access*.
2. In Group I-3 occupancies and in mental health areas in Group I-2 occupancies, access to portable fire extinguishers shall be permitted to be locked or to be located in staff locations provided that the staff has keys.

**C** Cabinets housing fire extinguishers must not be locked in order to provide quick access in an emergency. Exception 1, however, allows the cabinets to be locked in occupancies where vandalism, theft or other malicious behavior is possible. Exception 2 also permits cabinets housing fire extinguishers to be locked or to be located in staff locations in Group I-3 occupancies and mental health areas in Group I-2 occupancies. Occupants in Group I-3 areas of jails, prisons or similar restrained occupancies should not have access to fire extinguishers because they could possibly be used as a weapon or be subject to vandalism. Staff members who have been adequately trained in the use of fire extinguishers are assumed to have ready access to the keys for the cabinets at all times.

**906.9 Extinguisher installation.** The installation of portable fire extinguishers shall be in accordance with Sections 906.9.1 through 906.9.3.

**C** This section introduces the installation criteria for portable fire extinguishers based on the weight of the unit.

**906.9.1 Extinguishers weighing 40 pounds or less.** Portable fire extinguishers having a gross weight not exceeding 40 pounds (18 kg) shall be installed so that their tops are not more than 5 feet (1524 mm) above the floor.

**C** Because of the varying height and physical strength levels of persons who might be called on to operate a portable fire extinguisher, the mounting height of the extinguisher must be commensurate with its weight so that it may be easily retrieved by anyone from its mounting location and placed into use.

**906.9.2 Extinguishers weighing more than 40 pounds.** Hand-held portable fire extinguishers having a gross weight exceeding 40 pounds (18 kg) shall be installed so that their tops are not more than 3.5 feet (1067 mm) above the floor.

**C** See the commentary to Section 906.9.1.

**906.9.3 Floor clearance.** The clearance between the floor and the bottom of installed hand-held portable fire extinguishers shall be not less than 4 inches (102 mm).

**C** The clearance between the floor and the bottom of installed hand-held extinguishers must not be less than 4 inches (102 mm) to facilitate cleaning beneath the unit and reduce the likelihood of the extinguisher becoming dislodged during cleaning operations (floor mopping, sweeping, etc.).

**906.10 Wheeled units.** Wheeled fire extinguishers shall be conspicuously located in a designated location.

**C** Wheeled fire extinguishers consist of a large-capacity (up to several hundred pounds of agent) fire extinguisher assembly (either stored-pressure or pressure-transfer type) equipped with a carriage and wheels and discharge hose. They are constructed so that one able-bodied person could move the unit to the fire area and begin extinguishment unassisted. Wheeled fire extinguishers are capable of delivering greater flow rates and stream range for various extinguishing agents than hand-held portable fire extinguishers. Wheeled fire extinguishers are generally more effective in high-hazard areas and, as with any extinguisher, must be readily available and stored in an approved location. The wheeled fire extinguisher should be located a safe distance from the hazard area so that it will not become involved in the fire or access to it compromised by a fire. These units are typically found at airport fueling ramps, refineries, bulk plants and similar locations where high-challenge fires may be encountered. The extinguishing agents available in wheeled units include carbon dioxide, dry chemical, dry powder and foam.

## SECTION 907—FIRE ALARM AND DETECTION SYSTEMS

**907.1 General.** This section covers the application, installation, performance and maintenance of fire alarm systems and their components in new and existing buildings and structures. The requirements of Section 907.2 are applicable to new buildings and structures. The requirements of Section 907.9 are applicable to existing buildings and structures.

**C** Fire alarm systems, which typically include manual fire alarm systems and automatic fire detection systems, must be installed in accordance with Section 907 and NFPA 72. As indicated in this section, only Section 907.9 is intended to be applicable to existing buildings and structures.

Manual fire alarm systems are installed in buildings to limit fire casualties and property losses. Fire alarm systems do this by promptly notifying the occupants of the building of an emergency, which increases the time available for evacuation. Similarly, when fire alarm systems are supervised, the fire department will be promptly notified and its response time relative to the onset of the fire will be reduced.

Automatic fire detection systems are required under certain conditions to increase the likelihood that a fire is detected and occupants are given an early warning. The detection system is a system of devices and associated hardware that activates the alarm system. The automatic detecting devices are to be smoke detectors, unless a condition exists that calls for the use of a different type of detector.

**907.1.1 Construction documents.** *Construction documents* for fire alarm systems shall be of sufficient clarity to indicate the location, nature and extent of the work proposed and show in detail that it will conform to the provisions of this code; the *International Building Code*; and relevant laws, ordinances, rules and regulations, as determined by the *fire code official*.

**C** Construction documents for fire alarm systems must be submitted for review to determine compliance with the code, the IBC and NFPA 72. All of the information required by this section may not be available during the design stage and initial permit process. Later submission of more detailed shop drawings may be required in accordance with Section 907.1.2. These provisions are intended to reflect the minimum scope of information needed to determine code compliance. When the work can be briefly described on the application form, the fire code official may utilize judgment in determining the need for more detailed documents.

**907.1.2 Fire alarm shop drawings.** Shop drawings for fire alarm systems shall be prepared in accordance with NFPA 72 and submitted for review and approval prior to system installation.

**C** Since the fire protection contractor(s) may not have been selected at the time a permit is issued for construction of a building, detailed shop drawings for fire alarm systems may not be available. Because they provide the information necessary to determine code compliance, as specified in this section, they must be submitted and approved by the fire code official before the contractor can begin installing the system.

**907.1.3 Equipment.** Systems and components shall be *listed* and *approved* for the purpose for which they are installed.

**C** The components of the fire alarm system must be approved for use in the planned system. NFPA 72 requires all devices, combinations of devices, appliances and equipment to be labeled for their proposed use. The testing agency will test the components for use in various types of systems and stipulate the use of the component on the label. Evidence of listing and labeling of the system components must be submitted with the shop drawings. In some instances, the entire system may be labeled.

At least one major testing agency, Underwriters Laboratories, Inc. (UL), has a program in which alarm installation and service companies are issued a certificate and become listed by the agency as being qualified to design, install and maintain local, auxiliary, remote station or proprietary fire alarm systems. The listed companies may then issue a certificate showing that the system is in compliance with Section 907. Terms of the company certification by UL include the company being responsible for keeping accurate system documentation, including as-built record drawings, acceptance test records and complete maintenance records on a given system. The company is also responsible for the required periodic inspection and testing of the system under contract with the owner. A similar program has been available for many years for central station alarm service, whereas the UL program is relatively new to the industry. Even though this company and system listing program is not required by the code or NFPA 72, it can be a valuable tool for the fire code official in determining compliance with the referenced standard.

Another issue that must be considered is the compatibility of the system components as required by NFPA 72. The labeling of system components discussed previously should include any compatibility restrictions for components. Compatibility is primarily an issue of the ability of smoke detectors and fire alarm control panels (FACPs) to function properly when interconnected and affects the two-wire type of smoke detectors, which obtain their operating power over the same pair of wires used to transmit signals to the FACP (the control unit initiating device circuits). Laboratories will test for component compatibility either by actual testing or by reviewing the circuit parameters of both the detector and the FACP. Generally, if both the two-wire detector and the FACP are of the same brand, there should not be a compatibility problem. Nevertheless, the fire code official must be satisfied that the components are listed as being compatible. Failure to comply with the compatibility requirements of NFPA 72 can lead to system malfunction or failure at a critical time.

**907.2 Where required—new buildings and structures.** An *approved fire alarm system* installed in accordance with the provisions of this code and NFPA 72 shall be provided in new buildings and structures in accordance with Sections 907.2.1 through 907.2.23 and provide occupant notification in accordance with Section 907.5, unless other requirements are provided by another section of this code.

Not fewer than one manual fire alarm box shall be provided in an *approved* location to initiate a fire alarm signal for fire alarm systems employing automatic fire detectors or waterflow detection devices. Where other sections of this code allow elimination of fire alarm boxes due to sprinklers, a single fire alarm box shall be installed.

**Exceptions:**

1. The manual fire alarm box is not required for fire alarm systems dedicated to elevator recall control and supervisory service.
2. The manual fire alarm box is not required for Group R-2 occupancies unless required by the *fire code official* to provide a means for fire watch personnel to initiate an alarm during a sprinkler system impairment event. Where provided, the manual fire alarm box shall not be located in an area that is open to the public.

**C** This section specifies the occupancies or conditions in new buildings or structures that require some form of fire alarm system that is either a manual fire alarm system (manual fire alarm boxes) or an automatic smoke detection system. These systems must, upon activation, provide occupant notification throughout the area protected by the system unless other alternative provisions are allowed by this section.

Manual fire alarm systems must be installed in certain occupancies, depending on the number of occupants, capabilities of the occupants and height of the building. An automatic smoke detection system must be installed in those occupancies and conditions where the need to detect the fire is essential to evacuation or protection of the occupants. The requirements for automatic smoke detection are generally based on the evacuation needs of the occupants and whether the occupancy includes sleeping accommodations.

Fire alarm systems must be installed in accordance with the code and NFPA 72. NFPA 72 identifies the minimum performance, location, mounting, testing and maintenance requirements for fire alarm systems. Smoke detectors must be used, except when ambient conditions would prohibit their use. In that case, other detection methods may be used. The manufacturer's literature will identify the limitations on the use of smoke detectors, including environmental conditions such as humidity, temperature and airflow.

Only certain occupancies are required to have either a manual fire alarm or automatic fire detection system installed (see Commentary Figure 907.2). The need for either system is generally determined by the number of occupants, the height of the building or the ability of the occupants for self-preservation.

Note that generally the fire alarm requirements are based on occupancy and not on fire area. Commentary Figure 907.2 contains the conditions that require where either system must be installed in a building. The extent that an alarm system must be installed in a building once it has been determined that such a system is required is based on several factors. One, if it is the only occupancy in the building, then it would be required throughout the building. Two, if the building is a mixed occupancy, it can either be separated or nonseparated. If the occupancy is separated in accordance with IBC Section 508.4, then the alarm system is only required within that separated portion of the building. If the building is considered a nonseparated, mixed-occupancy building, then IBC Section 508.3.1 states that the code applies to each portion of the building based on the occupancy classification of that space and that the most restrictive provisions of Chapter 9 shall apply to the building or portion thereof in which the nonseparated occupancies are located. Therefore, if you have a Group A occupancy in a nonseparated mixed occupancy (containing other occupancies such as Groups B and M) where the Group A occupancy exceeds an occupant load of 300, then the entire nonseparated mixed occupancy would require the alarm system. Note that IBC Section 508.3.1 focuses on each space to determine occupancy and requirements. Once the occupant load is determined, then any requirements, such as fire alarms, would be required throughout.

The code does not address whether a nonseparated mixed occupancy has a completely independent means of egress such as in a strip mall. Additionally, in a building containing primarily Group A occupancies, the code does not clearly address whether such occupancies within a building should be looked at as an aggregate or individual space. This issue was clarified in the 2012 edition of the code through the use of the fire area concept for Group A occupancies, but only for the basic manual fire alarm requirements in Section 907.2.1. The emergency voice/alarm communication requirements in Section 907.2.1.1 provide a criterion of 1,000 or more occupants in a Group A occupancy. A building with multiple Group A occupancies without separation of egress paths would need to be reviewed in aggregate. Fire area separation could not be used to provide separation of occupancies in this case.

Commentary Figure 907.2 contains the threshold requirements for where a manual fire alarm system or an automatic fire detection system is required based on the occupancy group. It is important to remember that although the requirement for manual pull stations may not apply (e.g., sprinklered buildings), alarm and occupant notification may still be required. Sections 907.2.11 through 907.2.23 contain additional requirements for fire alarm systems, depending on special occupancy conditions such as atriums, high-rise buildings or covered mall buildings.

The single manual fire alarm box required by this section is needed to provide a means of manually activating a fire alarm system that only contains automatic devices such as sprinkler waterflow switches or smoke detectors. Its primary use is for alarm system maintenance technicians to be able to manually activate the fire alarm system in the event of a fire during the time the system or portions of the system are down for maintenance. Note that this requirement is not subject to any of the exceptions in Sections 907.2.1 through 907.2.23 that might waive the need for manual fire alarm boxes in certain buildings.

Exception 1 recognizes the specialized nature of fire alarm systems installed only for emergency elevator control and supervision.

Exception 2 waives the single manual fire alarm box but gives the fire code official authority to require it in sprinklered buildings for use by fire watch personnel or sprinkler maintenance personnel to be able to manually activate the fire alarm system in the event of a fire during the time the sprinkler system is down for maintenance.

**Commentary Figure 907.2—Summary of Manual Fire Alarm and Automatic Smoke Detection System Thresholds**

| <b>MANUAL FIRE ALARM SYSTEM</b>                    |                                                                                                                                                                                                                                                                                              |
|----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Occupancy Group(s)</b>                          | <b>Threshold</b>                                                                                                                                                                                                                                                                             |
| Assembly (A-1, A-2, A-3, A-4, A-5)                 | All with an occupant load of $\geq 300$ ; or, $> 100$ above or below the lowest level of exit discharge (907.2.1)                                                                                                                                                                            |
| Business (B)                                       | Total Group B occupant load of $\geq 500$ ; or, $> 100$ above or below the lowest level of exit discharge; or, in Group B fire areas containing an ambulatory care facility (907.2.2)                                                                                                        |
| Educational (E)                                    | $> 50$ occupants (several exceptions for manual fire alarm box placement) (907.2.3)                                                                                                                                                                                                          |
| Factory (F-1, F-2)                                 | $\geq 2$ stories with occupant load of $\geq 500$ above or below the lowest level of exit discharge (exception for sprinklers) (907.2.4)                                                                                                                                                     |
| High hazard (H)                                    | Group H-5 occupancies and occupancies used for manufacture of organic coatings (907.2.5)                                                                                                                                                                                                     |
| Institutional (I-1, I-2, I-3, I-4)                 | All (exceptions for I-1 and I-2 manual fire alarm box placement and private mode signaling) (907.2.6)                                                                                                                                                                                        |
| Mercantile (M)                                     | Total Group M occupant load of $\geq 500$ ; or, occupant load of $> 100$ above or below the lowest level of exit discharge (907.2.7)                                                                                                                                                         |
| Hotels (R-1)                                       | All (exceptions for $\leq 2$ stories with sleeping units having exit directly to exterior; sprinklers) (907.2.8.1)                                                                                                                                                                           |
| Multiple family (R-2)                              | If units $\geq 3$ stories above lowest level of exit discharge; or, $> 1$ story below the highest level of exit discharge; or, $> 16$ units (exceptions for $\leq 2$ stories with sleeping units having exit directly to exterior; sprinklers; open ended corridors/no corridor) (907.2.9.1) |
| Residential care/assisted living (R-4)             | All (exceptions for sprinklers, manual fire alarm boxes at staff locations, direct exit to exterior, $< 2$ stories) (907.2.10.2)                                                                                                                                                             |
| <b>AUTOMATIC SMOKE DETECTION SYSTEM</b>            |                                                                                                                                                                                                                                                                                              |
| Business (B) Ambulatory care facilities            | Facility plus public use areas outside of it including public corridors and elevator lobbies (exception for sprinklers) (907.2.2.1)                                                                                                                                                          |
| High hazard (H)                                    | Highly toxic gases, organic peroxides, oxidizers (907.2.5)                                                                                                                                                                                                                                   |
| Institutional (I-1, I-2, I-3)                      | All, in specific areas by occupancy (907.2.6.1, 907.2.6.2, 907.2.6.3.3)                                                                                                                                                                                                                      |
| Hotels (R-1)                                       | All, in interior corridors (exception for buildings without interior corridors and with sleeping units having exit directly to exterior) (907.2.8.2)                                                                                                                                         |
| Residential care/assisted living (R-4)             | All, in corridors, waiting areas open to corridors, nonsleeping area habitable spaces and kitchens (907.2.10.2)                                                                                                                                                                              |
| College and university buildings (R-2 dormitories) | Common spaces; laundry, mechanical and storage rooms; interior corridors (exception for no interior corridors and each room has direct exit) (907.2.9.3)                                                                                                                                     |

**907.2.1 Group A.** A manual *fire alarm system* that activates the occupant notification system in accordance with Section 907.5 shall be installed in Group A occupancies where the *occupant load* due to the assembly occupancy is 300 or more, or where the Group A *occupant load* is more than 100 persons above or below the lowest *level of exit discharge*. Group A occupancies not separated from one another in accordance with Section 707.3.10 of the *International Building Code* shall be considered as a single occupancy for the purposes of applying this section. Portions of Group E occupancies occupied for assembly purposes shall be provided with a *fire alarm system* as required for the Group E occupancy.

**Exceptions:**

1. Manual fire alarm boxes are not required where the building is equipped throughout with an *automatic sprinkler system* installed in accordance with Section 903.3.1.1 and the occupant notification appliances will activate throughout the notification zones upon sprinkler water flow.
2. Manual fire alarm boxes and the associated occupant notification system or emergency voice/alarm communication system are not required for Group A-5 outdoor bleacher-type seating having an occupant load of greater than or equal to 300 and less than 15,000 occupants, provided that all of the following are met:
  - 2.1. A public address system with standby power is provided.
  - 2.2. Enclosed spaces attached to or within 5 feet (1524 mm) of the outdoor bleacher-type seating compose, in the aggregate, a maximum of 10 percent of the overall area of the outdoor bleacher-type seating or 1,000 square feet (92.9 m<sup>2</sup>), whichever is less.
  - 2.3. Enclosed accessory spaces under or attached to the outdoor bleacher-type seating shall be separated from the bleacher-type seating in accordance with Section 1030.1.1.1.
  - 2.4. All means of egress from the bleacher-type seating are open to the outside.
3. Manual fire alarm boxes and the associated occupant notification system or emergency voice/alarm communication system are not required for temporary Group A-5 outdoor bleacher-type seating, provided that all of the following are met:
  - 3.1. There are no enclosed spaces under or attached to the outdoor bleacher-type seating.
  - 3.2. The bleacher-type seating is erected for a period of less than 180 days.
  - 3.3. Evacuation of the bleacher-type seating is included in an *approved* fire safety plan.

**C** Group A occupancies are typically occupied by a significant number of people who are not completely familiar with their surroundings. The provisions of this section address three separate situations regarding the application of the alarm requirements for Group A occupancies. The three situations addressed by the provisions are: (1) where an assembly occupancy and another occupancy are involved; (2) where multiple assembly areas exist in a building; and (3) where the assembly use occurs in and is a part of a Group E occupancy.

In situations where an assembly area and another occupancy are involved, the code specifies that it is the occupant load “due to the assembly occupancy” that would need to be 300 or more before the manual alarm system is required. For example, if the building is constructed with an assembly occupancy, such as a restaurant, with an occupant load of 250 and an adjacent office area with an occupant load of 100, the assembly space would not require an alarm system because the occupant load “due to the assembly occupancy” is less than 300. This is simply a clarification of the way the provisions were intended to be applied. This would be the intended way to apply the provision whether the building was constructed using the accessory-, separated- or nonseparated-occupancy requirements of IBC Chapter 5.

In buildings that contain multiple assembly areas, the second portion of the code text requires that the aggregate occupant load of the assembly areas is used unless the spaces are separated as required for fire areas in IBC Section 707.3.10. Consider two examples to address this portion of the requirements. In a multiple-theater complex, the auditoriums are generally not separated from each other by the 2-hour fire-resistance rating that IBC Table 707.3.10 would require; therefore, the aggregate occupant load of all of the assembly spaces would be used to determine if the occupant load was 300 or more. If it was, then the manual alarm would be required in all of the assembly spaces. As another example, consider a strip mall shopping center with a restaurant at one end of the building with an occupant load of 200 and a different restaurant with an occupant load of 150 at the other end of the building. Even though these are two completely separate establishments and have an amount of retail occupancy between them, the occupant load of the assembly areas does exceed 300. Therefore, unless a 2-hour fire-resistance-rated separation complying with IBC Section 707.3.10 is provided somewhere between the two restaurants to separate them into different fire areas, a manual fire alarm would be required in the Group A occupancies. If a complying separation is provided at some point in the building, then each assembly space can be reviewed independently and would not require the installation of the alarm system. Be aware that the separation of assembly spaces or the need to aggregate the occupant loads from them could occur not only on the same floor within a building, but also to assembly uses located on different stories.

Exception 1 allows the omission of manual fire alarm boxes in buildings equipped throughout with an automatic sprinkler system if activation of the sprinkler system will activate the building evacuation alarms associated with the manual fire alarm system. Exception 2 applies to permanent (erected for 180 days or more) Group A-5 outdoor bleacher-type seating that has an occupant load of greater than or equal to 300 and less than 15,000 occupants. Exception 3 applies to all

temporary Group A-5 outdoor bleacher-type seating that is erected for a period of less than 180 days. These exceptions both recognize that these occupancies are outdoors and not in a typical building with walls and ceilings that allow for the installation and operation of manual fire alarm boxes, occupant notification appliances and emergency voice/alarm communication systems. In place of this, all the listed alternate requirements for either Exception 2 or 3, as applicable, are to be met. Note that the requirements in these two exceptions are similar to the exceptions in Section 309.1 of ICC 300.

This section also permits assembly-type areas in Group E occupancies to comply with Section 907.2.3 instead of the requirements of this section. A typical high school, for example, contains many areas used for assembly purposes such as a gymnasium, cafeteria, auditorium or library; however, they all exist to serve as an educational facility as their main function. The exception does not eliminate the fire alarm system and occupant notification system, but rather permits them to be initiated automatically by the sprinkler waterflow switch(es) instead of by the manual fire alarm boxes. It also reduces the possibility of mischievous or malicious false alarms being turned on by way of manual fire alarm boxes in venues where large numbers of people congregate.

**907.2.1.1 System initiation in Group A occupancies with an occupant load of 1,000 or more.** Activation of the fire alarm in Group A occupancies with an *occupant load* of 1,000 or more shall initiate a signal using an emergency voice/alarm communications system in accordance with Section 907.5.2.2.

**Exception:** Where *approved*, the prerecorded announcement is allowed to be manually deactivated for a period of time, not to exceed 3 minutes, for the sole purpose of allowing a live voice announcement from an *approved*, constantly attended location.

**C** In order to afford authorized personnel, the ability to selectively evacuate or manage occupant relocation in large assembly venues, this section requires the fire alarm system to operate through an emergency voice/alarm communications system. The exception allows the automatic alarm signals to be overridden for live voice instructions if the live voice instructions do not exceed 3 minutes. The location from which the live voice announcement originates must be constantly attended and approved by the fire code official (see also commentary, Section 907.5.2.2). In terms of the applicability of this section, it is not as specific as Section 907.2.1. More specifically, the concept of fire areas does not apply. Credit is not given to reduce the occupant load through the use of the fire area concept (see commentary, Section 907.2).

**907.2.1.2 Emergency voice/alarm communication system captions.** Stadiums, arenas and *grandstands* required to caption audible public announcements shall be in accordance with Section 907.5.2.2.4.

**C** A 2008 US Federal Court case prompted a change to the code and the IBC. The court ruled that persons with hearing impairments who attend events at stadiums, grandstands and arenas require a means of equivalent communications in lieu of the public address system. Providing occupant notification in these structures is challenging because of the building area and the number and diversity of occupants. Provisions were added in the code to require captioned messages in these buildings and grandstands when public address (PA) systems are prescribed by the accessibility requirements.

IBC Section 1109.2.7.3 sets forth requirements for audible PA systems in stadiums, arenas and grandstands. It requires that equivalent text information be provided to the audience and that the delivery time for these messages be the same as those broadcast from the PA system. The requirements apply to prerecorded and real-time messages. IBC Section 1109.2.7.3 also requires captioning of messages in stadiums, arenas and grandstands that have more than 15,000 fixed seats.

Because messages being broadcast can include instructions to building or site occupants explaining the actions they need to take in the event of an emergency, the requirements of NFPA 72 are applicable for captioning systems. Such a system falls within the scope of Chapter 24 of NFPA 72 entitled "Emergency Communication Systems." NFPA 72 defines an emergency communications system (ECS) as a system designed for life safety that indicates the existence of an emergency and communicates the appropriate response and action. The ECS is required to be classified as either a one- or two-way path system. It can be within a building or over a wide area or can be targeted to a particular group of recipients. The messages that will be broadcast are based on an emergency response plan developed during a risk analysis by the project stakeholders and approved by the fire code official.

The NFPA 72 ECS requirements are based on in-building or wide-area occupant notification. Wide-area systems could include the entire area of a jurisdiction. For stadium, arena and grandstand captioning, NFPA 72 defines these as mass notification systems (MNS). In the context of the NFPA 72 requirements, this particular code change requires a one-way MNS where instructions are broadcasted by personnel authorized to distribute messages. This could include firefighters during a fire event; also, the system could be used by law enforcement officers during a domestic terrorism incident or a weather event like a tornado warning.

The design of the compliant MNS Chapter 24 of NFPA 72 is not prescriptive—an MNS is a performance-based design. Accordingly, fire code officials should require their design to be sealed by a registered professional engineer. Section 24.7 of the standard requires the preparation of a risk analysis based on the nature and anticipated risks of the facility. The risk analysis is part of the design brief, which will serve as the basis of the system design and is a required design document. NFPA 72 requires the following elements be included in the risk analysis:

- The number of persons within the building, area, space, campus or region.
- The character of the occupancy, such as unique hazards and the rate at which the hazard can develop.

- The anticipated threats, including natural, technological and intentional events.
- The reliability and performance of the MNS.
- Security of the MNS and its components.
- How the building or staff implements the risk analysis, including the use of the MNS.
- How emergency services, such as the fire service and law enforcement agencies, can employ the MNS.

In a stadium or arena, the captioning system is required to be a component of the emergency voice/alarm communications system (EV/ACS). Such a system is required by Section 907.2.1.1 in Group A occupancies with an occupant load of 1,000 or more.

The requirement in Section 907.5.2.2.4 specifies the captioning system would be connected to the EV/ACS. The fire alarm control unit will require a listed interface unit capable of displaying text messages. Textual visible appliances are allowed by NFPA 72 when used in conjunction with audible, visual or both types of notification appliances. In the public mode, textual visible appliances are located to ensure readability by the building occupants. Such a system can display messages using televisions or light emitting diode (LED) marquee signs.

The design concept of MNS is relatively new in the design community. Captioning systems might utilize components that are not listed for fire alarm service so the design will be required to comply with Chapter 24 of NFPA 72 for textual visible notification appliances. Emergency textual messages take precedence over any nonemergency text messages. Under NFPA 72, these devices require a primary and secondary power supply. If the devices are not monitored for integrity or loss of communications by an autonomous control or a fire alarm control unit, the appliance must clearly display its status. The size of characters displayed must comply with the requirements in NFPA 72. The NFPA 72 size, character and font requirements are based on the location of the display in relation to the height and distance from the persons viewing it.

**907.2.2 Group B.** A manual fire alarm system, which activates the occupant notification system in accordance with Section 907.5, shall be installed in Group B occupancies where one of the following conditions exists:

1. The combined Group B *occupant load* of all floors is 500 or more.
2. The Group B *occupant load* is more than 100 persons above or below the lowest *level of exit discharge*.
3. The *fire area* contains an ambulatory care facility.

**Exception:** Manual fire alarm boxes are not required where the building is equipped throughout with an *automatic sprinkler system* installed in accordance with Section 903.3.1.1 and the occupant notification appliances will activate throughout the notification zones upon sprinkler water flow.

**C** Group B occupancies generally involve individuals or groups of people in separate office areas. As a result, the occupants are not necessarily aware of what is going on in other parts of the building. Group B buildings with large occupant loads, even in single-story buildings or where a substantial number of occupants are above or below the level of exit discharge, increase the difficulty of alerting the occupants of a fire. This is especially true in nonsprinklered buildings with given occupant load thresholds. Group B occupancies include a specific use called ambulatory care facilities, which present a higher level of life hazard than the typical Group B occupancy. The fact that the care recipients of such facilities may be rendered incapable of self-preservation for limited periods of time makes the need for a fire alarm system critical. Section 907.2.2 requires a manual alarm system any time a fire area contains an ambulatory care facility. See the commentary to “Ambulatory care facility” in Section 202 and the commentary to Section 907.2.2.1.

The exception does not eliminate the fire alarm system, but rather permits it to be initiated automatically by the sprinkler waterflow switch(es) instead of by the manual fire alarm boxes.

**907.2.2.1 Ambulatory care facilities.** *Fire areas* containing ambulatory care facilities shall be provided with an electronically supervised *automatic smoke detection system* installed within the ambulatory care facility and in public use areas outside of tenant spaces, including public *corridors* and elevator lobbies.

**Exception:** Buildings equipped throughout with an *automatic sprinkler system* in accordance with Section 903.3.1.1 provided that the occupant notification appliances will activate throughout the notification zones upon sprinkler water flow.

**C** Years ago, few surgical procedures were performed outside of a hospital. Today, complex outpatient surgeries conducted outside of a hospital are commonplace. They are performed in facilities often called “day surgery centers” or “ambulatory surgical centers” because patients are able to walk in and walk out the same day. Procedures render care recipients temporarily incapable of self-preservation by application of nerve blocks, sedation or anesthesia; however, they do typically recover quickly.

The code identifies health care Group I occupancies as including a 24-hour stay. Without a 24-hour stay, these surgery centers were classified as Group B, which allowed the care providers to render an unlimited number of people incapable of self-preservation with no more protection than a business office. Since these types of facilities contain distinctly different hazards to life safety than other Group B occupancies, they are now required to have a higher level of life safety and fire protection as evidenced by the requirements of this section as well as Section 903.2.2 and the construction provisions of the code.

This section more specifically states that any time a fire area contains an ambulatory care facility, the fire area should be provided with a supervised smoke detection system in the ambulatory care facility and in public use areas outside of tenant spaces. Therefore, in a medical office building, for example, the ambulatory care facility contained within would have a full coverage system. The other offices in the building would not require smoke detection in the individual tenant spaces, but instead in the public areas, such as lobby or lounge areas.

The exception does not eliminate the fire alarm system, but rather permits it to be initiated automatically by the sprinkler waterflow switch(es) instead of by the smoke detection system.

**907.2.2.2 Laboratories involving research and development or testing.** A *fire alarm system* activated by an air-sampling-type smoke detection system or a radiant-energy-sensing detection system shall be installed throughout the entire *fire area* utilized for the research and development or testing of lithium-ion or lithium metal batteries.

**C** This section requires that all laboratories, regardless of the size, that are used for the research, development or testing of lithium-ion or lithium metal batteries have a fire alarm system activated by one of the detection systems listed and installed throughout the fire area where it occurs. These uses can have an increased risk of thermal runaway, which is intentionally caused in some cases.

**907.2.3 Group E.** A manual fire alarm system that initiates the occupant notification signal utilizing an emergency voice/alarm communication system meeting the requirements of Section 907.5.2.2 and installed in accordance with Section 907.6 shall be installed in Group E occupancies. Where *automatic sprinkler systems* or smoke detectors are installed, such systems or detectors shall be connected to the building *fire alarm system*.

**Exceptions:**

1. A manual *fire alarm system* shall not be required in Group E occupancies with an *occupant load* of 50 or less.
2. Emergency voice/alarm communication systems meeting the requirements of Section 907.5.2.2 and installed in accordance with Section 907.6 shall not be required in Group E occupancies with *occupant loads* of 100 or less, provided that activation of the manual *fire alarm system* initiates an *approved* occupant notification signal in accordance with Section 907.5.
3. Manual fire alarm boxes shall not be required in Group E occupancies where all of the following apply:
  - 3.1. Interior *corridors* are protected by smoke detectors.
  - 3.2. Auditoriums, cafeterias, gymnasiums and similar areas are protected by *heat detectors* or other *approved* detection devices.
  - 3.3. Shops and laboratories involving dusts or vapors are protected by *heat detectors* or other *approved* detection devices.
  - 3.4. Manual activation is provided from a normally occupied location.
4. Manual fire alarm boxes shall not be required in Group E occupancies where all of the following apply:
  - 4.1. The building is equipped throughout with an *approved automatic sprinkler system* installed in accordance with Section 903.3.1.1.
  - 4.2. The emergency voice/alarm communication system will activate on sprinkler water flow.
  - 4.3. Manual activation is provided from a normally occupied location.

**C** Section 404.2.3 addresses the development and implementation of lockdown plans. These requirements were developed to ensure that the level of life safety inside of the building is not reduced or compromised during a lockdown. In order for a building to safely function in a lockdown condition, the code requires a means of communication between the established central location and each secured area. Section 404.2.3.1 does not prescribe the means of communication, which could include the use of text messages to cell phones/mobile devices, email messages or the use of preestablished audio or visual signals. The provisions in Section 404.2.3 are not specific to Group E occupancies—they are applicable to all occupancies that develop and implement lockdown plans.

Because of concerns of school campus safety serving kindergarten through 12th grade students, specific requirements were put into the 2012 edition of the code and the IBC for enhanced communication between the school administrators, teachers and students when a lockdown plan is activated in Group E occupancies. As a result, emergency voice/alarm communications systems (EV/ACS) are prescribed in Group E occupancies. Previously, the code permitted the manual fire alarm system to use audible and visible alarm notification appliances but did not require the added capabilities that an EV/ACS provides.

Section 907.2.3 sets forth the requirements for automatic fire alarm and detection system requirements in Group E occupancies and prescribes the installation of an EV/ACS as opposed to a traditional horn/strobe occupant notification system.

Exception 1 exempts Group E occupancies from requiring a fire alarm system when the occupant load is 50 or less. This would exempt small day care centers that serve children older than 2 1/2 years of age or a small Sunday school classroom at a place of religious worship.

Exception 2 provides relief for smaller schools. If the occupant load is 100 or less, notification is not required to be via an emergency voice/alarm communications system. A school with 100 occupants only has a couple of classrooms of children. Communication is simplified and an emergency voice/alarm communications system is considered to be excessive.

Exception 3 exempts manual fire alarm boxes in interior corridors, laboratories, auditoriums, cafeterias, gymnasiums and similar spaces where heat/smoke detectors are installed or where manual activation is provided at normally occupied locations. This is not an exception from the EV/ACS but simply an exemption of locations requiring manual fire alarm boxes. The applicability of Exception 2 is independent of whether an automatic sprinkler system is installed. If an automatic smoke detection system is installed, it must be connected to the building fire alarm system.

Exception 4 allows the omission of the manual fire alarm boxes in Group E occupancies equipped throughout with an automatic sprinkler system if the actuation of the sprinkler system will activate the EV/ACS. See Section 903.2.3 for sprinkler requirements in Group E buildings.

**907.2.4 Group F.** A manual *fire alarm system* that activates the occupant notification system in accordance with Section 907.5 shall be installed in Group F occupancies where both of the following conditions exist:

1. The Group F occupancy is two or more stories in height.
2. The Group F occupancy has a combined *occupant load* of 500 or more above or below the lowest *level of exit discharge*.

**Exception:** Manual fire alarm boxes are not required where the building is equipped throughout with an *automatic sprinkler system* installed in accordance with Section 903.3.1.1 and the occupant notification appliances will activate throughout the notification zones upon sprinkler water flow.

**C** This section is intended to apply to large multiple-story manufacturing facilities. For this reason, a manual fire alarm system would be required only if the building were at least two stories in height and had 500 or more occupants above or below the level of exit discharge. An unlimited area, two-story Group F occupancy complying with IBC Section 507.5 would be indicative of an occupancy requiring a manual fire alarm system.

Buildings in compliance with IBC Section 507.5, and large manufacturing facilities in general, however, must be fully sprinklered and would thus be eligible for the exception. The exception does not eliminate the fire alarm system but rather permits it to be initiated automatically by the sprinkler system waterflow switch(es) instead of by the manual fire alarm boxes.

**907.2.4.1 Manufacturing involving lithium-ion or lithium metal batteries.** A *fire alarm system* activated by an air-sampling-type smoke detection system or a radiant-energy-sensing detection system shall be installed throughout the entire *fire area* where lithium-ion or lithium metal batteries are manufactured; and where the manufacturer of vehicles, energy storage systems or equipment containing lithium-ion or lithium metal batteries where the batteries are installed as part of the manufacturing process.

**C** This section requires that the manufacturer of lithium-ion or lithium metal batteries, or where the manufacturer includes them as a component of the process, have a fire alarm system activated by one of the detection systems listed and installed throughout the fire area where manufacturing occurs. This requirement includes the manufacture of vehicles, ESS and equipment where the battery is installed as part of the manufacturing process. This is due to the greater potential for a fire event and the additional increased risk of thermal runaway as a result of the manufacturing process.

**907.2.5 Group H.** A manual *fire alarm system* that activates the occupant notification system in accordance with Section 907.5 shall be installed in Group H-5 occupancies and in occupancies used for the manufacture of organic coatings. An *automatic smoke detection system* shall be installed for highly toxic gases, *organic peroxides* and *oxidizers* in accordance with Chapters 60, 62 and 63, respectively.

**C** Because of the nature and potential quantity of hazardous materials in Group H-5 occupancies, a manual means of activating an occupant notification system is essential for the safety of the occupants. In accordance with Section 2703.11, the activation of the alarm system must initiate a local alarm and transmit a signal to the emergency control station. The manual fire alarm system requirement for the building is in addition to the emergency alarm requirements in Section 2703.12 (see also Section 908.2).

Occupancies involved in the manufacture of organic coatings present special hazardous conditions because of the unstable character of the materials, such as nitrocellulose. Good housekeeping and control of ignition sources is critical. Chapter 29 contains additional requirements for organic coating manufacturing processes.

This section also requires an automatic smoke detection system in certain occupancy conditions involving either highly toxic gases or organic peroxides and oxidizers. The need for the automatic smoke detection system may depend on the class of materials and additional levels of fire protection provided. This requirement also assumes the quantity of materials is in excess of the maximum allowable quantities shown in Tables 5003.1.1(1) and 5003.1.1(2).

**907.2.6 Group I.** A manual *fire alarm system* that activates the occupant notification system in accordance with Section 907.5 shall be installed in Group I occupancies. An *automatic smoke detection system* that activates the occupant notification system in accordance with Section 907.5 shall be provided in accordance with Sections 907.2.6.1, 907.2.6.2 and 907.2.6.3.3.

**Exceptions:**

1. Manual fire alarm boxes in *sleeping units* of Group I-1 and I-2 occupancies shall not be required at *exits* if located at all care providers' control stations or other constantly attended staff locations, provided that such manual fire alarm boxes are visible and provided with *ready access*, and the distances of travel required in Section 907.4.2.1 are not exceeded.
2. Occupant notification systems are not required to be activated where private mode signaling installed in accordance with NFPA 72 is *approved* by the *fire code official* and staff evacuation responsibilities are included in the fire safety and evacuation plan required by Section 404.

**C** Because the protection and possible evacuation of the occupants in Group I occupancies are most often dependent on the response by care providers, occupancies in Group I must be protected with a manual fire alarm system and in certain instances, as described in Sections 907.2.6.1, 907.2.6.2 and 907.2.6.3, an automatic smoke detection system. In Group I-1, smoke alarms are also required in accordance with Section 907.2.6.1.1.

It is not the intent of this section to require a smoke detection system throughout all Group I occupancies. Smoke detectors are only generally required in the corridors and in waiting rooms that are open to corridors, unless noted otherwise. IFC Committee Interpretation No. 36-03 makes it clear that the Group I provisions only require a manual fire alarm system with smoke detectors in selected areas. To reduce the potential for unwanted alarms, manual fire alarm boxes may be located at the care providers' control stations or another constantly attended location.

Exception 1 reduces the likelihood of accidental or malicious false alarm system activations by manual means by allowing the pull stations to be located in a more controlled area. It assumes the approved location is always accessible by care providers and within a distance of travel of 200 feet (60 960 mm).

Exception 2 allows the common practice in Group I occupancies of notifying only the care providers instead of all building occupants in the event of a fire, subject to the approval of the fire code official. In order to have confidence that the actions taken will be appropriate, the code also requires that the responsibilities of the staff be documented in the fire safety and evacuation plan for the facility. These requirements are found in Section 404. This will increase the likelihood for any staff training to be linked to the allowance of private mode signaling.

**907.2.6.1 Group I-1.** An *automatic smoke detection system* shall be installed in *corridors*, waiting areas open to *corridors* and *habitable spaces* other than *sleeping units* and kitchens. The system shall be activated in accordance with Section 907.5.

**Exceptions:**

1. For Group I-1, Condition 1 occupancies, smoke detection in *habitable spaces* is not required where the facility is equipped throughout with an *automatic sprinkler system* installed in accordance with Section 903.3.1.1.
2. Smoke detection is not required for exterior balconies.

**C** Occupancies in Group I-1 tend to be compartmentalized into small rooms so that a fire in one area of the building would not easily be noticed by occupants in another part of the building. Therefore, smoke detection is required in areas other than sleeping units and kitchens. Sleeping units are required by Section 907.2.6.1.1 to be equipped with single- and multiple-station smoke alarms in accordance with Section 907.2.11.1.

Given that Group I-1 occupancies may not be supervised by care providers, and to reduce the likelihood that a fire within a waiting area open to a corridor or within the corridor itself could develop beyond the incipient stage, thereby jeopardizing the building egress, these areas must be equipped with automatic smoke detection.

Exception 1 allows smoke detectors to be eliminated from habitable spaces of Group I-1, Condition 1 occupancies if the building is equipped throughout with an NFPA 13 automatic sprinkler system. The sprinkler system should control any fire and perform occupant notification through actuation of the waterflow switch and subsequent activation of the building alarm notification appliances. A sprinkler system is required for all Group I occupancies in accordance with Section 903.2.6. It should be noted that Group I-1, Condition 1 is the lower risk Group I-1 occupancy where residents are able to evacuate without assistance. See the commentary to IBC Section 308.3.

Exception 2 allows for omitting smoke detectors from exterior balconies for environmental reasons and does not require the installation of an alternative type of detector. The exterior balconies are assumed to be sufficiently open to the atmosphere to readily allow the dissipation of smoke and hot gases.

**907.2.6.1.1 Smoke alarms.** Single- and multiple-station smoke alarms shall be installed in accordance with Section 907.2.11.

**C** As with dwelling units or sleeping units in any occupancy, this section requires that single- and multiple-station smoke alarms be installed in accordance with Section 907.2.11.1. Section 907.2.11.2 deals specifically with the requirements for Group I-1.

**907.2.6.2 Group I-2.** An *automatic smoke detection system* shall be installed in *corridors* in Group I-2, Condition 1 facilities and spaces permitted to be open to the *corridors* by Section 407.2 of the *International Building Code*. The system shall be activated in

accordance with Section 907.4. Group I-2, Condition 2 occupancies shall be equipped with an *automatic smoke detection system* as required in Section 407 of the *International Building Code*.

**Exceptions:**

1. *Corridor* smoke detection is not required in *smoke compartments* that contain *sleeping units* where such units are provided with smoke detectors that comply with UL 268. Such detectors shall provide a visual display on the *corridor* side of each *sleeping unit* and shall provide an audible and visual alarm at the care providers' station attending each unit.
2. *Corridor* smoke detection is not required in *smoke compartments* that contain *sleeping units* where *sleeping unit* doors are equipped with automatic door-closing devices with integral smoke detectors on the unit sides installed in accordance with their listing, provided that the integral detectors perform the required alerting function.

**C** Automatic smoke detection is required in areas permitted to be open to corridors in occupancies classified as Group I-2, Condition 2 and corridors in Group I-2, Condition 1 occupancies (e.g., nursing homes, long-term care facilities and detoxification facilities). In recognition of quick-response sprinkler technology and the fact that the sprinkler system is electronically supervised, and because the doors to care recipients' sleeping units are continuously supervised by care providers when in the open position, smoke detectors are not required for adequate fire safety in care recipient sleeping units.

In Group I-2, Condition 1 occupancies (nursing homes, long-term care facilities and detoxification facilities), some redundancy is appropriate because such facilities typically have less control over furnishings and personal items, thereby resulting in a less predictable and usually higher fire hazard load than Group I-2, Condition 2 occupancies (hospitals). Also, there is generally less care provider's supervision in these facilities than in other health care facilities and thus less control over care recipient smoking and other fire causes. Therefore, to provide additional protection against fires spreading from the room of origin, smoke detection is required in corridors of nursing homes, long-term care facilities and detoxification facilities.

Smoke detection is not required in corridors of Group I-2, Condition 2 occupancies except where otherwise specifically required in the code. Similarly, because areas open to the corridor very often are the room of fire origin, and such areas are no longer required by the code to be under visual supervision by care providers, some redundancy to protection by the sprinkler system is requested. Accordingly, all areas open to corridors must be protected by an automatic smoke detection system. This requirement provides an additional level of protection against sprinkler system failures or lapses in care provider supervision.

These requirements are not applicable to Group I-2, Condition 2 (hospitals). The scope of this section clearly indicates that its provisions are only applicable to detoxification facilities and nursing homes (Group I-2, Condition 1). Hospitals are noted as being subject to the provisions in IBC Section 407.2. IFC Committee Interpretation No. 37-03 addresses this issue. IBC Section 407.2 notes that smoke detection is only required for spaces open to corridors, such as waiting areas and mental health treatment areas where patients are not capable of self-preservation (see commentary, IBC Section 407.2).

There are two exceptions to the requirement for an automatic fire detection system in corridors of nursing homes, long-term care facilities and detoxification facilities. Both exceptions provide an alternative method for redundant protection in care recipient sleeping units. For this reason, they provide either a backup to the notification of a fire or containment of fire in the room of origin.

Exception 1 requires smoke detectors in sleeping units that activate both a visual display on the corridor side of the care recipient sleeping unit and visual and audible alarms at the care provider's station serving the room. Detectors complying with UL 268 are intended for open area protection and for connection to a normal power supply or as part of a fire alarm system. This exception, however, is specifically designed to not require the detectors to activate the building fire alarm system where approved care recipient sleeping unit smoke detectors are installed and where visual and audible alarms are provided. This is in response to the concern over unwanted alarms. The required alarm signals will not necessarily indicate to care providers that a fire emergency exists because the care provider call system may typically be used to identify numerous conditions within the room.

Exception 2 addresses the situation where smoke detectors are incorporated within automatic door-closing devices. The units are acceptable as long as the required alarm functions are still provided. Such units are usually listed as combination door closer and hold-open devices.

**907.2.6.3 Group I-3 occupancies.** Group I-3 occupancies shall be equipped with a manual *fire alarm system* and *automatic smoke detection system* installed for alerting staff.

**C** Because of the evacuation difficulties associated with Group I-3 occupancies and the dependence on adequate staff response, a manual fire alarm system and an automatic smoke detection system are required subject to the special occupancy conditions in Sections 907.2.6.3.1 through 907.2.6.3.3. This section recognizes that the evacuation of Group I-3 occupancies depends on an effective staff response. Chapter 4 and specifically Section 403.7.3 contain the requirements for an emergency plan, including employee training, staff availability, the need for occupants to notify staff and the need for the proper keys for unlocking doors for staff in Group I-3 occupancies.

**907.2.6.3.1 System initiation.** Actuation of an *automatic fire-extinguishing system*, *automatic sprinkler system*, a manual fire alarm box or a fire detector shall initiate an *approved* fire alarm signal that automatically notifies staff.

**C** This section specifies the systems that, upon activation, must initiate the required alarm signal immediately and automatically to the staff so that staff will respond in a timely manner.

**907.2.6.3.2 Manual fire alarm boxes.** Manual fire alarm boxes are not required to be located in accordance with Section 907.4.2 where the fire alarm boxes are provided at staff-attended locations having direct supervision over areas where manual fire alarm boxes have been omitted.

**C** Because of the potential for intentional false alarms and the resulting disruption to the facility, manual fire alarm boxes in Group I-3 occupancies may be either locked or made inaccessible to the occupants.

**907.2.6.3.2.1 Manual fire alarms boxes in detainee areas.** Manual fire alarm boxes are allowed to be locked in areas occupied by detainees, provided that staff members are present within the subject area and have keys readily available to operate the manual fire alarm boxes.

**C** The locking of manual fire alarm boxes is permitted only in areas where staff members are present and keys are readily available to them to unlock the boxes, or where the alarm boxes are located in a manned staff location that has direct supervision of the Group I-3 area.

**907.2.6.3.3 Automatic smoke detection system.** An *automatic smoke detection system* shall be installed throughout resident housing areas, including *sleeping units* and contiguous day rooms, group activity spaces and other common spaces normally open to residents.

**Exceptions:**

1. Other *approved* smoke detection arrangements providing equivalent protection, including, but not limited to, placing detectors in exhaust ducts from cells or behind protective guards *listed* for the purpose, are allowed where necessary to prevent damage or tampering.
2. *Sleeping units* in Use Conditions 2 and 3 as described in Section 308 of the *International Building Code*.
3. Smoke detectors are not required in *sleeping units* with four or fewer occupants in *smoke compartments* that are equipped throughout with an *automatic sprinkler system* installed in accordance with Section 903.3.1.1.

**C** Evacuation of Group I-3 facilities is impractical because of the need to maintain security. An automatic smoke detection system is therefore required to provide early warning of a fire.

As indicated in Exception 1, the installation of automatic smoke detectors must take into account the need to protect the detector from vandalism by residents. As a result, detectors may have to be located in return air ducts or be protected by a substantial physical barrier.

Since occupants in Use Condition 2 or 3 are not locked in their sleeping units, Exception 2 reduces the need for smoke detection.

Exception 3 allows smoke detectors to be omitted in sleeping units housing no more than four occupants on the basis that in a building that is protected throughout with an approved automatic sprinkler system, the system will provide both detection and suppression functions. Group I facilities are assumed to be fully sprinklered throughout in accordance with NFPA 13 as required by Section 903.2.6. The limitation of four occupants reduces the potential fuel load (mattresses, clothes, etc.) and the likelihood of involvement over an extended area.

**907.2.7 Group M.** Fire alarm systems shall be required in Group M occupancies in accordance with Sections 907.2.7.1 and 907.2.7.2.

**C** The section lists the conditions and types of Group M occupancies that are required to have a fire alarm system.

**907.2.7.1 Occupant load.** A manual *fire alarm system* that activates the occupant notification system in accordance with Section 907.5 shall be installed in Group M occupancies where one of the following conditions exists:

1. The combined Group M *occupant load* of all floors is 500 or more persons.
2. The Group M *occupant load* is more than 100 persons above or below the lowest *level of exit discharge*.

**Exceptions:**

1. A manual *fire alarm system* is not required in covered or open mall buildings complying with Section 402 of the *International Building Code*.
2. Manual fire alarm boxes are not required where the building is equipped throughout with an *automatic sprinkler system* installed in accordance with Section 903.3.1.1 and the occupant notification appliances will automatically activate throughout the notification zones upon sprinkler water flow.

**C** Group M occupancies have the potential for large numbers of occupants who may not be familiar with their surroundings. The installation of a fire alarm system increases the ability to alert the occupants of a fire. Note that the occupant thresholds must be considered independently. If the total occupant load is 500 or more persons, a manual fire alarm system is required. If more than 100 persons are above or below the level of exit discharge, a manual fire alarm system is required.

This section also specifies that the manual fire alarm boxes must, upon activation, provide occupant notification throughout the Group M occupancy.

The extent of fire alarm application is based on the area in which the Group M occupancy is located. If the building is considered as a separated mixed occupancy, then the fire alarm system is only required in the individual occupancy in which the occupant load exceeds the threshold quantity. The rest of the building would not require a fire alarm system. This approach is noted in IBC Section 508.4.1, which states that each separated space must comply with the code based on the occupancy classification of that portion of the building. If the Group M occupancy was part of a nonseparated mixed-use building, then the alarm system would be required in the entire building in accordance with IBC Section 508.3.1. The determination as to when such a system is required would be based solely on the Group M occupant load.

Exception 1 recognizes the increased level of fixed automatic protection inherently required in covered mall buildings, including an automatic sprinkler system and possibly a smoke control system. Covered mall buildings are also required to contain an emergency voice communication system (see Section 907.2.20).

Exception 2 does not eliminate the fire alarm system, but rather allows it to be initiated automatically by sprinkler system waterflow switch(es) instead of by manual fire alarm boxes. Buildings with a fire area containing a Group M occupancy in excess of 12,000 square feet (1115 m<sup>2</sup>) must be equipped with an automatic sprinkler system complying with Section 903.2.7.

**907.2.7.1.1 Occupant notification.** During times that the building is occupied, the initiation of a signal from a manual fire alarm box or from a waterflow switch shall not be required to activate the alarm notification appliances when an alarm signal is activated at a constantly attended location from which evacuation instructions shall be initiated over an emergency voice/alarm communication system installed in accordance with Section 907.5.2.2.

**C** Occupants in a mercantile occupancy may assume the alarm is a false alarm or act inappropriately and thus delay evacuation of the building. To prevent such a dangerous situation, the manual fire alarm system may be part of an EV/ACS. The signal is to be sent to a constantly attended location on-site from which evacuation instructions can be given.

It should be noted that, although the alarm notification alternative allows for the manual use of an EV/ACS, the alternative does not remove the requirement for audible and visual notification devices.

**907.2.7.2 Storage of lithium-ion or lithium metal batteries.** A *fire alarm system* activated by an air-sampling-type smoke detection system or a radiant-energy-sensing detection system shall be installed in a room or space within a Group M occupancy where required for the storage of lithium-ion or lithium metal batteries in accordance with Section 320.

**C** This section references Section 320 for Group M occupancies that are used for the storage of lithium-ion or lithium metal batteries. Section 320 requires these occupancies to have a fire alarm system that is activated by one of the types of detection systems listed and installed throughout the room or space where required.

**907.2.8 Group R-1.** Fire alarm systems and smoke alarms shall be installed in Group R-1 occupancies as required in Sections 907.2.8.1 through 907.2.8.3.

**C** Because residents of Group R-1 occupancies may be asleep and are usually transients who are unfamiliar with the building, and because such buildings contain numerous small rooms where occupants may not notice a fire in another part of the building, occupancies in Group R-1 must have a manual fire alarm system and an automatic smoke detection system installed throughout. Requirements for single- or multiple-station smoke alarms in sleeping units are contained in Section 907.2.11.1.

**907.2.8.1 Manual fire alarm system.** A manual *fire alarm system* that activates the occupant notification system in accordance with Section 907.5 shall be installed in Group R-1 occupancies.

**Exceptions:**

1. A manual *fire alarm system* is not required in buildings not more than two stories in height where all individual *sleeping units* and contiguous attic and crawl spaces to those units are separated from each other and public or common areas by not less than 1-hour *fire partitions* and each individual *sleeping unit* has an *exit* directly to a *public way, egress court* or yard.
2. Manual fire alarm boxes are not required throughout the building where all of the following conditions are met:
  - 2.1. The building is equipped throughout with an *automatic sprinkler system* installed in accordance with Section 903.3.1.1 or 903.3.1.2.
  - 2.2. The notification appliances will activate upon sprinkler water flow.
  - 2.3. Not fewer than one manual fire alarm box is installed at an *approved* location.

**C** This section is specific to manual fire alarm systems and requires such systems in all Group R-1 occupancies, with two exceptions.

Exception 1 eliminates the requirement for a manual fire alarm system if the sleeping units have an exit discharging directly to a public way, exit court or yard. Even though the building may be two stories in height, the sleeping units on each floor must have access directly to an approved exit at grade level. The use of an exterior exit access balcony with exterior

stairs serving the second floor does not constitute an exit directly at grade. The minimum 1-hour fire-resistance rating required for adequate separation of the sleeping units must be maintained.

Exception 2 does not omit the fire alarm system but rather permits it to be initiated automatically by sprinkler system waterflow switch(es) in lieu of manual fire alarm boxes. The sprinkler system must activate the occupant notification system, and at least one manual fire alarm box shall be installed at an approved location. See the commentary to Section 907.2 for a discussion of the single manual fire alarm box.

The exceptions do not affect the independent provision in Section 907.2.11 for single- or multiple-station smoke alarms.

**907.2.8.2 Automatic smoke detection system.** An *automatic smoke detection system* that activates the occupant notification system in accordance with Section 907.5 shall be installed throughout all interior *corridors* serving *sleeping units*.

**Exception:** An *automatic smoke detection system* is not required in buildings that do not have interior *corridors* serving *sleeping units* and where each *sleeping unit* has a *means of egress* door opening directly to an *exit* or to an exterior *exit access* that leads directly to an *exit*.

**C** This section requires an automatic smoke detection system within interior corridors. Such systems make use of smoke detectors for alarm initiation in accordance with Section 907.2, with one exception.

The exception provides that automatic fire detectors are not required in motels and hotels that do not have interior corridors and in which sleeping units have a door opening directly to an exterior exit access that leads directly to the exits. The intent of the exception is that the exit access from the sleeping unit door be exterior and not require reentering the building prior to entering the exit. Since the exit access is outside, the need for detectors other than the smoke alarms required by Section 907.2.8.3 in sleeping units is greatly reduced.

**907.2.8.3 Smoke alarms.** Single- and multiple-station smoke alarms shall be installed in accordance with Section 907.2.11.

**C** The actual requirements for single- and multiple-station smoke alarms are located in Section 907.2.11. That section requires that the single- and multiple-station smoke alarms within sleeping units be connected to the emergency electrical system. Automatic activation of the fire alarm system is avoided to reduce unnecessary alarms within such buildings.

**907.2.9 Group R-2.** Fire alarm systems and smoke alarms shall be installed in Group R-2 occupancies as required in Sections 907.2.9.1 and 907.2.9.3.

**C** This section introduces the fire alarm system and smoke alarm requirements for Group R-2 occupancies. This includes Group R-2 occupancies in general and also Group R-2 college and university buildings.

**907.2.9.1 Manual fire alarm system.** A manual *fire alarm system* that activates the occupant notification system in accordance with Section 907.5 shall be installed in Group R-2 occupancies where any of the following conditions apply:

1. Any *dwelling unit* or *sleeping unit* is located three or more stories above the lowest *level of exit discharge*.
2. Any *dwelling unit* or *sleeping unit* is located more than one story below the highest *level of exit discharge* of *exits* serving the *dwelling unit* or *sleeping unit*.
3. The building contains more than 16 *dwelling units* or *sleeping units*.

**Exceptions:**

1. A *fire alarm system* is not required in buildings not more than two stories in height where all *dwelling units* or *sleeping units* and contiguous attic and crawl spaces are separated from each other and public or common areas by not less than 1-hour *fire partitions* and each *dwelling unit* or *sleeping unit* has an *exit* directly to a *public way*, *egress court* or yard.
2. Manual fire alarm boxes are not required where the building is equipped throughout with an *automatic sprinkler system* installed in accordance with Section 903.3.1.1 or 903.3.1.2 and the occupant notification appliances will automatically activate throughout the notification zones upon a sprinkler water flow.
3. A *fire alarm system* is not required in buildings that do not have interior *corridors* serving *dwelling units* and are protected by an *approved automatic sprinkler system* installed in accordance with Section 903.3.1.1 or 903.3.1.2, provided that *dwelling units* either have a *means of egress* door opening directly to an exterior *exit access* that leads directly to the *exits* or are served by open-ended *corridors* designed in accordance with Section 1027.6, Exception 3.

**C** The occupants of Group R-2 occupancies are not considered to be as transient as those of Group R-1, which increases the probability that residents can more readily notify each other of a fire. Therefore, Group R-1 occupancies must have a manual fire alarm system with audible and visual notification appliances subject to the exceptions in Section 907.2.8.1, whereas Group R-2 occupancies are required to have only a manual fire alarm system as stipulated in one of the three listed conditions. The threshold conditions are meant to be applied independently of each other.

Exception 1 eliminates the requirement for a manual fire alarm system if the sleeping units have an exit discharging directly to a public way, exit court or yard. Even though the building may be two stories in height, the sleeping units on each floor must have access directly to an approved exit at grade level. The use of an exterior exit access balcony with exterior stairs serving the second floor does not constitute an exit directly at grade. The minimum 1-hour fire-resistance rating required for adequate separation of the sleeping units must be maintained.

Exception 2 does not omit the fire alarm system but rather permits it to be initiated automatically by sprinkler system waterflow switch(es) in lieu of manual fire alarm boxes. The sprinkler system must activate the occupant notification system. This exception does not affect the independent provisions of Section 907.2.11.

Exception 3 allows the omission of a fire alarm system in fully sprinklered buildings (NFPA 13 or 13R) with no interior corridors and that exit directly to an exterior exit access or have open-ended corridors. The important thing to note is that the sprinkler system is not required to activate alarm notification appliances since a fire alarm system would not be required. Only the sprinkler alarms required by Section 903.4 would be required.

**907.2.9.2 Smoke alarms.** Single- and multiple-station smoke alarms shall be installed in accordance with Section 907.2.11.

**C** The actual requirements for single- and multiple-station smoke alarms are located in Section 907.2.11. That section requires that the single- and multiple-station smoke alarms within sleeping units be connected to the emergency electrical system. Automatic activation of the fire alarm system is avoided to reduce unnecessary alarms within such buildings.

**907.2.9.3 Group R-2 college and university buildings.** An *automatic smoke detection system* that activates the occupant notification system in accordance with Section 907.5 shall be installed in Group R-2 occupancies operated by a college or university for student or staff housing in all of the following locations:

1. Common spaces outside of *dwelling units* and *sleeping units*.
2. Laundry rooms, mechanical equipment rooms and storage rooms.
3. All interior *corridors* serving *sleeping units* or *dwelling units*.

**Exception:** An *automatic smoke detection system* is not required in buildings that do not have interior *corridors* serving *sleeping units* or *dwelling units* and where each *sleeping unit* or *dwelling unit* either has a *means of egress* door opening directly to an exterior *exit access* that leads directly to an *exit* or a *means of egress* door opening directly to an *exit*.

Required smoke alarms in *dwelling units* and *sleeping units* in Group R-2 occupancies operated by a college or university for student or staff housing shall be interconnected with the *fire alarm system* in accordance with NFPA 72.

**C** This section requires an automatic smoke detection system be provided in Group R-2 occupancies operated by a college or university for student or staff housing. It also requires the smoke alarms in individual units to be interconnected with the fire alarm system. This interconnection is only for the purpose of making occupants within each unit aware of the fire alarm activation in the building. The intent is not to activate the building fire alarm system by smoke alarms in each unit. This is more restrictive than a Group R-2 occupancy in general, as typically the requirements are limited to a manual fire alarm system and smoke alarms in the individual sleeping or dwelling units.

The smoke detection system is focused on common areas, such as interior corridors, lounge areas, laundry rooms, and areas such as mechanical rooms, which could be the source of a fire, especially in these specific types of Group R-2 occupancies.

In a study completed by the New York State Governor's Task Force on Campus Fire Safety, it was cited that 43 percent of fires in college dormitories are located in dorm rooms or kitchens, leaving the other 57 percent located in areas that would not require smoke detection under the current code. The study also showed that there were approximately 300 fires on college campuses over a 3-year period, while only 160 were reported to the fire department. The Center for Campus Fire Safety states that 132 deaths have been "reported" in fires in student housing since January 2000.

An NFPA study on student housing showed an annual average of 4,100 structural fires in dormitories, fraternities, sororities and barracks between 2011 and 2015. The 4,180 reported structure fires in the dormitory occupancy group in 2015 represented a 31 percent increase from the 3,200 structure fires in 1980. Fires ranged from 2,300 to 2,700 from 1982 through 1995, and then declined from 1996 to 1998. Since 2003, annual estimates have ranged from a low of 3,350 to a high of 4,230 in 2014.

The requirements in this section are very similar to the recommendations of the study done in New York State. It is important to note that the recommendations for that study were specifically aimed at the properties of the colleges and universities, so it was not the study's intent to cover off-campus housing in this particular regard. There were recommendations for off-campus housing, such as sororities and fraternities, to have annual inspections. This particular distinction was not addressed in detail during the code development process initially; however, since this is somewhat of a continuation of requirements that were added into Chapter 4 of the 2006 edition of the code for emergency preparedness and planning, and those requirements were intended to deal with buildings that were college or university property, it seemed reasonable to interpret that this requirement is also limited to the buildings that are college or university property and does not apply to Group R-2 occupancies that are not college or university property. This has since been clarified through the code development process and the section now specifically notes that the Group R-2 occupancies are specifically operated by a college or university.

The exception allows for the elimination of the smoke detection system in the specific situation where there are no interior corridors and the occupants essentially exit directly to the outside. The lack of interior corridors and exterior exits reduces the amount of smoke one unit will expose to another. Note that smoke alarms within the units are still required by

Section 907.2.11.2. Note also that it is entirely possible that some areas that are required to have smoke detection may have ambient conditions that warrant a different type of alarm-initiating device. Section 907.4.3 addresses this concern.

**907.2.10 Group S.** A fire alarm system shall be installed in a Group S occupancy as required by Sections 907.2.10.1 and 907.2.10.2.

**C** The section lists the conditions and types of Group S occupancies that are required to have a fire alarm system.

**907.2.10.1 Public- and self-storage occupancies.** A manual fire alarm system that activates the occupant notification system in accordance with Section 907.5 shall be installed in Group S public- and self-storage occupancies three stories or greater in height for interior corridors and interior common areas. Visible notification appliances are not required within storage units.

**Exception:** Manual fire alarm boxes are not required where the building is equipped throughout with an automatic sprinkler system installed in accordance with Section 903.3.1.1, and the occupant notification appliances will activate throughout the notification zones upon sprinkler water flow.

**C** The occupants of Group S public and self-storage buildings are not expected to have familiarity with the layout of these multistory buildings that have interior corridors or common areas. Thus, a manual fire alarm system that activates the occupant notification system must be provided for them. As noted, since individual self-storage units are typically not large in size, additional visible notification appliances are not required to be provided within each unit. It is expected that the door will be open when a unit is occupied, and the devices located in the interior corridors leading to the units will be visible at all times to occupants.

Note that Section 903.2.9.4 requires that an automatic sprinkler system be provided throughout a Group S-1 fire area where the area used for the storage of upholstered furniture or mattresses exceeds 2,500 square feet.

**907.2.10.2 Storage of lithium-ion or lithium metal batteries.** A fire alarm system activated by an air-sampling-type smoke detection system or a radiant-energy-sensing detection system shall be installed throughout the entire fire area where required for the storage of lithium-ion batteries or lithium metal batteries in accordance with Section 320.

**C** The section references Section 320 for Group S occupancies that are used for the storage of lithium-ion or lithium metal batteries. In accordance with Section 320, these occupancies must have a fire alarm system that is activated by one of the types of detection systems listed and installed throughout the entire fire area where required.

**907.2.11 Single- and multiple-station smoke alarms.** Listed single- and multiple-station smoke alarms complying with UL 217 shall be installed in accordance with Sections 907.2.11.1 through 907.2.11.7, NFPA 72 and the manufacturer's instructions.

**C** Single- and multiple-station smoke alarms have evolved as one of the most important fire safety features in residential and similar occupancies having sleeping occupants. The value of early fire warning in these occupancies has been repeatedly demonstrated in fires involving both successful and unsuccessful smoke alarm performance.

For successful smoke alarm operation and performance, single- and multiple-station smoke alarms must be listed in accordance with UL 217 and installed to comply with the code, the manufacturer's instructions, and Chapter 29 of NFPA 72, which contains the minimum requirements for the selection, installation, operation and maintenance of fire warning equipment for use in family living units. These devices are called "smoke alarms" rather than "smoke detectors" because they are independent of a fire alarm system and include an integral alarm notification device.

**907.2.11.1 Group R-1.** Single- or multiple-station smoke alarms shall be installed in all of the following locations in Group R-1:

1. In sleeping areas.
2. In every room in the path of the *means of egress* from the sleeping area to the door leading from the *sleeping unit*.
3. In each story within the *sleeping unit*, including *basements*. For *sleeping units* with split levels and without an intervening door between the adjacent levels, a smoke alarm installed on the upper level shall suffice for the adjacent lower level provided that the lower level is less than one full story below the upper level.

**C** Because the occupants of a sleeping unit or suite may be asleep and unaware of a fire developing in the room or in the egress path, single- or multiple-station smoke alarms must be provided in the sleeping unit and in any intervening room between the sleeping unit and the exit access door from the room. If the sleeping unit or suite involves more than one level, a smoke alarm must also be installed on every level. See the commentary to "Sleeping unit" in Section 202.

Smoke alarms are required in split-level arrangements, except those that meet the conditions described in Item 3. In accordance with Section 907.2.11.5, all smoke alarms within a sleeping unit or suite must be interconnected so that actuation of one alarm will actuate all smoke alarms within the sleeping unit or suite.

**907.2.11.2 Groups R-2, R-3, R-4 and I-1.** Single- or multiple-station smoke alarms shall be installed and maintained in Groups R-2, R-3, R-4 and I-1 regardless of *occupant load* at all of the following locations:

1. On the ceiling or wall outside of each separate sleeping area in the immediate vicinity of bedrooms.
2. In each room used for sleeping purposes.
3. In each story within a *dwelling unit*, including *basements* but not including crawl spaces and uninhabitable attics. In *dwelling units* or *dwelling units* with split levels and without an intervening door between the adjacent levels, a smoke alarm installed

on the upper level shall suffice for the adjacent lower level provided that the lower level is less than one full story below the upper level.

**C** Because the occupants of a dwelling unit may be asleep and unaware of a fire developing in the room or in an area within the dwelling unit that will affect their ability to escape, single- or multiple-station smoke alarms must be installed in every bedroom, in the vicinity of all bedrooms (e.g., hallways leading to the bedrooms) and on each story of the dwelling unit (see Commentary Figure 907.2.11.2 and the commentary to “Dwelling unit” in Section 202).

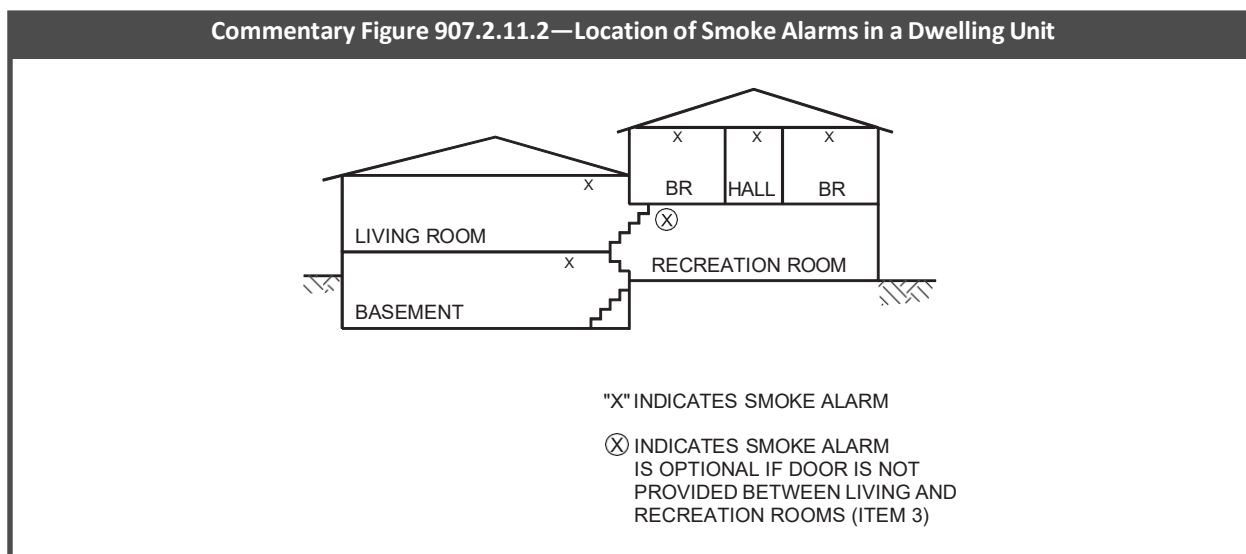
If a sprinkler system is installed throughout the building in accordance with NFPA 13, 13R or 13D, if applicable, smoke alarms would still be required in the bedrooms even if residential sprinklers are used.

Smoke alarms are required in split-level arrangements. As required by Section 907.2.11.5, all smoke alarms within a dwelling unit must be interconnected so that actuation of one alarm will actuate the alarms in all detectors within the dwelling unit.

These provisions do not apply to one- and two-family dwellings and multiple single-family dwellings (townhouses) not more than three stories in height with a separate means of egress that are regulated by the IRC. The IRC is intended to be a stand-alone document, but if the residential units do not fall within the scope of the IRC or for other reasons are intended to be subject to this code, then the requirements of this section would apply. IFC Committee Interpretation No. 42-03 addresses this condition and contains additional explanatory information about the IRC and its relationship to the other I-Codes.

Although the occupants of a sleeping unit in a Group I-1 occupancy may be asleep, they are still considered capable of self-preservation. Regardless, smoke alarms are required in sleeping units. The exception allows single- or multiple-station smoke alarms to be eliminated in the room if an automatic fire detection system that includes in-room system smoke detectors is installed as required by Section 907.2.6.

Commentary Figure 907.2.11.2—Location of Smoke Alarms in a Dwelling Unit



**907.2.11.3 Installation near cooking appliances.** Smoke alarms shall be installed not less than 10 feet (3048 mm) horizontally from a permanently installed cooking appliance.

**Exception:** Smoke alarms shall be permitted to be installed not less than 6 feet (1829 mm) horizontally from a permanently installed cooking appliance where necessary to comply with Section 907.2.11.1 or 907.2.11.2.

**C** This requirement is intended to reduce nuisance alarms attributed to locating smoke alarms in close proximity to cooking appliances and bathrooms in which steam is produced. These provisions are aligned with the 8th edition of UL 217 and are consistent with similar requirements in NFPA 72. Section 29.11.3.4(4)(2) of NFPA 72 requires smoke alarms to be listed for resistance to common nuisance sources from cooking in accordance with the 8th edition of UL 217 or subsequent editions. UL smoke alarm and detector standards have new performance tests to reduce the frequency of unwanted alarm activation from normal cooking activities such as pan frying, sauteing or baking. The new cooking resistance tests are necessary because normal cooking activities are the leading cause of unwanted alarm activations that result in homeowners removing or deactivating their smoke alarms.

**907.2.11.4 Installation near bathrooms.** Smoke alarms shall be installed not less than 3 feet (914 mm) horizontally from the door or opening of a bathroom that contains a bathtub or shower unless this would prevent placement of a smoke alarm required by Section 907.2.11.1 or 907.2.11.2.

**C** See the commentary to Section 907.2.11.3. Sections 907.2.11.3 and 907.2.11.4 are provided to reduce nuisance alarms.

**907.2.11.5 Interconnection.** Where more than one smoke alarm is required to be installed within an individual *dwelling unit* or *sleeping unit* in Group R or I-1 occupancies, the smoke alarms shall be interconnected in such a manner that the activation of one alarm will activate all of the alarms in the individual unit. Physical interconnection of smoke alarms shall not be required where *listed* wireless alarms are installed and all alarms sound upon activation of one alarm. The alarm shall be clearly audible in all bedrooms over background noise levels with all intervening doors closed.

**C** The installation of smoke alarms in areas remote from the sleeping area will be of minimal value if the alarm is not heard by the occupants. Interconnection of multiple smoke alarms within an individual dwelling unit or sleeping unit is required in order to alert a sleeping occupant of a remote fire within the unit before the combustion products reach the smoke alarm in the sleeping area and thus provide additional time for evacuation.

The term “interconnection” refers to either hardwired systems or listed wireless systems. UL has listed smoke detectors that use this technology. It is presumed that on safely evacuating the unit or room of fire origin, an occupant will notify other occupants by actuating the manual fire alarm system or using other available means. Section 907.7.1 addresses the testing of the smoke alarms to demonstrate that interconnection of such smoke alarms is properly functioning.

Similar requirements can now be found in the IRC, IBC and *International Existing Building Code*® (IEBC®) for both new and existing buildings to allow interconnection with wireless technology of smoke alarms. All wirelessly interconnected smoke alarms are listed to UL 217 and are classified by NFPA 72 as low-power systems.

**907.2.11.6 Power source.** In new construction, required smoke alarms shall receive their primary power from the building wiring where such wiring is served from a commercial source and shall be equipped with a battery backup. Smoke alarms with integral strobes that are not equipped with battery backup shall be connected to an emergency electrical system in accordance with Section 1203. Smoke alarms shall emit a signal when the batteries are low. Wiring shall be permanent and without a disconnecting switch other than as required for overcurrent protection.

**Exception:** Smoke alarms are not required to be equipped with battery backup where they are connected to an emergency electrical system that complies with Section 603.

**C** Smoke alarms are required to use AC as a primary power source and battery power as a secondary source to improve their reliability. For example, during a power outage, the probability of fire is increased because of the use of candles or lanterns for temporary light. Required backup battery power is intended to provide continued functioning of the smoke alarms. Smoke alarms are commonly designed to emit a recurring signal when batteries are low and need to be replaced.

Certain occupancies may already have an emergency electrical system in the building to monitor other building system conditions. The emergency electrical system provides a level of reliability equivalent to battery backup.

**907.2.11.7 Smoke detection system.** Smoke detectors *listed* in accordance with UL 268 and provided as part of the building *fire alarm system* shall be an acceptable alternative to single- and multiple-station *smoke alarms* and shall comply with the following:

1. The *fire alarm system* shall comply with all applicable requirements in Section 907.
2. Activation of a smoke detector in a *dwelling unit* or *sleeping unit* shall initiate alarm notification in the *dwelling unit* or *sleeping unit* in accordance with Section 907.5.2.
3. Activation of a smoke detector in a *dwelling unit* or *sleeping unit* shall not activate alarm notification appliances outside of the *dwelling unit* or *sleeping unit*, provided that a supervisory signal is generated and monitored in accordance with Section 907.6.6.

**C** This section specifically allows the use of an automatic smoke detection system as an alternative to smoke alarms. In the past, when this concept was proposed, it was only allowed through an alternative method and materials approach (see Section 104.7.4 of this code), even though, in concept, it provided the same level of protection. Such systems provide the same safety features necessary for occupants but are simply part of a fire alarm system. Note that if a detector activates within a sleeping or dwelling unit, the occupant notification system is not intended to activate. This is consistent with the operation of smoke alarms. Item 3 specifically requires the notification to be only to occupants of the sleeping unit or dwelling unit.

**907.2.12 Special amusement areas.** Fire detection and alarm systems shall be provided in *special amusement areas* in accordance with Section 914.7.

**C** Special amusement areas are areas in which the means of egress is not readily apparent, is intentionally confounded or is not readily available. Special amusement areas must also comply with the provisions of IBC Section 411.

The approved automatic smoke detection system is required to provide early warning of a fire. The detection system is required regardless of the presence of staff in the building.

**907.2.13 High-rise buildings.** High-rise buildings shall be provided with an *automatic smoke detection system* in accordance with Section 907.2.13.1, a fire department communication system in accordance with Section 907.2.13.2 and an emergency voice/alarm communication system in accordance with Section 907.5.2.2.

**Exceptions:**

1. Airport traffic control towers in accordance with Section 907.2.22 of this code and Section 412 of the *International Building Code*.

2. Open parking garages in accordance with Section 406.5 of the *International Building Code*.
3. Buildings with an occupancy in Group A-5 in accordance with Section 303.1 of the *International Building Code*.
4. Low-hazard special occupancies in accordance with Section 503.1.1 of the *International Building Code*.
5. Buildings with an occupancy in Group H-1, H-2 or H-3 in accordance with Section 415 of the *International Building Code*.
6. In Group I-1 and I-2 occupancies, the alarm shall sound at a constantly attended location and occupant notification shall be broadcast by the emergency voice/alarm communication system.

**C** High-rise buildings require additional fire protection systems because of the difficulties with smoke movement, egress time and fire department access. As a result, this section requires both an automatic fire alarm system and an emergency voice/alarm communications system (see commentary, Section 907.5.2.2). Exceptions 1 through 5 are the same as those in IBC Section 403.1 regarding the applicability of the high-rise provisions.

Exception 1 addresses airport traffic control towers and is based on the limited fuel load and the limited number of persons occupying the tower.

Open parking garages and places of outdoor assembly (Group A-5) are exempted by Exceptions 2 and 3, respectively, because of the free ventilation to the outside that exists in such structures.

In Exception 4, low-hazard special industrial occupancies may be exempted when approved by the fire code official. Such buildings should be evaluated based on the occupant load and the hazards of the occupancy and its contents to determine whether the protection features required by IBC Section 403 are necessary.

Buildings with occupancies in Groups H-1, H-2 and H-3 are excluded from the requirements of this section by Exception 5 because the fire hazard characteristics of these occupancies have not yet been considered in high-rise buildings.

Exception 6 recognizes the supervised environment typical of institutional uses and the reliance placed on staff to act appropriately in an emergency. As is the case for most voice alarms, the key is in being able to deliver specific information to the people who can affect a safe egress—whether it be the public, employees or both.

**907.2.13.1 Automatic smoke detection.** Automatic smoke detection in high-rise buildings shall be in accordance with Sections 907.2.13.1.1 and 907.2.13.1.2.

**C** This section simply introduces the fire alarm and detection system requirements for high-rise buildings.

**907.2.13.1.1 Area smoke detection.** Area smoke detectors shall be provided in accordance with this section. Smoke detectors shall be connected to an automatic *fire alarm system*. The activation of any detector required by this section shall activate the emergency voice/alarm communication system in accordance with Section 907.5.2.2. In addition to smoke detectors required by Sections 907.2.1 through 907.2.9, smoke detectors shall be located as follows:

1. In each mechanical equipment, electrical, transformer, telephone equipment or similar room that is not provided with sprinkler protection.
2. In each elevator machine room, machinery space, control room and control space and in elevator lobbies.

**C** Automatic smoke detectors are required in all high-rise buildings in certain locations so that a fire will be detected in its early stages of development. The detectors must be connected to the automatic fire alarm system and be capable of initiating operation of the EV/ACS.

This section divides the automatic smoke detection requirement into two categories. Smoke detectors must be installed in rooms that are not typically occupied. This includes rooms used for mechanical equipment, electrical equipment, transformer equipment and telephone equipment where such rooms do not have automatic sprinkler protection. In most cases, these rooms will have sprinkler protection by virtue of being in a high-rise building and will therefore not require smoke detectors. However, in elevator machine rooms, machinery spaces, control rooms, control spaces and elevator lobbies, smoke detectors are required regardless of sprinkler protection.

Note that smoke detection and smoke alarms may be required based on occupancy-related requirements elsewhere in Section 907.2.

**907.2.13.1.2 Duct smoke detection.** Duct smoke detectors complying with Section 907.3.1 shall be located as follows:

1. In the main return air and exhaust air plenum of each air-conditioning system having a capacity greater than 2,000 cubic feet per minute (cfm) (0.94 m<sup>3</sup>/s). Such detectors shall be located in a serviceable area downstream of the last duct inlet.
2. At each connection to a vertical duct or riser serving two or more stories from a return air duct or plenum of an air-conditioning system. In Group R-1 and R-2 occupancies, a smoke detector is allowed to be used in each return air riser carrying not more than 5,000 cfm (2.4 m<sup>3</sup>/s) and serving not more than 10 air-inlet openings.

**C** Smoke detectors must be installed in the main return air and exhaust air plenum of each air-conditioning system having a design capacity exceeding 2,000 cubic feet per minute (cfm) (0.94 m<sup>3</sup>/s). Systems with design capacities equal to or less than 2,000 cfm (0.94 m<sup>3</sup>/s) are exempt from this requirement because their small size limits their capacity for spreading smoke to parts of the building not already involved with fire.

The area that could be served by a 2,000-cfm (0.94 m<sup>3</sup>/s) system (approximately 5 tons of cooling capacity) is comparatively small; therefore, the distribution of smoke in a system of that size would be minimal. Smoke detectors must be located so that they monitor the total airflow within the system. If a single detector is unable to sample the total airflow at all times, then multiple detectors are required. The smoke detectors must be made accessible for maintenance and inspection. Many failures and false alarms are caused by a lack of maintenance and cleaning of the smoke detectors.

Consistent with IMC Section 606.2.3, return air risers serving two or more stories must have smoke detectors installed at each story. Item 2 allows the use of a single listed smoke detector in each return air riser in a Group R-1 or R-2 occupancy if the capacity of each riser does not exceed 5,000 cfm (2.4 m<sup>3</sup>/s) and does not serve more than 10 air-inlet openings. This alternative allows smaller buildings that have residential occupancies to only monitor the return air from each return air riser.

**907.2.13.2 Fire department communication system.** Where a wired communication system is *approved* in lieu of an in-building, two-way emergency responder communication coverage system in accordance with Section 510, the wired fire department communication system shall be designed and installed in accordance with NFPA 72 and shall operate between a *fire command center* complying with Section 508, elevators, elevator lobbies, emergency and standby power rooms, fire pump rooms, areas of refuge and inside *interior exit stairways*. The fire department communication device shall be provided at each floor level within the *interior exit stairway*.

**C** High-rise buildings have posed a challenge to the traditional communication systems used by the fire service for fire-to-ground communications to assist fire ground officers in communicating with the firefighters working in various areas of the building. Where testing of the in-building two-way emergency responder radio coverage system required by IBC Section 403.4.5 shows that the signal strengths are not satisfactory, Section 510.1, Exception 1, allows for the alternative of installation of a wired communication system designed in accordance with this section. The system must be capable of operating between the fire command center and every elevator, elevator lobby, emergency/standby power room, fire pump room, area of refuge and interior exit stairway. Note that this section does not offer specific criteria as to what constitutes an acceptable wired communication system or its components. It could be a component of an emergency voice/alarm communications system that complies with Section 907.5.2.2 or a building's telephone system. In any event, when applying Section 510.1, Exception 1, and this section, the concurrent approval of the fire and building code officials is required.

**907.2.13.3 Multiple-channel voice evacuation.** In buildings with an occupied floor more than 120 feet (36 576 mm) above the lowest level of fire department vehicle access, voice evacuation systems for high-rise buildings shall be multiple-channel systems.

**C** This section requires multiple-channel voice evacuation systems in high-rise buildings when the buildings have an occupied floor more than 120 feet above the lowest level of fire department vehicle access. Having a multiple-channel system allows the emergency responders to deliver different live messages to various areas of the building at one time, which can lead to more detailed and more efficient emergency communications to the occupants.

For example, if a fire occurs on the sixth floor of a high-rise building, a multiple-channel system can be used by the emergency responders to direct the occupants of the fifth, sixth, and seventh floors to immediately egress via the nearest exit stairway, while at the same time separately informing the occupants of the eighth and higher levels of the situation and directing them to stand by. This type of ability can help the emergency responders control crowding within the exit stairways. Additionally, the benefits of the multiple-channel system are not limited to fire evacuations only, as these systems can also be used effectively in active-shooter and other emergencies.

**907.2.14 Atriums connecting more than two stories.** A *fire alarm system* shall be installed in occupancies with an atrium that connects more than two stories, with smoke detection in locations required by a rational analysis in Section 909.4 and in accordance with the system operation requirements in Section 909.17. The system shall be activated in accordance with Section 907.5. Such occupancies in Group A, E or M shall be provided with an emergency voice/alarm communication system complying with the requirements of Section 907.5.2.2.

**C** Buildings containing an atrium that connects more than two stories are to be equipped with a fire alarm system that can be used to notify building occupants to begin evacuating in case of a fire. The other critical part of such fire alarm systems is to activate the smoke control system. The system is to be activated by smoke detection designed and installed in accordance with the rational analysis as required in Section 909.4. More specifically, smoke control systems are engineered systems that are activated by carefully placed and zoned smoke detection. If improperly designed and installed, the system may not be effective. For instance, wrongly placed or inappropriate smoke detection technology may not activate fast enough and the smoke control system would be overwhelmed. This section goes on to state that the alarm system must be initiated in accordance with Section 907.5, which requires that the alarm system is to be initiated by the sprinkler system and any automatic or manual fire alarm-initiating devices in the building. It does not intend to require certain features to be installed within the atrium but rather is simply requiring that any such features present initiate the occupant notification system. It would not necessarily be appropriate to also initiate the smoke control system on activation of the alarm system within a building containing an atrium (see Section 909.12.3). The alarm system needs to be carefully zoned in such buildings to avoid an inappropriate activation of the smoke control system from a space not associated with the atrium.

Groups A, E and M must have an emergency voice/alarm communications system that complies with Section 907.5.2.2 because of the number of persons to be evacuated and the lack of familiarity with the location of exits that is typical of occupants in Groups A and M.

**907.2.15 High-piled combustible storage areas.** An *automatic smoke detection system* shall be installed throughout *high-piled combustible storage areas* where required by Section 3206.5.

**C** Section 3206.5 requires an automatic fire detection system in high-piled combustible storage areas, depending on the commodity class, the size of the high-piled storage area and the presence of an automatic sprinkler system. High-piled storage is the storage of Class I through IV commodities in piles, bin boxes, on pallets or in racks more than 12 feet (3658 mm) high or for high-hazard commodities stored higher than 6 feet (1829 mm). Chapter 32 and NFPA 13 contain additional requirements for all high-piled storage conditions.

**907.2.16 Aerosol storage uses.** Aerosol product rooms and general-purpose warehouses containing aerosol products shall be provided with an *approved manual fire alarm system* where required by this code.

**C** Chapter 32 and NFPA 30B contain additional guidance on the storage of and fire protection requirements for aerosol products. The requirements for storing the various levels of aerosol products are dependent on the level of sprinkler protection, the type of storage and the quantity of aerosol products. Although aerosol product fires generally involve property loss as opposed to loss of life, installation of a manual fire alarm system could aid in the prompt evacuation of the occupants. Fires involving aerosol products can spread rapidly through a building that is not properly protected and controlled.

**907.2.17 Lumber, wood structural panel and veneer mills.** Lumber, wood structural panel and veneer mills shall be provided with a *manual fire alarm system*.

**C** Any facility using mechanical methods to process wood into finished products produces debris and the potential for combustible dust. Such facilities include mills that produce solid wood lumber and wood veneers as well as those that manufacture structural wood panels such as waferboard, oriented strandboard, composite wood panels or plywood. Good housekeeping and control of ignition sources are therefore essential. To aid in the quick evacuation of occupants in an emergency, Section 2804.2.1 requires a manual fire alarm system in lumber, wood structural panel and veneer mills that contain product dryers because of their potential as a source of ignition. A manual fire alarm system is not required, however, if the dryers and all other potential sources of ignition are protected by a supervised automatic sprinkler system.

**907.2.18 Underground buildings with smoke control systems.** Where a smoke control system is installed in an underground building in accordance with the *International Building Code*, automatic smoke detectors shall be provided in accordance with Section 907.2.18.1.

**C** As indicated in IBC Section 405.5.2, each compartment of an underground building must have a smoke control/exhaust system that can be activated both automatically and manually. Floor levels more than 60 feet (18 288 mm) below the lowest level of exit discharge must be compartmented. Compartmentation is a key element in the egress and fire access plan for floor areas in an underground building. The smoke control system must not only facilitate egress during a fire, but also improve fire department access to the fire source by maintaining visibility that would be otherwise impaired given the inability of the fire service to manually ventilate the underground portion of the building (see commentary, IBC Section 405.4.1).

**907.2.18.1 Smoke detectors.** Not fewer than one smoke detector *listed* for the intended purpose shall be installed in all of the following areas:

1. Mechanical equipment, electrical, transformer, telephone equipment, elevator machine or similar rooms.
2. Elevator lobbies.
3. The main return and exhaust air plenum of each air-conditioning system serving more than one story and located in a serviceable area downstream of the last duct inlet.
4. Each connection to a vertical duct or riser serving two or more floors from return air ducts or plenums of heating, ventilating and air-conditioning systems, except that in Group R occupancies, a *listed* smoke detector is allowed to be used in each return air riser carrying not more than 5,000 cfm (2.4 m<sup>3</sup>/s) and serving not more than 10 air inlet openings.

**C** Automatic smoke detectors are required in certain locations in all underground buildings so that a fire will be detected in its early stages of development. Underground buildings are similar to high-rise buildings in that they present an unusual hazard by being virtually inaccessible to exterior fire department suppression and rescue operations with the increased potential to trap occupants inside the structure. For this reason, the smoke detector location requirements for underground buildings are similar to those in Section 907.2.13.1 for high-rise buildings (see commentary, Section 907.2.13.1).

The requirement for a smoke detector in the main return and exhaust air plenum of an air-conditioning system in an underground building, however, differs from that of a high-rise building in that it is not a function of capacity [2,000 cfm (0.94 m<sup>3</sup>/s)] but rather a function of whether the system serves more than one floor level. There is more concern over the threat of smoke movement from floor to floor because the products of combustion cannot be vented directly to the atmosphere.

**907.2.18.2 Alarm required.** Activation of the smoke control system shall activate an audible alarm at a constantly attended location.

**C** The audible alarm is required to notify qualified personnel immediately that the smoke control system has been activated and to put emergency procedures into action quickly.

**907.2.19 Deep underground buildings.** Where the lowest level of a structure is more than 60 feet (18 288 mm) below the finished floor of the lowest *level of exit discharge*, the structure shall be equipped throughout with a manual fire alarm system, including an emergency voice/alarm communication system installed in accordance with Section 907.5.2.2.

**C** The ability to communicate and offer warning of a fire can increase the time available for egress from the building. Underground structures located more than 60 feet (18 288 mm) below the level of exit discharge must therefore have a manual fire alarm system. An emergency voice/alarm communications system is also required as part of this system (see commentary, Section 907.5.2.2).

**907.2.20 Covered and open mall buildings.** Where the total floor area exceeds 50,000 square feet (4645 m<sup>2</sup>) within either a covered mall building or within the perimeter line of an open mall building, an emergency voice/alarm communication system shall be provided. Access to emergency voice/alarm communication systems serving a mall, required or otherwise, shall be provided for the fire department. The system shall be provided in accordance with Section 907.5.2.2.

**C** Because of the potentially large number of occupants and their unfamiliarity with their surroundings, an EV/ACS, accessible by the fire department, is required to aid in evacuation of covered mall buildings exceeding 50,000 square feet (4645 m<sup>2</sup>) in total floor area or an open mall exceeding 50,000 square feet (4645 m<sup>2</sup>) measured within the perimeter lines of the open mall. Anchor stores are not included as part of the covered or open mall building (see commentary, Section 202 for “Covered mall building”).

**907.2.21 Residential aircraft hangars.** Not fewer than one single-station smoke alarm shall be installed within a residential aircraft hangar as defined in Chapter 2 of the *International Building Code* and shall be interconnected into the residential smoke alarm or other sounding device to provide an alarm that will be audible in all sleeping areas of the *dwelling*.

**C** Residential aircraft hangars are assumed to be on the same property as a one- or two-family dwelling. IBC Section 412.5 contains additional requirements for the construction of residential aircraft hangars. The hangar could be located immediately adjacent to the dwelling unit if it is separated by 1-hour fire-resistance-rated construction. Because of the potentially close proximity of the aircraft and its flammability and fuel source, at least one smoke alarm is required in the hangar that is interconnected to the residential smoke alarms. It should be noted, however, that the requirement for a smoke alarm is also applicable to residential aircraft hangars that are detached from the dwelling unit. Because a minimum separation distance is not specified, a fire in the hangar could still present a serious fire hazard to the dwelling unit.

**907.2.22 Airport traffic control towers.** An *automatic smoke detection system* that activates the occupant notification system in accordance with Section 907.5 shall be provided in airport control towers in accordance with Sections 907.2.22.1 and 907.2.22.2.

**Exception:** Audible appliances shall not be installed within the control tower cab.

**C** Airport traffic control towers must be designed to comply with IBC Section 412.3. These structures are unique in that they can be built to excessive heights and are often permitted to have one exit stairway. IBC Section 412.3 requires that airport traffic control towers with an occupied floor more than 35 feet above fire department vehicle access be equipped throughout with an automatic sprinkler system. The requirements for detection systems and associated occupant notification are addressed based on whether the airport traffic control tower is equipped throughout with an automatic sprinkler system and whether multiple exits are provided.

The exception recognizes the sensitive nature of the operations that take place in the cab located at the top of the tower and prohibits the installation of audible alarm notification devices there. Notification of occupants within the cab is to be by visual notification appliances only.

**907.2.22.1 Airport traffic control towers with multiple exits and automatic sprinklers.** Airport traffic control towers with multiple exits and equipped throughout with an *automatic sprinkler system* in accordance with Section 903.3.1.1 shall be provided with smoke detectors in all of the following locations:

1. Airport traffic control cab.
2. Electrical and mechanical equipment rooms.
3. Airport terminal radar and electronics rooms.
4. Outside each opening into *interior exit stairways*.
5. Along the single *means of egress* permitted from observation levels.
6. Outside each opening into the single *means of egress* permitted from observation levels.

**C** This section addresses airport traffic control towers that are equipped throughout with both an automatic sprinkler system and multiple exits. The requirements are less restrictive than Section 907.2.22.2 because two important safety aspects are provided.

The first three items address occupiable or equipment-related rooms where the fires are more likely to start. The last two items address the paths of egress. Item 4 requires one detector outside each entrance to the interior stairway. Item 5 is specific to providing detection along the entire means of egress path from the observation levels.

**907.2.22.2 Other airport traffic control towers.** Airport traffic control towers with a single *exit* or where sprinklers are not installed throughout shall be provided with smoke detectors in all of the following locations:

1. Airport traffic control cab.
2. Electrical and mechanical equipment rooms.
3. Airport terminal radar and electronics rooms.
4. Office spaces incidental to the tower operation.
5. Lounges for employees, including sanitary facilities.
6. *Means of egress.*
7. Utility shafts where *access* to smoke detectors can be provided.

**C** This section addresses airport traffic control towers that have only a single exit or where an automatic sprinkler system is not provided. Essentially all of the items addressed are the areas permitted in an airport traffic control tower in accordance with IBC Section 412.3. Items 1 through 3 are the same as Section 907.2.22.1. These are the critical occupiable spaces and equipment spaces where fires have the greatest effect on the operation of airport traffic control towers. Since this is a single exit tower and possibly not equipped throughout with an automatic sprinkler system, other occupiable areas such as offices and lounges for employees must also provide smoke detection. Since there are limited exits or no systems able to control the fire, early warning of a fire becomes more critical. Item 6 requires smoke detection along the means of egress path. The intent is to address the exit access path leading to the interior exit stairway. Finally, Item 7 addresses the potential for fires in any utility shaft that may be accessible to building occupants.

**907.2.23 Energy storage systems.** An *automatic smoke detection system* or radiant-energy detection system shall be installed in rooms, areas and walk-in units containing energy storage systems as required in Section 1207.5.4.

**C** Stationary lead-acid battery systems are commonly used for standby power, emergency power or uninterrupted power supplies. The release of hydrogen gas during battery system operation is usually minimal. Adequate ventilation will disperse the small amounts of liberated hydrogen. Because standby power and emergency power systems control many important building emergency systems and functions, a supervised automatic smoke detection system or radiant-energy detection system is required for early warning notification of a hazardous condition. Section 1207 contains additional requirements, including the need for safety venting, room enclosure requirements, spill control and neutralization provisions, ventilation criteria, signage and seismic protection. IBC Section 509 also requires that such rooms in certain occupancies be separated by 1-hour construction.

**907.3 Fire safety functions.** Automatic fire detectors utilized for the purpose of performing fire safety functions shall be connected to the building's fire alarm control unit where a *fire alarm system* is required by Section 907.2. Detectors shall, upon actuation, perform the intended function and activate the alarm notification appliances or activate a visible and audible supervisory signal at a constantly attended location. In buildings not equipped with a *fire alarm system*, the automatic fire detector shall be powered by normal electrical service and, upon actuation, perform the intended function. The detectors shall be located in accordance with NFPA 72.

**C** When the code requires installation of automatic fire detectors to perform a specific function, such as elevator recall or smokeproof enclosure ventilation, or when detectors are installed to comply with a permitted alternative, such as door-closing devices, these detectors must be connected to the building's automatic fire alarm system if the building is required by the code to have such a system.

In addition to performing its intended function (for example, closing a door), if a detector is activated, it must also activate either the building alarm devices (if present) or a supervisory signal at a constantly attended location. This requirement recognizes that these detectors and the devices they control are part of the building fire protection system and are expected to perform as designed. If they are connected to a fire alarm system, they will have the supervision necessary for operational reliability. If they are not connected to and supervised by a fire alarm system, they still must be supervised through the constantly attended location.

For buildings not required to have a fire alarm system, the fire safety function detectors must be powered by the building electrical system and be located as required by NFPA 72. Without this stipulation, these detectors could not be expected to perform as intended because there would be no power supply.

**907.3.1 Duct smoke detectors.** Smoke detectors installed in ducts shall be *listed* for the air velocity, temperature and humidity present in the duct. Duct smoke detectors shall be connected to the building's fire alarm control unit where a *fire alarm system* is required by Section 907.2. Activation of a duct smoke detector shall initiate a visible and audible supervisory signal at a *constantly attended location* and shall perform the intended fire safety function in accordance with this code and the *International Mechanical*

**Code.** In facilities that are required to be monitored by a supervising station, duct smoke detectors shall report only as a supervisory signal and not as a fire alarm. They shall not be used as a substitute for required open area detection.

**Exceptions:**

1. The supervisory signal at a *constantly attended location* is not required where duct smoke detectors activate the building's alarm notification appliances.
2. In occupancies not required to be equipped with a *fire alarm system*, actuation of a smoke detector shall activate a visible and an audible signal in an *approved location*. Smoke detector trouble conditions shall activate a visible or audible signal in an *approved location* and shall be identified as air duct detector trouble.

**C** It is not the intent of this section to send a signal to the fire department or to activate the alarm notification devices within a building. Instead, this section requires that a supervisory signal be sent to a constantly attended location. Smoke detectors must be connected to a fire alarm system where such systems are installed. Connection to the fire alarm system will activate a visible and audible supervisory signal at a constantly attended location, which will alert building supervisory personnel that a smoke alarm has activated and will also provide electronic supervision of the duct detectors, thereby indicating any problems that may develop in the detector system circuitry or power supply.

Exception 1 allows activation of the building alarm notification appliances in place of a supervisory signal. Causing the occupant notification system to sound would alert the occupants of the building that an alarm condition exists within the air distribution system, thereby performing the same function as a supervisory signal sent to a constantly attended location.

Exception 2 recognizes that not all buildings are required to have a fire alarm system. A visible and audible signal must be activated at an approved location that will alert building supervisory personnel to take action. Additionally, the duct smoke detectors must be electronically supervised to indicate trouble (system fault) in the detector system circuitry or power supply. A trouble condition must activate a distinct visible or audible signal at a location that will alert the responsible personnel.

**907.3.2 Special locking systems.** Where special locking systems are installed on *means of egress* doors in accordance with Section 1010.2.13 or 1010.2.12, an automatic detection system shall be installed as required by that section.

**C** This section alerts the code user to additional requirements in Sections 1010.2.12 and 1010.2.13 that tie the operation of egress doors into the activation of an automatic fire detection system. A smoke or heat detection system is required to unlock delayed egress locks upon activation. The heat detection system can be the sprinkler system. For example, a similar requirement is found in Section 1010.3.3 that requires horizontal sliding doors used as a component of the means of egress, where required to be rated, to be self-closing or automatic-closing upon smoke detection. Also, electrically locked egress doors in occupancies as required by Section 1010.2.12 must be capable of being automatically unlocked by activation of an automatic fire detection system, if one is installed.

**907.3.3 Elevator emergency operation.** Automatic fire detectors installed for elevator emergency operation shall be installed in accordance with the provisions of ASME A17.1/CSA B44 and NFPA 72.

**C** This section provides correlation with Section 604.1 by making it clear that automatic fire detection devices used to initiate Phase I emergency recall of elevators are to be installed in accordance with both ASME A17.1 and NFPA 72.

**907.3.4 Wiring.** The wiring to the auxiliary devices and equipment used to accomplish the fire safety functions shall be monitored for integrity in accordance with NFPA 72.

**C** In order to provide a reasonable level of integrity and reliability to the installation of automatic fire detection devices and related equipment installed to perform various fire safety functions in accordance with Section 907.3, this section requires that all wiring interconnecting such devices and equipment be monitored for integrity in accordance with NFPA 72.

**907.4 Initiating devices.** Where a *fire alarm system* is required by another section of this code, occupant notification in accordance with Section 907.5 shall be initiated by one or more of the following. Initiating devices shall be installed in accordance with Sections 907.4.1 through 907.4.3.1.

1. Manual fire alarm boxes.
2. Automatic fire detectors.
3. *Automatic sprinkler system* waterflow devices.
4. *Automatic fire-extinguishing systems*.

**C** This section introduces Sections 907.4.1 through 907.4.3.1, which contain requirements for the various types of manual or automatic fire alarm-initiating devices. In addition, the required occupant notification must be initiated by one or more of the four listed devices or systems. Note that manual fire alarm boxes are the only allowable manual means of initiation. The other three—fire detectors, sprinkler systems and fire-extinguishing systems—are all required to be automatic in operation and initiation of the fire alarm occupant notification. Activation of any of these must result in occupant notification by itself independently or in combination with the others.

**907.4.1 Protection of fire alarm control unit.** In areas that are not continuously occupied, a single smoke detector shall be provided at the location of each fire alarm control unit, notification appliance circuit power extenders and supervising station transmitting equipment.

**Exception:** Where ambient conditions prohibit installation of a smoke detector, a *heat detector* shall be permitted.

**C** This section requires a smoke detector at the fire alarm control unit. This is consistent with Section 10.4.4 of NFPA 72. This smoke detector will activate the fire alarm control unit and allow it to either notify occupants or transmit a signal to a remote monitoring location before the fire impairs the fire alarm control unit. The exception parallels Section 907.4.3 by allowing a heat detector to be installed in lieu of a smoke detector in areas where the ambient environment is hostile to smoke detectors and could lead to unwanted alarm activations. This exception is also allowed by NFPA 72.

**907.4.2 Manual fire alarm boxes.** Where a manual *fire alarm system* is required by another section of this code, it shall be activated by fire alarm boxes installed in accordance with Sections 907.4.2.1 through 907.4.2.6.

**C** This section specifies the requirements for manual fire alarm boxes that are part of a required manual fire alarm system.

**907.4.2.1 Location.** Manual fire alarm boxes shall be located not more than 5 feet (1524 mm) from the entrance to each *exit*. In buildings not protected by an *automatic sprinkler system* in accordance with Section 903.3.1.1 or 903.3.1.2, additional manual fire alarm boxes shall be located so that the distance of travel to the nearest box does not exceed 200 feet (60 960 mm).

**C** Manual fire alarm boxes must be located in the path of egress and be readily accessible to the occupants. They must be located within 5 feet (1524 mm) of the entrance to each exit on every story of the building. This would include the need to locate manual fire alarm boxes near each horizontal exit, as well as entrances to stairs and exit doors to the exterior.

Manual fire alarm boxes are located near exits so that an adequate number of devices are available in the path of egress to transmit an alarm in a timely manner. These locations also encourage the actuation of a manual fire alarm box on the fire floor prior to entering the stair, resulting in the alarm being received from the actual fire floor and not another floor along the path of egress.

The location also presumes that individuals will be evacuating the area where the fire originated. When evacuation of the fire area is unlikely, consideration could be given to putting manual fire alarm boxes in more convenient places. Examples of such instances would be officer stations in Group I-3 occupancies and care provider's stations in Group I-2 occupancies.

The 200-foot (60 960 mm) exit access travel distance limitation is consistent with the exit access travel distance permitted for most nonsprinklered occupancies. If the 200-foot (60 960 mm) travel distance to a manual fire alarm box is exceeded in a nonsprinklered building, additional manual fire alarm boxes would be required.

**907.4.2.2 Height.** The height of the manual fire alarm boxes shall be not less than 42 inches (1067 mm) and not more than 48 inches (1372 mm) measured vertically, from the floor level to the activating handle or lever of the box.

**C** Manual fire alarm boxes must be reachable by the occupants of the building. They must also be mounted high enough to reduce the likelihood of damage or false alarms from something accidentally striking the device. Therefore, manual fire alarm boxes must be mounted a minimum of 42 inches (1067 mm) and a maximum of 48 inches (1219 mm) above the floor level. The 48-inch (1219 mm) measurement corresponds to the maximum unobstructed side-reach height by a person in a wheelchair.

**907.4.2.3 Color.** Manual fire alarm boxes shall be red in color.

**C** Manual fire alarm boxes are to be painted or manufactured in a distinctive and traditional red color to provide a visual cue to help building occupants identify the device.

**907.4.2.4 Signs.** Where fire alarm systems are not monitored by an *approved* supervising station in accordance with Section 907.6.6, an *approved* permanent sign shall be installed adjacent to each manual fire alarm box that reads: "WHEN ALARM SOUNDS—CALL FIRE DEPARTMENT."

**Exception:** Where the manufacturer has permanently provided this information on the manual fire alarm box.

**C** This section has limited application because, as indicated in Section 907.6.6, fire alarm systems generally must be monitored by an approved supervising station. Where a system is not monitored, such as with a fire alarm system that is not required by code, adequate signage must be displayed to tell occupants what response actions must be taken. Most building occupants assume that when an alarm device is activated, the fire department will automatically be notified as well. The sign must be conspicuously located next to the manual fire alarm box unless it is mounted on the manual fire alarm box itself by the manufacturer.

**907.4.2.5 Protective covers.** The *fire code official* is authorized to require the installation of *listed* manual fire alarm box protective covers to prevent malicious false alarms or to provide the manual fire alarm box with protection from physical damage. The protective cover shall be transparent or red in color with a transparent face to permit visibility of the manual fire alarm box. Each cover

shall include proper operating instructions. A protective cover that emits a local alarm signal shall not be installed unless *approved*. Protective covers shall not project more than that permitted by Section 1003.3.3.

**C** Although manual fire alarm boxes must be readily available to all occupants in buildings required to have a manual fire alarm system, this section allows the use of protective covers if they are approved by the fire code official. Protective covers are commonly used to reduce either the potential for intentional false alarms or vandalism. They also provide protection in locations where the manual fire alarm boxes may be exposed to physical damage, such as in gymnasiums, indoor tennis courts and the like.

**907.4.2.6 Unobstructed and unobscured.** Manual fire alarm boxes shall be provided with *ready access*, unobstructed, unobscured and visible at all times.

**C** This section addresses the concern that manual fire alarm boxes be kept clear and unobstructed. It is recommended that a minimum of 3 feet (914 mm) be kept clear but more may be needed. NFPA 72 addresses the need for manual fire alarm boxes to be unobstructed in Section 17.14.8.2 and states that manual fire alarm boxes be conspicuous, unobstructed and accessible. This requirement will assist during the design, construction, inspection and future maintenance of manual fire alarm boxes, helping ensure they are located where they will be provided with enough space for access and will not be obstructed.

**907.4.3 Automatic smoke detection.** Where an *automatic smoke detection system* is required, it shall utilize smoke detectors unless ambient conditions prohibit such an installation. In spaces where smoke detectors cannot be utilized due to ambient conditions, *approved automatic heat detectors* shall be permitted.

**C** Smoke detectors must be used, except when ambient conditions would prohibit their use. This section would allow a heat detector to be installed in lieu of a smoke detector in areas where the ambient environment is hostile to smoke detectors and could lead to unwanted alarm activations. The smoke detector manufacturer's literature will identify the limitations on the use of smoke detectors, including environmental conditions such as humidity, temperature and airflow.

**907.4.3.1 Automatic sprinkler system.** For conditions other than specific fire safety functions noted in Section 907.3, in areas where ambient conditions prohibit the installation of smoke detectors, an *automatic sprinkler system* installed in such areas in accordance with Section 903.3.1.1 or 903.3.1.2 and that is connected to the *fire alarm system* shall be *approved* as automatic heat detection.

**C** This section states that automatic heat detection is not required when buildings are fully sprinklered in accordance with NFPA 13 or 13R. The presence of a sprinkler system exempts areas where a heat detector can be installed in place of a smoke detector, such as in storage or furnace rooms. The sprinkler head in this case essentially acts as a heat detection device. Note that this provision does not apply to the fire safety functions indicated in Section 907.3.

**907.5 Occupant notification.** Occupant notification by fire alarms shall be in accordance with Sections 907.5.1 through 907.5.2.3.3. Occupant notification by smoke alarms in Group R-1 and R-2 occupancies shall comply with Section 907.5.2.1.3.2.

**C** This section introduces the occupant notification requirements in the following sections and also provides the section reference for occupant notification by smoke alarms in Group R-1 and R-2 occupancies, which include a low-frequency signal level of 520 Hz.

**907.5.1 Alarm activation and annunciation.** Upon activation, fire alarm systems shall initiate occupant notification and shall annunciate at the fire alarm control unit, or where allowed elsewhere by Section 907, at a *constantly attended location*.

**C** This section makes it clear that fire alarm system activation begins first by activating the fire alarm control unit, then by notifying the occupants of an alarm condition.

In addition, there is a recognition that there are places in the code where an alternative to occupant notification is an alarm notification at a constantly attended location. This is intended to clarify the code so that there is no question as to whether this general provision for alarm activation is superseded by the other sections addressing the alarm notification at a constantly attended location.

**907.5.1.1 Presignal feature.** A presignal feature shall only be provided where *approved*. The presignal shall be annunciated at an *approved, constantly attended location*, having the capability to activate the occupant notification system in the event of fire or other emergency.

**C** A presignal feature on a fire alarm system allows the occupant notification devices to activate in selected, constantly attended locations only and from which human intervention is required to activate a general occupant notification signal. Alternatively, this feature can be programmed to delay the general alarm notification for more than 1 minute before it will automatically be activated by the control panel. In either presignal scenario, remote transmission of the alarm signal to the fire department is immediate. See NFPA 72 for additional information on the presignal feature.

Improper use of the presignal feature has been a contributing factor in several multiple-death fire incidents. In most instances, the staff failed to activate the general alarm quickly and the occupants of the building were unaware of the fire. Therefore, the use of a presignal feature is discouraged by the code. A presignal feature may be used only if it is approved by the fire code official and the fire department.

**907.5.2 Alarm notification appliances.** Alarm notification appliances shall be provided and shall be *listed* for their purpose.

**C** The code requires that fire alarm systems be equipped with approved alarm notification appliances so that in an emergency, the fire alarm system will notify the occupants of the need for evacuation or implementation of the fire emergency plan. Alarm notification devices required by the code are of two general types: visible and audible. Except for voice/alarm signaling systems, once the system has been activated, all visible and audible alarms are required to activate. Voice/alarm signaling systems are special signaling systems that are activated selectively in response to specific emergency conditions.

**907.5.2.1 Audible alarms.** Audible alarm notification appliances shall be provided and emit a distinctive sound that is not to be used for any purpose other than that of a fire alarm.

**Exceptions:**

1. Audible alarm notification appliances are not required in critical care areas of Group I-2, Condition 2 occupancies that are in compliance with Section 907.2.6, Exception 2.
2. A visible alarm notification appliance installed in a nurses' control station or other continuously attended staff location in a Group I-2, Condition 2 suite shall be an acceptable alternative to the installation of audible alarm notification appliances throughout a suite or unit in Group I-2, Condition 2 occupancies that are in compliance with Section 907.2.6, Exception 2.
3. Where provided, audible notification appliances located in each enclosed occupant evacuation elevator lobby in accordance with Section 3008.9.1 of the *International Building Code* shall be connected to a separate notification zone for manual paging only.

**C** To attract the attention of building occupants, audible alarms must be distinctive, using a sound that is unique to the fire alarm system and used for no other purpose than alerting occupants to a fire emergency. Other emergencies, such as tornadoes, must be signaled by another sound different from the fire signal.

Exception 1 recognizes that the occupants in critical care areas of Group I-2 occupancies are usually incapacitated. The audible alarms may have the effect of unnecessarily disrupting the care recipients who are most likely not capable of self-preservation. Likewise, audible alarms in operating theaters of hospitals could be hazardous because an alarm activation could startle a surgeon during a delicate procedure. Critical care areas are also assumed to be adequately staffed at all times. Section 907.2.6, Exception 2 allows the use of private mode signaling in accordance with NFPA 72 and also requires that staff evacuation responsibilities be included in the fire safety and evacuation plan. See the commentary to Section 907.2.6. In private mode, as permitted by Section 907.2.6, there is still a requirement for an audible alarm notification from appliances, though at a much lower decibel level meant to alert staff of the alarm activation. Allowing the audible alarm to be eliminated from critical care areas (operating rooms) in exchange for a visual notification device is also not appropriate since the visual signal device also creates a distraction in critical care areas that may not be able to immediately stop a patient procedure. The emergency action plan would include provisions for alerting critical area staff and initiating the actions to be taken.

Exception 2 allows hospital care suites to eliminate audible alarms where visible alarm notification appliances are located at a continuously attended staff location or nurses control station. In a suite arrangement, the "control area" is the centrally manned location for staff monitoring patients in the separate rooms. An alarm indicator at this location will alert staff for response in a more effective and efficient manner. Similar to Exception 1, audible alarms can unnecessarily disrupt the care recipients. In addition, simply providing visible alarm notification throughout the suite is not appropriate. The patients' evacuation depends on staff since they are not capable of self-preservation. As with Exception 1, compliance with Exception 2 to Section 907.2.6 would be required in order to take advantage of this exception.

Exception 3 is intended to address the concern that automatic emergency voice/alarm messages do not interfere with operation of the two-way communication associated with the occupant evacuation elevators. Live voice messages would be appropriate in the lobbies.

**907.5.2.1.1 Average sound pressure.** The audible alarm notification appliances shall provide a sound pressure level of 15 decibels (dBA) above the average ambient sound level or 5 dBA above the maximum sound level having a duration of not less than 60 seconds, whichever is greater, in every occupiable space within the building.

**C** To attract the attention of building occupants, this section requires that the distinctive audible alarms must be capable of being heard above the ambient noise level in a space. The indicated levels are considered the minimum pressure differential that will be perceivable by most people. It prescribes that the sound pressure level (SPL) for notification appliances shall be a minimum of 15 decibels measured in the A-scale (dBA) above the ambient SPL or 5 dBA above the maximum SPL in every space that can be occupied in a building. These SPLs are based on a minimum 1-minute measurement period. SPLs for Group R and I-1 occupancies, mechanical rooms and other occupancies are no longer stipulated as they had been in previous editions of the code.

The values mandated in Section 907.5.2.1.1 in previous editions of the code were not consistent with the notification appliance SPL requirements in NFPA 72. NFPA 72 requirements for the audible notification appliances are based on if the devices emit alert or evacuation tones, voice messages or audible notifications for exit markings. The provisions in Section

907.5.2.1.1 would apply to all notification appliances designed to operate in either public or private mode. In sleeping areas, the minimum SPL is no longer specified in Section 907.5.2.1.1; however, for smoke alarms, NFPA 72 requires a minimum 75 dBA SPL at the pillow.

Also note that the 2010 Americans with Disabilities Act, *Standards for Accessible Design*, has an exception for medical care facilities following industry practice that will allow a dependence on staff. The activation of either audible or visible alarms could be detrimental to the care recipients in locations like operating rooms and intensive or critical care units.

**907.5.2.1.2 Maximum sound pressure.** The total sound pressure level produced by combining the ambient sound pressure level with all audible notification appliances operating shall not exceed 110 dBA at the minimum hearing distance from the audible appliance. Where the average ambient noise is greater than 105 dBA, visible alarm notification appliances shall be provided in accordance with NFPA 72 and audible alarm notification appliances shall not be required.

**C** In no case may the sound pressure level exceed 110 dBA at the minimum hearing distance from the audible appliance. This is consistent with Americans with Disabilities Act (ADA) requirements. Sound pressures above that level can cause pain or even permanent hearing loss. In such cases, audible alarms are not required to be installed but visual alarms would be necessary to compensate for the lack of audibility.

In alignment with Section 18.4.1.2 of NFPA 72, this section requires that the total sound pressure level (produced by combining the ambient sound pressure level with all audible notification appliances operating) shall not exceed 110 dBA.

It should also be noted that in certain work areas, the Occupational Safety and Health Administration (OSHA) requires employees to wear hearing protection, possibly preventing them from hearing an audible alarm. Additionally, the noise factor in these areas is high enough that an audible alarm may not be discernible. In these areas, as well as in others, the primary method of indicating a fire can be by a visible signal. Employees must be capable of identifying such a signal as indicating a fire.

**907.5.2.1.3 Audible alarm signal frequency in Group R-1, R-2 and I-1 sleeping rooms.** Audible alarm signal frequency in Group R-1, R-2 and I-1 occupancies shall be in accordance with Sections 907.5.2.1.3.1 and 907.5.2.1.3.2.

**C** In new Group R-1, R-2 and I-1 sleeping rooms, this section requires an audible low-frequency fire alarm signal of 520 Hz, the frequency considered most effective for waking high-risk population segments such as the hearing and alcohol impaired. The intent is to get the low-frequency signal in buildings where the technology is commercially available and avoid requiring the low-frequency signal in buildings where the technology is not currently available in the stream of commerce. This section does not require the low-frequency technology in buildings without a fire alarm system because there are no smoke alarms currently available with an integral sounder capable of producing the low-frequency signal. However, it does not prohibit their installation if the product becomes available in the future. This section does require the low-frequency signal in sleeping areas of buildings with a fire alarm system because there are numerous manufacturers of system-connected smoke detectors with an integral sounder that produces the 520 Hz low-frequency signal.

There are several product solutions currently available that are capable of providing the 520 Hz low-frequency signal, including:

1. Fire alarm system horns and horn/strobes.
2. Smoke detectors with integral sounder bases.
3. Speakers connected to an EV/ACS.

**907.5.2.1.3.1 Fire alarm system audible signal.** In sleeping rooms of Group R-1, R-2 and I-1 occupancies, the audible alarm signal activated by a *fire alarm system* shall be a 520-Hz low-frequency signal complying with NFPA 72.

**C** Peer-reviewed research has concluded the 520 Hz low frequency is six times more effective than the standard 3 KHz signal at waking high-risk segments of the population (people over 65, people who are hard of hearing, school-age children and people who are alcohol impaired). The standard 3 KHz audible alarm signal has been used in the majority of fire alarm horns and smoke alarms for over 30 years.

**907.5.2.1.3.2 Smoke alarm signal in sleeping rooms.** In sleeping rooms of Group R-1, R-2 and I-1 occupancies that are required by Section 907.2.8 or 907.2.9 to have a *fire alarm system*, the audible alarm signal activated by single- or multiple-station smoke alarms in the *dwelling unit* or *sleeping unit* shall be a 520-Hz signal complying with NFPA 72.

Where a sleeping room smoke alarm is unable to produce a 520-Hz signal, the 520-Hz alarm signal shall be provided by a *listed* notification appliance or a smoke detector with an integral 520-Hz sounder.

**C** NFPA 72 requires both the 520 Hz and 3 KHz signal in the sleeping rooms of hotels, dormitories, apartment buildings and Institutional Group I occupancies where smoke alarms are installed. Specifically, Chapter 18 of NFPA 72 requires audible notification appliances (horns, speakers or smoke detectors with an integral sounder base) to produce the 520 Hz low-frequency signal in all sleeping rooms of buildings with a protected premises fire alarm system. Chapter 29 of NFPA 72 requires only smoke alarms to produce the 520 Hz low-frequency signal for people with hearing loss or to be provided voluntarily for those with hearing loss. The different requirements within NFPA 72 present a life safety issue because the effec-

tiveness of the 520 Hz low frequency is superior to a 3 KHz audible alarm signal in awakening high-risk segments of the population. The low-frequency signal needs to be provided in areas intended for sleeping for people over 65, people who are hard of hearing, school-age children and people who are alcohol impaired.

**907.5.2.2 Emergency voice/alarm communication systems.** Emergency voice/alarm communication systems required by this code shall be designed and installed in accordance with NFPA 72. The operation of any automatic fire detector, sprinkler waterflow device or manual fire alarm box shall automatically sound an alert tone followed by voice instructions giving *approved* information and directions for a general or staged evacuation in accordance with the building's fire safety and evacuation plans required by Section 404. In high-rise buildings, the system shall operate on at least the alarming floor, the floor above and the floor below. Speakers shall be provided throughout the building by paging zones. At a minimum, paging zones shall be provided as follows:

1. Elevator groups.
2. *Interior exit stairways.*
3. Each floor.
4. *Areas of refuge* as defined in Chapter 2.

**Exception:** In Group I-1 and I-2 occupancies, the alarm shall sound in a constantly attended area and a general occupant notification shall be broadcast over the overhead page.

**C** The primary purpose of an emergency voice/alarm communications system is to provide dedicated manual and automatic facilities for the origination, control and transmission of information and instructions pertaining to a fire alarm emergency to the occupants of a building. This section identifies that notification speakers are required throughout the building with a minimum of one speaker in each paging zone when an emergency voice/alarm communications system is required. The system may sound a general alarm or be a selective system in which only certain areas of the building receive the alarm indication for staged evacuation. See Chapter 4 for evacuation plan requirements. The intent is to provide the capability to send out selective messages to individual areas; however, it does not prohibit the same message to be sent to all areas. In high-rise buildings, a minimum area of notification must include the alarming floor and one floor above and one floor below it.

This section also identifies the minimum paging zone arrangement. This does not preclude further zone divisions for logical staged evacuation in accordance with an approved evacuation plan.

This section also indicates that the emergency voice/alarm system is to be initiated as all other fire alarm systems are initiated. The functional operation of the system begins with an alert tone (usually 3 to 10 seconds in duration) followed by the evacuation signal (message). It is important to remember that the voice alarm system is not an "audible alarm." It has its own specific criteria for installation and approval according to NFPA 72. Consequently, the sound pressure requirements for audible alarms do not apply to voice alarm systems. For voice alarm systems, the intent is communication and an understanding of what is being said, not volume.

The exception is similar to Section 907.5.2.1, Exception 2 and recognizes the supervised environment typical of institutional uses and the reliance placed on staff to act appropriately in an emergency. As is the case for most voice alarms, the key is in being able to deliver specific information to the people who can affect a safe egress—whether this is the public, employees or both.

**907.5.2.2.1 Manual override.** A manual override for emergency voice communication shall be provided on a selective and all-call basis for all paging zones.

**C** The intent of this section is to provide the ability to transmit live voice instructions over any previously initiated signals or prerecorded messages for all zones. This would include the ability to override the voice message at once throughout the building or to be able to select individual paging zones for the message override.

**907.5.2.2.2 Live voice messages.** The emergency voice/alarm communication system shall have the capability to broadcast live voice messages by paging zones on a selective and all-call basis.

**C** This would include the ability to provide the live voice message at once throughout the building or to be able to select individual paging zones to receive the message. Speakers used for background music must not be used unless specifically listed for fire alarm system use. NFPA 72 has additional requirements for the placement, location and audibility of speakers used as part of an emergency voice/alarm communications system.

**907.5.2.2.3 Alternative uses.** The emergency voice/alarm communication system shall be allowed to be used for other announcements, provided that the manual fire alarm use takes precedence over any other use.

**C** In certain circumstances that should be approved by the fire code officials, the emergency voice/alarm communications system could be used to convey information other than fire alarm-related items. This could include severe weather warnings that might require evacuation or relocation, lockdown instructions (see commentary, Section 404.2.3) and similar approved messages. In the event of such usage, the system must respond immediately to manual fire alarm box activations.

**907.5.2.2.4 Emergency voice/alarm communication captions.** Where stadiums, arenas and *grandstands* have 15,000 fixed seats or more and provide audible public announcements, the emergency/voice alarm communication system shall provide prerecorded or real-time captions. Prerecorded or live emergency captions shall be from an *approved* location constantly attended by personnel trained to respond to an emergency.

**C** This provision links the EV/ACS with the requirements for captioning in IBC Section 1109.2.7.3. IBC Section 1109.2.7.3 requires that stadiums, arenas and grandstands with 15,000 or more fixed seats provide captioning for audible announcements (see commentary, Section 907.2.1.2).

**907.5.2.2.5 Standby power.** Emergency voice/alarm communications systems shall be provided with standby power in accordance with Section 1203.

**C** Because the emergency voice/alarm communications system is a critical aid in evacuating a building, the system must be connected to an approved standby power source complying with Section 1203.

**907.5.2.3 Visible alarms.** Visible alarm notification appliances shall be provided in accordance with Sections 907.5.2.3.1 through 907.5.2.3.3.

**Exceptions:**

1. Visible alarm notification appliances are not required in *alterations*, except where an existing *fire alarm system* is upgraded or replaced, or a new *fire alarm system* is installed.
2. Visible alarm notification appliances shall not be required in *exits* as defined in Chapter 2.
3. Visible alarm notification appliances shall not be required in elevator cars.
4. Visual alarm notification appliances are not required in critical care areas of Group I-2, Condition 2 occupancies that are in compliance with Section 907.2.6, Exception 2.
5. A visible alarm notification appliance installed in a nurses' control station or other continuously attended staff location in a Group I-2, Condition 2 suite shall be an acceptable alternative to the installation of visible alarm notification appliances throughout the suite or unit in Group I-2, Condition 2 occupancies that are in compliance with Section 907.2.6, Exception 2.

**C** This section contains alarm system requirements for occupants who are hearing impaired. Visible alarm notification appliances are to be installed in conjunction with the audible devices and located and oriented so that they will display alarm signals throughout a space. It is not the intent of the code to offer visible alarm signals as an option to audible alarm signals. Both are required. However, the code acknowledges conditions when audible alarms may be of little or no value, such as when the ambient sound level exceeds 105 dBA. In such cases, Section 907.5.2.1.2, similar to NFPA 72, allows for visible alarm notification appliances in the area.

Exception 1 states that visible alarm devices are not required in previously approved existing fire alarm systems or as part of minor alterations to existing fire alarm systems. Extensive modifications to an existing fire alarm system such as an upgrade or replacement would require the installation of visible alarm devices even if the previous existing system neither had them nor required them. The main reason is a combination of simple economics and practical application. Many existing systems that do not have visible signal devices do not have the wiring capability to include such devices. To make the necessary changes to the existing system, a total replacement of the existing system may need to take place. In many cases, this is cost prohibitive. Thus, if the alteration is small, the system can be left as is, without the visual devices. The second consideration is scope. If the alteration involves only a limited area, it could be confusing to have part of the area equipped with visual devices and part without. This is not good practice, as the alarm could be confusing. If an entire floor is being altered, then it becomes subject to consideration for an upgrade to an alarm system with visual devices. If only an office is being remodeled, then the implication is that the upgrade to visual devices may not be warranted. This determination will be subjective in many cases and should be applied based on the life safety benefit and financial expenses involved and whether adequate audible devices are present for full coverage.

In Exception 2, visible alarm devices are not required in exit elements because of the potential distraction during evacuation. Exits, as defined in Chapter 2, could include interior exit stairways or exit passageways but not exit access corridors. In tall buildings, exiting may be phased based on alarm zone. If the alarm floor and adjacent floors are notified of the emergency but the remainder of the building is not, then a visual device in the stairway would be confusing to those people who may not be coming from the alarm floor.

Previously, some jurisdictions were requiring visible alarm notification appliances to be installed in elevator cars since there was no exception in the code or NFPA 72 to allow omission of this type of notification appliance in elevator cars. Exception 3 eliminates any confusion regarding the need to install visible notification appliances in elevator cars. The rationale for not installing visible notification appliances in elevator cars is the same as for interior exit stairways; high light intensity from these notification appliances may cause confusion and disorientation. Also, elevator passengers are "captive" in that they cannot respond to such devices until the elevator arrives at its destination or is recalled by the Phase I emergency recall feature, which could lead to passenger panic.

Exception 4 was added to eliminate visual alarm notification appliances in critical care areas of Group I-2, Condition 2 occupancies (hospitals) because of the hazards they may pose to the occupants. Such occupancies already have staff

procedures in place during fires that more than compensate for visual alarm notification appliances. In addition, in these areas of the hospital, the patients are not typically ambulatory. This allowance includes specific requirements for detailing of the staff evacuation responsibilities based on the requirements in Section 907.2.6. See also the commentary to Exception 1 of Section 907.5.2.1.

Exception 5 allows for both visible and audible alarms in hospitals. Visible alarms can cause harmful distractions similar to audible distractions when delivering care. For example, delicate surgery such as neurological (brain/spine) surgery, cardiac surgery or any invasive procedure adjacent to arteries or veins requires precision. A visual flash or sudden alarm can cause a surgeon to startle, which could have devastating consequences to the patient. To mitigate this situation in the surgery environment, the circulating or charge nurse in the surgical area is responsible for being aware of the alarms in the surgical suite and is at a continuously monitored control station that serves the unit or suite. This staff is directly involved in the emergency response and informs the surgical team of the situation in a manner appropriate for the level of care. This eliminates the need for the startling alarms in the operation room, and the team can react to the situation by safely closing the patient.

**907.5.2.3.1 Public use areas and common use areas.** Visible alarm notification appliances shall be provided in *public use areas* and *common use areas*.

**Exception:** Where employee work areas have audible alarm coverage, the notification appliance circuits serving the employee work areas shall be initially designed with not less than 20 percent spare capacity to account for the potential of adding visible notification appliances in the future to accommodate hearing-impaired employee(s).

**C** Visible alarm notification appliances must provide coverage in all areas open to the public (use areas) as well as all shared or common use areas (e.g., corridors, public restrooms, shared offices, classrooms, medical exam rooms, etc.). Areas where visible alarm notification appliances are not required include private offices, mechanical rooms or similar spaces. The intent with this section is to replicate the provisions included in the Americans with Disabilities Act, *Standards for Accessible Design* (ADASAD).

The exception allows employee work areas to provide only for spare capacity on notification circuits to allow for those with hearing impairments to be accommodated as necessary. This spare capacity is intended to eliminate the potential for overloading notification circuits when a hearing-impaired person is hired and needs to be accommodated but reduces the initial construction cost as such alarm notification appliances may not be necessary in every situation.

**907.5.2.3.2 Groups I-1 and R-1.** *Habitable spaces in dwelling units and sleeping units* in Group I-1 and R-1 occupancies in accordance with Table 907.5.2.3.2 shall be provided with visible alarm notification. Visible alarms shall be activated by the in-room smoke alarm and the building fire alarm system.

**C** Fire alarm systems in Group I-1 and R-1 sleeping accommodations must be equipped with visible alarms to the extent stated in Table 907.5.2.3.2. The visible alarm notification devices in these rooms are to be activated by both the required in-room smoke alarm and the building fire alarm system. All visible alarm notification appliances in a building, however, need not be activated by individual room smoke alarms. It is not a requirement that the accessible sleeping units be provided with visible alarm notification appliances even though some elderly patients or residents may be both mobility and hearing impaired.

**TABLE 907.5.2.3.2—VISIBLE ALARMS**

| NUMBER OF SLEEPING UNITS | SLEEPING ACCOMMODATIONS WITH VISIBLE ALARMS |
|--------------------------|---------------------------------------------|
| 6 to 25                  | 2                                           |
| 26 to 50                 | 4                                           |
| 51 to 75                 | 7                                           |
| 76 to 100                | 9                                           |
| 101 to 150               | 12                                          |
| 151 to 200               | 14                                          |
| 201 to 300               | 17                                          |
| 301 to 400               | 20                                          |
| 401 to 500               | 22                                          |
| 501 to 1,000             | 5% of total                                 |
| 1,001 and over           | 50 plus 3 for each 100 over 1,000           |

**C** This table specifies the minimum number of sleeping units that are to be equipped with visible and audible alarms. The numbers are based on the total number of sleeping accommodations in the facility. The requirements in this table are intended to be consistent with the ADASAD.

**907.5.2.3.3 Group R-2.** In Group R-2 occupancies required by Section 907 to have a *fire alarm system*, each *story* that contains *dwelling units* and *sleeping units* shall be provided with the capability to support future visible alarm notification appliances in accordance with Chapter 11 of ICC A117.1. Such capability shall accommodate wired or wireless equipment.

**C** Group R-2 occupancies with a fire alarm system are required to have the capability to support visual alarm notification appliances in accordance with Chapter 11 of ICC A117.1. This requirement has been in the code, and the language is intended to provide more specific guidance as to what is meant by “capability.” Note that this requirement includes all dwelling and sleeping units, not just those classified as either Type A or B. Sections 1106.2 through 1106.4.4 of ICC A117.1 address smoke and fire alarm requirements as they pertain to accessible communication features. More specifically, Section 1106.2 states that where unit smoke detection is provided, it shall provide audible notification in compliance with NFPA 72. Section 1106.3 focuses on buildings where fire alarm systems are provided. If a fire alarm system is provided in the building, ICC A117.1 requires that the wiring be extended to a point within the unit in the vicinity of the smoke detection system. Based on the type of unit and the strategy used by the designer, this location may vary. Section 1106.4 addresses the visible alarm requirements specifically and has various issues it addresses, as follows:

1. Complies with Section 702 of ICC A117.1, which focuses on the requirements of NFPA 72, and that such notification devices be hardwired.
2. Addresses the fact that all visible notification devices be activated within the unit, either when the smoke alarms in the unit activate or when that portion of the building fire alarm system in that portion of the building activates.
3. Allows the same visible notification for the smoke alarms in the unit and the building fire alarm system.
4. Prohibits the use of the visible notification for anything other than the operation of the smoke alarms in the unit or the building fire alarm system.

**907.5.2.3.3.1 Wired equipment.** Where wired equipment is used to comply with the future capability required by Section 907.5.2.3.3, the system shall include one of the following capabilities:

1. The replacement of audible appliances with combination audible/visible appliances or additional visible notification appliances.
2. The future extension of the existing wiring from the unit smoke alarm locations to required locations for visible appliances.
3. For wired equipment, the fire alarm power supply and circuits shall have not less than 5 percent excess capacity to accommodate the future addition of visible alarm notification appliances, and a single access point to such circuits shall be available on every *story*. Such circuits shall not be required to be extended beyond a single access point on a *story*. The *fire alarm system* shop drawings required by Section 907.1.2 shall include the power supply and circuit documentation to accommodate the future addition of visible notification appliances.

**C** In terms of the specific capability requirements, this section provides direction as to what may be meant by bringing the wiring to the unit. There has been confusion in the past when it was interpreted that all units are required to be prewired for visible appliances, which was not the intent of ICC A117.1. More specifically, the requirements provide three options for future capability, as follows:

1. Replacement of audible appliances with combination audible/visible appliances or additional visible appliances.
2. Extension of wiring from the unit smoke alarm locations to the required locations of visible appliances.
3. Excess power supply and circuit capacity for potential connection to visible alarm notification appliances. Only a single access point for each circuit is required on each *story*. As noted with this option, the fire alarm system shop drawings required by Section 907.1.2 shall include the power supply and circuit documentation to accommodate the future addition of visible notification appliances.

It is important to remember that the location of visible notification devices, if installed, are driven by the requirements of NFPA 72, which may vary the approach taken based on the configuration of the space.

**907.6 Installation and monitoring.** A *fire alarm system* shall be installed and monitored in accordance with Sections 907.6.1 through 907.6.6.3 and NFPA 72.

**C** This section specifies the requirements for fire alarm system installation and monitoring and also references the installation requirements of NFPA 72.

**907.6.1 Wiring.** Wiring shall comply with the requirements of NFPA 70 and NFPA 72. Wireless protection systems utilizing radio-frequency transmitting devices shall comply with the special requirements for supervision of low-power wireless systems in NFPA 72.

**C** Wiring for fire alarm systems must be installed so that it is secure and will function reliably in an emergency. The code requires that the wiring for fire alarm systems meet the requirements of NFPA 70 and NFPA 72. This requirement is in addition to the general requirements for electrical installations set forth in Chapter 27 of the IBC. For reliability, systems that use radio-frequency transmitting devices for signal transmission are required to have supervised transmitting and receive-

ing equipment that conforms to the special requirements contained in NFPA 72. This requirement is in addition to the general requirements for supervision in Section 907.6.6.

**907.6.2 Power supply.** The primary and secondary power supply for the *fire alarm system* shall be provided in accordance with NFPA 72.

**Exception:** Backup power for single-station and multiple-station smoke alarms as required in Section 907.2.11.6.

**C** The operation of fire alarm systems is essential to life safety in buildings and must be reliable in the event the normal power supply fails. For proper operation of fire alarm systems, this section requires that the primary and secondary power supplies comply with NFPA 72. This is in addition to the general requirements for electrical installations in Chapter 27. NFPA 72 offers three alternatives for secondary supply: a 24-hour storage battery, storage batteries with a 4-hour capacity and a generator or multiple generators.

NFPA 72 requires that the primary and secondary power supplies for remotely located control equipment essential to the system operation must conform to the requirements for primary and secondary power supplies for the main system. Also, NFPA 72 contains requirements for monitoring the integrity of primary power supplies and requires a backup power supply.

**907.6.3 Initiating device identification.** The *fire alarm system* shall identify the specific initiating device address, location, device type, floor level where applicable and status including indication of normal, alarm, trouble and supervisory status, as appropriate.

**Exceptions:**

1. Fire alarm systems in single-story buildings less than 22,500 square feet (2090 m<sup>2</sup>) in area.
2. Fire alarm systems that only include manual fire alarm boxes, waterflow initiating devices and not more than 10 additional alarm-initiating devices.
3. Special initiating devices that do not support individual device identification.
4. Fire alarm systems or devices that are replacing existing equipment.

**C** Current technology makes identification of alarm-initiating devices much easier. This section takes advantage of this technology to improve the ability of emergency responders to rapidly identify the location and status of initiating devices at the time of an emergency. It will also help identify problematic alarm-initiating devices and thus reduce nuisance alarms. It also eliminates the requirements for providing zone indication of system status. This is considered particularly important in high-rise buildings, where the number of initiating devices and the geometry of the building warrant a need for point monitoring of individual devices, which is not currently accommodated by single floor zones.

This section allows the building official the flexibility to not require individual detection device identification in smaller buildings where the source of alarm and trouble signals can be more easily determined.

The 22,500-square-foot limitation noted in Exception 1 relates to the size of a typical fire alarm zone (see Section 907.6.4) and represents a small building. Exception 2 addresses manual fire alarm systems in which the location of initiation may not be an indicator of where the fire actually is or an automatic sprinkler waterflow alarm-initiating device that could be annunciating an entire building, along with a very limited number of other alarm initiating devices. Exception 3 recognizes that some devices will not work with this requirement. Finally, Exception 4 provides flexibility to existing system replacement. Replacement should be encouraged, and requiring identification of alarm-initiating device locations is considered onerous.

**907.6.3.1 Annunciation.** The initiating device status shall be annunciated at an *approved* on-site location.

**C** This section specifically notes that the alarm-initiating device status of trouble versus alarm needs to be provided in an approved location to enable rapid identification of problematic devices or alarm conditions by first responders.

**907.6.4 Zones.** Each floor shall be zoned separately and a zone shall not exceed 22,500 square feet (2090 m<sup>2</sup>). The length of any zone shall not exceed 300 feet (91 440 mm) in any direction.

**Exception:** *Automatic sprinkler system* zones shall not exceed the area permitted by NFPA 13.

**C** Since the fire alarm system also aids emergency personnel in locating the fire, the system must be zoned to shorten response time to the fire location. Zoning is also critical if the fire alarm system initiates certain other fire protection systems or control features, such as smoke control systems.

At a minimum, each floor of a building must constitute one zone of the system. If the floor area exceeds 22,500 square feet (2090 m<sup>2</sup>), additional zones per floor are required. The maximum length of a zone is 300 feet (91 440 mm).

The exception states that NFPA 13 defines the maximum areas to be protected by one sprinkler system and that the sprinkler system need not be designed to meet the 22,500-square-foot (2090 m<sup>2</sup>) area limitations for a fire alarm system zone. For example, NFPA 13 permits a sprinkler system riser in a light-hazard occupancy to protect an area of 52,000 square feet (4831 m<sup>2</sup>) per floor. In accordance with the exception, a single waterflow switch, and consequently a single fire alarm system zone, would be acceptable. If other alarm-initiating devices are present on the floor, they would need to be zoned separately to meet the 22,500-square-foot (2090 m<sup>2</sup>) limitation.

It is not intended that this section apply to sprinkler systems. This section only applies where a fire alarm system is required in accordance with Section 907. Unless the building is categorized as a high rise and must comply with Section 907.6.4.2, the code does not mandate the zoning of sprinkler systems per floor. With today's fully addressable fire alarm systems, each detector effectively becomes its own zone. The intent with zoning is to identify and limit the search area for fire alarm systems. Addressable devices will indicate the precise location of the alarm condition, thereby eliminating the need for the zoning contemplated by this section when approved by the fire code official in accordance with Section 104.2.3.

**907.6.4.1 Zoning indicator panel.** A zoning indicator panel and the associated controls shall be provided in an *approved* location. The visual zone indication shall lock in until the system is reset and shall not be canceled by the operation of an audible alarm-silencing switch.

**C** The zoning indicator panel, which can be the fire alarm control unit or a separate fire alarm annunciator panel (FAAP), must be installed in a location approved by the fire code official. One of the key considerations in determining panel placement is whether the panel is located to permit ready access by emergency responders. Once an alarm-initiating device within a zone has been activated, the annunciation of the zone must lock in until the system is reset.

**907.6.4.2 High-rise buildings.** In high-rise buildings, a separate zone by floor shall be provided for each of the following types of alarm-initiating devices where provided:

1. Smoke detectors.
2. Sprinkler waterflow devices.
3. Manual fire alarm boxes.
4. Other *approved* types of automatic fire protection systems.

**C** High-rise buildings must have a separate zone by floor for each indicated type of alarm-initiating device. Although this feature may be desirable in all buildings, the incremental cost difference is substantially higher in low-rise buildings in which basic fire alarm systems are installed. State-of-the-art fire alarm systems installed in high-rise buildings are addressable and by their nature automatically provide this minimum zoning.

**907.6.5 Access.** Access shall be provided to each fire alarm device and notification appliance for periodic inspection, maintenance and testing.

**C** Automatic fire detectors, especially smoke detectors, require periodic cleaning to reduce the likelihood of malfunction. Section 907.8 and NFPA 72 require inspection and testing at regular intervals. Access to perform the required inspections, necessary maintenance and testing is a particularly important consideration for those detectors that are installed within a concealed space, such as an air duct.

**907.6.6 Monitoring.** Fire alarm systems required by this chapter or by the *International Building Code* shall be monitored by an *approved* supervising station in accordance with NFPA 72.

**Exception:** Monitoring by a supervising station is not required for:

1. Single- and multiple-station smoke alarms required by Section 907.2.11.
2. Smoke detectors in Group I-3 occupancies.
3. *Automatic sprinkler systems* in one- and two-family dwellings.

**C** Fire alarm systems required by Section 907 are required to be electrically supervised by one of the methods prescribed in NFPA 72.

Exception 1 exempts single- and multiple-station smoke alarms from being supervised because of the potential for unwanted false alarms.

Exception 2 recognizes a similar problem in Group I-3 occupancies. Accordingly, because of the concern over unwanted alarms, smoke detectors in Group I-3 occupancies need only sound an approved alarm signal that automatically notifies staff (see Section 907.2.6.3.1). Smoke detectors in such occupancies are typically subject to misuse and abuse, and frequent unwanted alarms would negate the effectiveness of the system.

Exception 3 clarifies that sprinkler systems in one- and two-family dwellings are not part of a dedicated fire alarm system and are typically designed in accordance with NFPA 13D, which does not require electrical supervision.

**907.6.6.1 Transmission of alarm signals.** Transmission of alarm signals to a supervising station shall be in accordance with NFPA 72.

**C** The communications industry has changed greatly in recent years and the rate of technology development is expected to continue escalating in the future. This change has impacted the transmission of alarm, trouble and supervisory signals from the protected premises to the supervising station. The technology used since the late 1980s used private copper telephone lines with power originating at the telephone central office. This technology is becoming a thing of the past because it is unsustainable by the telephone companies. The automatic dialer devices are no longer used or approved within NFPA 72. Thus, this section provides a direct reference to NFPA 72, which covers the advancements in technologies as they move forward.

**907.6.6.2 MIY monitoring.** Direct transmission of alarms associated with monitor it yourself (MIY) transmitters to a public safety answering point (PSAP) shall not be permitted unless *approved by the fire code official*.

**C** Do it yourself (DIY) fire alarm devices and monitor it yourself (MIY) technology practices are currently being used and are expected to increase in use. The fire and life safety concern is that smoke detector/fire alarm systems that currently can easily communicate with our phones, tablets, watches and laptops will also be able to directly contact the Emergency Communication Center (911) without human interface. This will lead to an increase in unwanted alarm dispatches and place our first responders at risk. This section would allow such direct contact only where the fire code official determines that the technology is acceptable.

**907.6.6.3 Termination of monitoring service.** Termination of fire alarm monitoring services shall be in accordance with Section 901.9.

**C** This is simply a cross link to a requirement in the code related to the termination of monitoring service. Section 901.9 requires the monitoring service itself to notify the fire code official of the service being terminated. Although the ultimate responsibility rests with the building owner, he or she is not cited in this section since if they discontinued the service, they would likely not understand the implications, and if they did, would have no incentive to contact the fire code official.

**907.7 Acceptance tests and completion.** Upon completion of the installation, the *fire alarm system* and all fire alarm components shall be tested in accordance with NFPA 72.

**C** A complete performance test of the fire alarm system must be conducted to determine that the system is operating as required by the code. The acceptance test must include a test of each circuit, alarm-initiating device, alarm notification appliance and any supplementary functions, such as activation of closers and dampers. The operation of the primary and secondary (emergency) power supplies must also be tested, as well as the supervisory function of the control panel. Section 901.5 assigns responsibility for conducting the acceptance tests to the owner or the owner's representative.

NFPA 72 contains specific acceptance test procedures. Additional guidance on periodic testing and inspection can also be obtained from Section 907.8 and NFPA 72.

**907.7.1 Single- and multiple-station alarm devices.** When the installation of the alarm devices is complete, each device and interconnecting wiring for multiple-station alarm devices shall be tested in accordance with the smoke alarm provisions of NFPA 72.

**C** To determine that smoke alarms have been properly installed and are ready to function as intended, they must be actuated during an acceptance test. The test also confirms that interconnected detectors will operate simultaneously as required. The responsibility for conducting the acceptance tests rests with the owner or the owner's representative as stated in Section 901.5.

**907.7.2 Record of completion.** A record of completion in accordance with NFPA 72 verifying that the system has been installed and tested in accordance with the *approved* plans and specifications shall be provided.

**C** In accordance with NFPA 72, this section requires a written statement from the installing contractor that the fire alarm system has been tested and installed in compliance with the approved plans and the manufacturer's specifications. Any deviations from the approved plans or the applicable provisions of NFPA 72 are to be noted in the record of completion.

**907.7.3 Instructions.** Operating, testing and maintenance instructions and record drawings ("as built") and equipment specifications shall be provided at an *approved* location.

**C** To permit adequate testing, maintenance and troubleshooting of the installed fire alarm system, an owner's manual with complete installation instructions must be kept on-site or in another approved location. The instructions include a description of the system, operating procedures, and testing and maintenance requirements.

**907.8 Inspection, testing and maintenance.** The maintenance and testing schedules and procedures for fire alarm and fire detection systems shall be in accordance with Sections 907.8.1 through 907.8.4 and NFPA 72. Records of inspection, testing and maintenance shall be maintained.

**C** Fire alarms and smoke detection systems are to be inspected, tested and maintained in accordance with Sections 907.8.1 through 907.8.4 and the applicable requirements of NFPA 72. It is the building owner's responsibility to keep these systems operable at all times.

**907.8.1 Maintenance required.** Where required for compliance with the provisions of this code, devices, equipment, systems, conditions, arrangements, levels of protection or other features shall thereafter be continuously maintained in accordance with applicable NFPA requirements or as directed by the *fire code official*.

**C** Periodic maintenance keeps systems in good working order or allows repair of defects discovered during inspections or testing. Because specialized tools and training are needed, only properly trained technicians or specialists should perform required periodic maintenance. Most maintenance is required only as needed, but many manufacturers suggest or require regular periodic replacement of parts subject to wear or abuse.

**907.8.2 Testing.** Testing shall be performed in accordance with the schedules in NFPA 72 or more frequently where required by the *fire code official*. Records of testing shall be maintained.

**Exception:** Devices or equipment that are inaccessible because of safety considerations shall be tested during scheduled shut-downs where *approved* by the *fire code official*, but not less than every 18 months.

**C** NFPA 72 includes schedules for testing frequencies of fire alarm and fire detection systems and their components. Periodic tests that follow standardized methods are intended to confirm the results of inspections, determine that all components function properly and determine that systems meet their original design specifications. Tools, devices or equipment are usually required to perform tests. Because tests are more detailed than inspections, they are usually done only once or twice per year in most cases. Some tests, however, may be required as frequently as bimonthly or quarterly. Because specialized knowledge and equipment are required, tests must usually be performed by technicians or specialists trained in the test methods involved.

Although Section 907.8.2 specifically addresses testing, NFPA 72 also contains schedules for visual inspection frequencies. An inspection consists of a visual check of a system or device to verify it is in operating condition and free from visible defects or damage.

Obvious damage and the general condition of the system must always be noted and recorded. Partly because of their cursory nature, inspections are conducted more frequently than tests and maintenance. Because special knowledge and tools are not required, inspections may be done by any competent person.

The exception recognizes the impracticality of testing every device or piece of equipment related to a fire alarm or fire detection system. Some devices may be inaccessible for safety considerations, such as those in continuous process operations. Testing, however, must be done during scheduled shutdowns.

**907.8.3 Smoke detector sensitivity.** Smoke detector sensitivity shall be checked within one year after installation and every alternate year thereafter. After the second calibration test, where sensitivity tests indicate that the detector has remained within its *listed* and marked sensitivity range (or 4 percent obscuration light gray smoke, if not marked), the length of time between calibration tests shall be permitted to be extended to not more than 5 years. Where the frequency is extended, records of detector-caused nuisance alarms and subsequent trends of these alarms shall be maintained. In zones or areas where nuisance alarms show any increase over the previous year, calibration tests shall be performed.

**C** Usually, changes in detector sensitivity are caused by inadequate maintenance. Regular sensitivity testing is intended to determine whether detectors require recalibration, maintenance or replacement. This section prescribes the intervals for testing smoke detector sensitivity. Where two successful tests have been conducted, the frequency of the calibration tests can be extended to a maximum of 5 years. This interval extension recognizes the stability of both the environment and the detector. However, if nuisance alarms occur during this time interval extension, calibration tests may be needed because of potential changes in the environment where the detector is located or in the performance of the detector itself.

**907.8.4 Inspection, testing and maintenance.** The building *owner* shall be responsible to maintain the fire and *life safety systems* in an operable condition at all times. Service personnel shall meet the qualification requirements of NFPA 72 for inspection, testing and maintenance of such systems. Records of inspection, testing and maintenance shall be maintained.

**C** This section clearly indicates that it is the responsibility of the building owner to maintain all fire alarm systems in proper working order. Often, an outside agency that employs adequately trained personnel will provide any maintenance and testing that is needed. NFPA 72 contains additional guidance on the qualifications for service personnel. Some examples include factory-trained and factory-certified individuals, individuals certified for fire alarm testing and inspection by NICET, or other individuals tested and certified by the local authority. Proper maintenance of fire alarm systems is essential so that the systems will perform as intended.

Inspection and test records provide a means for determining compliance with the requirements of the code. Inspectors should be prepared to determine that inspection, test and maintenance logs are accurate and complete. Records must include the nature of the activity or service performed, when the activity occurred, who performed the activity and who witnessed testing or approved the work upon completion.

**907.9 Where required in existing buildings and structures.** An *approved fire alarm system* shall be provided in existing buildings and structures where required in Chapter 11.

**C** Retroactive requirements for fire alarms are found in Section 1103.7. All the retroactive code requirements are found in Chapter 11 for convenience and to help with consistent enforcement.

**907.10 Smoke alarm maintenance.** Smoke alarms shall be tested and maintained in accordance with the manufacturer's instructions and this code.

**C** This section clarifies that smoke alarms are required to be tested and maintained according to both the manufacturer's instructions and also the requirements of the code.

**907.10.1 Smoke alarm replacement.** Smoke alarms shall be replaced when any of the following apply:

1. The smoke alarm fails to respond to operability tests.

2. The smoke alarm exceeds 10 years from the date of manufacture marked on the unit, unless an earlier replacement is specified in the manufacturer's instructions.
3. The smoke alarm end-of-life signal is sounded.
4. The smoke alarm date of manufacture cannot be determined.

Where the replacement of smoke alarms is required by this section, smoke alarms shall not be required to include the 520-Hz signal unless the smoke alarms to be replaced include that signal.

**C** Smoke alarms shall be replaced if any of the following conditions apply: they fail to respond to operability tests, they exceed 10 years from the date of manufacture (unless an earlier replacement is specified in the manufacturer's published instructions), the smoke alarm end of life signal is sounded, or the smoke alarm date of manufacturer cannot be determined. The replacement of smoke alarms when they exceed 10 years from the date of manufacture is considered to be an operational and maintenance requirement for new and existing smoke alarms, not a construction requirement. It is the intent of this section to maintain a fully operational smoke alarm system, and the 10-year replacement criterion is the maximum equipment lifespan.

If smoke alarms are required to be replaced by any of the conditions of Section 907.10.1, they are not required to be replaced with a low frequency smoke alarm.

## SECTION 908—EMERGENCY ALARM SYSTEMS

**908.1 Group H occupancies.** Emergency alarms for the detection and notification of an emergency condition in Group H occupancies shall be provided as required in Chapter 50.

**C** Emergency alarm systems provide indication and warning of emergency situations involving hazardous materials. An emergency alarm system is required in all Group H occupancies as indicated in Sections 5004.9 and 5005.4.4, as well as Group H-5 HPM facilities, as indicated in Section 908.2. The Group H occupancy classification assumes the storage or use of hazardous materials exceeds the maximum allowable quantities specified in Tables 5003.1.1(1) and 5003.1.1(2).

An emergency alarm system should include an emergency alarm-initiating device outside each interior door of hazardous material storage areas, a local alarm device and adequate supervision.

Even though ozone gas generator rooms (Section 6005.3.2), repair garages (Section 2311.8.9) and refrigeration systems (Section 608.9) are not typically classified as Group H occupancies, the potential hazards associated with these occupancy conditions are great enough to require additional means of early warning detection.

**908.2 Group H-5 occupancy.** Emergency alarms for notification of an emergency condition in an HPM facility shall be provided as required in Section 2703.12.

**C** In addition to hazardous material storage areas as regulated by Section 5004.9, Section 2703.12.1 also requires emergency alarms for service corridors, exit access corridors and interior exit stairways because of the potential transport of hazardous materials through these areas. Section 2703.13 requires a continuous gas detection system for early detection of leaks in areas where HPM gas is used. Gas detection systems are required to initiate a local alarm and transmit a signal to the emergency control station upon detection (see commentary, Sections 2703.12 and 2703.13).

**908.3 Fire alarm system interface.** Where an *emergency alarm system* is interfaced with a building's *fire alarm system*, the signal produced at the fire alarm control unit shall be a supervisory signal.

**C** This section provides the requirement for how an emergency alarm system is to interface with a fire alarm system. The emergency alarm system may or may not require evacuation but should not activate the fire alarm system. It is important to ensure the integrity of the fire alarm system and to ensure that the fire alarm signal be reserved for fire emergencies. Where interface between different alarm systems is desired, such as may occur when providing off-site monitoring services, it is important to ensure that unwanted activations of the fire alarm system are avoided. By requiring the supervisory signal, the code clarifies what is intended where systems are interfaced and provides an appropriate level of signal to monitoring facilities to ensure that corrective action is taken.

## SECTION 909—SMOKE CONTROL SYSTEMS

**909.1 Scope and purpose.** This section applies to mechanical or passive smoke control systems where they are required for new buildings or portions thereof by provisions of the *International Building Code* or this code. The purpose of this section is to establish minimum requirements for the design, installation and acceptance testing of smoke control systems that are intended to provide a tenable environment for the evacuation or relocation of occupants. These provisions are not intended for the preservation of contents, the timely restoration of operations or for assistance in fire suppression or overhaul activities. Smoke control systems regulated by this section serve a different purpose than the smoke- and heat-removal provisions found in Section 910. Mechanical smoke control systems shall not be considered exhaust systems under Chapter 5 of the *International Mechanical Code*.

**C** This section is clarifying the intent of smoke control provisions, which is to provide a tenable environment to occupants during evacuation and relocation and not to protect the contents, enable timely restoration of operations or facilitate fire

suppression and overhaul activities. It should also be understood that a smoke control system cannot maintain tenable conditions within the immediate area of fire origin. There are provisions for high-rise buildings in IBC Section 403.4.7 that are focused on the removal of smoke for post-fire and overhaul operations, which is very different than the smoke control provisions in Section 909. Another element addressed in this section is that smoke control systems serve a different purpose than smoke and heat vents (see Section 910). This eliminates any confusion that smoke and heat vents can be used as a substitution for smoke control. Additionally, a clarification is provided to note that smoke control systems are not considered an exhaust system in accordance with Chapter 5 of the IMC. This is because such systems are unique in their operation and are not necessarily designed to exhaust smoke but are focused on tenability for occupants during egress. It should be noted that the smoke control provisions are duplicated in Chapter 5 of the IMC.

It is important to note that these provisions only apply when smoke control is required by other sections of the code. The code requires smoke management within atrium spaces (see IBC Section 404.5) and underground buildings (see IBC Section 405.5). High-rise facilities require smokeproof exit enclosures in accordance with Sections 909.20 and 1023.11 (see IBC Section 403.5.4). Also, covered mall buildings that contain atriums that connect more than two stories require smoke control (see IBC Section 402.7.2). Underground buildings with floors used for human occupancy that are greater than 30 feet below the floor of the level of exit discharge are required to be provided with smoke control (see IBC Section 405.5). I-3 detention occupancies that are in windowless buildings are also required to be provided with smoke control compliant with Section 909 (see IBC Section 408.9).

Section 909 focuses primarily on mechanical smoke control systems, but there are many instances within the code where smoke is required to be managed in a passive way through the use of concepts such as smoke compartments. Smoke compartments are formed through the use of smoke barriers in accordance with IBC Section 709. Smoke barriers can be used simply as a passive smoke management system or can be a design component of a mechanical smoke control system in accordance with Section 909. Some examples of occupancies requiring passive systems include hospitals, nursing homes and similar facilities (Group I-2 occupancies) and detention facilities (Group I-3 occupancies) (see IBC Sections 407.5 and 408.6).

In some cases, mechanical smoke control in accordance with Section 909 is allowed as an option for compliance. More specifically, if a Group I-3 contains windowless areas of the facility, natural or mechanical smoke management is required (see IBC Section 408.9).

Over time, smoke control provisions have become more complex. The reason is related to the fact that smoke is a complex problem. There are numerous parameters that need to be considered when designing a smoke control system, including factors such as buoyancy; expansion of gases; wind; the geometry of the space and of communicating spaces; the dynamics of the fire, including heat-release rate; the production and distribution of smoke; and the interaction of the building systems. The means of egress from the space is also a major consideration and time to egress is a factor in designing for tenability. The historical cookbook approach of providing six air changes or four air changes based on volume of the communicating space does not take these factors into consideration.

Smoke control systems can be either passive or active. Active systems are sometimes referred to as mechanical. Passive smoke control systems take advantage of smoke barriers surrounding the zone in which the fire event occurs or high bay areas that act as reservoirs to control the movement of smoke to other areas of the building. Active systems utilize pressure differences to contain smoke within the event zone or exhaust flow rates sufficient to slow the descent of the upper-level smoke accumulation to some predetermined position above necessary means of egress paths through the zone of fire origin. On rare occasions, there is also a possibility of controlling the movement of smoke horizontally by opposed airflow.

Essentially, there are three methods of mechanical or active smoke control that can be used separately or in combination within a design: pressurization, exhaust and opposed airflow.

Of course, all of these active approaches can be used in combination with the passive method.

Typically, the mechanical pressurization method is used in high-rise buildings when pressurizing shafts (interior exit stairways or elevator hoistways) and for zoned smoke control. Pressurization is not practical in large open spaces such as atriums or malls, since it is difficult to develop the required pressure differences due to the large volume of the space.

The exhaust method is typically used in large open spaces such as atriums and malls. The opposed airflow method, which basically uses a velocity of air horizontally to slow the movement of smoke, is typically applied in combination with either a pressurization method or exhaust method within hallways or openings into atriums and malls.

The application of each of these methods will be dependent on the specifics of the building design. Smoke control within a building is fundamentally an architecturally driven problem. Different architectural geometries first dictate the need, or lack thereof, for smoke control, and then define the bounds of available solutions to the problem.

**909.2 General design requirements.** Buildings, structures, or parts thereof required by the *International Building Code* or this code to have a smoke control system or systems shall have such systems designed in accordance with the applicable requirements of Section 909 and the generally accepted and well-established principles of engineering relevant to the design. The *construction documents* shall include sufficient information and detail to describe adequately the elements of the design necessary for the proper

implementation of the smoke control systems. These documents shall be accompanied with sufficient information and analysis to demonstrate compliance with these provisions.

**C** This section simply states that when smoke control systems are required by the code, the design is required to be in accordance with the provisions of this section. As noted in the commentary to Section 909.1, there are instances within the code that have smoke management systems that are purely passive in nature and do not reference Section 909.

This section stresses that designs in accordance with this section need to follow “generally accepted and well-established principles of engineering relevant to the design,” essentially requiring a certain level of qualifications in the applicable areas of engineering to prepare such designs. The primary engineering disciplines tend to be fire engineering and mechanical engineering. It should be noted that each state in the United States typically requires minimum qualifications to undertake engineering design. Two important resources when designing smoke control systems are the International Code Council’s (ICC) *A Guide to Smoke Control in the 2006 IBC®* and American Society of Heating, Refrigerating and Air-Conditioning Engineers’ (ASHRAE) *Handbook of Smoke Control Engineering*. Additionally, Section 909.8 requires the use of NFPA 92 for the design of smoke control systems using the exhaust method. This standard has many relevant aspects beyond the design of exhaust method systems that are also beneficial to the system designer. In particular, Annex B provides resources in terms of determination of fire size for design. ICC’s *A Guide to Smoke Control in the 2006 IBC®* also provides guidance on design fires.

A key element covered in this section is the need for detailed and clear construction documents so that the system is installed correctly. In most complex designs, the key to success is appropriate communication to the contractors as to what needs to be installed. The more complex a design becomes, the more likely there is to be construction errors. Most smoke control systems are complex, which is why special inspections in accordance with Section 909.3 and Chapter 17 of the IBC are critical for smoke control systems. Additionally, in order for the design to be accepted, analyses and justifications need to be provided in enough detail to evaluate for compliance. Adequate documentation is critical to the commissioning, inspection, testing and maintenance of smoke control systems and significantly contributes to the overall reliability and effectiveness of such systems.

**909.3 Special inspection and test requirements.** In addition to the ordinary inspection and test requirements that buildings, structures and parts thereof are required to undergo, smoke control systems subject to the provisions of Section 909 shall undergo special inspections and tests sufficient to verify the proper commissioning of the smoke control design in its final installed condition. The design submission accompanying the *construction documents* shall clearly detail procedures and methods to be used and the items subject to such inspections and tests. Such commissioning shall be in accordance with generally accepted engineering practice and, where possible, based on published standards for the particular testing involved. The special inspections and tests required by this section shall be conducted under the same terms as in Section 1704 of the *International Building Code*.

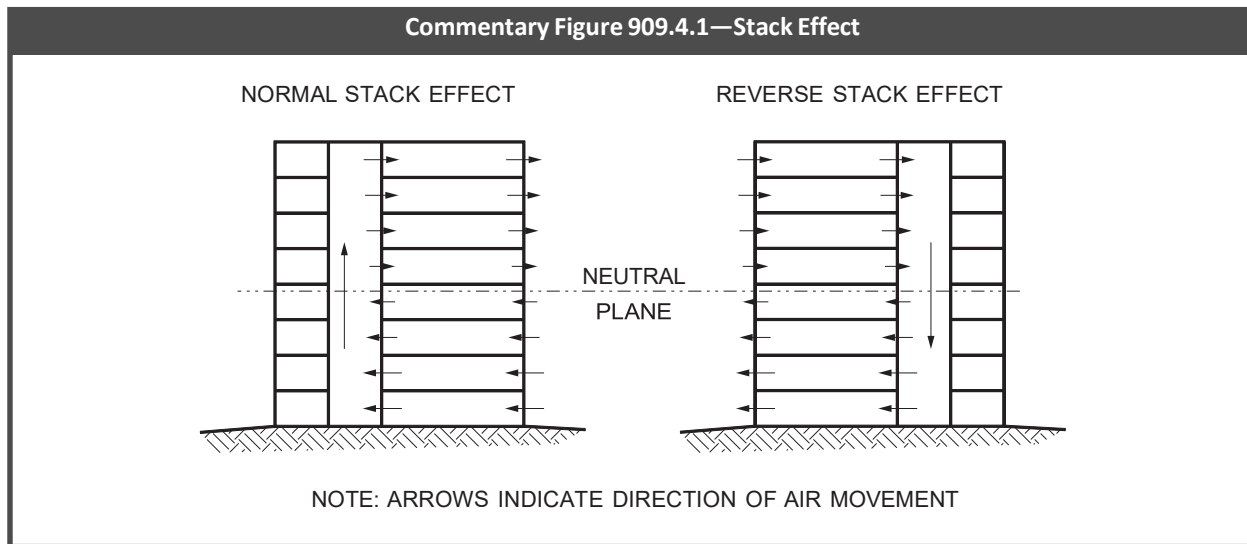
**C** Because of the complexity and uniqueness of each design, special inspection and testing must be conducted. The designer needs to provide specific recommendations for special inspection and testing within his or her documentation. In fact, in IBC Chapter 17, the code specifies that special inspection agencies for smoke control have expertise in fire protection engineering, mechanical engineering and certification as air balancers. Since the designs are unique to each building, there probably will not be a generic approach available to inspect and test such systems. The designer can and should, however, use any available published standards or guides when developing the special inspection and testing requirements for that particular design. ICC’s *A Guide to Smoke Control in the 2006 IBC®* provides some background on such inspections. Also, ASHRAE Guideline 5 is a good starting place, but only as a general outline. In addition, NFPA 92 also has extensive testing, documentation and maintenance requirements that may be a good resource. NFPA 92 is referenced in Sections 909.7 and 909.8 for the design of smoke control systems using the airflow method and exhaust method. Each system will require a unique commissioning plan that can be developed only after careful and thoughtful examination of the final design and all of its components and interrelationships. Generally, these provisions may be included in design standards or engineering guides.

**909.4 Analysis.** A rational analysis supporting the types of smoke control systems to be employed, the methods of their operations, the systems supporting them and the methods of construction to be utilized shall accompany the *construction documents* submission and include, but not be limited to, the items indicated in Sections 909.4.1 through 909.4.7.

**C** This section indicates that simply determining airflow, exhaust rates and pressures to maintain tenable conditions is not adequate. There are many factors that could alter the effectiveness of a smoke control system, including stack effect, temperature effect of fire, wind effect, heating, ventilating and air-conditioning (HVAC) system interaction and climate, as well as the placement, quantity of inlets/outlets and velocity of supply and exhaust air, changes in building occupancy, placement of fuel loads, and type of fuels may have significant impact on the suitability of a smoke control system. The underlying physics related to how these factors impact the design of a smoke control system are discussed in the *SFPE Handbook of Fire Protection Engineering*, 5th edition, Chapters 50 and 51. These factors are addressed in the sections that follow. Additionally, the duration of operation of any smoke control system is mandated at a minimum of 20 minutes or 1.5 times the egress time, whichever is less. The code cannot reasonably anticipate every conceivable building arrangement and condition the building may be subject to over its life and must depend on such factors being addressed through a rational analysis.

**909.4.1 Stack effect.** The system shall be designed such that the maximum probable normal or reverse stack effect will not adversely interfere with the system's capabilities. In determining the maximum probable stack effect, altitude, elevation, weather history and interior temperatures shall be used.

**C** Stack effect is the tendency for air to rise within a heated shaft when the temperature is colder on the exterior of the building. Air in the building has a buoyant force because it is warmer and less dense than the outside air. This buoyant force causes air to rise within the shafts of buildings. Reverse stack effect is the tendency for air to flow downward within a building when the interior is cooler than the exterior of the building. Stack effect can have a significant impact on smoke flow during building fires and can affect the intended operation of a smoke control system. If stack effect is great enough, it may overcome the pressures determined during the design analyses and allow smoke to enter areas outside the zone of origin (see Commentary Figure 909.4.1).



**909.4.2 Temperature effect of fire.** Buoyancy and expansion caused by the design fire in accordance with Section 909.9 shall be analyzed. The system shall be designed such that these effects do not adversely interfere with the system's capabilities.

**C** This section requires that the design account for the effect temperature may have on the success of the system. When air or any gases are heated, they will expand. This expansion makes the gases lighter and therefore more buoyant. The buoyancy of hot gases is important when the design is to exhaust such gases from a location in or close to the ceiling; therefore, if sprinklers are part of the design, as required by Section 909, the gases may be significantly cooler than an unsprinklered fire, making it more difficult to remove the smoke and alter the plume dynamics. The fact that air expands when heated needs to be accounted for in the design.

When using the pressurization method, the expansion of hot gases needs to be accounted for, since it will take a larger volume of air to create the necessary pressure differences to maintain the area of fire origin in negative pressure. The expansion of the gases has the effect of pushing the hot gases out of the area of fire origin. Since sprinklers will tend to cool the gases, the effect of expansion is lower. The pressure difference required in Section 909.6.1 is specifically based on a sprinklered building. If the building is nonsprinklered, higher pressure differences may be required. The minimum pressure difference for certain unsprinklered ceiling height buildings is as follows:

| Ceiling height<br>(feet) | Minimum pressure difference<br>(inch water gage) |
|--------------------------|--------------------------------------------------|
| 9                        | 0.10                                             |
| 15                       | 0.14                                             |
| 21                       | 0.18                                             |

This is a very complex issue that needs to be part of the design analysis. It needs to address the type and reaction of the fire protection systems, ceiling heights and the size of the design fire.

**909.4.3 Wind effect.** The design shall consider the adverse effects of wind. Such consideration shall be consistent with the wind-loading provisions of the *International Building Code*.

**C** The effect of wind on a smoke control system within a building is very complex. It is generally known that wind exerts a load on a building and that wind has a pronounced impact on smoke movement within a building. The loads are looked at as

windward (positive pressure) and leeward (negative pressure). The velocity of winds will vary based on the terrain and the height above grade; therefore, the height of the building and surrounding obstructions will have an effect on these velocities. These pressures alter the operation of fans, especially propeller fans, thus altering the pressure differences and airflow direction in the building. There is not an easy solution to dealing with these effects. In fact, little research has been done in this area.

It should be noted that in larger buildings a wind study is normally undertaken for the structural design. The data from those studies can be used in the analysis of the effects on the pressures and airflow within the building with regard to the performance of the smoke control system.

**909.4.4 Systems.** The design shall consider the effects of the heating, ventilating and air-conditioning (HVAC) systems on both smoke and fire transport. The analysis shall include all permutations of systems status. The design shall consider the effects of the fire on the heating, ventilating and air-conditioning systems.

**C** If not properly configured to shut down or be included as part of the design, the HVAC system can alter the smoke control design. More specifically, if dampers are not provided between smoke zones within the HVAC system ducts, smoke could be transported from one zone to another. Additionally, if the HVAC system places more supply air than assumed for the smoke control system design, the velocity of the air may adversely affect the fire plume or a positive pressure may be created. Generally, an analysis of the smoke control design and the HVAC system in all potential modes should occur and be noted within the design documentation as well as incorporated into inspection, testing and maintenance procedures. This is critical as these systems need to be maintained and tested to help ensure that they operate and shut down systems as required.

**909.4.5 Climate.** The design shall consider the effects of low temperatures on systems, property and occupants. Air inlets and exhausts shall be located so as to prevent snow or ice blockage.

**C** This section is focused on properly protecting equipment from weather conditions that may affect the reliability of the design. For instance, extremely cold or hot air may damage critical equipment within the system when pulled directly from the outside. Some listings of duct smoke detectors are for specific temperature ranges; therefore, placing such detectors within areas exposed to extreme temperatures may void the listing. Also, the equipment and air inlets and outlets should be designed and located so as to not collect snow and ice that could block air from entering or exiting the building.

**909.4.6 Duration of operation.** All portions of active or engineered smoke control systems shall be capable of continued operation after detection of the fire event for a period of not less than either 20 minutes or 1.5 times the calculated egress time, whichever is greater.

**C** The intent of the smoke control provisions is to provide a tenable environment for occupants to either evacuate or relocate to a safe place. Evacuation and relocation activities include a preevacuation period (notifying occupants, seeking information, decision time, preparing to evacuate) and the movement period. In order to achieve this goal, the code has established 20 minutes or 1.5 times the calculated egress time, whichever is greater, as a minimum time for evacuation or relocation. Basically, this allows a designer to undertake an analysis to more closely determine the necessary time required. The code provides a safety factor of 1.5 times the egress time to account for uncertainty related to human behavior. It is stressed that the 20-minute duration as well as the calculated egress time, whichever approach is chosen, begins after the detection of the fire event and notification to the building occupants to evacuate have occurred, since occupants need to be alerted before evacuation can happen. The calculation of evacuation time needs to include delays with notification and the start of evacuation (i.e., preevacuation time). It is stressed that the code states 20 minutes or 1.5 times the egress time, whichever is greater (i.e., 20 minutes is a minimum). Egress of occupants can be addressed through hand calculations or through the use of computerized evacuation models. Some of the more advanced models can address a variety of factors, including the building layout, different sizes of people, different movement speeds and different egress paths available. With these types of models, the actual time can be even more precisely calculated. Of course, it is cautioned that in many cases these models provide the optimal time for egress. The safety factor of 1.5 within the code is intended to address many of these uncertainties.

Note that this section applies to all types of smoke control designed in accordance with Section 909. Also, most smoke control systems will typically have the ability to run for much longer than 20 minutes as they are on standby power and may be able to continue to achieve the tenability goals.

System response as required in Section 909.17 needs to be accounted for when determining the ability of the smoke control system to keep the smoke layer interface at the appropriate level (see commentary, Section 909.17).

**909.4.7 Smoke control system interaction.** The design shall consider the interaction effects of the operation of multiple smoke control systems for all design scenarios.

**C** The focus of this section is related to the interaction of multiple mechanical smoke control systems by asking for a specific analysis of the interaction of such systems similar to that required for the interaction of HVAC systems. Where hoistway pressurization is chosen as an option for compliance with the hoistway opening protection requirements, the potential

exists for a pressurized stair system to also be present. These two systems need to be able to operate at the same time without a negative impact on either system. It is also possible that an atrium with a smoke control system is located in a building containing a stair pressurization system.

**909.5 Smoke barrier construction.** *Smoke barriers* required for passive smoke control and a smoke control system using the pressurization method shall comply with Section 709 of the *International Building Code*. The maximum allowable leakage area shall be the aggregate area calculated using the following leakage area ratios:

1. Walls:  $A/A_w = 0.00100$
2. Interior exit stairways and ramps and exit passageways:  $A/A_w = 0.00035$
3. Enclosed exit access stairways and ramps and all other shafts:  $A/A_w = 0.00150$
4. Floors and roofs:  $A/A_f = 0.00050$

where:

$A$  = Total leakage area, square feet ( $m^2$ ).

$A_f$  = Unit floor or roof area of barrier, square feet ( $m^2$ ).

$A_w$  = Unit wall area of barrier, square feet ( $m^2$ ).

The leakage area ratios shown do not include openings due to gaps around doors and operable windows. The total leakage area of the *smoke barrier* shall be determined in accordance with Section 909.5.1 and tested in accordance with Section 909.5.2.

**C** Part of the strategy of smoke control systems, particularly smoke control systems using the pressurization method (often termed “zoned smoke control”), is the use of smoke barriers to divide a building into separate smoke zones (or compartments). This strategy is used in both passive and mechanical systems. It should be noted that not all walls, ceilings or floors would be considered smoke barriers. Only walls that designate separate smoke zones within a building need to be constructed as smoke barriers. This section is simply providing requirements for walls, floors and ceilings that are used as smoke barriers. It should be noted that it is possible that a smoke control system utilizing the exhaust method may not need to utilize a smoke barrier to divide the building into separate smoke zones; therefore, the evaluation of barrier construction and leakage area may not be necessary and is primarily focused on designs using a passive approach or the pressurization method.

In order for smoke to not travel from one smoke zone to another, specific construction requirements are necessary in accordance with the code. It should be noted that openings such as doors and windows are dealt with separately within Section 909.5.3 from openings such as cracks or penetrations.

**909.5.1 Total leakage area.** Total leakage area of the barrier is the product of the *smoke barrier* gross area multiplied by the allowable leakage area ratio, plus the area of other openings such as gaps around doors and operable windows.

**C** It is impossible for walls and floors to be constructed that are completely free from openings that may allow the migration of smoke; therefore, leakage needs to be compensated for within the design by calculating the leakage area of walls, ceilings and floors. The factors provided in Section 909.5, which originate from ASHRAE’s provisions on leaky buildings, are used to calculate the total leakage area. The total leakage area is then used in the design process to determine the proper amount of air to create the required pressure differences across these surfaces that form smoke zones.

Additionally, Section 909.5 provides ratios to determine the maximum allowable leakage in walls, interior exit stairways, shafts, floors and roofs. These leakage areas are critical in determining whether the proper pressure differences are provided when utilizing the pressurization method of smoke control. Pressure differences will decrease as the openings get larger.

**909.5.2 Testing of leakage area.** Compliance with the maximum total leakage area shall be determined by achieving the minimum air pressure difference across the barrier with the system in the smoke control mode for mechanical smoke control systems utilizing the pressurization method. Compliance with the maximum total leakage area of passive smoke control systems shall be verified through methods such as door fan testing or other methods, as *approved by the fire code official*.

**C** These leakage criteria need to be evaluated through testing. For the case of a pressurization system, pressure differences need to be verified when the system is in smoke control mode. In the case of passive smoke control systems, pressure testing through tests such as the door fan method is necessary.

**909.5.3 Opening protection.** Openings in *smoke barriers* shall be protected by automatic-closing devices actuated by the required controls for the mechanical smoke control system. Door openings shall be protected by fire door assemblies complying with Section 716 of the *International Building Code*.

**Exceptions:**

1. Passive smoke control systems with automatic-closing devices actuated by spot-type smoke detectors *listed* for releasing service installed in accordance with Section 907.3.
2. Fixed openings between smoke zones that are protected utilizing the airflow method.
3. In Group I-1, Condition 2; Group I-2; and ambulatory care facilities, where a pair of opposite-swinging doors are installed across a *corridor* in accordance with Section 909.5.3.1, the doors shall not be required to be protected in accordance

with Section 716 of the *International Building Code*. The doors shall be close-fitting within operational tolerances and shall not have a center mullion or undercuts in excess of  $\frac{3}{4}$ -inch (19.1 mm) louvers or grilles. The doors shall have head and jamb stops and astragals or rabbets at meeting edges and, where permitted by the door manufacturer's listing, positive-latching devices are not required.

4. In Group I-2 and ambulatory care facilities, where such doors are special-purpose horizontal sliding, accordion or folding door assemblies installed in accordance with Section 1010.3.3 and are automatic closing by smoke detection in accordance with Section 716.2.6.6 of the *International Building Code*.
5. Group I-3.
6. Openings between smoke zones with clear ceiling heights of 14 feet (4267 mm) or greater and bank-down capacity of greater than 20 minutes as determined by the design fire size.

**C** Similar to concerns of smoke leakage between smoke zones, openings may compromise the necessary pressure differences between smoke zones. Openings in smoke barriers, such as doors and windows, must be either constantly or automatically closed when the smoke control system is operating. This section requires that doors be automatically closed through the activation of an automatic-closing device linked to the smoke control system. Essentially, when the smoke control system is activated, all openings are automatically closed. This most likely would mean that the mechanism that activates the smoke control system would also automatically close all openings. The smoke control system will be activated by a specifically zoned smoke detection or sprinkler system as required by Sections 909.12.3 and 909.12.4.

In terms of actual opening protection, Section 909.5.3 is simply referring the user to IBC Section 716 for specific construction requirements for doors located in smoke barriers. Note that smoke barriers are different from fire barriers, since the intended measure of performance is different. One is focused on fire spread from the perspective of heat, the other from the perspective of smoke passage. Smoke barriers do require a 1-hour fire-resistance rating.

There are several exceptions to this particular section. Exception 1 is specifically for passive systems. Passive systems, as noted, are systems in which there is no use of mechanical systems. Instead, the system operates primarily on the configuration of barriers and layout of the building to provide smoke control. Passive systems can use spot-type detectors to close doors that constitute portions of a smoke barrier. Essentially, this means a full fire alarm system would not be required. Instead, single station detectors would be allowed to close the doors. Such doors would need to fail in the closed position if power is lost. The specifics as to approved devices would be found in NFPA 72.

Exception 2 is based on the fact that some systems take advantage of the opposed airflow method such that smoke is prevented from migrating past the doors. Therefore, since the design already accounts for potential smoke migration at these openings through the use of air movement, it is unnecessary to require the barrier to be closed.

Exception 3 is specifically related to the unique requirements for Group I-1, Condition 2; I-2 occupancies; and ambulatory care occupancies. Essentially, a very specific alternative, which meets the functional needs of these occupancy types, is provided. Opposite-swinging doors are allowed without meeting the specific requirements of IBC Section 716. Note that Group I-2 and ambulatory care facilities utilize the concept of smoke compartments, which is a form of a passive smoke control system. The requirements of this section are focused on openings in smoke barriers that need to close upon activation of a pressurization system.

Exception 4 provides a specific allowance for horizontal sliding doors in Group I-2, Condition 2 occupancies and ambulatory care occupancies. These doors are commonly used in such occupancies and have very specific installation requirements in Section 1010.3.3. These doors are an alternative to pivoted or side-hinged, swinging-type doors. This exception requires compliance with IBC Section 716.2.6.6 for automatic closing upon detection of smoke.

Exception 5 allows an exemption from the automatic-closing requirements for all Group I-3 occupancies. This is because facilities that have occupants under restraint or with specific security restrictions have unique requirements in accordance with IBC Section 408. These requirements accomplish the intent of providing reliable barriers between each smoke zone since, for the most part, such facilities will have a majority of doors closed and in a locked position because of the nature of the facility. The staff very closely controls these types of facilities.

Exception 6 relates to the behavior of smoke. The assumption is that smoke rises because of the buoyancy of hot gases, and if the ceiling is sufficiently high, the smoke layer will be contained for a longer period of time before it begins to move into the next smoke zone. Therefore, it is not as critical that the doors automatically close. This allowance is dependent on the specific design fire for a building. See Section 909.9 for more information on design fire determination. Different size design fires create different amounts of smoke that, depending on the layout of the building, may migrate in different ways throughout the building. This section mandates that smoke cannot begin to migrate into the next smoke zone for at least 20 minutes. This is consistent with the 20-minute minimum duration of operation of smoke control systems required in Section 909.4.6. It should be noted that a minimum of 14-foot (4267 mm) ceilings are required to take advantage of this exception. This exception would require an engineering analysis.

**909.5.3.1 Group I-1, Condition 2; Group I-2; and ambulatory care facilities.** In Group I-1, Condition 2; Group I-2; and *ambulatory care facilities*, where doors are installed across a *corridor*, the doors shall be automatic closing by smoke detection in accordance

with Section 716.2.6.6 of the *International Building Code* and shall have a vision panel with fire-protection-rated glazing materials in fire-protection-rated frames, the area of which shall not exceed that tested.

**C** Part of the alternative allowed for horizontal sliding doors in Exception 3 of Section 909.5.3 for Group I-1, Condition 2; Group I-2; and ambulatory care facilities is the requirement that vision panels be provided. These vision panels need to be approved fire-protection-rated glazing and be within frames of a size that does not exceed the frame size that was used for testing of the glazing.

**909.5.3.2 Ducts and air transfer openings.** Ducts and air transfer openings are required to be protected with a minimum Class II, 250°F (121°C) smoke damper complying with Section 717 of the *International Building Code*.

**C** Another factor that adds to the reliability of smoke barriers is the protection of ducts and air transfer openings within smoke barriers. Left open, these openings may allow the transfer of smoke between smoke zones. These ducts and air transfer openings most often are part of the HVAC system. Damper operation and the reaction with the smoke control system will be evaluated during acceptance testing. It should be noted that there are duct systems used within a smoke control design that are controlled by the smoke control system and should not automatically close upon detection of smoke via a smoke damper.

It should be noted that a smoke damper works differently than a fire damper. Fire dampers react to heat via a fusible link, while smoke dampers activate upon the detection of smoke. The smoke dampers used should be rated as Class II, 250°F (121°C). The class of the smoke damper refers to its level of performance relative to leakage. The temperature rating is related to its ability to withstand the heat of smoke resulting from a fire. It should be noted that although smoke barriers are only required to utilize smoke dampers, there may be many instances where a fire damper is also required. For instance, the smoke barrier may also be used as a fire barrier. Also, IBC Section 717.5.3 would require penetration of shafts to contain both a smoke damper and a fire damper. Therefore, in some cases both a smoke damper and fire damper would be required. There are listings specific to combination smoke and fire dampers. IBC Section 717.5.3, Exception 1.3 recognizes that dampers may inhibit the performance of a smoke control system. This exception allows duct penetrations without dampers if they will inhibit the performance of the smoke control system.

More specific requirements about dampers can be found in Chapter 7 of the IBC and Chapter 6 of the IMC.

**909.6 Pressurization method.** The primary mechanical means of controlling smoke shall be by pressure differences across *smoke barriers*. Maintenance of a tenable environment is not required in the smoke-control zone of fire origin.

**C** There are several methods or strategies that may be used to control smoke movement. One of these methods is pressurization, wherein the system primarily utilizes pressure differences across smoke barriers to control the movement of smoke. Basically, if the area of fire origin maintains a negative pressure, then the smoke will be contained to that smoke zone. A typical approach used to obtain a negative pressure is to exhaust the fire floor. This is a fairly common practice in high-rise buildings. Interior exit stairways also utilize the concept of pressurization by keeping the interior exit stairways under positive pressure. The pressurization method in large open spaces, such as malls and atria, is impractical since it would take a large quantity of supply air to create the necessary pressure differences. It should be noted that pressurization is mandated as the primary method for mechanical smoke control design, but this is related to the primary methods historically used for smoke control in high-rise buildings. Currently high-rise buildings do not require smoke control. Airflow and exhaust methods are only allowed when appropriate. The exhaust method is the most commonly applied method based on the use of the atrium provisions in IBC Section 404.5.

The pressurization method does not require that tenable conditions be maintained in the smoke zone where the fire originates. Maintaining this area tenable would be impossible, because pressures from the surrounding smoke zones would be placing a negative pressure within the zone of origin to keep the smoke from migrating.

Pressurization is used often with interior exit stairways. This method provides a positive pressure within the interior exit stairways to resist the passage of smoke. Stair pressurization is one method of compliance for stairways in high-rise or underground buildings where the floor surface is located more than 75 feet (22 860 mm) above the lowest level of fire department vehicle access or more than 30 feet (9144 mm) below the floor surface of the lowest level of exit discharge. It should be noted that there are two methods found in the code that address smoke movement—smokeproof enclosures or pressurized stairs. A smokeproof enclosure requires a certain fire-resistance rating along with access through a ventilated vestibule or an exterior balcony. The vestibule can be ventilated in two ways: using natural ventilation or mechanical ventilation as outlined in Sections 909.20.3 and 909.20.5. The pressurization method requires a sprinklered building and a minimum pressure difference of 0.15 inch of water (37 Pa) and a maximum of 0.35 inch of water (87 Pa). These pressure differences are to be available with all doors closed under maximum stack pressures (see Sections 909.20 and 1023.12 for more details).

As noted, the pressurization method utilizes pressure differences across smoke barriers to achieve control of smoke. Sections 909.6.1 and 909.6.2 provide the criteria for smoke control design in terms of minimum and maximum pressure differences.

In summary, the pressurization method is used in two ways. The first is through the use of smoke zones where the zone of origin is exhausted, creating a negative pressure. The second is stair pressurization that creates a positive pressure within the stair to avoid the penetration of smoke. Note that the code allows the use of a smokeproof enclosure instead of pressurization.

**909.6.1 Minimum pressure difference.** The pressure difference across a *smoke barrier* used to separate smoke zones shall be not less than 0.05-inch water gage (0.0124 kPa) in buildings equipped throughout with *automatic sprinkler systems*.

In buildings permitted to be not equipped throughout with *automatic sprinkler systems*, the smoke control system shall be designed to achieve pressure differences not less than two times the maximum calculated pressure difference produced by the design fire.

**C** The minimum pressure difference is established as 0.05-inch water gage (12 Pa) in buildings equipped throughout with automatic sprinkler systems. This particular criterion is related to the pressures needed to overcome buoyancy and the pressures generated by the fire, which include expansion. It is based on a building equipped throughout with automatic sprinkler systems. The pressure difference would need to be higher in a building that is not sprinklered. Additionally, the pressure difference needs to be provided based on the possible stack and wind effects present.

**909.6.2 Maximum pressure difference.** The maximum air pressure difference across a *smoke barrier* shall be determined by required door-opening or closing forces. The actual force required to open exit doors when the system is in the smoke control mode shall be in accordance with Section 1010.1.3. Opening and closing forces for other doors shall be determined by standard engineering methods for the resolution of forces and reactions. The calculated force to set a side-hinged, swinging door in motion shall be determined by:

$$\text{Equation 9-1} \quad F = F_{dc} + K(WA\Delta P)/2(W - d)$$

where:

$A$  = Door area, square feet (m<sup>2</sup>).

$d$  = Distance from door handle to latch edge of door, feet (m).

$F$  = Total door opening force, pounds (N).

$F_{dc}$  = Force required to overcome closing device, pounds (N).

$K$  = Coefficient 5.2 (1.0).

$W$  = Door width, feet (m).

$\Delta P$  = Design pressure difference, inches of water (Pa).

**C** The maximum pressure difference is based primarily on the force needed to open and close doors. The code establishes maximum opening forces for doors. This maximum opening force cannot be exceeded, taking into account the pressure differences across a doorway in a pressurized environment. Essentially, based on the opening force requirements of Section 1010.1.3, the maximum pressure difference can be calculated in accordance with Equation 9-1. In accordance with Chapter 10, the maximum opening force of a door has three components, including:

Door latch release: Maximum of 15 pounds (67 N)

Set door in motion: Maximum of 30 pounds (134 N)

Swing to full-open position: Maximum of 15 pounds (67 N)

Equation 9-1 is used to calculate the total force to set the door into motion when in the smoke control mode; therefore, the limiting criteria would be 30 pounds (134 N). It should be noted that although the accessibility requirements related to door opening force are more restrictive in Section 404.2.8 of ICC A117.1, fire doors do not require compliance with these requirements.

**909.6.3 Pressurized stairways and elevator hoistways.** Where *stairways* or elevator hoistways are pressurized, such pressurization systems shall comply with Section 909 as smoke control systems, in addition to the requirements of Sections 909.20 and 909.21.

**C** The purpose of this section is to clarify that pressurized stairways and pressurized hoistways are smoke control systems and must be addressed in the same way that a pressurization system in accordance with Section 909.6 addresses such systems. This would require compliance with various sections but in particular the requirements for a rational analysis in accordance with Section 909.4.

**909.7 Airflow design method.** Where *approved* by the *fire code official*, smoke migration through openings fixed in a permanently open position, which are located between smoke control zones by the use of the airflow method, shall be permitted. The design airflow shall be in accordance with this section. Airflow shall be directed to limit smoke migration from the fire zone. The geometry of openings shall be considered to prevent flow reversal from turbulent effects. Smoke control systems using the airflow method shall be designed in accordance with NFPA 92.

**C** This method is allowed only where approved by the building official. As the title states, this method utilizes airflow to avoid the migration of smoke across smoke barriers. This has been referred to as opposed airflow. Specifically, this method is suited for the protection of smoke migration through doors and related openings fixed in a permanently open position.

This method consists of providing a particular velocity of air based on the temperature of the smoke and the height of the opening. The temperature of the smoke will depend on the design fire that is established for the particular building. The higher the temperature of the smoke and the larger the opening, the higher the velocity necessary to maintain the smoke from migrating into the smoke zone. It should be noted that the airflow method seldom works for large openings, since the velocity to oppose the smoke becomes too high. This method tends to work better for smaller openings, such as pass-through windows. Reference is made to NFPA 92 to determine the minimum velocity required to limit smoke spread.

**909.7.1 Prohibited conditions.** This method shall not be employed where either the quantity of air or the velocity of the airflow will adversely affect other portions of the smoke control system, unduly intensify the fire, disrupt plume dynamics or interfere with exiting. Airflow toward the fire shall not exceed 200 feet per minute (1.02 m/s). Where the calculated airflow exceeds this limit, the airflow method shall not be used.

**C** The airflow method has a limitation on maximum velocity. This limitation is because air may distort the flame and cause additional entrainment and turbulence; therefore, having a high velocity of air entering the zone of fire origin has the potential of increasing the amount of smoke produced. The velocity may also interact with other portions of the smoke control design. For instance, the pressure differences in other areas of the building may be altered, which may exceed the limitations of Sections 909.6.1 and 909.6.2. This section requires that when a velocity of over 200 feet per minute (1.02 m/s) is calculated at the design fire, the airflow method is not allowed. The solution may result in requiring a barrier such as a wall or door.

If the airflow design method is chosen to protect areas communicating with an atrium, the air added to the smoke layer needs to be accounted for in the exhaust rate.

**909.8 Exhaust method.** Where *approved by the fire code official*, mechanical smoke control for large enclosed volumes, such as in atriums or malls, shall be permitted to utilize the exhaust method. Smoke control systems using the exhaust method shall be designed in accordance with NFPA 92.

**C** This method is allowed only where approved by the fire code official. The primary application of the exhaust method is in large volume spaces, such as atriums, arenas, convention centers, airport terminals and malls, and is the most widely used method in the IBC. The strategy of this method is to maintain tenable conditions in a large-volume space. This is primarily accomplished through exhausting smoke. The amount of exhaust depends on the design fire [see Commentary Figure 909.8(1)] and the geometry of the space under consideration. Essentially, fires produce different amounts and properties of smoke based on the material being burned, and size and placement of the fire; therefore, NFPA 92 is referenced for the design of such systems. NFPA 92 presents several ways to address the control of smoke, which includes the use of the following tools:

- Scale Modeling (Small-scale testing)—Utilizes the concept of scaling to allow small-scale tests to be conducted to understand smoke movement within a space.
  - o Benefits—More realistic understanding of smoke movement in spaces with unusual configurations or projections than algebraic calculations.
  - o Disadvantages—Expensive and the application of results is limited to the uniqueness of the space being analyzed.
- Algebraic Calculations—Empirically derived (based on testing) modeling in its simplest form.
  - o Benefits—Simple, cost-effective analysis.
  - o Disadvantages—Limited applicability because of the range of values they were derived from. Only appropriate with certain types of design fires, typically over conservative outputs that increase equipment needs and equipment costs and can impact aesthetics and architectural design.
- Computer Modeling [Computational Fluid Dynamics (CFD) or zone models]—Combination of theory and empirical values to determine the smoke movement and fire-induced conditions within a space and effectiveness of the smoke control system.
  - o Benefits—More realistic understanding of smoke movement in spaces with unusual configurations or projections and less expensive than scale modeling. Helps significantly in designing smoke control systems tailored to spaces and achieving cost-effective designs and can help limit the impact to architectural design.
  - o Disadvantages—Computing time and cost can be greater than algebraic calculations but benefits typically outweigh this disadvantage. Early planning is important and can limit these adverse impacts.

In terms of computer models, as noted, there are essentially two methods that include zone models and CFD models. Generally, zone models are based on the unifying assumption that in any compartment where the effects of the fire are present, there are distinct control volumes (hot upper layer, cool lower layer). In real life, such distinct layers do not exist. An example of a zone model used in such applications include the Consolidated Model of Fire Growth and Smoke Transport (CFAST). See Chapter 31 of the *SFPE Handbook of Fire Protection Engineering, 5th edition* for further information on zone models. CFD models take this much further and actually divide the space into thousands or millions of interconnected

“control volumes” (“three-dimensional cells” or “fields”). The model then yields time-varying predictions of temperature, gas velocity, gas species concentrations, etc. in each individual cell and how it interacts with those adjacent to it. The use of such models becomes more accurate with more numerous and smaller cells, but the computing power and expertise required is much higher than for zone models. See Chapter 32 of the *SFPE Handbook of Fire Protection Engineering, 5th edition* for further information of CFD models. As noted, the use of either type of model can be advantageous, but such use must be undertaken by someone qualified. Proper review and verification of the input and output is critical. As stated in the SFPE Engineering Guide, *Guidelines for Substantiating a Fire Model for a Given Application*, determining the suitability of a fire model is part of a five-step process that includes:

1. Defining the problem of interest.
2. Selecting a candidate model.
3. Verification and validation.
4. User effect.
5. Documentation.

A commonly used CFD model is the Fire Dynamics Simulator (FDS) developed by NIST. Other models such as ANSYS Fluent are sometimes used (Fluent Inc.).

Depending on the space being evaluated, some design strategies may provide a better approach than others. In past editions of the code, smoke control provisions for the exhaust method mandated the use of the algebraic method with a steady fire. This also mandated that a mechanical system be used, whereas NFPA 92 allows an overall review of smoke layer movement and whether the design goals, which in this case are mandated by the code, can be met. Therefore, if it can be shown that the smoke layer interface can be held at 6 feet (1829 mm) above the highest level of egress as mandated in Section 909.8.1 for the design operation time required by Section 909.4.6 without mechanical ventilation, then the space would comply with Section 909. NFPA 92 presents several design approaches. This allows more flexibility in design than that found in previous editions of the code.

NFPA 92 as a standard does not set the minimum smoke layer interface height or duration for system operation. Such criteria are found within Sections 909.8.1 and 909.4.6, respectively. See the commentary to those sections.

If the algebraic approach is used, consideration of three types of fire plumes may be required to determine which one is the most demanding in terms of smoke removal needs based on the space being assessed. They include:

Axisymmetric plumes—Smoke rises unimpeded by walls, balconies or similar projections [see Commentary Figure 909.8(2)].

Balcony spill plumes—Smoke flows under and around edges of a horizontal projection [see Commentary Figure 909.8(3)].

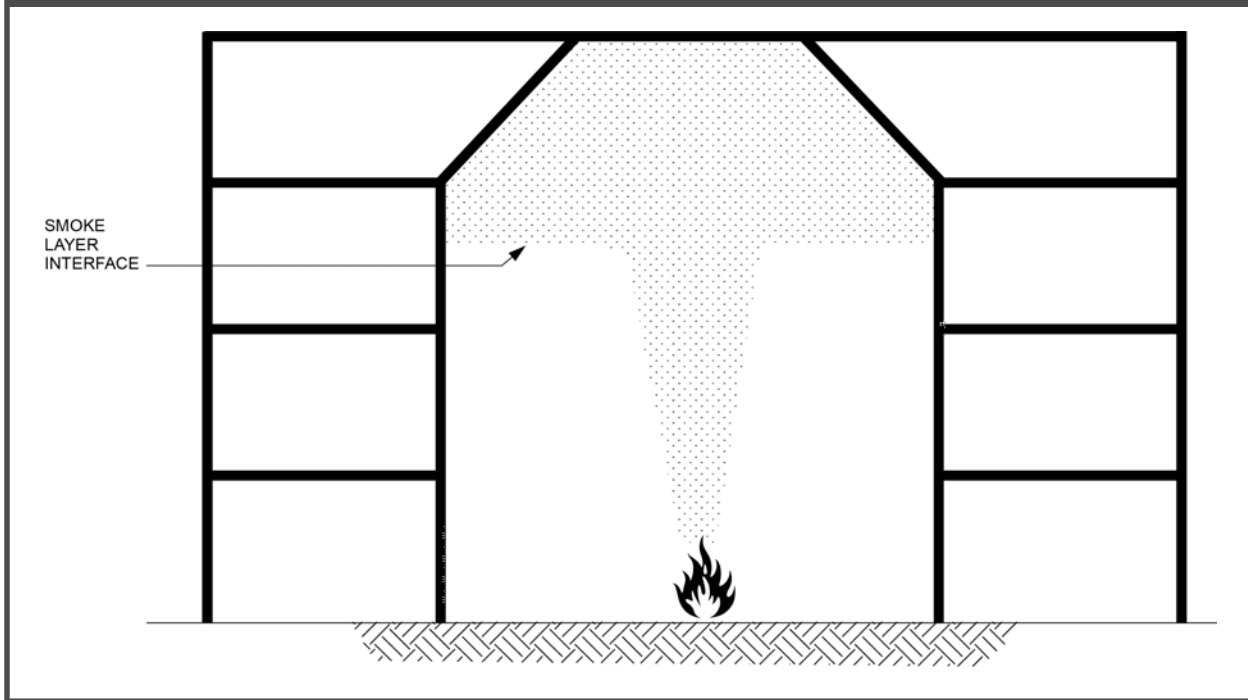
Window plumes—Smoke flows through an opening into a large-volume space [see Commentary Figure 909.8(4)].

It should be noted that prior to the reference to NFPA 92 in the code, the balcony spill and window plume calculations had been eliminated from the smoke control requirements of the code because of concerns with the applicability of those calculations. The major difference is that NFPA 92 does not mandate the use of such equations as did previous editions of the IBC. The use of such equations will depend on the design fires agreed upon for the particular design and whether an algebraic approach is chosen. These equations are used to determine a mass flow rate of smoke to ultimately determine the required exhaust volume for that space. If the potential for balcony or window spill plumes is known to exist within the space, then appropriate measures need to be taken to address these, as they typically result in more onerous exhaust and supply requirements. Part of the reason for the initial deletion of these equations was the fact that such scenarios are not as likely or their impact is significantly reduced in sprinklered buildings. There is also some concern with the applicability of the balcony spill plume equation in a variety of applications. These potential fire scenarios and resulting plumes may further the need to undertake a CFD analysis to address such hazards more appropriately and effectively.

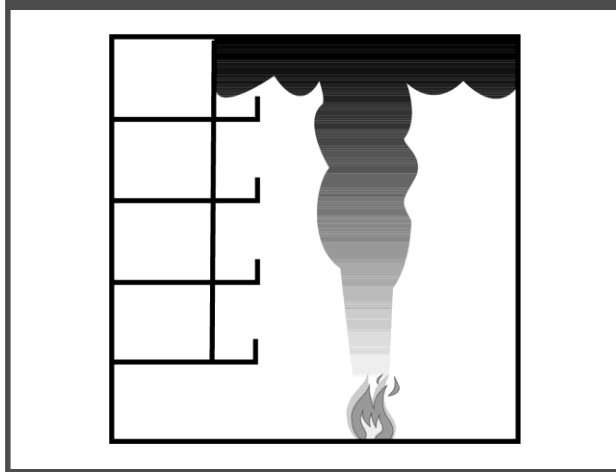
Another key aspect that NFPA 92 included within the algebraic method is equations to determine that a minimum number of exhaust inlets are available to prevent plugholing. Plugholing occurs when the exhaust capacity at a single point is sufficiently large enough to draw air from the lower layer in addition to smoke. As such, less smoke is removed by the exhaust fans and a deeper layer results. If plugholing occurs, some of the fan capacity is used to exhaust air rather than smoke and thus can affect the ability to maintain the smoke layer at or above the design height. Scale modeling and computer fire modeling would demonstrate these potential problems during the testing and analysis, respectively [see Commentary Figure 909.8(5)].

It should be noted that this section specifically references NFPA 92 for the design of smoke control using the exhaust method. Therefore, the requirements in NFPA 92 related to testing, documentation and maintenance would not be applicable though they may be a good resource. Equipment and controls would be part of the design; therefore, related provisions of NFPA 92 would apply. Generally, the code addresses equipment and controls in a similar fashion.

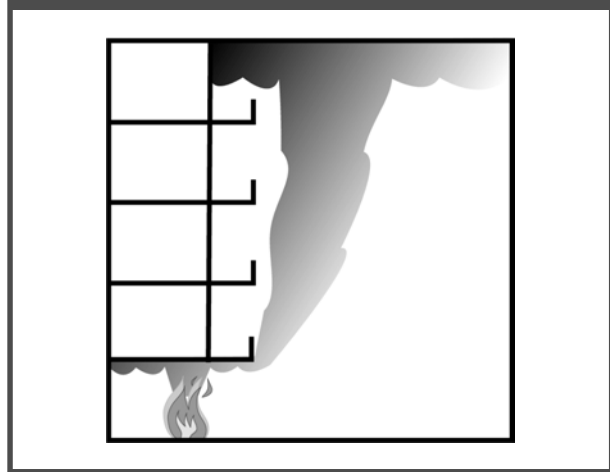
Commentary Figure 909.8(1)—Smoke Layer Interface in an Atrium



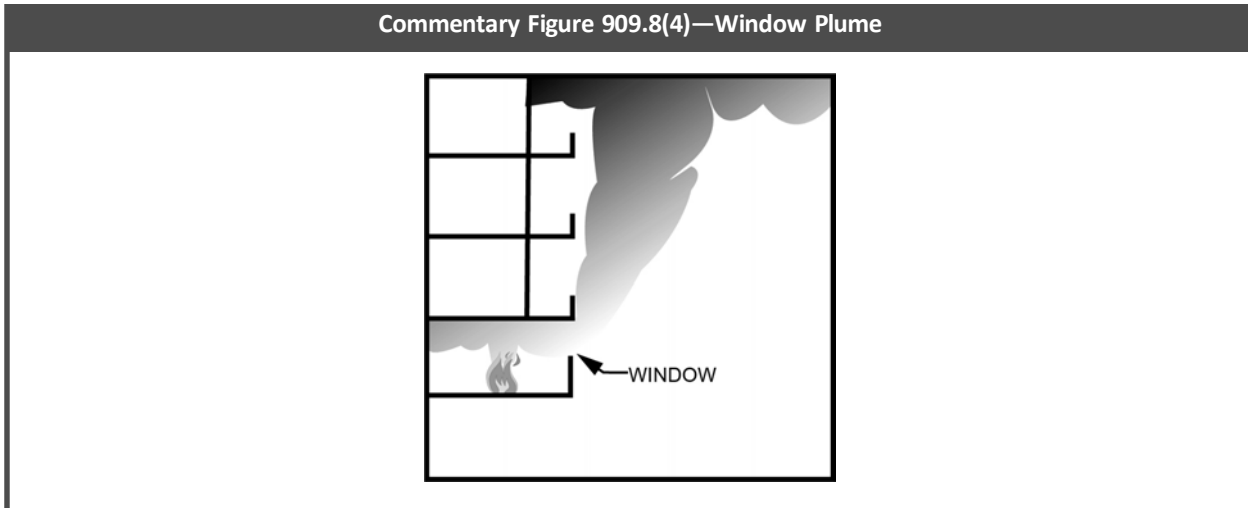
Commentary Figure 909.8(2)—Axisymmetric Plume

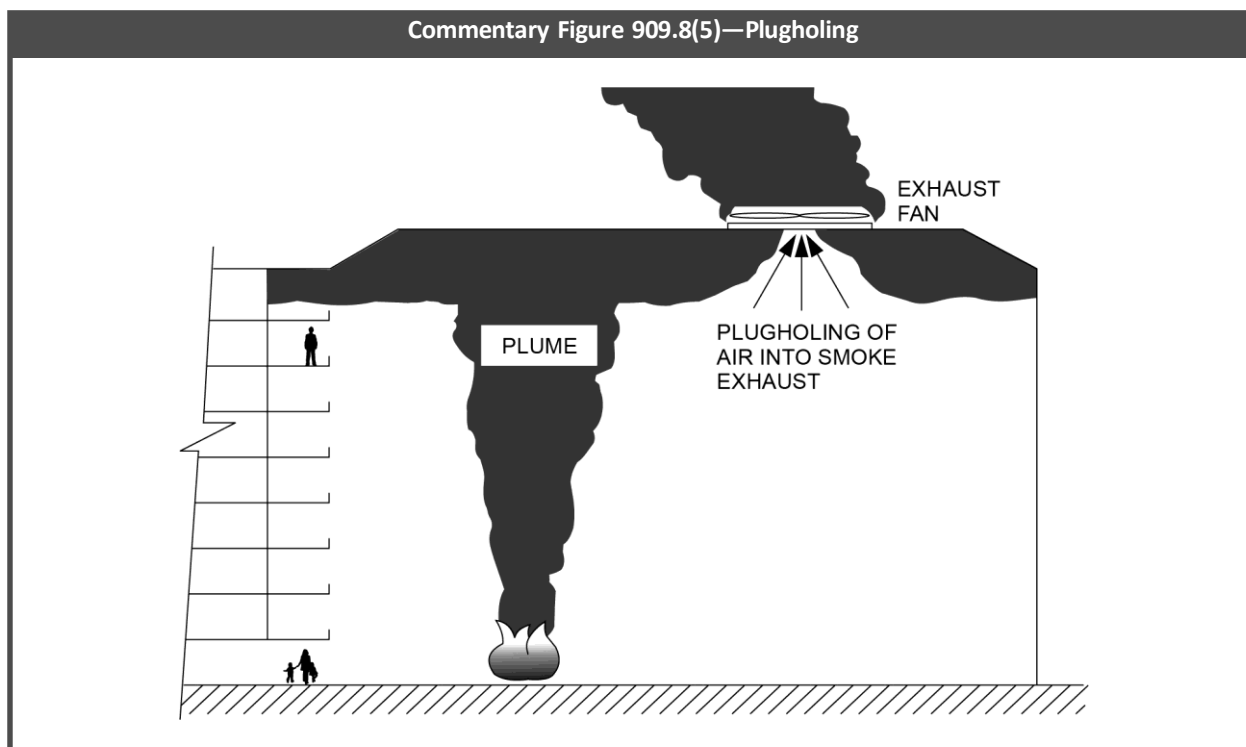


Commentary Figure 909.8(3)—Balcony Spill Plume



Commentary Figure 909.8(4)—Window Plume





**909.8.1 Smoke layer.** The height of the lowest horizontal surface of the smoke layer interface shall be maintained not less than 6 feet (1829 mm) above a walking surface that forms a portion of a required egress system within the smoke zone.

**C** The design condition to be used when applying NFPA 92 is to maintain the smoke layer interface at least 6 feet (1829 mm) above any walking surface that is considered part of the required egress within the particular smoke zone, such as an atrium, for 20 minutes or 1.5 times the calculated egress time (see Section 909.4.6). Chapter 10 considers the majority of occupiable space as part of the means of egress system. Also keep in mind that the criteria of 6 feet (1829 mm) does not apply just to the main floor surface of the mall or atrium but to any level where occupants may be exposed (for example, balconies) [see Commentary Figure 909.8.1(1)].

The code uses the terminology “lowest horizontal surface of the smoke layer interface.” NFPA 92 has several definitions related to smoke layer, which include the following:

**Smoke layer.** The accumulated thickness of smoke below a physical barrier.

**Smoke layer interface.** The theoretical boundary between a smoke layer and the smoke-free air. (Note: This boundary is at the beginning of the transition zone.)

**First indication of smoke.** The boundary between the transition zone and the smoke-free air.

**Transition zone.** The layer between the smoke layer interface and the first indication of smoke in which the smoke layer temperature decreases to ambient. The transition zone may be several feet thick (large open space) or may barely exist (small area with intense fire).

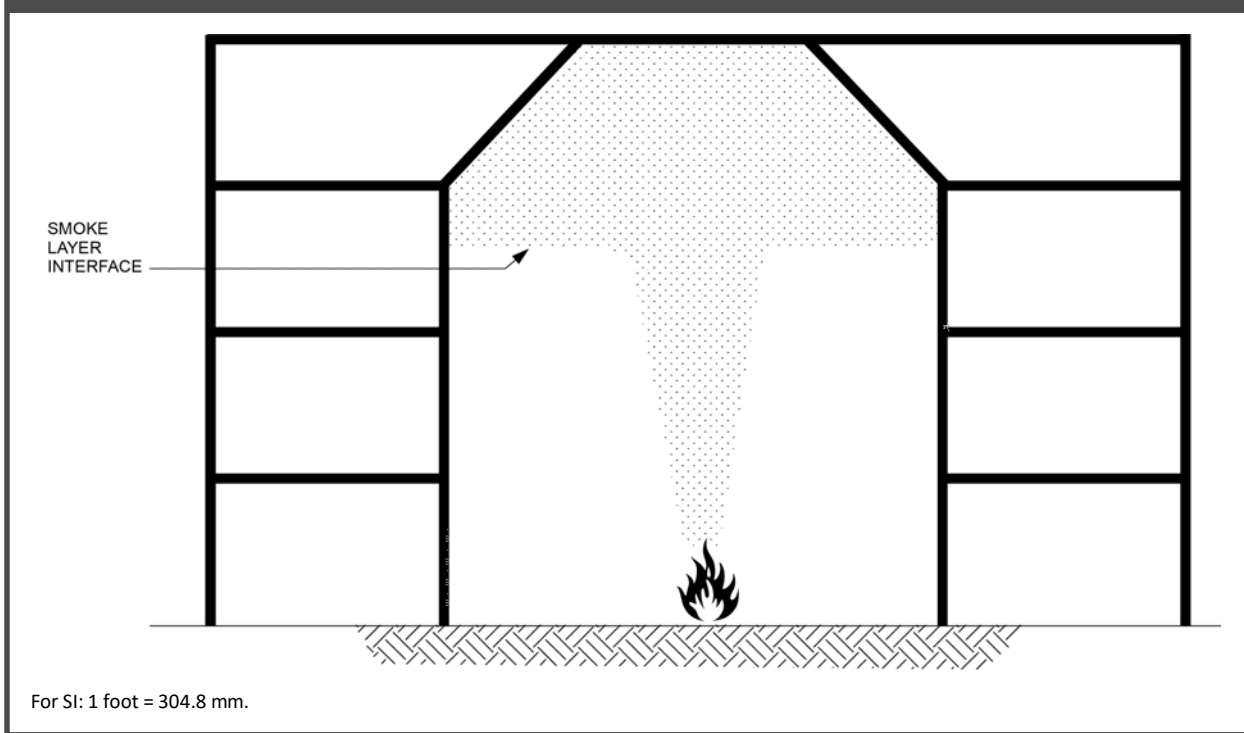
See also Commentary Figure 909.8.1(2).

NFPA 92 provides algebraic equations to determine the first indication of smoke but is limited to very specific conditions such as a uniform cross section, specific aspect ratios, steady or unsteady fires and no smoke exhaust operating. When using algebraic equations for smoke layer interface by looking at different types of plumes, the smoke layer interface terminology is used and the user enters the desired smoke layer interface height. Zone models use simplifying assumptions so the layers are distinct from one another. In contrast, when CFD or scale modeling is used, the data must be analyzed to verify that the smoke layer interface is located at or above 6 feet (1829 mm) during the event. This is not a simple analysis, as CFD and scale modeling provide more detail on actual smoke behavior; therefore, the location of the smoke layer interface may not be initially clear without some level of analysis. Again, it depends on the depth of the transition layer. This may require reviewing tenability within the transition zone. Tenability limits need to be agreed upon by the stakeholders involved. The SFPE Handbook of Fire Protection Engineering, 5th edition, Chapters 61 and 62 provide guidance on how to define tenability limits. Using CFD or scale modeling would likely need to occur through the alternative methods and materials section (see Section 104.2.3) because of the need to review tenability limits. It should be noted that NFPA 92 Annex A suggests that there are methods to determine where the smoke layer interface and first indication of smoke are located when undertaking CFD and scale modeling using a limited number of point measurements.

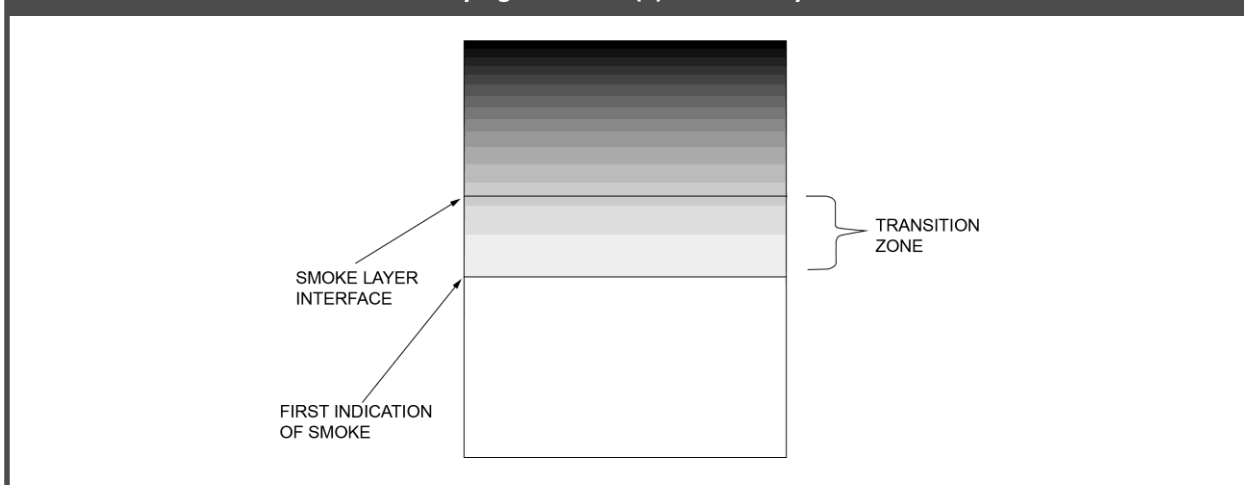
Also, Section 909.8.1 specifies a minimum distance for the smoke layer interface from any walking surface whereas Section 4.5.4.2 of NFPA 92 has provisions that simply allow the analysis to demonstrate tenability regardless of where the layer height is located above the floor. Defining tenability can be more difficult as there is not a standard definition. Any design using that approach would need to be addressed through Section 104.2.3.

Note that the response time of the system components (detection, activation, ramp-up time, shutting down HVAC, opening/closing doors and dampers, etc.) needs to be accounted for when analyzing the location of the smoke layer interface in relation to the duration of operations stated in Section 909.4.6 (see commentary, Section 909.17).

Commentary Figure 909.8.1(1)—Example of Critical Level of Smoke



Commentary Figure 909.8.1(2)—Smoke Layer Interface



**909.9 Design fire.** The design fire shall be based on a rational analysis performed by the *registered design professional* and *approved* by the *fire code official*. The design fire shall be based on the analysis in accordance with Section 909.4 and this section.

**C** Estimating the design fire is the most critical element in the smoke control system design; several design fire types and/or locations should be considered in the design of most systems. The fire is what produces the smoke to be controlled by the system; thus, the size of the fire directly impacts the quantity of smoke being produced. This section ensures that the design fire be determined through a rational analysis by a registered design professional with knowledge in this area. Such professionals should have experience in the area of fire dynamics, fire engineering and general building design, including mechan-

ical systems. When determining the design fire, the designer should work with various stakeholders to determine the types of hazards and combustible materials (fire scenarios) on a permanent as well as temporary basis (i.e., Christmas/holiday decorative materials or scenery, temporary art exhibits) that may be present throughout the use of the building once occupied. Fire scenarios describe a sequence of possible events and set of conditions that describe the development of fire and the spread of combustion products throughout a building or part of a building. Those hazards then need to be translated to potential design fires to be used when determining the smoke layer interface height for the duration as determined by Section 909.4.6. See the commentary to Section 909.9.3 for potential sources when determining design fires.

This section also does not mandate the type of fire (i.e., steady versus unsteady). A steady fire assumes a constant heat-release rate over a period of time, where unsteady fires do not. An unsteady fire includes the growth and decay phases of the fire, as well as the peak heat-release rate. An unsteady fire will hit a peak heat-release rate when burning in the open, like an axisymmetric fire. An unsteady fire is a more realistic view of how fires actually burn. It should be noted that fires can be a combination of unsteady and steady fires when sufficient fuel is available. In other words, the fire initially grows (unsteady) then reaches a steady state and burns for some time at a particular heat-release rate before decay occurs.

Design fire information should therefore typically include growth rate, peak heat-release rate, duration and decay as well as information related to fire locations and products of combustion yield (CO, smoke, etc.) that are produced by the various design fires that are deemed credible for the space.

To provide an order of magnitude of fire sizes obtained from various combustibles, the following data from fire tests are provided. The following heat-release rates, found in Section 3, Chapter 3-1 of the 5th edition of the *SFPE Handbook of Fire Protection Engineering*, are peak heat-release rates:

- Plastic trash bags/paper trash: Approximately 120–350 kW
- Latex foam pillow: Approximately 120 kW
- Dry Douglas fir Christmas tree: Approximately 3000 kW at 20% moisture
- Polyurethane foam (PVC Ticking material) mattress: Approximately 2630 kW
- Plywood wardrobe: Approximately 2900–6400 kW

**909.9.1 Factors considered.** The engineering analysis shall include the characteristics of the fuel, fuel load, effects included by the fire and whether the fire is likely to be steady or unsteady.

**C** This section simply provides more detail on the factors that should be taken under consideration when determining the design fire size. To determine the appropriate fire size, an engineering analysis is necessary that takes into account the following elements: fuel (potential burning rates), fuel load (how much), effects included by the fire (smoke particulate size and density), steady or unsteady (burn steadily or simply peak and dissipate), and likelihood of sprinkler activation (based on height and distance from the fire).

**909.9.2 Design fire fuel.** Determination of the design fire shall include consideration of the type of fuel, fuel spacing and configuration.

**C** The design fire size may also be affected by surrounding combustibles, which may have the effect of increasing the fire size. More specifically, there is concern that if sufficient separation is not maintained between combustibles, then a larger design fire is likely. The code does not provide extensive detail on this as such determination is left to the rational analysis undertaken by the design professional. NFPA 92 provides one method in which to determine the critical separation distance, *R*. This is based on fire size and the critical radiant heat flux for nonpiloted ignition. Nonpiloted ignition means the radiated heat from the fire without direct flame contact will ignite adjacent combustibles.

**909.9.3 Heat-release assumptions.** The analysis shall make use of best available data from *approved* sources and shall not be based on excessively stringent limitations of combustible material.

**C** This section stresses that data obtained for use in a rational analysis need to come from relevant and appropriate sources. Data can be obtained from groups such as NIST or from Annex B of NFPA 92. Data from fire tests are available and are a good resource for such analysis. As noted earlier, such data are not prevalent (see also Chapter 8, Analysis of Design Fires, from *A Guide to Smoke Control in the 2006 IBC®* and Chapter 26 of the *SFPE Handbook of Fire Protection Engineering, 5th edition*).

**909.9.4 Sprinkler effectiveness assumptions.** A documented engineering analysis shall be provided for conditions that assume fire growth is halted at the time of sprinkler activation.

**C** This section raises a few questions regarding activation of sprinklers and their impact on the fire both in terms of their ability to “control” as well as “extinguish” a fire. The first is concerning an assumption that sprinklers will immediately control the fire as soon as they are activated (i.e., control results in limiting further growth and maintaining the heat-release rate at approximately the same fire size as when the sprinklers activated). This assumption may be true in some cases, but for high ceilings, the sprinkler may not activate or may be ineffective. Sprinklers may be ineffective in high spaces, since by the time they are activated, the fire is too large to control. Essentially, the fire plume may push away and evaporate the water before it actually reaches the seat of the fire. In addition, the fire may be shielded from sprinkler spray so that insufficient quanti-

ties of water reach the fuel. These are common problems with high-piled storage as well as other fires, including retail, and has been shown in actual tests. Also, if the fire becomes too large before the sprinklers are activated, the available water supply and pressure for the system may be compromised. Additionally, based on the layout of the room and the movement of the fire effluents, the wrong sprinklers could be activated, which leads to a larger fire size and depletion of the available water supply and pressure.

Another issue is whether the sprinklers “control” or “extinguish” the fire. Typical sprinklers are assumed only to control fires as opposed to extinguishing them. Sprinklers may be able to extinguish the fire, but it should not automatically be assumed. A fire that is controlled will achieve steady state and maintain a certain fire size, which is very different from a fire that is actually extinguished.

Based on these concerns, each scenario needs to be looked at individually to determine whether sprinklers would be effective in halting the growth or extinguishing the fire. More specifically, the evaluation should include droplet size, density and area of coverage and should also be based on actual test results.

**909.10 Equipment.** Equipment including, but not limited to, fans, ducts, automatic dampers and balance dampers shall be suitable for their intended use, suitable for the probable exposure temperatures that the rational analysis indicates, and as *approved* by the *fire code official*.

**C** Section 909.10 and subsequent sections are primarily related to the reliability of the system components to provide a smoke control system that works according to the design. One of the largest concerns when using smoke control provisions is the overall reliability of the system. Such systems have many different components, such as smoke and fire dampers, fans, ducts and controls associated with such components. The more components a system has, the less reliable it becomes. In fact, one approach in providing a higher level of reliability is utilizing the normal building systems such as the HVAC to provide the smoke control system. Basically, systems used every day are more likely to be working appropriately, since they are essentially being tested daily; however, there are many components that are specific to the smoke control system, such as exhaust fans in an atrium or the smoke control panel.

Also, there is not a generic prescriptive set of requirements as to how all smoke control system elements should operate, since each design may be fairly unique. The specifics on operation of such a system need to be included within the design and construction documents. Most components used in smoke control systems are elements used in many other applications such as HVAC systems; therefore, the basic mechanisms of a fan used in a smoke control system may not be different, although they may be applied differently.

**909.10.1 Exhaust fans.** Components of exhaust fans shall be rated and certified by the manufacturer for the probable temperature rise to which the components will be exposed. This temperature rise shall be computed by:

**Equation 9-2**  $T_s = (Q_c/mc) + (T_a)$

where:

$c$  = Specific heat of smoke at smoke layer temperature, Btu/lb°F (kJ/kg × K).

$m$  = Exhaust rate, pounds per second (kg/s).

$Q_c$  = Convective heat output of fire, Btu/s (kW).

$T_a$  = Ambient temperature, °F (K).

$T_s$  = Smoke temperature, °F (K).

**Exception:** Reduced  $T_s$  as calculated based on the assurance of adequate dilution air.

**C** Fans used for smoke control systems must be able to tolerate the possible elevated temperatures to which they will be exposed. Again, like many other factors, this depends on the specifics of the design fire. Essentially, Equation 9-2 requires the calculation of the potential temperature rise. The exhaust fans must be specifically rated and certified by the manufacturer to be able to handle these rises in temperature. There is an exception that allows reduction of the temperature if it can be shown that adequate temperature reduction will occur. In many cases, if the exhaust fans are near the ceiling, the smoke will be much cooler than the value resulting from Equation 9-2 since the smoke may cool considerably by the time it reaches the ceiling. Also, sprinkler activation will assist in cooling the smoke further.

**909.10.2 Ducts.** Duct materials and joints shall be capable of withstanding the probable temperatures and pressures to which they are exposed as determined in accordance with Section 909.10.1. Ducts shall be constructed and supported in accordance with the *International Mechanical Code*. Ducts shall be leak tested to 1.5 times the maximum design pressure in accordance with nationally accepted practices. Measured leakage shall not exceed 5 percent of design flow. Results of such testing shall be a part of the documentation procedure. Ducts shall be supported directly from *fire-resistance-rated* structural elements of the building by substantial, noncombustible supports.

**Exception:** Flexible connections, for the purpose of vibration isolation, complying with the *International Mechanical Code* and that are constructed of *approved fire-resistance-rated* materials.

**C** The next essential component of a smoke control system is the integrity of the ducts to transport supply and exhaust air. The integrity of ducts is also important for an HVAC system but is more critical in this case since it is not simply a comfort

issue but one of life safety. The key concern with ducts in smoke control systems is that they can withstand elevated temperatures and that there will be minimal leakage. The concern with leakage is the potential of leaking smoke into another smoke zone or not providing the proper amount of supply air to support the system.

More specifically, all ducts need to be leak tested to 1.5 times the maximum static design pressure. The leakage resulting should be no more than 5 percent of the design flow. For example, a duct that has a design flow of 300 cubic feet per minute (cfm) (0.141 m<sup>3</sup>/s) would be allowed 15 cfm (0.007 m<sup>3</sup>/s) of leakage when exposed to a pressure equal to 1.5 times the design pressure for that duct. The tests should be in accordance with nationally accepted practices. This criterion will often limit ductwork for smoke control systems to lined systems, since the amount of leakage in such systems is much less.

As part of the concern for possible exposure to fire and fire products, the ducts are required to be supported by way of substantial noncombustible supports connected to the fire-resistance-rated structural elements of the building. As noted, the system needs to be able to run for 20 minutes starting from the detection of the fire. The exception to this section is really more of an acknowledgment that flexible connections for vibration isolation are acceptable when constructed of approved fire-resistance-rated materials. More specifically, it is often necessary to use such connections for connecting the duct to the fan. These connections cannot necessarily meet the requirements of the main section but are a minimal part of the ductwork and as long as they perform adequately with regard to fire resistance, they are permitted. Note that the term “approved” is used to determine the required fire resistance; therefore, flexibility is provided. The code does not specifically address this determination but perhaps a relationship to the duration or operation and these flexible connections could be made to determine the necessary performance.

**909.10.3 Equipment, inlets and outlets.** Equipment shall be located so as to not expose uninvolved portions of the building to an additional fire hazard. Outside air inlets shall be located so as to minimize the potential for introducing smoke or flame into the building. Exhaust outlets shall be so located as to minimize reintroduction of smoke into the building and to limit exposure of the building or adjacent buildings to an additional fire hazard.

**C** The intent of this section is to minimize the likelihood of smoke being reintroduced into the building due to poorly placed outdoor air inlets and exhaust air outlets; therefore, placing one right next to the other on the exterior of the building would be inappropriate. In addition, wind and other adverse conditions should be considered when choosing locations for these inlets and outlets. Particular attention should be paid to introducing exhausted smoke into another smoke zone. Also, smoke should be exhausted in a direction that will not introduce it into surrounding buildings or facilities. Within the building itself, the supply air and exhaust outlets should also be strategically located. The exhaust inlets and supply air should be evenly distributed to reduce the likelihood of a high velocity of air that may disrupt the fire plume and also push smoke back into occupied areas. See the commentary to Section 909.8 for discussion on avoiding plugholing.

**909.10.4 Automatic dampers.** Automatic dampers, regardless of the purpose for which they are installed within the smoke control system, shall be *listed* and conform to the requirements of *approved* recognized standards.

**C** This section addresses the reliability of any dampers used within a smoke control system. This particular provision requires that the dampers be listed and conform to the appropriate recognized standards. More specifically, IBC Section 717 contains more detailed information on the specific requirements for smoke and fire dampers. Smoke and fire dampers should be listed in accordance with UL 555S and 555, respectively. Also, remember that each smoke control design is unique, and the sequence and methods used to activate the dampers may vary from design to design. This information needs to be addressed in the construction documents.

Another factor to take into account, with regard to timing of the system, is the fact that some dampers react more quickly than others, simply due to the particular smoke damper characteristics. Additionally, during the commissioning of the system, the damper is going to be exposed to many repetitions. These repetitions need to be accounted for in the overall reliability of the system.

**909.10.5 Fans.** In addition to other requirements, belt-driven fans shall have 1.5 times the number of belts required for the design duty with the minimum number of belts being two. Fans shall be selected for stable performance based on normal temperature and, where applicable, elevated temperature. Calculations and manufacturer’s fan curves shall be part of the documentation procedures. Fans shall be supported and restrained by noncombustible devices in accordance with the structural design requirements of Chapter 16 of the *International Building Code*.

Motors driving fans shall not be operated beyond their nameplate horsepower (kilowatts) as determined from measurement of actual current draw and shall have a minimum service factor of 1.15.

**C** Part of the overall reliability requires that fans used to provide supply air and exhaust capacity will be functioning when necessary; therefore, a safety factor of 1.5 is placed on the required belts for fans. All fans used as part of a smoke control system must provide 1.5 times the number of required belts with a minimum of two belts for all fans.

This section also points out that the fan chosen should fit the specific application. It should be able to withstand the temperature rise as calculated in Section 909.10.1 and generally be able to handle typical exposure conditions, such as location and wind. For instance, propeller fans are highly sensitive to the effects of wind. When located on the windward side of a building, wall-mounted, nonhooded propeller fans are not able to compensate for wind effects. Additionally, even

hooded propeller fans located on the leeward side of the building may not adequately compensate for the decrease in pressure caused by wind effects. In general, when designing a system, it should be remembered that field conditions might vary from the calculations; therefore, flexibility should be built into the design that would account for things such as variations in wind conditions.

Finally, this section stresses that fan motors not be operated beyond their rated horsepower.

**909.11 Standby power.** Smoke control systems shall be provided with standby power in accordance with Section 1203.

**C** This section references Chapter 12 for the specifics as to what is required for standby power. As with any life safety system, a level of redundancy with regard to power supply is required to enable the functioning of the system during a fire. The primary source is the building's normal power system. The secondary power system is by means of standby power. One of the key elements is that standby power systems are intended to operate within 60 seconds of loss of primary power. It should be noted that the primary difference between standby power and emergency power is that emergency power must operate within 10 seconds of loss of primary power versus 60 seconds.

**909.11.1 Equipment room.** The standby power source and its transfer switches shall be in a room separate from the normal power transformers and switch gears and ventilated directly to and from the exterior. The room shall be enclosed with not less than 1-hour fire barriers constructed in accordance with Section 707 of the *International Building Code* or horizontal assemblies constructed in accordance with Section 711 of the *International Building Code*, or both.

**C** This section requires isolation from normal building power systems via a 1-hour fire barrier, 1-hour horizontal assembly, or both, depending on the location within the building. This increases the reliability and reduces the likelihood that a single event could remove both power supplies. The intent of the ventilation is focused on the proper function of the standby power source in terms of engine-driven generators having appropriate cooling air and combustion air. The requirement that it be from the outside is related to the protection of such ventilation from the effects of fire.

**909.11.2 Power sources and power surges.** Elements of the smoke control system relying on volatile memories or the like shall be supplied with uninterruptible power sources of sufficient duration to span 15-minute primary power interruption. Elements of the smoke control system susceptible to power surges shall be suitably protected by conditioners, suppressors or other *approved* means.

**C** Smoke control systems have many components, sometimes highly sensitive electronics, that are adversely affected by any interruption in or sudden surges of power. Therefore, Section 909.11.2 requires that any components of a smoke control system, such as volatile memories, be supplied with an uninterruptible power system for the first 15 minutes of loss of primary power. Volatile memory components will lose memory upon any loss of power, no matter how short the time period. Once the 15 minutes elapse, these elements can be transitioned to the already operating standby power supply.

With regard to components sensitive to power surges, they need to be provided with surge protection in the form of conditioners, suppressors or other approved means.

**909.12 Detection and control systems.** Fire detection systems providing control input or output signals to mechanical smoke control systems or elements thereof shall comply with the requirements of Section 907. Such systems shall be equipped with a control unit complying with UL 864 and *listed* as smoke control equipment.

**C** This section is focused on proper monitoring of the fire detection systems that activate the smoke control system through compliance with Section 907 and UL 864. This requires a specific listing of the fire alarm control unit as smoke control equipment. UL 864 has a subcategory (UUKL) specific to fire alarm control panels for smoke control system applications.

**909.12.1 Verification.** Control systems for mechanical smoke control systems shall include provisions for verification. Verification shall include positive confirmation of actuation, testing, manual override and the presence of power downstream of all disconnects. A preprogrammed weekly test sequence shall report abnormal conditions audibly, visually and by printed report. The preprogrammed weekly test shall operate all devices, equipment and components used for smoke control.

**Exception:** Where verification of individual components tested through the preprogrammed weekly testing sequence will interfere with, and produce unwanted effects to, normal building operation, such individual components are permitted to be bypassed from the preprogrammed weekly testing, where *approved* by the *fire code official* and in accordance with both of the following:

1. Where the operation of components is bypassed from the preprogrammed weekly test, presence of power downstream of all disconnects shall be verified weekly by a *listed* control unit.
2. Testing of all components bypassed from the preprogrammed weekly test shall be in accordance with Section 909.22.6.

**C** This section addresses the function of the mechanical elements of the smoke control system once the system is activated. In particular, there is a focus on verification of activities. Verification would include the following two aspects:

1. The system is able to verify actuations, testing, manual overrides and the presence of power downstream. This would require information reported back to the smoke control panel, which can be accomplished via the weekly test sequence or through full electronic monitoring of the system.

2. Conduct a preprogrammed weekly test that simulates an actual (smoke) event to test the components of the system. These components would include elements such as smoke dampers, fans and doors. Abnormal conditions need to be reported in three ways:
  - a. Audibly.
  - b. Visually.
  - c. Printed report.

It should be noted that electrical monitoring of the control components is not required (supervision). Such supervision verifies integrity of the conductors from a fire alarm control unit to the control system input. The weekly test is considered sufficient verification of system performance and is often termed “end-to-end verification.” In other words, the control system input provides the expected results. Verification can be accomplished through any sensor that is calibrated to distinguish the difference between proper operation and a fault condition. For fans, proper operation means that the fan is moving air within the intent of its design. Fault conditions include power failure, broken fan belts, adverse wind effects, a locked rotor condition and/or filters or large ducts that are blocked, causing significantly reduced airflow. In addition to differential pressure transmitters and sail switches, this can be accomplished by state-of-the-art current sensors. There is more discussion on verification for elements such as ducts and fire doors in Chapter 9 of *A Guide to Smoke Control in the 2006 IBC®*.

Also, the fact that a smoke control system is nondedicated (integrated with an HVAC system) does not mean that it is automatically being tested on a daily basis. It is cautioned that simply depending on occupant discomfort, for example, is sometimes an insufficient indicator of a fully functioning smoke control system. There may be various modes in which the HVAC system could operate that may not exercise the smoke control features and the sequence in which the system should operate. An example is an air-conditioning system operating only in full recirculating mode versus exhaust mode. This failure will likely not affect occupants and will not exercise the exhaust function. Plus, doors, which may be part of the smoke barrier, may not need to be closed in normal building operations but would need to be closed during smoke control system operation. This is why this section does not necessarily differentiate between dedicated and nondedicated smoke control systems and requires the system components to be tested.

It is important to note that this weekly test sequence is not an actual smoke event and is only intended to activate the system to ensure that the components are working correctly.

The exception addresses the impracticality of requiring a weekly test for many buildings. For many systems, the weekly test requires the introduction of untreated air into the smoke zone. This can be impractical in areas with cold or hot climates and for buildings that require close control of temperature and humidity, such as art museums and similar facilities. The introduction of the untreated air can also result in wasting energy to reheat, recool, humidify, or dehumidify the smoke control zone.

The intent of the exception is to provide means to verify that the required systems will be available when needed. The code requires control units to comply with UL 864; thus, all components of the control system will be supervised. The exception includes requirements for verification of the power downstream of all disconnects, such as power breakers, power disconnects, automatic transfer switches, motor starters, and motor controls. This will provide reasonable assurance that power will be available for all smoke control components, such as fans, dampers, doors, and windows. The exception also adds the semiannual requirement for a complete system test by reference to Section 909.22.6. This allows the building owner to schedule complete system testing on days that will reduce the impact to the building and energy needs. The combination of additional supervision and additional testing provides a reasonable alternative to weekly testing.

**909.12.2 Wiring.** In addition to meeting requirements of NFPA 70, all wiring, regardless of voltage, shall be fully enclosed within continuous raceways.

**C** Wiring is required to be placed within continuous raceways, which provides an additional level of reliability for the system. The definition of the term “raceway” in NFPA 70 lists several acceptable types of complying raceway that can be used; however, manufactured cable assemblies such as metal-clad cable (Type MC) or armored cable (Type AC) are not included.

**909.12.3 Activation.** Smoke control systems shall be activated in accordance with this section.

**C** The activation of a smoke control system is dependent on when such a system is required. Mechanical smoke control systems, which could include pressurization, airflow or exhaust methods, require an automatic activation mechanism. When using a passive system that depends on compartmentation, spot-type detectors are acceptable for the release of door closers and similar openings. With more complex mechanical systems, such activation needs to go beyond single-station detectors and be part of an automatic coordinated system.

**909.12.3.1 Pressurization, airflow or exhaust method.** Mechanical smoke control systems using the pressurization, airflow or exhaust method shall have completely automatic control.

**C** Automatic activation of such systems is especially critical as tenability is much more difficult to achieve if a delay occurs waiting for manual activation of the system. See Sections 909.6 for the pressurization method, 909.7 for the airflow design method and 909.8 for the exhaust method.

**909.12.3.2 Passive method.** Passive smoke control systems actuated by *approved* spot-type detectors *listed* for releasing service shall be permitted.

**C** This section recognizes that a passive system does not address smoke containment through mechanical means; therefore, it does not need to be “automatically activated” except in cases where smoke barriers have openings. These openings would be required to have smoke detectors to close openings where required by the design. Although spot-type detectors are technically automatic, they are not part of a more coordinated system of activation as needed for mechanical smoke control systems. Such detectors are simply standalone devices that fail in the fail-safe position. In other words, if the power were lost, a door on a magnetic hold would simply close.

**909.12.4 Automatic control.** Where completely automatic control is required or used, the automatic-control sequences shall be initiated from an appropriately zoned *automatic sprinkler system* complying with Section 903.3.1.1, manual controls provided with *ready access* for the fire department and any smoke detectors required by the engineering analysis.

**C** Where automatic activation is required, it must be accomplished by a properly zoned automatic sprinkler system and/or smoke detection where required by this code, or other initiation methods if the engineering analysis requires them. Manual control for the fire department needs to be provided. An important point with this particular requirement is that smoke control systems are engineered systems, and a prescribed smoke detection system may not fit the needs of the specific design. Other types of detectors, such as beam detectors (within an atrium), may be used and could be more useful and more practical from a maintenance standpoint. Also, it may not be practical or appropriate for the building’s fire alarm system to activate such systems, as it may alter the effectiveness of the system by pulling smoke through the building versus removing or containing the smoke. For example, a building with an atrium may have several floors below the space. If a fire occurs in one of the floors not associated with the atrium, the atrium smoke control system could possibly pull smoke throughout the building if the detection is zoned incorrectly.

**909.13 Control air tubing.** Control air tubing shall be of sufficient size to meet the required response times. Tubing shall be flushed clean and dry prior to final connections and shall be adequately supported and protected from damage. Tubing passing through concrete or masonry shall be sleeved and protected from abrasion and electrolytic action.

**C** Control tubing is a method that uses pneumatics to operate components such as the opening and closing of dampers. Because of the sophistication of electronic systems today, control tubing is becoming less common.

These particular requirements provide the criteria for properly designing and installing control tubing. Essentially, it is up to the design professional to determine the size requirements and to properly design appropriate supports. This information needs to be detailed within the construction documents. Additionally, because of the effect of moisture and other contaminants on control tubing, it must be flushed clean and then dried before installation.

**909.13.1 Materials.** Control air tubing shall be hard drawn copper, Type L, ACR in accordance with ASTM B42, ASTM B43, ASTM B68/B68M, ASTM B88, ASTM B251 and ASTM B280. Fittings shall be wrought copper or brass, solder type, in accordance with ASME B16.18 or ASME B16.22. Changes in direction shall be made with appropriate tool bends. Brass compression-type fittings shall be used at final connection to devices; other joints shall be brazed using a BCuP5 brazing alloy with solidus above 1,100°F (593°C) and liquidus below 1,500°F (816°C). Brazing flux shall be used on copper-to-brass joints only.

**Exception:** Nonmetallic tubing used within control panels and at the final connection to devices, provided that all of the following conditions are met:

1. Tubing shall comply with the requirements of Section 602.3.5 of the *International Mechanical Code*.
2. Tubing and the connected device shall be completely enclosed within a galvanized or paint-grade steel enclosure having a minimum thickness of 0.0296 inch (0.7534 mm) (No. 22 gage). Entry to the enclosure shall be by copper tubing with a protective grommet of neoprene or Teflon or by suitable brass compression to male-barbed adapter.
3. Tubing shall be identified by appropriately documented coding.
4. Tubing shall be neatly tied and supported within the enclosure. Tubing bridging cabinets and doors or moveable devices shall be of sufficient length to avoid tension and excessive stress. Tubing shall be protected against abrasion. Tubing connected to devices on doors shall be fastened along hinges.

**C** This section addresses the materials allowed for control air tubing along with approved methods of connection. All of this information needs to be documented, as it will be subject to review by the special inspector.

**909.13.2 Isolation from other functions.** Control tubing serving other than smoke control functions shall be isolated by automatic isolation valves or shall be an independent system.

**C** This section requires separation of control tubing used for other functions through the use of isolation valves or a completely separate system. This is necessary due to the difference in requirements for control tubing used in a smoke control system versus other building systems. The isolation of the control air tubing for a smoke control system needs to be specifically noted on the construction documents.

**909.13.3 Testing.** Control air tubing shall be tested at three times the operating pressure for not less than 30 minutes without any noticeable loss in gauge pressure prior to final connection to devices.

**C** As part of the acceptance testing of the smoke control system, the control air tubing will be pressure tested three times the operating pressure for 30 minutes or more. The performance criterion as to whether the control tubing is considered a failure is when there is any noticeable loss in gauge pressure prior to final connection of devices during the 30-minute duration test.

**909.14 Marking and identification.** The detection and control systems shall be clearly marked at all junctions, accesses and terminations.

**C** This section requires that all portions of the fire detection system that activate the smoke control system be marked and identified appropriately. This includes all applicable fire alarm-initiating devices, the respective junction boxes, all data-gathering panels and fire alarm control panels. Additionally, all components of the smoke control system, which are not considered a fire detection system, are required to be properly identified and marked. This would include all applicable junction boxes, control tubing, temperature control modules, relays, damper sensors, automatic door sensors and air movement sensors.

**909.15 Control diagrams.** Identical control diagrams showing all devices in the system and identifying their location and function shall be maintained current and kept on file with the *fire code official*, the fire department and in the *fire command center* in a format and manner *approved* by the *fire code official*.

**C** The purpose of control diagrams is to provide consistent information on the system in several key locations, including the building department, the fire department and the fire command center. If a fire command center is not required or provided, the diagrams need to be located such that they can be readily accessed during an emergency. Some possible locations may be the security office, the building manager's office or, if possible, within the smoke control panel. This information is intended to assist in the use and operation of the smoke control system. The format of the control diagram is as approved by the fire chief. This is necessary since the fire department is the agency that will be using such a system during a fire and when the system is tested in the future. The more clearly the information is communicated, the more effective the smoke control system will be.

It should be noted that the fire department may want all smoke control systems within a jurisdiction to follow a particular protocol for control diagrams. Generally, the control diagrams should indicate the required reaction of the system in all scenarios. The status or position of every fan and damper in every scenario must be clearly identified.

**909.16 Firefighter's smoke control panel.** A firefighter's smoke control panel for fire department emergency response purposes only shall be provided and shall include manual control or override of automatic control for mechanical smoke control systems. The panel shall be located in a *fire command center* complying with Section 508 in high-rise buildings or buildings with smoke-protected assembly seating. In all other buildings, the firefighter's smoke control panel shall be installed in an *approved* location adjacent to the fire alarm control panel. The firefighter's smoke control panel shall comply with Sections 909.16.1 through 909.16.3.

**C** One of the elements that makes a smoke control system effective is that its activity is successfully communicated to the fire department and the fire department is able to manually operate the system. The following sections provide requirements for a control panel specifically for smoke control systems. This panel is required to be located within a fire command center when it is located in a high-rise building or there is smoke-protected seating. Section 508.1 would require a fire command center for high-rise buildings. Smoke-protected seating does not require a fire command center in Chapter 10, but this provision would ensure that one exists and contains the smoke control panel. Facilities with smoke-protected seating tend to be larger facilities that, at the very least, would already have a central security center if not a fire command center as required by the jurisdiction. All other locations would only need to provide the panel in an approved location as long as it is located with the fire alarm panel. The specific location will depend on the needs of the fire department in that jurisdiction. The reason not all firefighter's smoke control panels need to be located in a fire command center is that many smoke control systems are located in a building containing an atrium that may only be three stories in height. A 200-square-foot (19 m<sup>2</sup>) fire command center would be excessive for such buildings. There are two components that include the requirements for the display and for the controls. This control panel will provide an ability to override any other controls, whether manual or automatic, within the building as they relate to the smoke control system.

Note that the publication *A Guide to Smoke Control in the 2006 IBC®* goes into more detail about the firefighter's smoke control panel requirements.

**909.16.1 Smoke control systems.** Fans within the building shall be shown on the firefighter's control panel. A clear indication of the direction of airflow and the relationship of components shall be displayed. Status indicators shall be provided for all smoke control equipment, annunciated by fan and zone and by pilot-lamp-type indicators as follows:

1. Fans, dampers and other operating equipment in their normal status—WHITE.
2. Fans, dampers and other operating equipment in their off or closed status—RED.
3. Fans, dampers and other operating equipment in their on or open status—GREEN.
4. Fans, dampers and other operating equipment in a fault status—YELLOW/AMBER.

**C** This section denotes what should be displayed on the control panel. The display is required to include all fans, an indication of the direction of airflow and the relationship of the components. Also, status lights are required, and this section sets out specific standardized colors to indicate normal status, closed status, open status and fault status. A standardized approach increases the likelihood that the fire department will be able to quickly become familiar with a system. Since the fire department has the ability to override the automatic functions of the system, this information is critical.

**909.16.2 Smoke control panel.** The firefighter's control panel shall provide control capability over the complete smoke control system equipment within the building as follows:

1. ON-AUTO-OFF control over each individual piece of operating smoke control equipment that can be controlled from other sources within the building. This includes *stairway* pressurization fans; smoke exhaust fans; supply, return and exhaust fans; elevator shaft fans; and other operating equipment used or intended for smoke control purposes.
2. OPEN-AUTO-CLOSE control over individual dampers relating to smoke control and that are controlled from other sources within the building.
3. ON-OFF or OPEN-CLOSE control over smoke control and other critical equipment associated with a fire or smoke emergency and that can only be controlled from the firefighter's control panel.

**Exceptions:**

1. Complex systems, where *approved*, where the controls and indicators are combined to control and indicate all elements of a single smoke zone as a unit.
2. Complex systems, where *approved*, where the control is accomplished by computer interface using *approved*, plain English commands.

**C** This section sets the requirements as to which controls need to be provided for the fire department on the control panel.

There are two aspects to the controls. The controls will include ON-AUTO-OFF and OPEN-AUTO-CLOSE settings or will be strictly ON-OFF or OPEN-CLOSE. If the system or component can be set on automatic (AUTO), it can be controlled from other locations beyond the fire command center. This would include control from an automatic smoke detection system or by manual activation. If a control contains only ON-OFF or OPEN-CLOSE settings, the only way the system component can be controlled is in the fire command center.

It should be noted that components such as fans are usually associated with ON-OFF-type controls, whereas components such as dampers are associated with OPEN-CLOSE-type controls.

**909.16.3 Control action and priorities.** The firefighter's control panel actions shall be as follows:

1. ON-OFF and OPEN-CLOSE control actions shall have the highest priority of any control point within the building. Once issued from the firefighter's control panel, automatic or manual control from any other control point within the building shall not contradict the control action. Where automatic means are provided to interrupt normal, nonemergency equipment operation or produce a specific result to safeguard the building or equipment including, but not limited to, duct freezestats, duct smoke detectors, high-temperature cutouts, temperature-actuated linkage and similar devices, such means shall be capable of being overridden by the firefighter's control panel. The last control action as indicated by each firefighter's control panel switch position shall prevail. Control actions shall not require the smoke control system to assume more than one configuration at any one time.

**Exception:** Power disconnects required by NFPA 70.

2. Only the AUTO position of each three-position firefighter's control panel switch shall allow automatic or manual control action from other control points within the building. The AUTO position shall be the NORMAL, nonemergency, building control position. Where a firefighter's control panel is in the AUTO position, the actual status of the device (on, off, open, closed) shall continue to be indicated by the status indicator described in Section 909.16.1. Where directed by an automatic signal to assume an emergency condition, the NORMAL position shall become the emergency condition for that device or group of devices within the zone. Control actions shall not require the smoke control system to assume more than one configuration at any one time.

**C** This section clarifies that when a component of the system is placed in an ON-OFF or OPEN-CLOSE configuration, no other control point in the building, whether automatic or manual, can override the action established in the fire command center. If a system component is configured in AUTO mode, it can be controlled from locations within the building beyond the fire command center. Some controls are specifically designed to only allow an action from the fire command center.

**909.17 System response time.** Smoke-control system activation shall be initiated immediately after receipt of an appropriate automatic or manual activation command. Smoke control systems shall activate individual components (such as dampers and fans) in the sequence necessary to prevent physical damage to the fans, dampers, ducts and other equipment. For purposes of smoke control, the firefighter's control panel response time shall be the same for automatic or manual smoke control action initiated from any other building control point. The total response time, including that necessary for detection, shutdown of operating equipment and smoke control system startup, shall allow for full operational mode to be achieved before the conditions in the space exceed the design smoke condition. Upon receipt of an alarm condition at the fire alarm control panel, fans, dampers and automatic doors shall have achieved their proper operating state and the final status shall be indicated at the smoke control panel within 90 seconds. The system response time for each component and their sequential relationships shall be detailed in the required rational analysis and verification of their installed condition reported in the required final report.

**C** This particular section provides the criteria as to when the smoke control system is required to begin operation. Whether or not the activation is manual or automatic, these criteria clarify that the system be initiated immediately. Also, it requires that components activate in a sequence that will not potentially damage the fans, dampers, ducts and other equipment. Unrealistic timing of the system has the potential of creating an unsuccessful system. Delays in the system can be seen in slow dampers, fans that ramp up or down, systems that poll slowly and intentional built-in delays. These factors can add significantly to the reaction time of the system and may hamper achieving the design goals.

The key element is that the system be fully operational before the smoke conditions exceed the design parameters. The design should include these possible delays when analyzing the smoke layer interface location. The sequence of events needs to be justified within the design analysis and described clearly in the construction documents.

Ninety seconds is the time in which the smoke control system must become fully operational from either manual or automatic initiation. Many time delays are inherent in activation components or programmed into the system. For instance, there is a delay between the activation of an initiating device and when the panel registers an alarm condition, and sprinkler systems are allowed to have a delay of up to 90 seconds to compensate for surges in the piping network. The 90-second delay referred to in this section covers the point from which the fire alarm control panel indicates an alarm condition until the smoke control system is fully operational. This 90-second delay is based on the time it takes smoke dampers to configure and annunciate their status (open/closed). Section 8.1 of UL 555S, which governs smoke dampers, requires them to fully configure (open or close) within 75 seconds. The additional time (15 seconds for a total of 90 seconds) allows for delays in data transfer and confirmation, as well as ample time for all other components to be fully operational (e.g., fans).

**909.18 Acceptance testing.** Devices, equipment, components and sequences shall be individually tested. These tests, in addition to those required by other provisions of this code, shall consist of determination of function, sequence and, where applicable, capacity of their installed condition.

**C** In order to achieve a certain level of performance, the smoke control system needs to be thoroughly tested. This section requires that all devices, equipment components and sequences be individually tested.

**909.18.1 Detection devices.** Smoke or fire detectors that are a part of a smoke control system shall be tested in accordance with Chapter 9 in their installed condition. Where applicable, this testing shall include verification of airflow in both minimum and maximum conditions.

**C** Detection devices are required to be tested in accordance with the fire protection requirements found in Chapter 9. Also, since such detectors may be subject to higher air velocities than typical detectors, their operation needs to be verified in the minimum and maximum anticipated airflow conditions.

**909.18.2 Ducts.** Ducts that are part of a smoke control system shall be traversed using generally accepted practices to determine actual air quantities.

**C** This section requires ducts that are part of the smoke control system to be tested to show that the proper amount of air is flowing. It should be noted that Section 909.10.2 requires that the ducts be leak tested to 1.5 times the maximum design pressure. Such leakage is not allowed to exceed 5 percent of the design flow.

**909.18.3 Dampers.** Dampers shall be tested for function in their installed condition in accordance with NFPA 80 and NFPA 105.

**C** This section requires that all dampers be tested to meet the function for which they are installed in accordance with the referenced standards. For instance, a damper that is to be open when the system is in smoke control mode should be verified to be open when testing the system. Also, a damper may have a specific timing associated with its operation that would need to be verified through testing.

**909.18.4 Inlets and outlets.** Inlets and outlets shall be read using generally accepted practices to determine air quantities.

**C** Similar to ducts, the appropriate amount of air that is entering or exiting the inlets and outlets, respectively, must be checked.

**909.18.5 Fans.** Fans shall be examined for correct rotation. Measurements of voltage, amperage, revolutions per minute and belt tension shall be made.

**C** This section requires the testing of fans for the following: correct rotation, voltage, amperage, revolutions per minute and belt tension. These features are key to having the system run as designed.

A common problem with fans is that they are often installed in the reverse direction. Also, to verify the reliability of the fans, elements such as the appropriate voltage and belt tension need to be tested.

**909.18.6 Smoke barriers.** Measurements using inclined manometers or other *approved* calibrated measuring devices shall be made of the pressure differences across *smoke barriers*. Such measurements shall be conducted for each possible smoke control condition.

**C** As discussed in Section 909.5.2, the testing of pressure differences across smoke barriers needs to be measured in the smoke control mode. As noted in this section, such testing is to be performed for every possible smoke control condition, and the measurements will be taken using an inclined manometer or other approved methods. Electronic devices are also available. Qualified individuals must calibrate these types of devices. Additionally, before using an alternative method of testing, the building official needs to approve such a method.

**909.18.7 Controls.** Each smoke zone equipped with an automatic-initiation device shall be put into operation by the actuation of one such device. Each additional device within the zone shall be verified to cause the same sequence without requiring the operation of fan motors in order to prevent damage. Control sequences shall be verified throughout the system, including verification of override from the firefighter's control panel and simulation of standby power conditions.

**C** This section requires the overall testing of the system. More specifically, each zone needs to individually initiate the smoke control system by the activation of an automatic initiation device. Once that has occurred, it needs to be verified that all other devices within each zone will activate the system, but to avoid damage, the fans do not need to be activated.

In addition to determining that all of the appropriate devices initiate the system, it must also be verified that all of the controls on the firefighter's control panel initiate the appropriate aspects of the smoke control system, including the override capability.

Finally, the initiation and availability of the standby power system need to be verified.

**909.18.8 Testing for smoke control.** Smoke control systems shall be tested by a special inspector in accordance with Section 1705.19 of the *International Building Code*.

**C** Smoke control systems require testing by a special inspector since they are unique and complex life safety systems. IBC Section 1705.19 provides the same requirements for testing as presented in Sections 909.18.8.1 and 909.18.8.2.

**909.18.8.1 Scope of testing.** Testing shall be conducted in accordance with the following:

1. During erection of ductwork and prior to concealment for the purposes of leakage testing and recording of device location.
2. Prior to occupancy and after sufficient completion for the purposes of pressure-difference testing, flow measurements, and detection and control verification.

**C** Special inspections need to occur at two different stages during construction to facilitate the necessary inspections. The first round of testing occurs before concealment of the ductwork or fire protection elements. The special inspector needs to verify the leakage in accordance with Section 909.10.2. Additionally, the location of all fire protection devices needs to be verified and documented at this time.

The second round of testing occurs just prior to occupancy. The testing includes the verification of pressure differences across smoke barriers, as required in Sections 909.5.2 and 909.18.6; the verification of appropriate volumes of airflow, as noted in the design; and finally, the verification of the appropriate operation of the detection and control mechanisms, as required in Sections 909.18.1 and 909.18.7. These tests need to occur just prior to occupancy, since the test results will more clearly represent actual conditions. This also makes a strong design and quality assurance during construction critical, as it is very costly and difficult in most cases to make changes at this stage. Note that the test does not actually place smoke into the space or demonstrate the smoke layer interface location. Instead, the testing is focused on all of the elements of the design such as airflow and duct closure as prescribed by the specific design.

**909.18.8.2 Qualifications.** *Approved agencies* for smoke control testing shall have expertise in fire protection engineering, mechanical engineering and certification as air balancers.

**C** As noted in Section 909.3, special inspections are required for smoke control systems. This means a certain level of qualification that would include the need for expertise in fire protection engineering, mechanical engineering and certification as air balancers.

**909.18.8.3 Reports.** A complete report of testing shall be prepared by the *approved agency*. The report shall include identification of all devices by manufacturer, nameplate data, design values, measured values and identification tag or mark. The report shall be

reviewed by the responsible *registered design professional* and, when satisfied that the design intent has been achieved, the responsible *registered design professional* shall sign, seal and date the report.

**C** Once the testing by the special inspector is complete, documentation of the activity is required. This documentation is to be prepared in the form of a report that identifies all devices by manufacturer, nameplate data, design values, measured values and identification or mark.

**909.18.8.3.1 Report filing.** A copy of the final report shall be filed with the *fire code official* and an identical copy shall be maintained in an *approved* location at the building.

**C** The report needs to be reviewed, approved and then signed, sealed and dated. This report is to be provided to the building official and a copy is also to remain in the building in an approved location. When a fire command center is required, this is the best location for such documents. Otherwise, a location such as the security office or building manager's office might be appropriate.

**909.18.9 Identification and documentation.** Charts, drawings and other documents identifying and locating each component of the smoke control system, and describing their proper function and maintenance requirements, shall be maintained on file at the building as an attachment to the report required by Section 909.18.8.3. Devices shall have an *approved* identifying tag or mark on them consistent with the other required documentation and shall be dated indicating the last time they were successfully tested and by whom.

**C** Additional documentation that needs to be maintained includes charts, drawings and other related documentation that assists in the identification of each aspect of the smoke control system. This documentation is where information, such as the last time a device or component was successfully tested and by whom, is recorded. This will serve as the main documentation for the system. Again, the fire command center, if required, is the most appropriate location for such information (see commentary, Section 909.18.8.3.1).

**909.19 System acceptance.** Buildings, or portions thereof, required by this code to comply with this section shall not be issued a certificate of occupancy until such time that the *fire code official* determines that the provisions of this section have been fully complied with and that the fire department has received satisfactory instruction on the operation, both automatic and manual, of the system and a written maintenance program complying with the requirements of Section 909.22.1 has been submitted and *approved* by the *fire code official*.

**Exception:** In buildings of phased construction, a temporary certificate of occupancy, as *approved* by the *fire code official*, shall be allowed, provided that those portions of the building to be occupied meet the requirements of this section and that the remainder does not pose a significant hazard to the safety of the proposed occupants or adjacent buildings.

**C** This section stipulates that the certificate of occupancy cannot be issued unless the smoke control system has been accepted. It is essential that the system be inspected and approved since it is a life safety system. There is an exception for buildings that are constructed in phases where a temporary certificate of occupancy is allowed. For example, a building where the portion requiring smoke control is not yet occupied so egress concerns through that space are not relevant. This space needs to be separated by smoke barriers (different smoke zone). The code also requires a maintenance program for smoke control systems since the long-term success of such systems depends heavily on proper maintenance in addition to rigorous acceptance testing. The IBC simply provides a reference to that section of the code.

**[BF] 909.20 Smokeproof enclosures.** Where required by Section 1023.12, a *smokeproof enclosure* shall be constructed in accordance with this section. A smokeproof enclosure shall consist of an *interior exit stairway* or *ramp* that is enclosed in accordance with the applicable provisions of Section 1023 and an open exterior balcony meeting the requirements of this section. Where access to the roof is required, such access shall be from the smokeproof enclosure where a *smokeproof enclosure* is required.

**C** In a building that serves stories where the floor surface is located more than 75 feet (22 860 mm) above the level of fire department access or more than 30 feet (9144 mm) below the level of exit discharge stairways, either a smokeproof enclosure or a pressurized stairway is required.

A smokeproof enclosure essentially takes advantage of fire-resistance-rated construction surrounding the interior exit stairway shaft and a buffer zone created by an open exterior balcony to avoid the accumulation of smoke within the interior exit stairway. The premise is that if access to the smokeproof enclosure provides a sufficient buffer to the effects of smoke via ventilation or simply being located outside, the smoke will not migrate into the interior exit stairway.

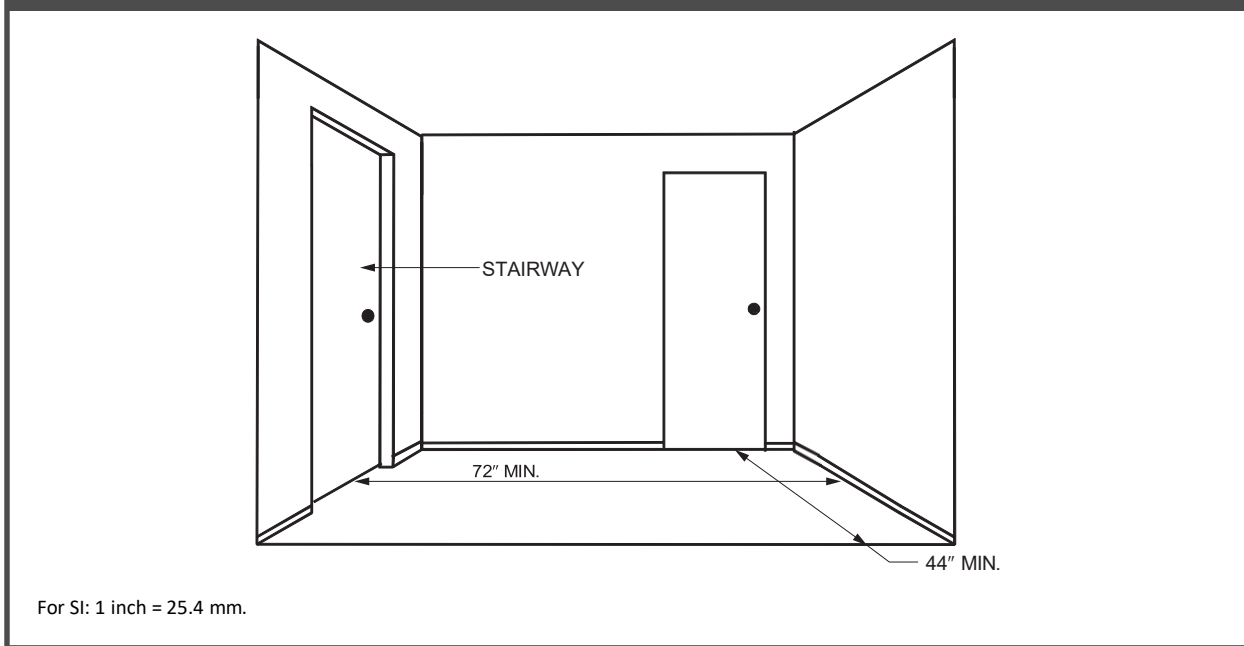
It should be noted that a smokeproof enclosure is not considered a pressurized stairway. The method of pressurizing stairways is recognized as an alternative to smokeproof enclosures in Section 909.20.4, but it is not required.

**[BF] 909.20.1 Access.** Access to the *stairway* or *ramp* shall be by way of a vestibule or an open exterior balcony. The minimum dimension of the vestibule shall be not less than the required clear width of the *corridor* leading to the vestibule but shall not have a width of less than 44 inches (1118 mm) and shall not have a length of less than 72 inches (1829 mm) in the direction of egress travel into the stairway, measured in a straight line between the centerline of the doorways into the vestibule and stairway.

**C** As noted, access to the stairway in a smokeproof enclosure is via an exterior balcony or a ventilated vestibule. This section provides the minimum dimensions for access to the smokeproof enclosure through a vestibule and then travel space to the

entrance into the stairway. If a vestibule is chosen, the minimum width is established as the larger of 44 inches (1118 mm) or the clear width of the corridor. Additionally, when entering a vestibule there needs to be at least 72 inches (1829 mm) in the direction of egress travel into the stairway, measured as specified (see Commentary Figure 909.20.1). It should be noted that the pressurized stairway alternative would not require a vestibule.

Commentary Figure 909.20.1—Dimensional Requirements for Smokeproof Enclosure Vestibule



**[BF] 909.20.2 Construction.** The *smokeproof enclosure* shall be separated from the remainder of the building by not less than 2-hour *fire barriers* constructed in accordance with Section 707 of the *International Building Code* or *horizontal assemblies* constructed in accordance with Section 711 of the *International Building Code*, or both. Openings are not permitted other than the required *means of egress* doors. The vestibule shall be separated from the *stairway* or *ramp* by not less than 2-hour *fire barriers* constructed in accordance with Section 707 of the *International Building Code* or *horizontal assemblies* constructed in accordance with Section 711 of the *International Building Code*, or both. The open exterior balcony shall be constructed in accordance with the *fire-resistance-rating* requirements for floor assemblies.

**C** This section sets the basic construction requirements for the stairway and the associated vestibule or exterior balcony. Essentially, a 2-hour fire-resistance-rated fire barrier or horizontal assembly is required between the enclosure and the rest of the building.

Additionally, vestibules need to be separated from the stairway by a 2-hour fire barrier or horizontal assembly. The exterior balcony, if used, would only be required to comply with the fire-rating requirements for floor construction. The opening protection for doors into the stairway from the vestibule is only required to be fire protection rated for 20 minutes (see Section 909.20.3.2).

**[BF] 909.20.2.1 Door closers.** Doors in a *smokeproof enclosure* shall be self-closing or automatic closing by actuation of a smoke detector in accordance with Section 716.2.6.6 of the *International Building Code* and shall be installed at the floor-side entrance to the *smokeproof enclosure*. The actuation of the smoke detector on any door shall activate the closing devices on all doors in the *smokeproof enclosure* at all levels. Smoke detectors shall be installed in accordance with Section 907.3.

**C** In order to maintain the separation between the smokeproof enclosure and the rest of the building, the doors need to be self-closing or automatic closing through the actuation of a smoke detector. Also, when a smoke detector is activated on any floor, doors at all levels are required to close for that particular smokeproof enclosure.

**[BF] 909.20.3 Natural ventilation alternative.** The provisions of Sections 909.20.3.1 through 909.20.3.3 shall apply to ventilation of *smokeproof enclosures* by natural means.

**C** Smoke and hot gases generated by fire may enter smokeproof enclosures through the doorways. This section provides for the diffusion of smoke and gases to the outside of the building by means of unenclosed openings to the outside.

**[BF] 909.20.3.1 Balcony doors.** Where access to the *stairway* or *ramp* is by way of an open exterior balcony, the door assembly into the enclosure shall be a fire door assembly in accordance with Section 716 of the *International Building Code*.

**C** Where an open exterior balcony is used for entry into the stairway, the doorways to the stairway are required to have a 1 1/2-hour fire protection rating as required for 2-hour fire-resistance-rated wall assemblies in accordance with the require-

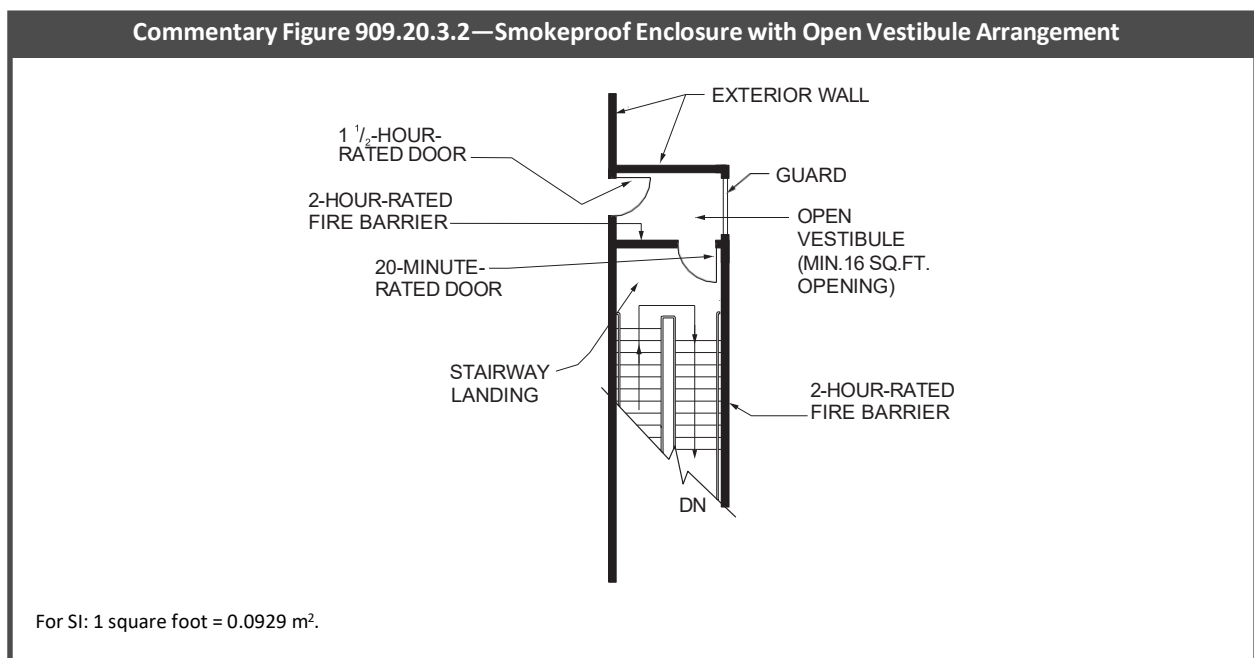
ments of IBC Section 716.1 If a vestibule is used, the rating of the door from the vestibule into the stairway would only be required to be fire protection rated for 20 minutes. The door into the vestibule would require a fire protection rating of 1 1/2 hours.

**[BF] 909.20.3.2 Vestibule doors.** Where access to the *stairway* or *ramp* is by way of a vestibule, the door assembly into the vestibule shall be a fire door assembly complying with Section 716 of the *International Building Code*. The door assembly from the vestibule to the *stairway* shall have not less than a 20-minute *fire protection rating* complying with Section 716 of the *International Building Code*.

**C** Where a smokeproof enclosure is of the type that employs a vestibule as the means of access to an enclosed exit stairway, the entry into the vestibule from the exit access must consist of a fire door assembly having a 1 1/2-hour fire protection rating. This requirement is commensurate with the 2-hour fire-resistance-rated smokeproof enclosure construction requirement and the associated fire protection ratings specified in IBC Table 716.1(2) Essentially, the fire door assemblies and the enclosure walls serve as the primary exit enclosure protection from fires occurring in adjacent spaces.

The entry from a vestibule into the stairway requires a 20-minute fire-protection-rated door assembly. These doorways are not normally subject to direct fire exposure and, therefore, only a nominal fire protection rating is required as a standard of construction that would minimize air leakage and thus reduce the possibilities of smoke and hot gases penetrating the stairway in amounts that could become a threat to life safety.

Door assemblies must comply with the requirements of IBC Section 716 (see Commentary Figure 909.20.3.2).



**[BF] 909.20.3.3 Vestibule ventilation.** Each vestibule shall have a minimum net area of 16 square feet (1.5 m<sup>2</sup>) of opening in a wall facing an outer *court*, *yard* or *public way* that is not less than 20 feet (6096 mm) in width.

**C** Natural ventilation is allowed for vestibules that have a minimum net opening of 16 square feet (1.5 m<sup>2</sup>) and open into an outer court, yard or public way that is at least 20 feet (6096 mm) in width. Basically, this option is intended to take advantage of the fact that smoke may disperse into an open area before entering the stair enclosure.

**[BF] 909.20.4 Stairway and ramp pressurization alternative.** Where the building is equipped throughout with an *automatic sprinkler system* in accordance with Section 903.3.1.1, the vestibule is not required, provided that each *interior exit stairway* or *ramp* is pressurized to not less than 0.10 inch of water (25 Pa) and not more than 0.35 inch of water (87 Pa) in the shaft relative to the building measured with all *interior exit stairway* and *ramp* doors closed under maximum anticipated conditions of stack effect and wind effect.

**C** This method is allowed only when the building is equipped throughout with an automatic sprinkler system. This is partially related to the fact that these pressure differences were developed based on a sprinklered fire. It should be noted that smokeproof enclosures are not required to be in buildings equipped throughout with an automatic sprinkler system, but the areas where smokeproof enclosures are required are often sprinklered buildings (i.e., high-rise buildings). This alternative would not require vestibules or an exterior exit balcony. The criteria for smoke control design are provided in terms of minimum and maximum pressure differences of 0.10 inch of water (37 Pa) and 0.35 inch of water (87 Pa), respectively, between the shaft and the building. This pressure difference is to be achieved when all doors are closed and maximum conditions of wind and stack effect have been taken into account. It should be noted that additional limitations may be placed on the maximum pressure differences for pressurized stairways due to the lower opening forces required in order to com-

ply with Section 1010.1.3. If the maximum pressure difference of 0.35 would exceed the requirements of Section 1010.1.3, the maximum pressure difference would need to be lowered. Also note that Section 404.2.8 of ICC A117.1 would not require opening forces to be lowered for accessibility purposes if the door is a fire door. Finally, as with all other smoke control systems addressed in Section 909, such systems need to be designed through a rational analysis, tested and documented as such. See also the commentary to Section 909.6.3.

**[BF] 909.20.5 Ventilating equipment.** The activation of ventilating equipment required by the alternative in Section 909.20.4 shall be by smoke detectors installed at each floor level at an *approved* location at the entrance to the *smokeproof enclosure*. When the closing device for the stairway and *ramp* shaft and vestibule doors is activated by smoke detection or power failure, the mechanical equipment shall activate and operate at the required performance levels. Smoke detectors shall be installed in accordance with Section 907.3.

**C** This section clarifies that the activation mechanism for both mechanical means of smoke management for interior exit stairways in Section 909.20.4 should be via a smoke detector located at each level outside the door leading into the vestibule and stairway, respectively. For systems that use automatic-closing devices on the doors, whether for vestibules in smokeproof enclosures or for pressurized stairways, if the door closes, the system must activate. This includes normal activation of the smoke detector or in the event of a power failure. Essentially, if there is a power failure, it will operate in a fail-safe manner even if there is not a fire.

**[BF] 909.20.5.1 Ventilation systems.** *Smokeproof enclosure* ventilation systems shall be independent of other building ventilation systems. The equipment, control wiring, power wiring and ductwork shall comply with one of the following:

1. Equipment, control wiring, power wiring and ductwork shall be located exterior to the building and directly connected to the *smokeproof enclosure* or connected to the *smokeproof enclosure* by ductwork enclosed by not less than 2-hour *fire barriers* constructed in accordance with Section 707 of the *International Building Code* or *horizontal assemblies* constructed in accordance with Section 711 of the *International Building Code*, or both.
2. Equipment, control wiring, power wiring and ductwork shall be located within the *smokeproof enclosure* with intake or exhaust directly from and to the outside or through ductwork enclosed by not less than 2-hour *fire barriers* constructed in accordance with Section 707 of the *International Building Code* or *horizontal assemblies* constructed in accordance with Section 711 of the *International Building Code*, or both.
3. Equipment, control wiring, power wiring and ductwork shall be located within the building if separated from the remainder of the building, including other mechanical equipment, by not less than 2-hour *fire barriers* constructed in accordance with Section 707 of the *International Building Code* or *horizontal assemblies* constructed in accordance with Section 711 of the *International Building Code*, or both.

**Exception:** Control wiring and power wiring located outside of a 2-hour *fire barrier* construction shall be protected using any one of the following methods:

1. Cables used for survivability of required *critical circuits* shall be *listed* in accordance with UL 2196 and shall have a *fire-resistance rating* of not less than 2 hours.
2. Where encased with not less than 2 inches (51 mm) of concrete.
3. Electrical circuit protective systems shall have a *fire-resistance rating* of not less than 2 hours. Electrical circuit protective systems shall be installed in accordance with their listing requirements.

**C** Smokeproof enclosures and pressurized stair shaft ventilation systems must be independent of other building ventilating systems. This section provides three options for the location of the ductwork associated with smokeproof enclosures and stair pressurization mechanical equipment. The three options include the following:

- Located on the exterior of the building and directly connected to the smokeproof enclosure.
- Within the smokeproof enclosure.
- Within the building but separated from the remainder of the building by 2-hour fire barriers or horizontal barriers, or both.

A question that often arises with the inclusion of the terms “control wiring” and “power wiring” is whether this addresses the smoke detection and fire alarm equipment associated with these ventilation systems. The original intent of the addition of these terms was focused primarily on the fans and related ductwork and not necessarily on the fire alarm system components related to the activation of such systems. This is not specifically discussed within the code itself however. Code change FS172-07/08 placed this language in the code.

There are three parts to the exception. Item 1 allows the use of 2-hour-rated cable. This provides the protection required in terms of rating but provides more flexibility in the placement of such wiring.

Item 2 allows a sufficient but not necessarily listed fire-resistance-rated approach by using concrete to encase the equipment or wiring.

Item 3 adds the option of using conventional cables with a protective material applied to them. These materials are called “electrical circuit protective systems” and are recognized by NFPA 70 for protection of fire pump control wiring, emergency system circuit wiring and critical operations power system circuit wiring.

**[BF] 909.20.5.2 Standby power.** Mechanical vestibule and *stairway* and *ramp* shaft ventilation systems and automatic fire detection systems shall be provided with standby power in accordance with Section 1203.

**C** This section requires standby power for mechanical systems for pressurized stair shafts, mechanically ventilated vestibules, stair shafts for smokeproof enclosures and any automatic fire detection systems related to the activation of such systems.

**[BF] 909.20.5.3 Acceptance and testing.** Before the mechanical equipment is *approved*, the system shall be tested in the presence of the building official to confirm that the system is operating in compliance with these requirements.

**C** This section requires verification of successful testing of the system witnessed by [or in the presence of] the building official. The requirements of Sections 909.18 and 909.19 should be addressed when testing smokeproof enclosures or pressurized stairs.

**[BF] 909.21 Elevator hoistway pressurization alternative.** Where elevator hoistway pressurization is provided in lieu of required enclosed elevator lobbies, the pressurization system shall comply with Sections 909.21.1 through 909.21.11. The design shall consider the interaction effects of the operation of multiple smoke control systems for all design scenarios in accordance with Section 909.4.7. All components or systems associated with the means of mitigating adverse interaction shall comply with the applicable subsections of Section 909.

**C** This section sets out the requirements for hoistway pressurization when used as a method to protect hoistway openings as required in IBC Section 3006. The minimum and maximum pressures are similar to those for pressurized stairs in Section 909.20.4. The provisions are located in Section 909 to group all the smoke control techniques and requirements into the same section. This promotes more consistency in code application when addressing smoke control.

Elevator hoistway pressurization systems will impact other smoke control systems serving high-rise buildings, including stair pressurization. Due to the high leakage factors of both open and closed elevator doors, a large volume of airflow must be introduced into the elevator hoistway to maintain the prescribed +0.10 inches of water (25 Pa) for elevator pressurization systems on all levels. Unless the air introduced into the building by the elevator pressurization is relieved to the exterior, the prescribed pressures cannot be maintained to the pressurized stair enclosures. Combined with the low exterior leakage factor of modern buildings to meet energy code requirements, this issue is further compounded. Relief dampers relieve air at the exterior of the building. The volume of air must be calculated using smoke control models such as CONTAM. While Section 909.4.7 specifically requires the engineer to consider the interaction of multiple systems, the need should be emphasized when the building utilizes elevator pressurization to protect the hoistway. Since the components and systems utilized to mitigate the impact are critical to the functionality of the system, this section clarifies that the entire system must comply with the provisions of Section 909.

**[BF] 909.21.1 Pressurization requirements.** Elevator hoistways shall be pressurized to maintain a minimum positive pressure of 0.10 inch of water (25 Pa) and a maximum positive pressure of 0.25 inch of water (67 Pa) with respect to adjacent occupied space on all floors. This pressure shall be measured at the midpoint of each hoistway door, with all elevator cars at the floor of recall and all hoistway doors on the floor of recall open and all other hoistway doors closed. The pressure differentials shall be measured between the hoistway and the adjacent elevator landing. The opening and closing of hoistway doors at each level must be demonstrated during this test. The supply air intake shall be from an outside, uncontaminated source located a minimum distance of 20 feet (6096 mm) from any air exhaust system or outlet.

**Exceptions:**

1. On floors containing only Group R occupancies, the pressure differential is permitted to be measured between the hoistway and a *dwelling unit or sleeping unit*.
2. Where an elevator opens into a lobby enclosed in accordance with Section 3007.6 or 3008.6 of the *International Building Code*, the pressure differential is permitted to be measured between the hoistway and the space immediately outside the door(s) from the floor to the enclosed lobby.
3. The pressure differential is permitted to be measured relative to the outdoor atmosphere on floors other than the following:
  - 3.1. The fire floor.
  - 3.2. The two floors immediately below the fire floor.
  - 3.3. The floor immediately above the fire floor.
4. The minimum positive pressure of 0.10 inch of water (25 Pa) and a maximum positive pressure of 0.25 inch of water (67 Pa) with respect to occupied floors is not required at the floor of recall with the doors open.

**C** This section states the minimum and the maximum positive pressure that must be achieved by the smoke control mechanical pressurization system. The minimum positive pressure is 0.10 inch of water column (25 Pa), the same as required for stairway pressurization in Section 909.20.4. The maximum pressure is 0.25 inch of water column (62 Pa), which is a little less than the maximum allowed for stairway pressurization. The minimum pressure is to ensure that the stack effect is overcome and the maximum pressure is an upper limit to ensure that the doors will operate properly. This section requires a test when the system is complete. The pressures are measured at the midpoint of each hoistway door with all elevator

cars at the recall floor and all the hoistway doors open on that level. This simulates the Phase I recall requirements in IBC Section 3003.2. Hoistway doors are then tested on each level to ensure proper operation.

The supply air intake for the pressurization system must be located at least 20 feet (6096 mm) away from any source of contamination to ensure that the hoistway remains tenable through the fire event or well into it before the elevators can no longer be used. Also, if smoke is drawn into the supply air, the system will only spread smoke and not prevent its spread.

There are four exceptions to the requirements of Section 909.21.1. The first two provide exceptions as to where the pressure differential can be measured. Exception 3 allows the pressure differences only to be measured at the fire floor and several designated floors. Finally, Exception 4 allows the pressure differential measurement to be omitted for the floor of recall.

The first three exceptions originated from the City of Seattle, Washington, which has had a long history of requiring pressurized hoistways in high-rise buildings to prevent smoke migration. In 2005, the City of Seattle Department of Planning & Development (DPD) convened a committee, which included representatives from industry, the Seattle Fire Department, and DPD, to decide whether to recommend changes to the high-rise smoke migration control requirements in place at that time. The committee also consulted with Dr. John Klote, who suggested the approach that Seattle eventually adopted with some small modifications. These requirements are an adaptation of the Seattle approach.

During the 2009/2010 code change cycle, a proposal was made to delete the hoistway pressurization requirements in the IBC without substitution (FS51-09/10), based on a study conducted by Drs. Miller and Beasley. This study showed that requiring the pressure differential of 0.10 inch of water column to be maintained at the recall floor with the elevator doors in the open position resulted in overpressurization of all the other floors—meaning the current standards in the code cannot be met. Based on further modeling by Dr. Miller, the proponent for FS51-09/10 submitted a public comment introducing Seattle's requirements into the IBC. The reason statement for the public comment stated Dr. Miller "concluded that the 'Seattle approach' does indeed meet all the prescriptive requirements of the IBC 2009." The proposal and its public comment were ultimately withdrawn by the proponent in anticipation of further review of the overall elevator lobby provisions.

The intent of the code is to keep smoke out of the hoistway, so the pressure should be measured between the elevator hoistway and the elevator landing/lobby. However, Exception 1 allows the pressure to be measured between the hoistway and sleeping or dwelling units in residential buildings, since they are highly compartmented. In addition, the fire source is most likely to be in the dwelling or sleeping unit, and providing positive pressure in the corridor/hallway outside the units (via leakage through the elevator hoistway doors) will help reduce the smoke migrating from the affected unit. Exception 2, which is specific to elevator lobbies associated with fire service access elevators (FSAE) and occupant evacuation elevators, allows the pressure to be measured between the hoistway and the space on the outside of the smoke barrier that forms the lobby. It should be noted that hoistway pressurization is not a design alternative to enclosed lobbies for FSAEs and occupant evacuation elevators. Enclosed elevator lobbies are always required for these types of elevators. This exception would only apply if such a system was provided.

Exception 3 allows the 0.10-inch-water-column pressure differential between the hoistway and the floor be met only on the four most critical floors—the floor of fire origin, the two floors immediately below, and one floor immediately above. For all other stories, the pressure differential is allowed to be measured between the hoistway and the outside of the building. The purpose of this requirement is to maintain a slightly positive pressure in the building relative to atmospheric, so as to lower the neutral pressure plane in the building, which then reduces the driving force of stack effect. This exception is intended to be permitted to be used in conjunction with Exceptions 1 and 2. The engineers who design this system begin by modeling one floor as the "notionalized" fire floor, and designing the system (fans, dampers, etc.) accordingly. Each floor is subsequently modeled as the notionalized fire floor, and the system is checked to make sure the maximum and minimum pressure differentials are met. (Note that actual models may not have to be run for each floor if it is clear the worst case has been covered.) Ultimately, the system will need to be designed so it will correctly configure itself for a fire originating on any floor in the building.

Exception 4 omits the need to measure the pressure differential at the floor of elevator recall where the door is typically open. Section 909.21.1 requires the pressure difference, required for the pressurization alternative, to be measured at the midpoint of each hoistway door, with all elevator cars at the floor of recall and all hoistway doors on the floor of recall open and all other hoistway doors closed.

Elevator hoistway pressurization is intended to minimize smoke movement into an elevator shaft when a lobby is not provided. Meeting the required pressure difference on the recall floor with the hoistway doors open is not necessary because the recall floor is protected by smoke detectors that will not allow the hoistway doors to open if smoke is present.

The pressurization method is based on using pressure differences produced by fans to minimize the spread of smoke across a barrier. A barrier will not exist on the recall floor when the hoistway doors are open, and smoke detectors used for elevator recall prevent the doors from opening when smoke is present.

**[BF] 909.21.1.1 Use of ventilation systems.** Ventilation systems, other than hoistway supply air systems, are permitted to be used to exhaust air from adjacent spaces on the fire floor, two floors immediately below and one floor immediately above the fire floor to the building's exterior where necessary to maintain positive pressure relationships as required in Section 909.21.1 during operation of the elevator shaft pressurization system.

**C** This section allows the use of the general building HVAC system to exhaust air to create/maintain the required pressure differential. It is to be noted that the requirements of the rest of Section 909.21—in particular, Section 909.21.10 regarding protection of equipment—would still apply to these components.

**[BF] 909.21.2 Rational analysis.** A rational analysis complying with Section 909.4 shall be submitted with the *construction documents*.

**C** Section 909.4 recognizes that there are many factors involved in a smoke control system, including stack effect due to height, temperature effect of fire, wind effect, interaction of the HVAC system, the weather, and the egress time, all of which must be evaluated. The report must be submitted with the permit documents. Most importantly, the duration of operation of the smoke control system is a function of 1.5 times the egress time or 20 minutes, whichever is less. More discussion on the duration of operation is found in the commentary to Section 909.4.6.

**[BF] 909.21.3 Ducts for system.** Any duct system that is part of the pressurization system shall be protected with the same *fire-resistance rating* as required for the elevator shaft enclosure.

**C** All ductwork necessary for hoistway pressurization must be protected from the effects of fire by enclosure in fire-resistance-rated construction equivalent to that required for the elevator hoistway shaft enclosure.

**[BF] 909.21.4 Fan system.** The fan system provided for the pressurization system shall be as required by Sections 909.21.4.1 through 909.21.4.4.

**C** This section details the requirements for the mechanical system used for pressurization of the hoistway enclosure.

**[BF] 909.21.4.1 Fire resistance.** Where located within the building, the fan system that provides the pressurization shall be protected with the same *fire-resistance rating* required for the elevator shaft enclosure.

**C** The only way to ensure that the mechanical pressurization system can operate during a fire is to locate it in a safe place. If located within the building, it must be in an enclosed room protected with the same fire-resistance-rated construction required for the hoistway enclosure.

**[BF] 909.21.4.2 Smoke detection.** The fan system shall be equipped with a smoke detector that will automatically shut down the fan system when smoke is detected within the system.

**C** The airflow must be free of smoke or it will only increase the likelihood of smoke spreading throughout the building. The smoke detector required by this section should be located on the intake side of the blower fan.

**[BF] 909.21.4.3 Separate systems.** A separate fan system shall be used for each elevator hoistway.

**C** This section requires that each hoistway enclosure have its own mechanical system. This provides a more redundant system and helps to increase the likelihood that fans will be operational during a fire.

**[BF] 909.21.4.4 Fan capacity.** The supply fan shall be either adjustable with a capacity of not less than 1,000 cfm (0.4719 m<sup>3</sup>/s) per door, or that specified by a *registered design professional* to meet the requirements of a designed pressurization system.

**C** The fan capacity should be as specified by the registered design professional to meet the operational ranges of pressure at each door or be adjustable with a capacity of at least 1,000 cfm (0.4719 m<sup>3</sup>/s) per hoistway door. In either case, it is subject to field testing and adjustments to meet the pressure ranges.

**[BF] 909.21.5 Standby power.** The pressurization system shall be provided with standby power in accordance with Section 1203.

**C** The elevator hoistway pressurization system is an emergency system and must have provisions for standby power like other emergency systems. IBC Section 2702 states the requirements that standby power systems must meet. It is consistent with other smoke control systems required by Section 909 in that such systems have standby power as they are life safety systems.

**[BF] 909.21.6 Activation of pressurization system.** The elevator pressurization system shall be activated upon activation of the elevator lobby smoke detectors.

**C** This section requires that the pressurization system will be activated when the elevator lobby smoke detectors are activated. All buildings using this pressurization option will be required to have them. Section 909.12 requires smoke detectors to activate the pressurization system if the design requires it to operate to remove the smoke. The fundamental intent of elevator hoistway pressurization is to provide an alternate means of opening protection for the elevator shaft enclosure and prevent smoke migration. The smoke detectors in the lobby will activate before any significant amount of smoke enters the hoistway. When these detectors initiate an alarm, the elevators will recall, and elevator pressurization system will activate to mitigate smoke migration in the hoistway.

**[BF] 909.21.7 Testing.** Testing for performance shall be required in accordance with Section 909.18.8. System acceptance shall be in accordance with Section 909.19.

**C** Testing will be required to evaluate the performance of the completed system (see commentary, Sections 909.18 and 909.19).

**[BF] 909.21.8 Marking and identification.** Detection and control systems shall be marked in accordance with Section 909.14.

**C** See the commentary to Section 909.14.

**[BF] 909.21.9 Control diagrams.** Control diagrams shall be provided in accordance with Section 909.15.

**C** See the commentary to Section 909.15.

**[BF] 909.21.10 Control panel.** A control panel complying with Section 909.16 shall be provided.

**C** See the commentary to Section 909.16.

**[BF] 909.21.11 System response time.** Hoistway pressurization systems shall comply with the requirements for smoke control system response time in Section 909.17.

**C** See the commentary to Section 909.17.

**909.22 Maintenance.** Smoke control systems shall be maintained to ensure to a reasonable degree that the system is capable of controlling smoke for the duration required. The system shall be maintained in accordance with the manufacturer's instructions and Sections 909.22.1 through 909.22.6.

**C** Routine maintenance and testing of smoke control systems is essential to ensure their performance, as designed, under fire conditions. Maintenance practices must be consistent with the manufacturer's recommendations and as indicated in Sections 909.22.1 through 909.22.6. Note that Section 909.12.1 requires weekly preprogrammed tests that report abnormal conditions.

**909.22.1 Schedule.** A routine maintenance and operational testing program shall be initiated immediately after the smoke control system has passed the acceptance tests. A written schedule for routine maintenance and operational testing shall be established.

**C** Operational testing and maintenance must be performed on the smoke control system periodically to verify that it still operates as required by the approved design. A written schedule must be maintained.

**909.22.2 Records.** Records of smoke control system testing and maintenance shall be maintained. The record shall include the date of the maintenance, identification of the servicing personnel and notification of any unsatisfactory condition and the corrective action taken, including parts replaced.

**C** This section prescribes the desired content of the written record for the smoke control testing and maintenance program. Test results and maintenance activities should be clearly documented. The written record should be available for inspection and reviewed by the fire code official.

**909.22.3 Testing.** Operational testing of the smoke control system shall include all equipment such as initiating devices, fans, dampers, controls, doors and windows.

**C** Smoke control systems are made up of components and equipment that are an integral part of other building systems such as fire alarm systems; heating, ventilating and air-conditioning (HVAC) equipment; and automatic sprinkler systems. For this reason, operational testing of all related system components must ensure that the system as a whole will perform as intended.

**909.22.4 Dedicated smoke control systems.** Dedicated smoke control systems shall be operated for each control sequence semiannually. The system shall be tested under standby power conditions.

**C** Because dedicated smoke control systems are designed for smoke control only, the operation of these systems does not adversely affect other building systems or operations. The control sequence for these systems must be tested semiannually to check for system component failures that may not get noticed because dedicated smoke control systems are independent of building HVAC systems.

**909.22.5 Nondedicated smoke control systems.** Nondedicated smoke control systems shall be operated for each control sequence annually. The system shall be tested under standby power conditions.

**C** Contrary to dedicated smoke control systems identified in Section 909.22.4, smoke control systems that are not dedicated share system components with other building systems, including the HVAC system. Consequently, testing of the control sequence of systems that are not dedicated can be done annually, rather than semiannually, because equipment failures related to other building systems would most likely be noticed and corrected when those other systems are tested or maintained.

Simply because a system is nondedicated does not guarantee that failures will be detected any easier. This relates to the fact that when a system is in smoke control mode, it may have very different demands than when simply operating as a traditional HVAC system (see commentary, Section 909.12).

**909.22.6 Components bypassing weekly test.** Where components of the smoke control system are bypassed by the preprogrammed weekly test required by Section 909.12.1, such components shall be tested semiannually. The system shall be tested under standby power conditions.

**C** This section is specifically linked to the exception in Section 909.12.1 that allows the elimination of the weekly test of the system. In order to allow the elimination of the weekly test, Section 909.12.1 provides additional monitoring features and requires more frequent testing of the overall system by a special inspector. This will primarily affect nondedicated systems that are currently only required to be tested annually. Dedicated systems are already required to be tested semiannually.

## SECTION 910—SMOKE AND HEAT REMOVAL

**910.1 General.** Where required by this code, smoke and heat vents or mechanical smoke removal systems shall conform to the requirements of this section.

**C** This section essentially requires either smoke and heat vents or a mechanical smoke removal system where required by Section 910.2. It should be noted that where high-piled combustible storage is involved, Chapter 32 of the IFC also applies.

The purpose of smoke and heat vents or smoke removal systems has historically been related to the needs of firefighters. More specifically, smoke and heat vents or smoke removal systems, when activated, have the potential effect of lifting the height of the smoke layer and providing more tenable conditions to undertake firefighting activities. Other potential benefits include a reduction in property damage and the creation of more tenable conditions for occupants.

These provisions are based on research on the interaction of sprinklers, roof vents and draft curtains funded by the National Fire Protection Research Foundation (NFPRF) and conducted at Underwriters Laboratories (UL) in 1997/1998. This research is summarized in a document referred to as *National Institute of Science and Technology Interagency Report* (NISTIR) 6196-1 dated September 1998. The current provisions were also based on the following:

- Provisions for the use of roof vents in sprinklered buildings included in the 2010 and 2013 editions of NFPA 13, including the substantiation statement for the NFPA 13 roof vent provisions.
- The capability of standard spray sprinklers to both control and extinguish a fire within 30 minutes of sprinkler operation without supplemental fire department activity has been documented.
- Recommendations contained in National Institute for Occupational Safety and Health (NIOSH) 2005-132, *Preventing Injuries and Deaths of Fire Fighters Due to Truss Systems*, and NIOSH 2010-153, *Preventing Deaths and Injuries of Fire Fighters Using Risk Management Principles at Structure Fires*.
- Recommendations contained in the Initial Report of the Federal Emergency Management Agency (FEMA)/National Fallen Firefighters Foundation (NFFF®) Firefighter Life Safety Summit held on April 14, 2004, in Tampa, Florida.

The primary purpose of smoke and heat removal from the perspective of the building code requirement is to assist firefighting operations after control of the fire has been achieved by the automatic sprinkler system. Automatic smoke and heat vents and automatic sprinkler systems were developed independently of one another and their interaction has been a concern for many years. Even today, there is no accepted method of analyzing their interaction and, therefore, the installation standards for each (NFPA 204 and NFPA 13, respectively) give cautions to the designers of buildings having both systems. Note that NFPA 204 is not referenced in Section 910.

Manually activated mechanical smoke removal systems can perform the same function as roof vents. Mechanical smoke removal systems as required in Section 910 provide fire-rated, grade-level enclosures for the control of the mechanical smoke removal system. This provides greater control of the system for the fire incident commander and reduces the need to place firefighters on roofs or in other hazardous situations to operate smoke and heat venting systems. This methodology is consistent with the latest recommendations from NIOSH and NFFF for firefighter safety, risk management and recommended firefighting tactics.

**910.2 Where required.** Smoke and heat vents or a mechanical smoke removal system shall be installed as required by Sections 910.2.1 and 910.2.2.

### Exceptions:

1. Frozen food warehouses used solely for storage of Class I and II commodities where protected by an *approved automatic sprinkler system*.
2. Smoke and heat removal shall not be required in areas of buildings equipped with early suppression fast-response (ESFR) sprinklers.
3. Smoke and heat removal shall not be required in areas of buildings equipped with control mode special application sprinklers with a response time index of  $50(m \times s)^{1/2}$  or less that are *listed* to control a fire in stored commodities with 12 or fewer sprinklers.

**C** Sections 910.2.1 and 910.2.2 provide the locations where such smoke removal or smoke and heat vents would be required. There are three overall exceptions to the application of Section 910.

Exception 1 recognizes the “building-within-a-building” nature of typical frozen food warehouses. As such, smoke from a fire within a freezer would be contained within the freezer, thus negating the usefulness of smoke and heat vents at the roof level.

Exception 2 recognizes the negative effect that smoke and heat vents can have on the operation of early suppression fast-response (ESFR) sprinklers. Those negative effects include diverting heat away from the sprinklers, which could delay their activation or result in the activation of more sprinklers in areas away from the source of the fire, which may overwhelm the system.

Exception 3 recognizes a new category of automatic sprinkler that shares the key characteristics of ESFR sprinklers, i.e., thermal elements that have a response time index (RTI) of 50 or less and are listed to protect a design area that involves 12 or fewer sprinklers. These control mode special application (CMSA) sprinklers, while not called ESFR, still require similar precautions to ESFR sprinklers with respect to not introducing unknowns, such as smoke and heat removal, that were not present in the full-scale fire tests that determined their listing parameters. Such unknowns can lead to sprinkler “skipping” and exceeding the 12-sprinkler design area, which was the exact concern that led to the ESFR-related provisions that are currently in this chapter and Chapter 32. Note that CMSA sprinklers must have both an RTI of 50 or less and be listed to control or suppress a fire with 12 or fewer sprinklers to qualify for this exception. Any sprinkler listed as “quick response” will satisfy the “50 RTI or less” criterion based on the definition of “quick response” in NFPA 13 Section 3.6.4.7. The number of operating sprinklers will be indicated in the listing criterion for each sprinkler.

**910.2.1 Group F-1 or S-1.** Smoke and heat vents installed in accordance with Section 910.3 or a mechanical smoke removal system installed in accordance with Section 910.4 shall be installed in buildings and portions thereof used as a Group F-1 or S-1 occupancy having more than 50,000 square feet (4645 m<sup>2</sup>) of undivided area. In occupied portions of a building equipped throughout with an *automatic sprinkler system* in accordance with Section 903.3.1.1, where the upper surface of the story is not a roof assembly, a mechanical smoke removal system in accordance with Section 910.4 shall be installed.

**Exception:** Group S-1 aircraft repair hangars.

**C** Large-area buildings with moderate to heavy fire loads present special challenges to the fire department in disposing of the smoke generated in a fire. In order to provide the fire department with the ability to rapidly and efficiently dispose of smoke in large-area Group F-1 and S-1 buildings exceeding 50,000 square feet (4645 m<sup>2</sup>) in undivided area without exposing personnel to the dangers associated with cutting ventilation holes in the roof, smoke and heat vents or a smoke removal system must be provided.

The code is not clear on what is meant by the term “undivided area.” However, the intent is to provide the ability to manage the smoke in large spaces. Draft curtains or potentially any physical separation (regardless of rating) would provide such division. Draft curtains are typically constructed of sheet metal, lath and plaster, gypsum board or other materials that resist the passage of smoke. Typically, draft curtains are at least 6 feet deep (1829 mm) from the ceiling. In keeping with the concern for managing smoke, the joints and connections should be smoke tight.

Based on the intent of “undivided area,” a fire barrier, smoke barrier, fire partition or smoke partition would be more than what is required and would therefore be an acceptable method of dividing the area.

This requirement is independent of the requirements related to high-piled storage in Section 910.2.2. High-piled combustible storage is not occupancy specific.

This section also addresses multiple-story buildings where the Group F-1 or S-1 occupancy is not the uppermost story and therefore would not have a roof in which to place smoke and heat vents. This section would require that a smoke removal system be installed.

**910.2.2 High-piled combustible storage.** Smoke and heat removal required by Table 3206.2 for buildings and portions thereof containing *high-piled combustible storage* shall be installed in accordance with Section 910.3 in unsprinklered buildings. In buildings and portions thereof containing *high-piled combustible storage* equipped throughout with an *automatic sprinkler system* in accordance with Section 903.3.1.1, a smoke and heat removal system shall be installed in accordance with Section 910.3 or 910.4. In occupied portions of a building equipped throughout with an *automatic sprinkler system* in accordance with Section 903.3.1.1 where the upper surface of the story is not a roof assembly, a mechanical smoke removal system in accordance with Section 910.4 shall be installed.

**C** This section requires smoke and heat removal as it is required in Chapter 32 for high-piled combustible storage. Specifically, Table 3206.2 sets out when smoke and heat removal is required. If Table 3206.2 does not require such protection, compliance with Section 910 is not necessary. The requirement in Table 3206.2 is based primarily on the size of the high-piled combustible storage area and whether the area is equipped with an automatic sprinkler system.

There are several requirements provided within this section. The first addresses nonsprinklered high-piled storage areas. Such areas are required to use smoke and heat vents as mechanical smoke removal systems are designed for use in sprinklered buildings. The rationale for this provision is that a mechanical smoke removal system capable of handling temperatures between 1,000°F and 2,000°F cannot be practically provided at a reasonable cost.

Where high-piled storage areas are equipped with an automatic sprinkler system, smoke and heat vents or a smoke removal system is required to comply with this section. However, if the high-piled storage area is located in a multiple-story building where the storage area is not located on the uppermost story, this section requires that a smoke removal system be used.

**910.3 Smoke and heat vents.** The design and installation of smoke and heat vents shall be in accordance with Sections 910.3.1 through 910.3.3.

**C** This section simply sets out the subsections that must be addressed to comply with the requirements for the installation of smoke and heat vents.

**910.3.1 Listing and labeling.** Smoke and heat vents shall be *listed* and *labeled* to indicate compliance with UL 793 or FM 4430.

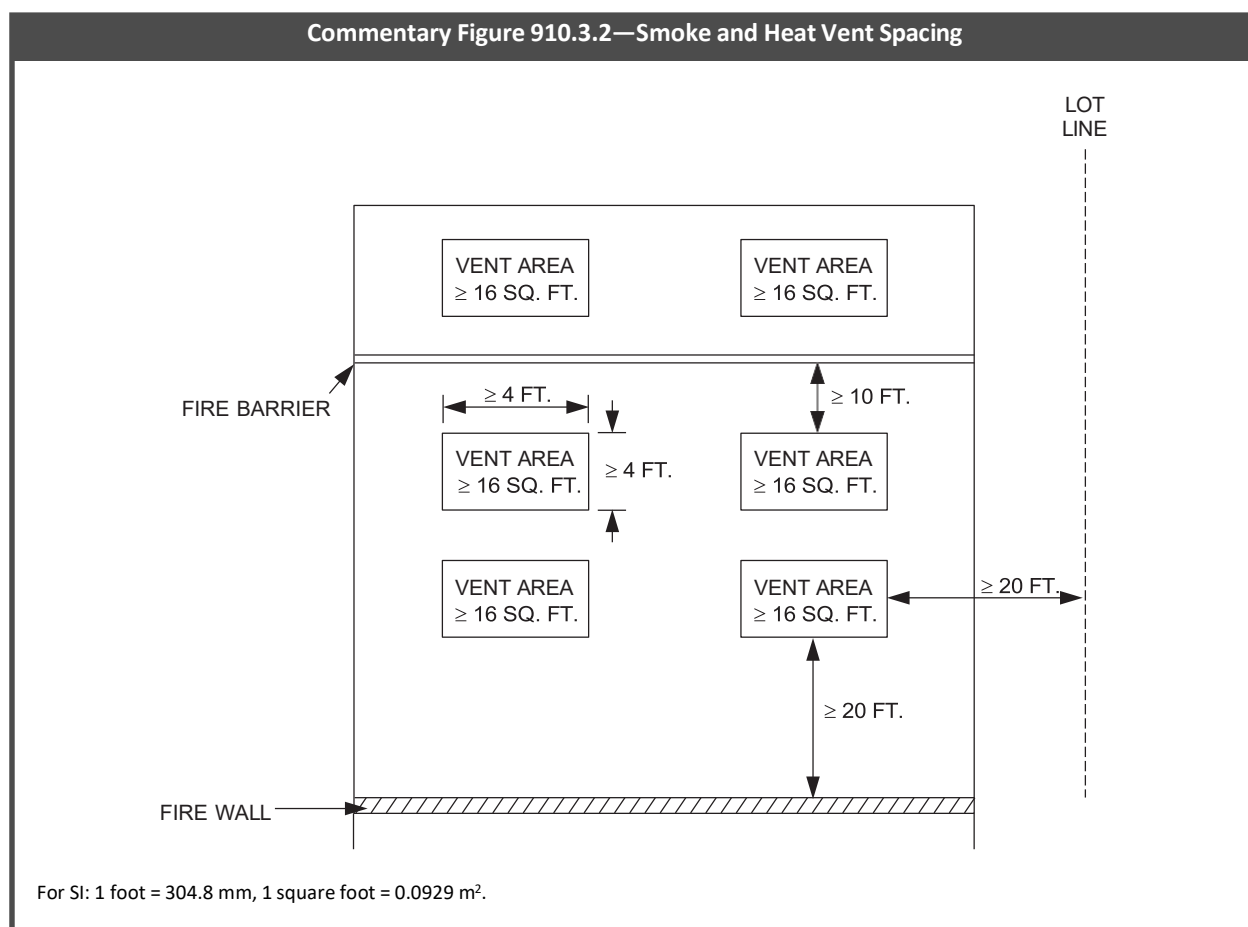
**C** This section specifically requires that all smoke and heat vents be both listed and labeled in accordance with UL 793 or FM 4430. This provides consistency and a level of quality where smoke and heat vents are required. The standard addresses smoke and heat vents that automatically operate during fires via nonelectrical means. Automatic vents listed and labeled to this standard can be operated both automatically and manually. There are two main mechanisms for activation that include a heat-responsive device or a plastic cover shrinking and falling out of place because of fire exposure.

**910.3.2 Smoke and heat vent locations.** Smoke and heat vents shall be located 20 feet (6096 mm) or more from adjacent *lot lines* and *fire walls* and 10 feet (3048 mm) or more from *fire barriers*. Vents shall be uniformly located within the roof in the areas of the building where the vents are required to be installed by Section 910.2, with consideration given to roof pitch, sprinkler location and structural members.

**C** This section has two functions, the first being to focus on hazards to adjacent buildings and the second being proper function of smoke and heat vents through proper placement.

In terms of adjacent properties, this section requires a minimum distance to lot lines and fire walls and then a minimum distance to fire barriers. The first set of distances focuses on separate buildings and exposures, whereas the distance to fire barriers is less restrictive since it focuses on different uses and occupancies within the same building (see Commentary Figure 910.3.2).

To enhance vent performance within the area containing the smoke and heat vents, such vents need to be uniformly spaced. Consideration of issues such as sprinkler location and roof pitch are also essential to proper vent location.



**910.3.3 Smoke and heat vents area.** The required aggregate area of smoke and heat vents shall be calculated as follows: For buildings equipped throughout with an *automatic sprinkler system* in accordance with Section 903.3.1.1:

**Equation 9-3**  $AVR = V/9000$

where:

$A_{VR}$  = The required aggregate vent area (ft<sup>2</sup>).

$V$  = Volume (ft<sup>3</sup>) of the area that requires smoke removal.

For unsprinklered buildings:

**Equation 9-4**  $AVR = A_{FA}/50$

where:

$A_{VR}$  = The required aggregate vent area (ft<sup>2</sup>).

$A_{FA}$  = The area of the floor in the area that requires smoke removal.

**C** This section provides the design criteria to determine the required area of smoke and heat vents. The requirements are based on whether the area requiring smoke and heat vents is equipped with an automatic sprinkler system.

The design of roof vents in buildings protected by an automatic sprinkler system require that the area of roof vents provide equivalent venting to that required for the mechanical smoke removal system (two air changes per hour) based on an assumption that each square foot of vent area will provide 300 cubic feet per minute (cfm) of ventilation. The reason for this requirement is that the roof vents should at least provide venting equivalent to the minimum venting provided by the mechanical smoke removal system. A factor of 300 cfm of venting per square foot of vent area was included in the 2012 edition of the code, although the use of this conversion factor is questionable at best. The actual ventilation provided by each square foot of vent area will depend on the temperature differential between ambient conditions and the smoke layer under the roof deck or the pressure achieved if positive pressure ventilation is utilized. If the prescribed value is not practical for a given building design, designers have the option of demonstrating other values that provide the same performance under Section 104.2.3 of the code, which allows alternative methods and designs.

The design of roof vents in buildings not protected by a sprinkler system requires that the ratio of the area of the vents to the floor area be a minimum of 1:50. The rationale is that the case where roof vents will be provided without sprinkler protection will be rare (e.g., buildings that contain high-piled storage with an area between 2,500 and 12,000 square feet). Given that this situation will be rare, a complex analysis to determine the required area of roof vents was believed to be unnecessary. The ratio of vent area to floor area of 1:50 is conservative based on the requirements that were included in the 2012 IBC and the code.

**910.3.4 Vent operation.** Smoke and heat vents shall be capable of being operated by *approved* automatic and manual means.

**C** While this requirement is found in Section 5.3.5 of NFPA 204, *Standard for Smoke and Heat Venting*, the standard is not referenced for installation. But even if it was referenced for installation, this requirement could still be missed since the UL test standard does not require a manual release. UL 793 states that IF a manual release is provided, it will be tested. However, there are UL-listed smoke and heat vents both with and without a manual release. This section brings the requirement to the attention of the designer, plan reviewer and inspector. The manual release is critical since the smoke and heat vent most likely will not open automatically in a sprinklered building if the sprinkler system is adequately cooling the ceiling temperatures. Therefore, the intent is that firefighters are to have the ability to open the smoke and heat vents manually.

**910.3.5 Fusible link temperature rating.** Where vents are installed in areas provided with automatic fire sprinklers and the vents operate by fusible link, the fusible link shall have a temperature rating of 360°F (182°C).

**C** There have been studies that show that draft curtains and smoke and heat vents can have negative consequences on fire sprinkler performance in storage buildings. This section provides the required temperature rating of fusible links used to operate smoke and heat vents. At least one legacy code, the *Uniform Fire Code*, required this link rating to be between 350°F and 400°F, and Factory Mutual guidance indicates 360°F is an appropriate temperature rating. In addition, the requirement of 360°F is in order to ensure that the fire sprinklers activate prior to opening of the smoke and heat vents.

**910.4 Mechanical smoke removal systems.** Mechanical smoke removal systems shall be designed and installed in accordance with Sections 910.4.1 through 910.4.7.

**C** Mechanical smoke removal systems are considered to be equivalent to smoke and heat vents in terms of code compliance. In multiple-story buildings, those areas requiring smoke and heat removal will require the use of a mechanical smoke removal system because of its location on stories other than the uppermost story. This section provides the various design requirements for such systems.

**910.4.1 Automatic sprinklers required.** The building shall be equipped throughout with an *approved automatic sprinkler system* in accordance with Section 903.3.1.1.

**C** The rationale for this provision is that a mechanical smoke removal system capable of handling temperatures between 1,000°F and 2,000°F cannot be practically provided at a reasonable cost. Therefore, in order to allow the use of a mechanical smoke removal system, the building is required to be equipped throughout with an automatic sprinkler system.

**910.4.2 Exhaust fan construction.** Exhaust fans that are part of a mechanical smoke removal system shall be rated for operation at 221°F (105°C). Exhaust fan motors shall be located outside of the exhaust fan airstream.

**C** This section requires exhaust fan motors to be located out of the exhaust stream to protect the mechanical equipment from excessive heat. Provisions for the mechanical smoke removal system permit the system to be designed to handle air at ambient temperature provided that the fan motors are located outside the air stream. The basis for this provision is the thermocouple temperature data for the large-scale fire tests conducted at UL in 1997/1998, specifically Tests P-1 and P-4. (In Tests P-1 and P-4, no vents opened so the ceiling temperatures recorded would be unaffected by the activation of vents. See pages 40 and 52 of the NISTIR 6196-1 report dated September 1998 for the thermocouple temperature data recorded as a function of time.)

The exposing temperatures and time periods were reviewed and not considered to pose a threat to the building structure, fans or power wiring.

The sprinkler activation times and ceiling temperature data for the five large-scale fire tests summarized in NISTIR 6196-1 indicate that the exposure of mechanical exhaust fans and ducts located at the ceiling to high temperatures will be relatively short. Since it is anticipated that the exhaust system will only be activated after the arrival of firefighters at the scene (estimated to be 7 minutes or longer after ignition), ceiling temperatures should be reduced sufficiently to allow fans rated for only ambient temperatures to be used for the exhaust system.

**910.4.3 System design criteria.** The mechanical smoke removal system shall be sized to exhaust the building at a minimum rate of two air changes per hour based on the volume of the building or portion thereof without contents. The capacity of each exhaust fan shall not exceed 30,000 cubic feet per minute (14.2 m<sup>3</sup>/s).

**C** Sections 910.4.3 and 910.4.3.1 specify the design requirements for the minimum number of air changes, maximum fan capacity, and the provision of makeup air.

These provisions require that the mechanical smoke removal system be sized to provide a minimum exhaust rate of two air changes per hour based on the enclosed volume of the building space to be exhausted, without any deductions for the space occupied by storage or equipment. An exhaust rate of two air changes per hour is based on an analysis assuming a conservative approach using a Factory Mutual Research Corporation Standard Plastic Commodity (polystyrene cups in compartmented cartons). This commodity is recognized to represent a severe fire hazard of high-density plastics.

In a calculation based on this commodity, a maximum of 68,960 cfm of smoke was generated by the design fire. Based on an empty building volume of 2.659 million cubic feet, the exhaust rate required to achieve two air changes per hour is 88,633 cfm. Because no single fan can exceed 30,000 cfm, this building required five fans, each exhausting 25,570 cfm for a total of 127,850 cfm. This exceeds the minimum two air changes per hour by more than 40 percent. Even at the minimum required rate of two air changes per hour, the calculation results show that the mechanical smoke removal system will be capable of removing the smoke from the building faster than it will be generated, ultimately removing smoke from the building.

**910.4.3.1 Makeup air.** Makeup air openings shall be provided within 6 feet (1829 mm) of the floor level. Operation of makeup air openings shall be manual or automatic. The minimum gross area of makeup air inlets shall be 8 square feet per 1,000 cubic feet per minute (0.74 m<sup>2</sup> per 0.4719 m<sup>3</sup>/s) of smoke exhaust.

**C** In order for a mechanical smoke removal system to work properly, makeup air at the proper location and volume needs to be provided. Generally, makeup air inlets need to be located much lower than where the smoke is exhausted to get the proper movement of air. In this case, the required location is within 6 feet of the floor.

The derivation of the gross vent area is based on NFPA 92—2012. Specifically, Annex Section A-4.4.4.1.4 states that the maximum air velocity through the makeup air inlet is 1 m/s or 200 ft/min. This is the same limitation found in Section 909.7.1, which is for the airflow method of smoke control. The area requirement is then derived as follows:

- Effective Vent Area = (1,000 ft<sup>3</sup>/min)/(200 ft/min) = 5 ft<sup>2</sup> per 1,000 cfm.
- Assume an orifice coefficient of 0.6.
- Gross Vent Area = 5 ft<sup>2</sup>/(0.6) = 8.33 ft<sup>2</sup> per 1,000 cfm, which is rounded down because of the conservative nature of the requirement.

The reason for this limitation is to prevent significant deflection of the plume, which will cause more air entrainment into the plume and more smoke production. This criterion is conservative as the requirement assumes an active fire and the design philosophy for this requirement is to provide post-fire smoke exhaust.

**910.4.4 Activation.** The mechanical smoke removal system shall be activated by manual controls only.

**C** This section requires that mechanical systems are to be activated manually so that the fire department is in control of the system. In some situations, automatic operation could cause a fire to grow or spread, opening an excessive number of sprinklers. Automatic operation of the mechanical smoke removal system could also be detrimental to the operation of the sprinkler system in a manner similar to draft curtains. The effect of the automatic mechanical smoke removal system on sprinkler operation would depend on when the system was activated. The sooner the system is automatically activated, the greater the detrimental effect. The fire department will retain the option to shut down the exhaust system.

**910.4.5 Manual control location.** Manual controls shall be located where they are able to be accessed by the fire service from an exterior door of the building and separated from the remainder of the building by not less than 1-hour *fire barriers* constructed in accordance with Section 707 of the *International Building Code* or *horizontal assemblies* constructed in accordance with Section 711 of the *International Building Code*, or both.

**C** This section establishes the required placement, access and protection of the manual controls to ensure that firefighters will have quick and protected access to them.

**910.4.6 Control wiring.** Wiring for operation and control of mechanical smoke removal systems shall be connected ahead of the main disconnect in accordance with Section 701.12E of NFPA 70 and be protected against interior fire exposure to temperatures in excess of 1,000°F (538°C) for a period of not less than 15 minutes.

**C** Unless the mechanical smoke removal system also functions as a component of a smoke control system, standby power is not specifically required (see commentary, Section 909.11 and IBC Section 2702). In order to provide an enhanced level of operational reliability, this section requires that the power supply to smoke exhaust fans must be provided from a circuit connected on the supply side (i.e., ahead) of the building's main electrical service disconnecting means. Note that this is one of the sources of standby power recognized by NFPA 70, Section 701.12(E). Such a circuit connected "ahead of the main" must still have its own approved overcurrent protection.

The provisions for the design of a mechanical smoke removal system indicate that wiring providing power to exhaust fans located in the interior of the building is to be protected by materials that will provide a 15-minute finish rating protection. The ceiling temperature data collected in the five large-scale fire tests summarized in NISTIR 6196-1 (cited earlier) show that temperatures at the ceiling will be far less than the exposure temperatures defined by the ASTM E119 time-temperature curve and that the ceiling temperatures will rapidly decrease once sprinklers activate. The ceiling temperature data included in NISTIR 6196-1 indicate that providing a 15-minute finish rating protection for the interior electrical power supply is more than adequate to prevent damage to the power supply wiring for the exhaust system.

**910.4.7 Controls.** Where building air-handling and mechanical smoke removal systems are combined or where independent building air-handling systems are provided, fans shall automatically shut down in accordance with the *International Mechanical Code*. The manual controls provided for the smoke removal system shall have the capability to override the automatic shutdown of fans that are part of the smoke removal system.

**C** This section requires that if a mechanical smoke removal system is integrated with a standard HVAC system, then the system must shut down upon detection of smoke as required by the IMC. This relates to the requirement in Section 910.4.4 that mechanical smoke removal systems shall be manually operated only.

The concern is that HVAC systems should not work against the intended operation of the smoke exhaust system. In some cases, the system may be a combination system where shutdown is not necessary or appropriate. It really depends on how the smoke exhaust system has been designed.

**910.5 Maintenance and testing.** Maintenance and testing of smoke and heat vents and mechanical smoke removal systems shall be in accordance with Sections 910.5.1 and 910.5.2. A written record of inspection, testing and maintenance that includes the date, identification of personnel involved, any unsatisfactory result, corrective action taken and replaced parts shall be maintained on the premises.

**C** Routine maintenance and testing of smoke removal systems is essential to ensure their performance, as designed, under fire conditions. Maintenance practices for these systems must be consistent with their design and any manufacturer's recommendations and as indicated in Section 910.5.1 or 910.5.2.

**910.5.1 Smoke and heat vents.** Smoke and heat vents shall be maintained in an operative condition. Inspection, testing and maintenance shall be in accordance with NFPA 204 except as follows:

1. Mechanically operated smoke and heat vents shall be inspected annually and operationally tested not less than every 5 years.
2. Gravity dropout smoke and heat vents shall be inspected annually.
3. Fused, damaged or painted fusible links shall be replaced.

**C** This section requires smoke and heat vents to be maintained in operating condition and incorporates NFPA 204 as the referenced standard for the maintenance of smoke and heat vents (see Section 901.6.1). Routine inspection, testing and maintenance of these devices are essential since they are typically only found in the largest commercial structures, and the

amount of fire loading is usually very high (i.e., high-piled combustible storage). Ensuring that these devices are inspected, tested and maintained in proper working order by the building's owner also has positive effects on firefighter safety. These benefits include:

- Easy identification of the location of the fire within the structure.
- Release of excess heat within the structure.
- Decreasing fire severity.
- Increased visibility for firefighters within the structure.
- Reduction of toxic products of combustion within the structure.

Additionally, the maintenance of these devices will have a mitigating effect on damage to the structure and/or its contents should a fire occur. These benefits include decreased likelihood of structural failure from heat retained within the structure and reduced damage to the structure and stored materials from smoke.

The sensing elements of fusible link devices installed as part of smoke and heat vents must be routinely visually inspected and replaced as needed to increase the likelihood that the vents will operate in a timely fashion when needed.

**910.5.2 Mechanical smoke removal systems.** Mechanical smoke removal systems shall be maintained in accordance with NFPA 204 and the equipment manufacturer's instructions except as follows:

1. Systems shall be inspected and operationally tested annually.
2. Testing shall include the operation of all system components, controls and ancillary equipment, such as makeup air openings.
3. A written schedule for routine maintenance and operational testing shall be established and testing shall be conducted in accordance with the schedule.

**C** This section references NFPA 204 with the exception of the three listed requirements. These have been added because it was determined that the maintenance requirements in NFPA 204 were not specific enough to ensure the necessary maintenance of the mechanical smoke removal systems.

Annual operational exercising of the entire mechanical smoke removal system must be performed to verify that it still operates as required by the approved design.

Because mechanical smoke removal systems are designed for smoke removal only, the operation of these systems does not adversely affect other building systems or operations. The control sequence for these systems must be included in the annual operational test to check for system component failures that may not get noticed because mechanical smoke removal systems are independent of building HVAC systems and operate only in emergency situations.

Routine operational testing and maintenance must be performed on the mechanical smoke removal system periodically to verify that it still operates as required by the approved design. A written schedule must be established and maintained.

This section prescribes the desired content of the written record for the mechanical smoke removal system testing and maintenance program. Test results and maintenance activities should be clearly documented.

The written record should be available for inspection and reviewed by the fire code official.

## SECTION 911—EXPLOSION CONTROL

**911.1 General.** Explosion control shall be provided in the following locations:

1. Where a structure, room or space is occupied for purposes involving explosion hazards as identified in Table 911.1.
2. Where quantities of hazardous materials specified in Table 911.1 exceed the maximum allowable quantities in Table 5003.1.1(1).

Such areas shall be provided with explosion (*deflagration*) venting, explosion (*deflagration*) prevention systems or *barricades* in accordance with this section and NFPA 68, NFPA 69 or NFPA 495 as applicable. *Deflagration* venting shall not be utilized as a means to protect buildings from *detonation* hazards.

**C** It is usually impractical to design a building to withstand the pressure created by an explosion. Therefore, this section requires an explosion relief system for structures, rooms or spaces with occupancies involving explosion hazards. Explosions may result from the overpressurization of a containing structure, by physical/chemical means or by a chemical reaction. During an explosion, a sudden release of a high-pressure gas occurs and the energy is dissipated in the form of a shock wave.

Structures, rooms or spaces with occupancies involving explosion hazards must be equipped with some method of explosion control as required by the material-specific requirements in the code. Table 911.1 also specifies where explosion control is required based on certain materials or occupancies where the quantities of hazardous materials involved exceed the maximum allowable quantities in Table 5003.1.1(1). Section 911 recognizes explosion (*deflagration*) venting and explosion (*deflagration*) prevention systems as acceptable methods of explosion control, where appropriate. The use of

barricades or other explosion protective devices, such as magazines, may be permitted as the means of explosion control where indicated in the code as an acceptable alternative and where approved by the fire code official.

**TABLE 911.1—EXPLOSION CONTROL REQUIREMENTS<sup>f</sup>**

| MATERIAL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | CLASS                     | EXPLOSION CONTROL METHODS |                                                                                 |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|---------------------------|---------------------------------------------------------------------------------|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                           | Barricade construction    | Explosion (deflagration) venting or explosion (deflagration) prevention systems |
| Hazard Category                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                           |                           |                                                                                 |
| Combustible dusts <sup>a</sup>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | —                         | Not required              | Required                                                                        |
| Cryogenic fluids                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Flammable                 | Not required              | Required                                                                        |
| Explosives                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Division 1.1              | Required                  | Not required                                                                    |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Division 1.2              | Required                  | Not required                                                                    |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Division 1.3              | Not required              | Required                                                                        |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Division 1.4 <sup>i</sup> | Not required              | Required                                                                        |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Division 1.5              | Required                  | Not required                                                                    |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Division 1.6              | Required                  | Not required                                                                    |
| Flammable gas                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Gaseous                   | Not required              | Required <sup>h</sup>                                                           |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Liquefied                 | Not required              | Required <sup>h</sup>                                                           |
| Flammable liquids                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | IA <sup>b</sup>           | Not required              | Required                                                                        |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | IB <sup>c</sup>           | Not required              | Required                                                                        |
| Organic peroxides                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Unclassified detonable    | Required                  | Not permitted                                                                   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | I                         | Required                  | Not permitted                                                                   |
| Oxidizer liquids and solids                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 4                         | Required                  | Not permitted                                                                   |
| Pyrophoric                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Gases                     | Not required              | Required                                                                        |
| Unstable (reactive)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 4                         | Required                  | Not permitted                                                                   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 3 detonable               | Required                  | Not permitted                                                                   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 3 nondetonable            | Not required              | Required                                                                        |
| Water-reactive liquids and solids                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 3                         | Not required              | Required                                                                        |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2 <sup>e</sup>            | Not required              | Required                                                                        |
| Special Uses                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                           |                           |                                                                                 |
| Acetylene generator rooms                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | —                         | Not required              | Required                                                                        |
| Electrochemical energy storage systems <sup>g</sup>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | —                         | Not required              | Required                                                                        |
| Energy storage systems <sup>g</sup>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | —                         | Not required              | Required                                                                        |
| Grain processing                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | —                         | Not required              | Required                                                                        |
| Liquefied petroleum gas distribution facilities                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | —                         | Not required              | Required                                                                        |
| Where explosion hazards exist <sup>d</sup>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Detonation                | Required                  | Not permitted                                                                   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Deflagration              | Not required              | Required                                                                        |
| For SI: 1 inch per second = 25.4 mm/s.<br>a. Combustible dusts where manufactured, generated or used in such a manner that the concentration and conditions create a fire or explosion hazard based on information prepared in accordance with Section 104.2.2. See definition of “Combustible dust” in Chapter 2.<br>b. Storage or use.<br>c. In open use or dispensing.<br>d. Rooms containing dispensing and use of hazardous materials where an explosive environment can occur because of the characteristics or nature of the hazardous materials or as a result of the dispensing or use process.<br>e. A method of explosion control shall be provided where Class 2 water-reactive materials can form potentially explosive mixtures.<br>f. Explosion venting is not required for Group H-5 Fabrication Areas complying with Chapter 27 and the <i>International Building Code</i> .<br>g. Where explosion control is required in Section 1207.6.3.<br>h. Not required for Category 1B Flammable Gases having a burning velocity not exceeding 3.9 inches per second.<br>i. Does not apply to consumer fireworks, 1.4G. |                           |                           |                                                                                 |

**C** This table designates where some methods of explosion control are required for specific material or special use conditions. This table applies where the quantities of hazardous materials involved exceed the maximum allowable quantities in Table 5003.1.1(1). Section 911.2 contains design criteria for explosion (deflagration) venting. Explosion prevention (suppression) systems, where used, must comply with NFPA 68 and 69. Barricade construction must be designed and installed to comply with NFPA 495. Chapters 50 through 67 of the code contain additional guidance on the applicability and design criteria for explosion control methods that depend on the specific type of hazardous material involved.

**911.2 Required deflagration venting.** Areas that are required to be provided with *deflagration* venting shall comply with the following:

1. Walls, ceilings and roofs exposing surrounding areas shall be designed to resist a minimum internal pressure of 100 pounds per square foot (psf) (4788 Pa). The minimum internal design pressure shall be not less than five times the maximum internal relief pressure specified in Item 5 of this section.
2. *Deflagration* venting shall be provided only in *exterior walls* and roofs.  
**Exception:** Where sufficient *exterior wall* and roof venting cannot be provided because of inadequate *exterior wall* or roof area, *deflagration* venting shall be allowed by specially designed shafts vented to the exterior of the building.
3. *Deflagration* venting shall be designed to prevent unacceptable structural damage. Where relieving a *deflagration*, vent closures shall not produce projectiles of sufficient velocity and mass to cause life threatening injuries to the occupants or other persons on the property or adjacent *public ways*.
4. The aggregate clear area of vents and venting devices shall be governed by the pressure resistance of the construction assemblies specified in Item 1 of this section and the maximum internal pressure allowed by Item 5 of this section.
5. Vents shall be designed to withstand loads in accordance with the *International Building Code*. Vents shall consist of any one or any combination of the following to relieve at a maximum internal pressure of 20 pounds per square foot (958 Pa), but not less than the loads required by the *International Building Code*:
  - 5.1. *Exterior walls* designed to release outward.
  - 5.2. Hatch covers.
  - 5.3. Outward swinging doors.
  - 5.4. Roofs designed to uplift.
  - 5.5. Venting devices *listed* for the purpose.
6. Vents designed to release from the *exterior walls* or roofs of the building when venting a *deflagration* shall discharge directly to the exterior of the building where an unoccupied space not less than 50 feet (15 240 mm) in width is provided between the *exterior walls* of the building and the *lot line*.  
**Exception:** Vents complying with Item 7 of this section.
7. Vents designed to remain attached to the building when venting a *deflagration* shall be so located that the discharge opening shall be not less than 10 feet (3048 mm) vertically from window openings and *exits* in the building and 20 feet (6096 mm) horizontally from *exits* in the building, from window openings and *exits* in adjacent buildings on the same lot and from the *lot line*.
8. Discharge from vents shall not be into the interior of the building.

**C** This section prescribes the basic design criteria necessary for deflagration venting.

Deflagration venting limits the deflagration pressure in a certain area so that, in case of an explosion, the damage to that enclosed area is minimized or eliminated. Because there are so many variables involved for adequate deflagration venting, the parameters for each design should fit the individual situation. NFPA 68 contains additional guidance on the design and use of deflagration venting systems.

The area of the vent must be adequate to relieve the pressure before it reaches a level in excess of what can be withstood by the weakest building member. The vent area, therefore, is dependent on the actual construction of the enclosed area and the anticipated pressure. The vent panel should be of lightweight construction so that it can easily release at low pressures. Because the lightweight panels have little structural strength, railings may be required along the floor edge to prevent people or objects from falling against the panel.

Item 5 indicates that the vents are to be designed to relieve at a maximum internal pressure of 20 pounds per square foot (psf) (958 Pa), but not less than the load design requirements in IBC Chapter 16. In areas commonly subject to high winds, the release pressure has to be increased accordingly to prevent the vents from being actuated by wind forces. Even though the release pressure should be as low as practical, it must always be higher than the external wind pressure.

Venting devices must be located to discharge directly to the open air or to an unoccupied space at least 50 feet (15 240 mm) in width on the same lot. To minimize damage and maintain the integrity of the existing system, window openings and egress facilities are not to be within 10 feet (3048 mm) vertically or 20 feet (6096 mm) horizontally of the vent. The spatial distance will permit the pressure to decrease and to not cause additional damage.

**911.3 Explosion prevention systems.** Explosion prevention systems shall be of an *approved* type and installed in accordance with the provisions of this code and NFPA 69.

**C** Depending on the conditions of the anticipated explosion hazard, the use of an explosion prevention system may be an effective means of explosion control. An explosion prevention system is most effective in confined spaces or enclosures in which combustible gases, mists or dusts are subject to deflagration in a gas-phase oxidant. Explosion prevention systems are intended to prevent an explosion hazard by combating the process of combustion in its incipient stage.

NFPA 69 contains further information on the installation, operation and design considerations for explosion prevention systems. Explosion prevention systems are commonly used to protect laboratory equipment, such as reactor vessels, mills and dust collectors.

**911.4 Deflagration venting.** *Deflagration* venting shall be of an *approved* type and installed in accordance with the provisions of this code and NFPA 68.

**C** This section requires that deflagration venting be acceptable to the fire code official and references NFPA 68, *Standard on Explosion Protection by Deflagration Venting*, for additional specific installation requirements.

**911.5 Barricades.** *Barricades* shall be designed and installed in accordance with NFPA 495.

**C** As indicated in Table 911.1, depending on the type of materials involved, barricade construction may be an acceptable method of explosion control. Barricade construction is an effective method of screening a building containing explosives from other buildings, magazines or public rights-of-way. The barricade could be either natural or artificial, where applicable, as specified in NFPA 495.

## SECTION 912—FIRE DEPARTMENT CONNECTIONS

**912.1 Installation.** Fire department connections shall be installed in accordance with the NFPA standard applicable to the system design and shall comply with Sections 912.2 through 912.7.

**C** An FDC is required as part of a water-based suppression system as the auxiliary water supply. These connections give the fire department the capability of supplying the necessary water to the automatic sprinkler or standpipe system at a sufficient pressure. The FDC also serves as an alternative source of water should a valve in the primary water supply be closed. A fire department connection does not, however, constitute an automatic water source. See Commentary Figure 903.3.1.1 for a typical FDC arrangement on a wet-pipe sprinkler system.

The requirements for the FDC depend on the type of sprinkler system installed and whether a standpipe system is installed. NFPA 13 and 13R, for example, include design considerations for FDCs that are an auxiliary water supply source for automatic sprinkler systems; NFPA 14 is the design standard to use for FDCs serving standpipe systems. Threads for FDCs to sprinkler systems, standpipes, yard hydrants or any other fire hose connection must be approved (NFPA 1963 may be utilized as part of the approval or as otherwise approved) and be compatible with the connections used by the local fire department (see commentary, Sections 903.3.6 and 905.1).

**912.2 Location.** With respect to hydrants, driveways, buildings and landscaping, fire department connections shall be so located that fire apparatus and hose connected to supply the system will not obstruct access to the buildings for other fire apparatus. The location of fire department connections shall be *approved* by the *fire code official*.

**C** This section specifies that the FDC must be located so that vehicles and hose lines will not interfere with access to the building for the use of other fire department apparatus. The location of potential connected hose lines to the FDC and hydrants must be preplanned with the fire department. Many fire departments have a policy restricting the distance that an FDC may be from a fire hydrant. Some also have policies that indicate the maximum distance from the nearest point of fire department vehicle access (often, the curb). Since fireground operations are based on local operational procedures, it is only reasonable that the fire chief of the jurisdiction has approval authority over the location of and access to the FDC.

Landscaping can also be a hindrance to fire department operations. Even where the FDC is visible, the extensive use of landscaping may make access difficult. Landscaping also changes over time. What may not have been an obstruction when it was planted can sometimes grow into an obstruction over time.

**912.2.1 Visible location.** Fire department connections shall be located on the street side of buildings or facing *approved* fire apparatus access roads, fully visible and recognizable from the street, fire apparatus access road or nearest point of fire department vehicle access or as otherwise *approved* by the *fire code official*.

**C** FDCs must be readily visible and easily accessed. A local policy constituting what is readily visible and accessible needs to be established. While the intent is clearly understandable, its application can vary widely. A precise policy is the best way to avoid ambiguous directives that result in inconsistent and arbitrary enforcement. Usually, the policy will address issues such as location on the outside of the building and proximity to fire hydrants.

Landscaping is often used to hide the FDCs from the public. This can greatly hamper the efforts of the fire department in staging operations and supplying water to the fire protection systems. Landscaping must be designed so that it does not obstruct the visibility of the FDC. Since fireground operations are based on local operational procedures, it is only reasonable that the fire chief of the jurisdiction has final approval authority over the visibility of and access to the FDC.

**912.2.2 Existing buildings.** On existing buildings, wherever the fire department connection is not visible to approaching fire apparatus, the fire department connection shall be indicated by an *approved* sign mounted on the street front or on the side of the building. Such sign shall have the letters "FDC" not less than 6 inches (152 mm) high and words in letters not less than 2 inches (51 mm) high or an arrow to indicate the location. Such signs shall be subject to the approval of the *fire code official*.

**C** The section acknowledges that FDCs on existing buildings may not always be readily visible from the street or nearest point of fire department vehicle access. In those instances, the location of the connection must be clearly marked with signage.

The FDC may be located on the side of the building or in an alley, not visible to arriving firefighting forces. A sign is necessary so that those driving the arriving apparatus know where to maneuver the vehicle to get close to the FDC.

**912.3 Fire hose threads.** Fire hose threads used in connection with standpipe systems shall be *approved* and shall be compatible with fire department hose threads.

**C** There are several sections in the code that contain requirements for fire department connections. This section simply correlates with those requirements by further clarifying that fire hose threads for standpipe systems must be approved.

**912.4 Access.** Immediate access to fire department connections shall be maintained at all times and without obstruction by fences, bushes, trees, walls or any other fixed or moveable object. Access to fire department connections shall be *approved* by the *fire code official*.

**Exception:** Fences, where provided with an access gate equipped with a sign complying with the legend requirements of Section 912.5 and a means of emergency operation. The gate and the means of emergency operation shall be *approved* by the *fire code official* and maintained operational at all times.

**C** The FDC must be readily accessible to firefighters and allow firefighting personnel an adequate area to maneuver a hose for the connection. Landscaping design must not block a clear view of the FDC from arriving fire department vehicles. Depending on the type of landscaping materials, an active maintenance program may be necessary to maintain ready access over time. This section also recognizes that the obstructing objects regulated here can be either fixed or moveable (such as outdoor furnishings, shopping cart queue areas, etc.). Note that no specific dimension is given as was the case in previous editions of the code. This performance language avoids previous misinterpretations that the code intended to allow obstructions to FDC access as long as they were kept 3 feet (914 mm) away. Since fireground operations are based on local operational procedures, it is only reasonable that the fire chief of the jurisdiction have final approval authority over the access to the FDC.

The exception recognizes the practical fact that sometimes security or other considerations make installation of a fence around a building necessary as long as the fence meets the stated criteria. The sign requirement intends to provide a visual location cue to approaching fire apparatus where the height of the fence may obscure the visibility of the FDC.

**912.4.1 Locking fire department connection caps.** The *fire code official* is authorized to require locking caps on fire department connections for water-based *fire protection systems* where the responding fire department carries appropriate key wrenches for removal.

**C** This section allows for the FDC caps to be equipped with locks as long as the fire departments that respond to that building or facility have the appropriate key wrenches. This avoids vandalism and affords a more functional FDC when needed. Locking caps, even more so than regular FDC caps, need proper maintenance so that they can be removed when required. Any time that an additional mechanical function is added to something that is exposed to the elements, it must be done with the understanding that the corrosive nature of the elements can place the FDC out of commission if the cap cannot be removed (see Commentary Figure 912.4.1).

Commentary Figure 912.4.1—Locking FDC Caps



Photo courtesy of Knox Company

**912.4.2 Clear space around connections.** A working space of not less than 36 inches (914 mm) in width, 36 inches (914 mm) in depth and 78 inches (1981 mm) in height shall be provided and maintained in front of and to the sides of wall-mounted fire department connections and around the circumference of free-standing fire department connections, except as otherwise required or *approved* by the *fire code official*.

**C** Care must be taken so that fences, utility poles, barricades and other obstructions do not prevent access to and use of FDCs. A clear space of 3 feet (914 mm) must be maintained in front of and to either side of wall-mounted FDCs and around free-standing FDCs to allow easy hose connections to the fitting and efficient use of spanner wrenches and other tools needed by the apparatus engineer.

Though not specifically mentioned in this section, it is also important that FDCs be installed with the hose connections well above adjoining grade to accommodate the free turning of a spanner wrench when connecting hoses to the FDC.

**912.4.3 Physical protection.** Where fire department connections are subject to impact by a motor vehicle, vehicle impact protection shall be provided in accordance with Section 312.

**C** Section 312 requires vehicle impact protection by placing steel posts filled with concrete around the FDC. Section 312 gives the specifications for the posts.

**912.5 Signs.** A metal sign with raised letters not less than 1 inch (25 mm) in size shall be mounted on all fire department connections serving automatic sprinklers, standpipes or fire pump connections. Such signs shall read: "AUTOMATIC SPRINKLERS," "STAND-PIPES," "TEST CONNECTION," "STANDPIPE AND AUTOSPKR" or "AUTOSPKR AND STANDPIPE," or a combination thereof as applicable.

**C** The purpose of the sign is to provide the responding firefighters with the correct information on which portions of a building are served by the fire department connection. They identify the type of system or zone served by a given FDC.

Signs may also distinguish FDCs from fire pump test headers. Usually, FDCs may be distinguished from fire pump test headers by the types of couplings provided. FDCs are customarily equipped with female couplings, while fire pump test headers usually have separately valved male couplings. Furthermore, fire pump test headers are equipped with one 2 1/2-inch (64 mm) outlet for each 250 gallons per minute (gpm) (16 L/s) of rated capacity.

Raised letters are required so that any repainting or fading of the colors on the sign will not affect its ability to be read. Each letter must be at least 1 inch (25 mm) in height so that the wording is clear. Often the wording may be abbreviated such that "AUTOMATIC SPRINKLERS" reads as "AUTO. SPKR." Existing signs may use language slightly different than that noted in the code. As long as the information is adequately communicated, there should be no reason to require new signage to replace existing ones (see Commentary Figure 912.5).

Commentary Figure 912.5—Fire Department Connection with Sign



**912.5.1 Lettering.** Each fire department connection (FDC) shall be designated by a sign with letters not less than 1 inch (25.4 mm) in height. For manual standpipe systems, the sign shall also indicate that the system is manual and that it is either wet or dry.

**C** This section includes the specific requirements for the sign letters. Each letter must be at least 1 inch (25 mm) in height so that the wording is clear and can be seen from a distance. In addition, the sign lettering for manual standpipe systems must indicate the type of system it serves so that it can be utilized quickly.

**912.5.2 Serving multiple buildings.** Where a fire department connection (FDC) services multiple buildings, structures or locations, a sign shall be provided indicating the building, structures or locations served. Where the FDC does not serve the entire building, a sign shall be provided indicating the portions of the building served.

**C** If the fire department connection serves multiple buildings, the lettering must indicate which buildings it is connected to so that it can be effectively utilized. Some buildings may have only a partial sprinkler system, such as rehabilitated buildings where a sprinkler system is only installed on certain floors or a building that only has basement sprinklers in accordance with Section 903.2.11.

**912.5.3 Multiple or combined systems.** Where combination or multiple system types are supplied by the fire department connection, the sign or combination of signs shall indicate both designated services.

**C** Many buildings include multiple sets of fire department connections that are not interconnected, such as separate connections for the building sprinkler system and the dry standpipe system in open parking structures. There may also be connections that serve multiple or combined systems. Thus, the signs for these connections are required to include lettering to clearly distinguish the specific system or systems it serves.

**912.5.4 Indication of pressure.** The sign also shall indicate the pressure required at the outlets to deliver the standpipe system demand.

**Exception:** Where the pressure required is 150 pounds per square inch (1034 kPa) or less.

**C** Standpipe sign lettering must indicate the required demand pressure of the system it is connected to so that it can be effectively utilized if the pumping capability is available. The exception clarifies that this requirement only applies to with pressure greater than 150 psi.

**912.6 Backflow protection.** The potable water supply to automatic sprinkler and standpipe systems shall be protected against backflow as required by the *International Plumbing Code*.

**C** IPC Section 608.17.4 requires all connections to automatic sprinkler systems and standpipe systems to be equipped with a means to protect the potable water supply. The means of backflow protection can be either a double check-valve assembly or a reduced-pressure-principle backflow preventer. This, in general, assumes an FDC is required. For example, a limited-area sprinkler system off the domestic supply does not necessarily require an FDC and would not require backflow protection.

**912.7 Inspection, testing and maintenance.** Fire department connections shall be periodically inspected, tested and maintained in accordance with NFPA 25. Records of inspection, testing and maintenance shall be maintained.

**C** Because FDCs are components of a water-based extinguishing system, NFPA 25 is applicable. Inspections must determine that connections are unobstructed, well-protected and in good working order. Plugs or covers must be installed to protect threads and pipe openings and must be easily removed to permit connection of a fire hose.

Caps or plugs must be kept in place whenever the connection is not in use to discourage the insertion of objects into the connection openings. The interior of piping behind connection clappers must be checked for foreign material and obstructions.

Threads must be compatible with local fire service hose couplings and free of burrs, depressions and other flaws. Couplings must spin freely. Clappers, if installed in the pipe openings, must open easily and automatically return to the closed position.

Exposed piping, fittings, valves and couplings must be free of water where subject to freezing. Defects must be corrected without delay. These and other maintenance features are addressed in NFPA 25. This section also requires that written inspection and maintenance records be kept. Such records should indicate the date and time and the name of the person conducting the inspection or maintenance. These records must be maintained by the owner and should be made available to the fire code official for review when requested. This requirement relieves the fire code official of the administrative burden of maintaining test records.

## SECTION 913—FIRE PUMPS

**913.1 General.** Where provided, fire pumps for *fire protection systems* shall be installed in accordance with this section and NFPA 20.

**Exception:** Pumps for *automatic sprinkler systems* installed in accordance with Section 903.3.1.3, or Section P2904 of the *International Residential Code*.

**C** This section contains specific installation requirements for fire pumps supplying water to fire protection systems. Inspection, testing and maintenance requirements comply with NFPA 20 unless noted otherwise. Applicable maintenance standards are also identified.

Fire pumps are installed in sprinkler and standpipe systems to pressurize the water supply for the minimum required sprinkler and standpipe operation. They are considered a design feature or component of the system. Fire pumps can improve only the pressure of the incoming water supply, not the volume of water available.

Where the volume from a water supply is not adequate to supply sprinkler or standpipe demand, water tanks for private fire protection, improvements in the size and capacity of fire mains or water distribution systems, or all of these for the installation of a fire pump are needed.

Where fire pumps are required to meet the pressure requirements of sprinkler and standpipe systems, they must be installed and tested in accordance with NFPA 20.

The exception recognizes that NFPA 13D does not require fire pumps or any pumps to follow NFPA 20. The intent is that this section does not require pumps used in NFPA 13D systems to be listed for fire protection.

**913.2 Protection against interruption of service.** The fire pump, driver and controller shall be protected in accordance with NFPA 20 against possible interruption of service through damage caused by explosion, fire, flood, earthquake, rodents, insects, windstorm, freezing, vandalism and other adverse conditions.

**C** This section lists hazards that must be taken into account when determining the extent of protection required for the fire pump and its auxiliary equipment. A pump room in a building that is protected against the listed hazards in compliance with the code would be considered in compliance. Because fire pumps are also typically located in separate detached structures, geographical and security issues must also be considered.

**913.2.1 Protection of fire pump rooms.** Rooms where fire pumps are located shall be separated from all other areas of the building in accordance with Section 913.2.1 of the *International Building Code*.

**C** The purpose of this section is to require indoor fire pump room separation by fire barriers and horizontal assemblies in accordance with the requirements in the IBC. See the commentary to IBC Section 913.2.1 for a complete discussion of the requirements.

**913.2.2 Circuits supplying fire pumps.** Cables used for survivability of circuits supplying fire pumps shall be protected using one of the following methods:

1. Cables used for survivability of required *critical circuits* shall be *listed* in accordance with UL 2196 and shall have a *fire-resistance rating* of not less than 1 hour.
2. Electrical circuit protective systems shall have a *fire-resistance rating* of not less than 1 hour. Electrical circuit protective systems shall be installed in accordance with their listing requirements.
3. Construction having a *fire-resistance rating* of not less than 1 hour.
4. The cable or raceway is encased in a minimum of 2 inches (51 mm) of concrete.

**Exception:** This section shall not apply to cables, or portions of cables, located within a fire pump room or generator room that is separated from the remainder of the occupancy with *fire-resistance-rated* construction.

**C** This section is provided to better protect cables used for survivability of circuits associated with fire pumps by referencing the appropriate standard. UL 2196 is the ANSI-approved standard for tests of fire-resistive cables. NFPA 20 includes selective survivability requirements to ensure integrity of certain critical circuits. NFPA 70 does not specify the applicable standard within its mandatory provisions but recognizes electrical circuit protective systems as an alternative to listed cables. An electrical circuit protective system is a field assembly of components that must be installed according to the listing requirements and manufacturer's instructions in order to maintain the listing for the system. There are more than two dozen electrical circuit protective systems listed in the UL Fire Resistance Directory.

IBC Section 913.2.1 requires that the fire pump room be separated by either 1-hour or 2-hour construction, depending on whether the building is high-rise. The exception clarifies that where the cables are located within the fire pump room or generator room, they are not required to be protected. The fire-resistance rating is not specified since it could be either 1-hour or 2-hour based on the requirements of other code sections.

**913.3 Temperature of pump room.** Suitable means shall be provided for maintaining the temperature of a pump room or pump house, where required, above 40°F (5°C).

**C** As previously noted for sprinkler systems, standpipe systems and other water-based fire protection systems, pump rooms or pump houses must be maintained at a temperature of 40°F (4°C) or above to prevent the system from freezing. This is consistent with Section 5.12.2.1 of NFPA 20.

**913.3.1 Engine manufacturer's recommendation.** Temperature of the pump room, pump house or area where engines are installed shall never be less than the minimum recommended by the engine manufacturer. The engine manufacturer's recommendations for oil heaters shall be followed.

**C** The engine manufacturer's recommendation must be compiled with where the recommended minimum temperature is higher than the minimum established in Section 913.3. Maintaining the desired engine temperature enhances the startability of the engine. Maintaining water heaters and oil heaters as required for diesel engines, for example, will improve the starting capabilities of the fire pump and reduce engine wear and the drain on batteries.

**913.4 Valve supervision.** Where provided, the fire pump suction, discharge and bypass valves, and isolation valves on the backflow prevention device or assembly shall be supervised open by one of the following methods:

1. Central-station, proprietary or remote-station signaling service.
2. Local signaling service that will cause the sounding of an audible signal at a *constantly attended location*.
3. Locking valves open.
4. Sealing of valves and *approved* weekly recorded inspection where valves are located within fenced enclosures under the control of the *owner*.

**C** As was the case with sprinkler systems, water control valves that are a part of the fire pump installation must be supervised in the open position so that the system is operational when needed and also to reduce the chance of a system failure (see commentary, Section 903.4.1). In most cases the required water-based extinguishing system, of which the fire pump is an integral component, will be electrically supervised. Locking or sealing valves open as the only means of supervision may not be permitted, depending on the type of valve. Section 903.4.1, for example, specifically exempts jockey pump control valves from being electrically supervised if they are sealed or locked in the open position.

**913.4.1 Test outlet valve supervision.** Fire pump test outlet valves shall be supervised in the closed position.

**C** Fire pump test outlet valves are for performance testing of the fire pump and do not control the available water supply to either a sprinkler system or a standpipe system. These valves are normally in a closed position and are supervised accordingly.

**913.5 Testing and maintenance.** Fire pumps shall be inspected, tested and maintained in accordance with the requirements of this section and NFPA 25. Records of inspection, testing and maintenance shall be maintained.

**C** Fire pumps require periodic maintenance so that they will perform as required. Monthly maintenance includes running the pump at churn to exercise the pump and driver. Pump packings and relief valve settings must be adjusted as needed.

Annually, the pump must be retested to verify its proper performance. Pressure, flow, revolutions per minute, voltage and, for electric motor-driven pumps, voltage and amperage readings must be recorded, plotted and compared with original design criteria. Upon completion of testing and maintenance, the pump must be left in the automatic-start condition, ready for service. Because a fire pump is a component of a water-based extinguishing system, NFPA 25 is applicable. If the fire pump is powered by a liquid fuel, such as diesel, it is important that the fuel supply be replenished as soon as possible after the test.

**913.5.1 Acceptance test.** Acceptance testing shall be done in accordance with the requirements of NFPA 20.

**C** Chapter 14 of NFPA 20 details the procedure for conducting a fire pump acceptance test. This test is run to determine that the installation matches the sprinkler or standpipe system design criteria, the approved shop drawings and the pump manufacturer's performance specifications. The test is to be conducted in the presence of the building official in accordance with Section 901.5 by the installing contractor and representatives of the pump manufacturer and the controller manufacturer. Where the pump engine and/or transfer switch are separately supplied components, their manufacturer representative must also be present.

**913.5.2 Generator sets.** Engine generator sets supplying emergency or standby power to fire pump assemblies shall be periodically tested in accordance with NFPA 110. Records of testing shall be maintained.

**C** This section does not require emergency or standby power for all fire pump installations, but rather requires the testing of on-site generator sets that are used for emergency or standby power to fire pump assemblies. The need for emergency or standby power is typically based on occupancy conditions as indicated in the IBC. IBC Section 403.4.8, for example, requires standby power for all electrically powered fire pumps in high-rise buildings. A generator set is recognized as a permissible standby power source. NFPA 110 prescribes the operational testing requirements, including load tests, as well as the periodic inspection and maintenance for generator sets.

**913.5.3 Transfer switches.** Automatic transfer switches shall be periodically tested in accordance with NFPA 110. Records of testing shall be maintained.

**C** Automatic transfer switches are self-acting equipment that is used to transfer power from a normal source of electrical supply to an alternative supply, such as an engine generator set. NFPA 110 requires a test on each automatic transfer switch that simulates failure of the normal power source. Upon failure, the automatic transfer switch must then automatically transfer the load to the emergency power supply. Manual transfer switches are not permitted as the only means to transfer power between the normal supply and the alternative supply to the fire pump controller.

**913.5.4 Pump room environmental conditions.** Tests of pump room environmental conditions, including heating, ventilation and illumination, shall be made to ensure proper manual or automatic operation of the associated equipment.

**C** Maintaining suitable environmental conditions is essential to the proper starting capability, performance and safe operation of fire pumps and associated emergency power supplies, where required. Adequate ventilation, for example, is needed to maintain the ambient temperature in the pump room within the range recommended by the manufacturer for the emergency power supply equipment.

**SECTION 914—FIRE PROTECTION BASED ON SPECIAL DETAILED REQUIREMENTS OF USE AND OCCUPANCY**

**914.1 General.** This section shall specify where *fire protection systems* are required based on the detailed requirements of use and occupancy of the *International Building Code*.

**C** This section is intended to be a duplication of the fire protection system requirements included in Chapter 4 of the IBC located here in a single section of the code for ease of access by the user. It is organized based on the type of facility rather than the type of fire protection system required. Frequently, reference is made to other sections of the code where the requirements are to be found. Occasionally, administrative requirements are noted that do not appear in other sections. In all cases, the referenced section must be reviewed carefully since this section is intended to be subordinate to the main requirements found elsewhere in the code.

**914.2 Covered and open mall buildings.** Covered and open mall buildings shall comply with Sections 914.2.1 through 914.2.4.

**C** Various fire protection-related code sections from IBC Section 402 are included in the code so that the fire code official has convenient access to the necessary tools to address conditions and monitor new construction requirements related to fire protection systems in covered and open mall buildings. Not all covered and open mall provisions are included here. Specific information regarding kiosk placement, types of signage and other concerns are located in the IBC and can be consulted when the fire code official reviews the covered or open mall building. Although those provisions are not included here, they address issues that can have an adverse effect on the safety of occupants in the covered or open mall building.

**914.2.1 Automatic sprinkler system.** Covered and open mall buildings and buildings connected shall be equipped throughout with an *automatic sprinkler system* in accordance with Section 903.3.1.1, which shall comply with the all of the following:

1. The *automatic sprinkler system* shall be complete and operative throughout occupied space in the mall building prior to occupancy of any of the tenant spaces. Unoccupied tenant spaces shall be similarly protected unless provided with *approved* alternative protection.
2. Sprinkler protection for the mall of a covered mall building shall be independent from that provided for tenant spaces or anchor buildings.
3. Sprinkler protection for the tenant spaces of an open mall building shall be independent from that provided for anchor buildings.
4. Sprinkler protection shall be provided beneath exterior circulation balconies located adjacent to an open mall.
5. Where tenant spaces are supplied by the same system, they shall be independently controlled.

**Exception:** An *automatic sprinkler system* shall not be required in spaces or areas of open parking garages separated from the covered or open mall in accordance with Section 402.4.2.3 of the *International Building Code* and constructed in accordance with Section 406.5 of the *International Building Code*.

**C** The requirement for an automatic sprinkler system is found in IBC Section 402.5. This reference provides the fire code official with the information and tools in order to work jointly with the building official in applying the sprinkler requirement for a covered and open mall building.

The covered or open mall building and connected buildings, such as anchor buildings, must be protected with an automatic sprinkler system to protect life and property effectively. As has been discussed throughout the section, numerous allowances (such as reduced tenant separations and elimination of area limitations) are based on the effectiveness of the automatic sprinkler system.

The sprinkler system is to be designed, installed, tested and maintained in accordance with Chapter 9 and NFPA 13. Additionally, the system must be installed such that any portion serving tenant spaces in a covered mall building may be shut down independently without affecting the operation of the systems protecting the mall area. This special feature is in recognition of the frequent need to shut down the system so that changes can be made to it as a result of tenant improvements and modifications.

In an open mall building, the sprinkler systems for the tenant spaces are to be separate from those provided for the anchor buildings. It is necessary that sprinkler protection for open mall buildings be extended to the underneath side of pedestrian walkways and exterior balconies used for circulation and egress purposes. Although the open mall itself does not require sprinkler protection, it is important that those areas below overhead walkways be provided with sprinklers.

Section 909.12.4 requires operation of the sprinkler system to automatically activate the mechanical smoke control system (where an automatic control system is utilized). It is imperative that the zoning of the sprinkler system match the zoning of the smoke control system. This is necessary so that the area where water flow has occurred will also be the area from which smoke is removed.

The exception clarifies that sprinkler protection need not be extended to complying open parking garages that are adequately separated from the covered mall building, open mall building or anchor store. The allowance is consistent with the automatic sprinkler provisions of Section 903.2 where no sprinkler protection is required in open parking garages based on a Group S-2 occupancy classification.

**914.2.2 Standpipe system.** The covered and open mall building shall be equipped throughout with a standpipe system as required by Section 905.3.3.

**C** The standpipe requirements for covered and open mall buildings exist in the code, as well as in IBC Section 402.7.1, which refers to Section 905 for the requirements. For additional information regarding the requirements, see the commentary to Section 905.3.3.

**914.2.3 Emergency voice/alarm communication system.** Where the total floor area exceeds 50,000 square feet (4645 m<sup>2</sup>) within either a covered mall building or within the perimeter line of an open mall building, an emergency voice/alarm communication system shall be provided. Access to emergency voice/alarm communication systems serving a mall, required or otherwise, shall be provided for the fire department. The system shall be provided in accordance with Section 907.5.2.2.

**C** This section is similar to IBC Section 402.7.4. In covered and open mall buildings, there is a need to be able to control the large number of people that may be present. This can best be done when specific instructions are conveyed. Covered and open malls that are less than 50,000 square feet (4645 m<sup>2</sup>) in area are not required to be provided with an emergency voice alarm system. If a public address system is provided in a smaller mall, it can serve the same purpose. See the commentary to Section 907.5.2.2 for additional information.

**914.2.4 Fire department access to equipment.** Rooms or areas containing controls for air-conditioning systems or *fire protection systems* shall be identified for use by the fire department.

**C** IBC Section 402.7.5 also states this requirement. The intent is that the fire department be able to access those systems that are critical to fire protection as well as those parts of the mechanical system that can aid in control of the ventilation system to use as conditions warrant. The text is included here so that the fire code official has a means for applying the requirement and approving the means by which identification of the rooms or areas will be provided.

**914.3 High-rise buildings.** High-rise buildings shall comply with Sections 914.3.1 through 914.3.7.

**C** High-rise buildings have unique challenges that set them apart from other buildings. IBC Section 403 establishes multiple fire protection requirements for such buildings.

**914.3.1 Automatic sprinkler system.** Buildings and structures shall be equipped throughout with an *automatic sprinkler system* in accordance with Section 903.3.1.1 and a secondary water supply where required by Section 914.3.2.

**Exception:** An *automatic sprinkler system* shall not be required in spaces or areas of telecommunications equipment buildings used exclusively for telecommunications equipment, associated electrical power distribution equipment, batteries and standby engines, provided that those spaces or areas are equipped throughout with an automatic fire detection system in accordance with Section 907.2 and are separated from the remainder of the building by not less than 1-hour *fire barriers* constructed in accordance with Section 707 of the *International Building Code* or not less than 2-hour *horizontal assemblies* constructed in accordance with Section 711 of the *International Building Code*, or both.

**C** Because of the difficulties associated with manual suppression of a fire in a high-rise building, all high-rise buildings regulated by this section are required to be protected throughout with an automatic sprinkler system. The sprinkler system is to be in accordance with Section 903.3.1.1, which requires compliance with NFPA 13.

In order to increase the reliability of the sprinkler system should an earthquake disable the primary water supply, a secondary water supply is required for buildings at sites with the specified seismic design category listed in IBC Section 403.3.3. The secondary water supply must be equal to the hydraulically calculated sprinkler demand and must have a duration of not less than 30 minutes.

The exception addresses cases where a telecommunications equipment facility is part of a high-rise building. The automatic sprinkler protection can be eliminated from certain areas where an automatic sprinkler fire detection system is provided and where the telecommunications equipment areas are separated from other building areas with fire-resistance-rated construction of the specified ratings. This same exception is provided in Section 903.2 but is restated here to emphasize its application to all buildings, including high-rise buildings. This exception is based on the need to maintain uninterrupted operation of telecommunications equipment, which frequently includes emergency telephone and similar communication services. The presence of an automatic sprinkler system or other automatic fire suppression system may be detrimental to public safety should disruption of emergency services occur. See the commentary to the same exception in Section 903.2.

**914.3.1.1 Number of sprinkler risers and system design.** The number of sprinkler risers and design shall comply with Section 914.3.1.1.1 or 914.3.1.1.2 based on building height.

**C** Due to the limits of pumps, gravity tanks and equipment working pressure, 420 feet was determined as the threshold for sprinkler system design requirements in Sections 914.3.1.1.1 and 914.3.1.1.2. This threshold also coincides with NFPA 14 and 20.

**914.3.1.1.1 Buildings 420 feet or less in height.** In buildings 420 feet (128 m) or less in height, sprinkler systems shall be supplied by a single standpipe or *sprinkler express riser* within each *vertical water supply zone*.

**C** Sprinkler systems in buildings up to 420 feet are capable of functioning with one sprinkler riser, while pressure limitations and design challenges for taller buildings need multiple risers to ensure sufficient supply to the sprinkler system on the higher levels. See the commentary to Sections 914.3.1.1 and 914.3.1.1.2.

**914.3.1.1.2 Buildings over 420 feet in height.** In buildings more than 420 feet (128 m) in height, not fewer than two standpipes or *sprinkler express risers* shall supply *automatic sprinkler systems* within each *vertical water supply zone*. Each standpipe or *sprinkler express riser* shall supply *automatic sprinkler systems* on alternating floors within the *vertical water supply zone* such that two adjacent floors are not supplied from the same riser.

**C** For all high-rise buildings with a height in excess of 420 feet (128 m), this section requires the sprinkler system to be designed with additional risers for each sprinkler zone. This redundancy provides increased reliability of the automatic sprinkler system. At least two risers must be provided for each sprinkler zone. Recommendation 12 of the NIST WTC Report called for the redundancy of active fire suppression systems to be increased, thereby accommodating the greater risks associated with increasing building height and population. Providing two risers in each zone mitigates the consequences of one riser being taken out of service. Should one fail, the other will be able to supply sprinklers on the floors above and below. This will impede any fire spread and allow the fire department time to respond and extinguish the fire.

**914.3.1.1.3 Riser location.** Standpipe or *sprinkler express risers* shall be placed in *interior exit stairways* and *ramps* that are remotely located in accordance with Section 1007.

**C** This section requires standpipes or sprinkler risers to be located in the interior exit stairways and ramps of buildings. Such interior exit stairways and ramps must be separated in accordance with the requirements of Section 1007.1. This separation reduces the possibility that one incident could incapacitate both risers and protects the rated enclosures. While the section references the separation requirements of Chapter 10, IBC Section 403.5.1 can result in an even greater separation of the exit enclosures in these taller high-rise buildings. Also, IBC Section 403.2.2, regarding the integrity of the interior exit stairways, is intended to ensure that the walls of the enclosures protecting the risers meet a minimum impact standard. This increased integrity is required both for buildings taller than 420 feet and high-rise buildings in Risk Category III or IV. See the commentary to Section 1007.

**914.3.1.2 Water supply to required fire pumps.** In all buildings that are more than 420 feet (128 m) in *building height*, and buildings of Type IVA and IVB construction that are more than 120 feet (36.6 m) in *building height*, required fire pumps shall be supplied by connections to not fewer than two water mains located in different streets. Separate supply piping shall be provided between each connection to the water main and the pumps. Each connection and the supply piping between the connection and the pumps shall be sized to supply the flow and pressure required for the pumps to operate.

**Exception:** Two connections to the same main shall be permitted provided that the main is valved such that an interruption can be isolated so that the water supply will continue without interruption through not fewer than one of the connections.

**C** Fire pumps are installed in sprinkler and standpipe systems to pressurize the water supply for the minimum required sprinkler and standpipe operation. Fire pumps are only “required” to meet the system needs. Therefore, whether a particular high-rise building includes fire pumps will depend on interaction of the height of the building, the local water system and the designs of the sprinkler and standpipe systems in the building (see Section 913 for more information on fire pumps).

Additional redundancy is required for buildings of extreme heights that can no longer be supported from the ground. This requirement applies at a lower height for Type IV-A and IV-B construction than for other types of construction. This relates to the potential contribution of the mass timber to the fuel load and limitations of the fire service apparatus to apply water at elevated floors.

Therefore, the requirements are applicable to buildings more than 420 feet (128 m) in height or Type IVA and IVB buildings more than 120 feet (36.6 m) in height. For such buildings, this section requires the fire pumps to be supplied by two water mains located on separate streets. Having two connections will greatly reduce the possibility of the loss of water due to a main break, given the valving that is a feature of a public water system. Each connection must be adequate to provide the flow and pressure needed for the fire pumps to operate. The exception is a performance-based provision that is not tied to any specific configuration.

**914.3.2 Secondary water supply.** An automatic secondary on-site water supply having a capacity not less than the hydraulically calculated sprinkler demand, including the hose stream requirement, shall be provided for high-rise buildings assigned to Seismic Design Category C, D, E or F as determined by the *International Building Code*. An additional fire pump shall not be required for the secondary water supply unless needed to provide the minimum design intake pressure at the suction side of the fire pump supplying the *automatic sprinkler system*. The secondary water supply shall have a duration of not less than 30 minutes as determined by the occupancy hazard classification in accordance with NFPA 13.

**C** The intent of this section is that a secondary water supply be provided on the high-rise building site in order to provide a high level of functional reliability for the fire protection systems if a seismic event disables the primary water supply for high-rise buildings assigned these seismic design categories. The categories are described in IBC Section 1613.

The text's specific wording that the secondary supply be "on-site" rather than "to the site" would preclude the use, for example, of a secondary connection to the municipal supply to achieve compliance with this requirement.

The required amount of water is equal to the hydraulically calculated sprinkler demand plus hose stream demand for a minimum 30-minute period, dependent on the appropriate occupancy hazard classification in NFPA 13.

Note that the beginning of Section 914.3.2 requires that the secondary water supply be automatic; in other words, switchover to the secondary water source cannot occur manually. This is consistent with the definitions of "Automatic sprinkler system" and "Standpipe, types of" in that both systems are required to be connected to a reliable water supply.

Generally, this section does not automatically require a second fire pump, but if necessary, an additional pump may be required. A second pump would be necessary if the secondary water supply does not provide the necessary water pressure for intake into the primary fire pump.

**914.3.3 Fire alarm system.** A *fire alarm system* shall be provided in accordance with Section 907.2.13.

**C** See the commentary to Section 907.2.13.

**914.3.4 Automatic smoke detection.** Smoke detection shall be provided in accordance with Section 907.2.13.1.

**C** Fire alarm systems must be provided in high-rise buildings in accordance with the provisions of Section 907.2.13. Although a fire alarm system is not specifically listed as a requirement in Section 907.2.13 Section 914.3.4 does require an automatic smoke detection system. Section 907.2.13.1.1 requires that the smoke detectors be connected to an automatic fire alarm system. Therefore, regardless of the occupancy classification(s) in the high-rise building, a fire alarm system must comply with the code and NFPA 72, as applicable.

**914.3.5 Emergency voice/alarm communication system.** An emergency voice/alarm communication system shall be provided in accordance with Section 907.5.2.2.

**C** One characteristic of high-rise buildings is longer evacuation times. As such, the traditional fire alarm system, which usually results in simultaneous total building evacuation, is not practical. Therefore, the alarm communication system should be able to:

- Direct the occupants of the fire zone to an area of refuge or exit.
- Notify the fire department of the existence of the fire.
- Sound no alarms outside the fire zone until deemed desirable.

The alarm signal to the fire zone may include continuous sounding devices (e.g., bells, horns, chimes), as well as voice direction, which may momentarily silence the continuous sounding devices so as to be heard clearly. In accordance with Section 907.5.2.3, visible alarm notification appliances are required in the common and public areas of the high-rise buildings. Visible alarm appliances are also required in certain dwelling units and sleeping units, such as hotel guestrooms. The message of the voice/alarm is to be predetermined but does not need to be a recorded message. It would typically indicate that a fire has been reported at a specific location and that occupants should await further instructions or evacuate.

**914.3.6 Emergency responder communication coverage.** In-building, two-way emergency responder communication coverage shall be provided in accordance with Section 510.

**C** High-rise buildings have posed a challenge to the traditional communication systems used by the fire service for fire-to-ground communications. Therefore, in-building, two-way emergency responder communication coverage complying with Section 510 must be provided. The coverage coordinates with the fire department radio system, allowing fireground officers to remain in communication with firefighters working in various areas of a building.

As facilities grow larger and more complex, they present further challenges for effective fire response. Other complications include the use of low-emission glass, which tends to block the ability of signals from outside the building. Therefore, more features must be provided within the building to enable adequate communication. While modeling and other techniques may provide a good prediction as to whether a building will interfere with the emergency responder communication systems, the reality is that it is unknown if a building will need to have an enhanced communication coverage system installed until after the building is constructed. The presumption is that high-rise buildings will more likely than not need to be provided with additional coverage to facilitate fire service and other emergency responder communications. Note that another term used to describe in-building, two-way communication coverage systems is "emergency responder communications enhancement systems" (ERCES). Section 510.1 offers four exceptions that may be considered by the building and fire code officials. See the commentary to Section 510 for complete information on this topic.

**914.3.7 Fire command.** A *fire command center* complying with Section 508 shall be provided in a location *approved* by the fire department.

**C** Fireground operations usually involve establishing an incident command post where the fire official can observe what is happening, control arriving personnel and equipment, and direct resources and firefighting operations effectively. Because of the difficulties in controlling a fire in a high-rise building, a separate room (enclosed in 1-hour-rated construction) within

the building must be established as a fire command center. The room must be provided at a location that is acceptable to the fire code official, often the fire chief. Such centers are typically located near the front of the building or near the main entrance. The room must contain equipment necessary to monitor or control fire protection and other building service systems. See the commentary to Section 508 for additional information.

**914.4 Atriums.** Atriums shall comply with Sections 914.4.1 and 914.4.2.

**C** Atriums are addressed in IBC Section 404, which identifies several fire protection systems that must be included in atriums. It is important to remember that these are requirements that are unique to the specific condition noted and the fire protection systems are a result of those conditions. These requirements are only partially related to the occupancies that may be present.

**914.4.1 Automatic sprinkler system.** An approved automatic sprinkler system shall be installed throughout the entire building.

**Exceptions:**

1. That area of a building adjacent to or above the atrium need not be sprinklered, provided that portion of the building is separated from the atrium portion by not less than a 2-hour *fire barrier* constructed in accordance with Section 707 of the *International Building Code* or *horizontal assemblies* constructed in accordance with Section 711 of the *International Building Code*, or both.
2. Where the ceiling of the atrium is more than 55 feet (16 764 mm) above the floor, sprinkler protection at the ceiling of the atrium is not required.

**C** One means of controlling the spread of fire and smoke through vertical openings is to control and extinguish the fire as early as possible. Therefore, all floor areas that are connected by the atrium are to be protected with an approved sprinkler system, including the atrium space itself. The system is to be designed, installed, tested and maintained in accordance with the provisions of Section 903.3. Since Chapter 9 requires that the sprinkler system be supervised, the reliability of the sprinkler system is improved.

Exception 1 clarifies that since the atrium protection is one of the various options for protecting vertical openings, typically an alternative to a shaft enclosure, the sprinkler system is not required in areas that are separated from the atrium by 2-hour fire barrier walls or horizontal assemblies that would otherwise be required for a shaft enclosure. Such fire barrier walls must conform to IBC Section 707, while the horizontal assemblies must comply with IBC Section 711.

Exception 2 permits the required sprinkler system to be deleted from the ceiling areas of atriums where the vertical distance between the atrium floor and atrium ceiling is greater than 55 feet (16 764 mm). A ceiling height of more than 55 feet (16 764 mm) is the height at which the system is no longer effective and installing such systems provides little benefit. This exception does not alter the use limitations of the atrium as stated in IBC Section 404.2, nor does it exempt any adjacent floor areas with lower ceiling heights that are included in the atrium boundary in accordance with Exception 1. It is important to note that where a smoke control system is required by IBC Section 404.5, the smoke control design should not take sprinkler activation into account for design fires that are located in areas not protected with sprinklers or in areas where sprinklers are installed but, due to their location, may not be able to control a fire.

**914.4.2 Fire alarm system.** A fire alarm system shall be provided where required by Section 907.2.14.

**C** Section 907.2.14 contains a requirement for a fire alarm system in atriums connecting more than two stories. In addition, where such atriums are located in Group A, E and M occupancies, an emergency voice/alarm communications system is required. See the commentary to Section 907.2.14.

**914.5 Underground buildings.** Underground buildings shall comply with Sections 914.5.1 through 914.5.5.

**C** IBC Section 405 contains requirements for underground buildings. The fire protection features necessary for this type of building are also found elsewhere in this chapter but are reiterated in the following sections. Underground buildings present unique hazards in that the path of egress is upward, in the same direction as vertical fire movement. Fires in such buildings are also more challenging for firefighters since there is no opportunity for an exterior fire attack. See the commentary to IBC Section 405 for additional information.

**914.5.1 Automatic sprinkler system.** The highest level of exit discharge serving the underground portions of the building and all levels below shall be equipped with an automatic sprinkler system installed in accordance with Section 903.3.1.1. Waterflow switches and control valves shall be supervised in accordance with Section 903.4.1.

**C** One of the most effective measures to prevent fire growth is the installation of an automatic sprinkler system. Because of the unique conditions for occupant egress and fire department access in the underground portion of a building, automatic sprinkler protection is required. The level of exit discharge and all floor levels below are required to be sprinklered throughout in accordance with Section 903.3.1.1. This section does permit a portion of a building to extend above the level of exit discharge and not be equipped with an automatic sprinkler system. Where, however, another code section (see Section 903) requires an automatic sprinkler system in the above-ground portion, such a requirement would still be applicable. Note that a smoke control system is required in accordance with IBC Section 405.5. Automatic sprinkler systems are essential elements of any smoke control system. Without suppression, the size of the fire or the resulting products of combustion

will rapidly overwhelm most mechanical smoke control systems. The pressurization system requirements in Section 909 are based on the use of sprinklers. Pressure differences necessary to contain the smoke to the area of origin would have to be greater in a nonsprinklered building.

**914.5.2 Smoke control system.** A smoke control system is required to control the migration of products of combustion in accordance with Section 909 and provisions of this section. Smoke control shall restrict movement of smoke to the general area of fire origin and maintain *means of egress* in a usable condition.

**C** One common application for an underground building is to use excavated caverns for storage. In such a condition, there is no roof from which the fire service can attempt to vent the “building.” Because occupants must egress in the same direction that smoke will vent itself, and because there are few, if any, openings by which to vent smoke to the exterior, the need for smoke control is greater than for traditional buildings. This section adds a degree of clarity to the design requirements of Section 909 in that the intent is to contain smoke to the general area of origin. Any method identified in Section 909 can be used, given that the intent is to allow for egress and limit smoke spread to areas of the building that are not involved in the fire event. See Section 909 for additional commentary on smoke control.

**914.5.3 Compartment smoke control system.** Where compartmentation is required by Section 405.4 of the *International Building Code*, each compartment shall have an independent smoke control system. The system shall be automatically activated and capable of manual operation in accordance with Section 907.2.18.

**C** Where buildings contain levels that are more than 60 feet (18 288 mm) below the lowest level of exit discharge, IBC Section 405.4.1 requires that there be at least two fire-resistance-rated compartments per floor and of approximately the same size. Because compartmentation exists, there is a need for additional safety with the smoke control system. Consequently, each compartment must have a smoke control system that is independent of that in the adjoining compartment. The word “independent” means that there can be no sharing of ductwork, mechanical equipment or power distribution. Connection to the standby system can be shared since that is consistent with the general intent of the smoke control system provisions in Section 909 and the IMC. See the commentary to IBC Section 405.4.

**914.5.4 Fire alarm system.** A *fire alarm system* shall be provided where required by Sections 907.2.18 and 907.2.19.

**C** The threshold for requiring a fire alarm system is the same as that for compartmentation—an occupied floor level more than 60 feet (18 288 mm) below the level of exit discharge. The fire alarm requirements include the requirements for an emergency voice/alarm communications system. See the commentary to Sections 907.2.18 and 907.2.19 for additional information.

**914.5.5 Standpipe system.** The underground building shall be provided throughout with a standpipe system in accordance with Section 905.

**C** Section 905.3.1 requires a standpipe system where the building’s lowest level is more than 30 feet (9144 mm) below the lowest level of fire department access. By definition, an underground building is one in which the lowest level is more than 30 feet (9144 mm) below the lowest level of exit discharge. One condition uses fire department vehicle access as the benchmark while the other uses the level of exit discharge. Care must be taken so that both conditions are addressed when determining standpipe system requirements for buildings that have levels below ground.

**914.6 Stages.** Stages shall comply with Section 914.6.1.

**C** IBC Section 410 contains definitions and requirements for both stages and platforms. A stage, however, is a special condition, according to the IBC. It does not mean, for example, that an elevated platform in the front of the banquet hall is intended to be regulated the same as a stage. The term “stage,” as used here, is traditionally considered a “legitimate stage”—one with scenery drops and a gridiron above. See the commentary to IBC Section 410 for further information.

**914.6.1 Automatic sprinkler system.** Stages shall be equipped with an *automatic sprinkler system* in accordance with Section 903.3.1.1. Sprinklers shall be installed under the roof and gridiron and under all catwalks and galleries over the stage. Sprinklers shall be installed in dressing rooms, performer lounges, shops and storerooms accessory to such stages.

**Exceptions:**

1. Sprinklers are not required under stage areas less than 4 feet (1219 mm) in clear height utilized exclusively for storage of tables and chairs, provided that the concealed space is separated from the adjacent spaces by Type X gypsum board not less than  $\frac{5}{8}$  inch (15.9 mm) in thickness.
2. Sprinklers are not required for stages 1,000 square feet (93 m<sup>2</sup>) or less in area and 50 feet (15 240 mm) or less in height where curtains, scenery or other combustible hangings are not retractable vertically. Combustible hangings shall be limited to a single main curtain, borders, legs and a single backdrop.
3. Sprinklers are not required within portable orchestra enclosures on stages.
4. Sprinklers are not required under catwalks and galleries where they are permitted to be omitted in accordance with Section 903.3.1.1.

**C** Stages contain significant quantities of combustible materials stored in, around and above the stage that are located in close proximity to large quantities of lighting equipment (i.e., scenery and lighting above the stage). There also is scenery

on the sides and rear of the stage; shops located along the back and sides of the stage; and storage, props, trap doors and lifts under the stage floor. This combination of fuel load and ignition sources increases the potential for a fire. As such, stages and accessory areas, such as dressing rooms, workshops and storerooms, are required to be protected with an automatic sprinkler system.

The reference to Chapter 9 indicates that the sprinkler system may be designed in accordance with NFPA 13 and Section 903.3.1.1 or, where not more than 20 sprinklers are required on any single connection, a limited area sprinkler system may be used (see Section 903.3.8).

Exception 1 applies to areas less than 4 feet (1219 mm) in clear height under stages that are used only for the storage of tables and chairs. Because of the limited fuel load present, such areas are not required to be protected with sprinklers, provided that the concealed space is separated from all adjacent spaces by not less than 5/8-inch (15.9 mm) Type X gypsum board. This level of separation is intended to provide fire containment in the absence of sprinkler protection. This arrangement is often found in educational buildings where the room is used for multiple purposes.

Exception 2 recognizes that stages 1,000 square feet (93 m<sup>2</sup>) or less in area and 50 feet (15 240 mm) or less in height represent a limited potential for combustibles that does not warrant the requirements for an automatic sprinkler system. The code further limits the amounts of combustible materials by not allowing any vertical retractable curtains, hangings and similar scenery elements. It should be noted, however, that although sprinkler protection is not required, the requirements of IBC Section 410.4 still apply.

Exception 3 acknowledges the limited hazards associated with portable orchestra enclosures. These elements are temporary in nature and are intended to improve the acoustics of the stage performances. These temporary enclosures do not lend themselves to temporary sprinkler heads; therefore, none are required.

Exception 4 clarifies that catwalks under 48 inches open on both sides or 36 inches when against a wall are similar to ducts and do not require sprinklers under them in accordance with NFPA 13 as referenced in Section 903.3.1.1.

**914.7 Special amusement areas.** Special amusement areas shall comply with Sections 914.7.1 and 914.7.2.

**Exceptions:**

1. Special amusement areas that are without walls or a roof and constructed to prevent the accumulation of smoke are not required to comply with this section.
2. Puzzle rooms provided with a means of egress that is unlocked, readily identifiable and always available are not required to comply with this section.

**C** A special amusement area is one in which the egress is not readily apparent, is intentionally confounded or is not readily available. Due to those characteristics, special fire protection features are required. See the commentary to IBC Section 411 for additional information about these types of areas.

**914.7.1 Automatic sprinkler system.** Buildings containing special amusement areas shall be equipped throughout with an *automatic sprinkler system* in accordance with Section 903.3.1.1. Where the special amusement area is temporary, the sprinkler water supply shall be of an *approved* temporary means.

**Exception:** Automatic sprinklers are not required where the total floor area of a temporary special amusement area is less than 1,000 square feet (93 m<sup>2</sup>) and the *exit access* travel distance from any point in the special amusement area to an *exit* is less than 50 feet (15 240 mm).

**C** Special amusement areas do not normally have a readily identifiable means of egress. Consequently, additional time for safe egress may be necessary. A sprinkler system will minimize the potential hazard to occupants by controlling fire development (see commentary, IBC Section 411).

**914.7.2 Detection and alarm systems.** Buildings containing *special amusement areas* shall be equipped throughout with an *automatic smoke detection system* and an emergency voice/alarm communications system in accordance with Section 907. Presignal alarms and alarm activation shall comply with Sections 914.7.2.1 and 914.7.2.2, and emergency voice/alarm communications systems shall comply with Section 914.7.2.3.

**C** An automatic smoke detection system and emergency voice/alarm communications system are required in special amusement areas to provide early warning of fire and audible instructions to occupants. The detection system is required regardless of the presence of staff in the building.

**914.7.2.1 Alarm presignal.** Activation of any single smoke detector, the automatic sprinkler system or any other single automatic fire detection device shall immediately initiate an audible and visible alarm at a constantly attended location at the special amusement area from which emergency action can be initiated, including the capability of manual initiation of requirements in Section 914.7.2.2.

**C** On activation of a smoke detector, automatic sprinkler system or other automatic fire-detection device, an alarm must initiate both an audible and visible signal at a constantly attended location. The staff at the location is then expected to be capable of providing the required egress illumination, stopping conflicting sounds and distractions, and activating the exit marking as required by Section 914.7.2.2. The staff is also expected to prevent additional people from entering the building.

**914.7.2.2 Alarm activation.** Activation of two or more smoke detectors, a single smoke detector equipped with an alarm verification feature, two or more other *approved* fire detection devices, the *automatic sprinkler system*, or a manual control located at the constantly attended station required by Section 914.7.2.1 shall automatically accomplish all of the following:

1. Illuminate the means of egress with an illumination level not less than 1 footcandle (11 lux) at the walking surface level.
2. Stop conflicting or confusing sounds and visual distractions.
3. Activate *approved* directional exit markings.
4. Activate a prerecorded message, audible throughout the special amusement area, instructing occupants to proceed to the nearest *exit*. Alarm signals used in conjunction with the prerecorded message shall produce a sound that is distinct from other sounds used during normal operation of the special amusement area.

**C** Once a fire has been detected, measures must be taken to stop the confusion or distractions. Additionally, the egress path must be illuminated and marked. These measures must occur automatically on detection of the fire or sprinkler water flow. A prerecorded message that can be heard throughout the building instructing the occupants to proceed to the nearest exit must be automatically activated. The message and alarm signals should be designed to prevent panic. The prerecorded message capability is in addition to the EV/ACS requirement of Section 914.7.2.3. The wiring of all devices must comply with NFPA 72.

**914.7.2.3 Emergency voice/alarm communications system.** An emergency voice/alarm communications system complying with Section 907.5.2.2 shall be installed in and audible throughout special amusement areas. The emergency voice/alarm communications system is allowed to also serve as a public address system.

**C** An integral part of the fire protection systems required for buildings with special amusement areas is an emergency voice/alarm communications system. This system can serve as a public address system to alert the building occupants of the fire emergency and provide them with the proper emergency instructions. The system must be installed in accordance with NFPA 72 and must be heard throughout the entire building. Upon activation, the system must sound an alert tone followed by the necessary voice instructions. These instructions can save valuable time in directing the building occupants to quickly and safely egress.

**914.8 Aircraft-related occupancies.** Aircraft-related occupancies shall comply with Sections 914.8.1 through 914.8.6.

**C** Aircraft related occupancies include air traffic control towers, residential and commercial hangars, and heliports and helistops. Hangars can be those for storage and “parking” of the aircraft or for repair and painting. All of these are addressed in IBC Section 412.

While the following provisions do not identify fire protection requirements for heliports and helistops, IBC Section 412.7 requires that these facilities be designed in accordance with NFPA 418. That standard contains various provisions for fire protection including a foam fire-extinguishing system for rooftop landing pads and portable fire extinguishers. Section 2007 of the code contains fire protection requirements for helistops and heliports. Section 905.3.5 requires a standpipe system for helistops and heliports. Additional requirements for portable fire extinguishers in aviation facilities can be found in Section 2005 of the code.

**914.8.1 Automatic smoke detection systems.** Airport traffic control towers shall be provided with an *automatic smoke detection system* installed in accordance with Section 907.2.22.

**C** Airport traffic control towers are required to have an automatic fire detection system. Because the tower is designed so that people are elevated in the air, the vertical supports generally are narrow and contain little room for separated stairways. An early notification of a fire hazard is important in alerting the occupants so that fire safety operations can begin as quickly as possible. An automatic detection system is therefore required. Section 907.2.22.1 applies specifically to towers where sprinklers are provided and there are at least two means of egress. Where there is only a single means of egress, Section 907.2.22.2 specifies the coverage of the fire detection systems. Section 907.2.22.2 also applies to towers that are not sprinkler protected, although only towers with an occupied floor at a height of 35 feet (10 668 mm) or less would be without sprinklers (see IBC Section 412.2.4). Where suppression is not required by the code, the detection system is vital for the safety of the occupants. No requirements are included in Chapter 11 of the code to address existing airport traffic control towers.

**914.8.2 Automatic sprinkler system for new airport traffic control towers.** Where an occupied floor is located more than 35 feet (10 668 mm) above the lowest level of fire department vehicle access, new airport traffic control towers shall be equipped with an *automatic sprinkler system* in accordance with Section 903.3.1.1.

**C** Under IBC Section 412.2.4, sprinkler systems are required for all towers being constructed once there is an occupied floor above 35 feet (10 668 mm). This height was selected because shorter towers should be reachable by ladders typically found at rural fire stations and smaller airports. These are also the types of airports that would have the shorter control towers. Due to the absence of aerial ladders or means to reach higher elevations at these locations, additional levels of fire protection are required. As such, life safety is positively affected by limiting the chance of smoke/fire spread and flashover in the facility where delayed evacuation of the cab may be required. In addition, property protection would allow earlier reuse of the structure.

**914.8.3 Fire suppression for aircraft hangars.** Aircraft hangars shall be provided with a fire suppression system designed in accordance with NFPA 409, based on the classification for the hangar given in Table 914.8.3.

**Exception:** Where a fixed base operator has separate repair facilities on-site, Group II hangars operated by a fixed base operator used for storage of transient aircraft only shall have a fire suppression system, but the system shall be exempt from foam requirements.

**C** To minimize the fire hazards associated with aircraft hangars, most hangars are required to be protected with a fire suppression system. Where required, the fire suppression system must be designed and installed in accordance with the referenced standard, NFPA 409, which requires fire suppression based on the type and construction and the activities in a given hangar.

In the standard, the suppression requirements are broken down based on three categories: Group I, Group II and Group III hangars. Table 914.8.3 designates which Group category applies to various sizes of fire areas within a hangar and the type of construction. For example, a hangar that is 28,000 square feet (2601 m<sup>2</sup>) in Type IIB construction would be a Group II hangar. Groups I and II hangars are required to have fire suppression as specified in NFPA 409. In general, Group III hangars are exempt from providing fire suppression unless one (or more) of the hazardous operations listed in Section 914.8.3.1 occurs within the hangar. In these situations, fire suppression based on the appropriate portion of the standard for either Group I or Group II is required.

The exception allows that the foam requirements for fire suppression do not need to be provided for Group II hangars if the hangar is essentially a parking garage for transient aircraft. The exception is only applicable where a larger airport facility has multiple hangars and separate hangars for repair operations.

**TABLE 914.8.3—HANGAR FIRE SUPPRESSION REQUIREMENTS<sup>a, b, c</sup>**

| MAXIMUM SINGLE FIRE AREA (square feet) | INTERNATIONAL BUILDING CODE TYPE OF CONSTRUCTION |           |           |           |           |           |           |           |           |
|----------------------------------------|--------------------------------------------------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|
|                                        | IA                                               | IB        | IIA       | IIB       | IIIA      | IIIB      | IV        | VA        | VB        |
| ≥ 40,001                               | Group I                                          | Group I   | Group I   | Group I   | Group I   | Group I   | Group I   | Group I   | Group I   |
| 40,000                                 | Group II                                         | Group II  | Group II  | Group II  | Group II  | Group II  | Group II  | Group II  | Group II  |
| 30,000                                 | Group III                                        | Group II  | Group II  | Group II  | Group II  | Group II  | Group II  | Group II  | Group II  |
| 20,000                                 | Group III                                        | Group III | Group II  | Group II  | Group II  | Group II  | Group II  | Group II  | Group II  |
| 15,000                                 | Group III                                        | Group III | Group III | Group II  | Group III | Group II  | Group III | Group II  | Group II  |
| 12,000                                 | Group III                                        | Group III | Group III | Group III | Group III | Group III | Group III | Group II  | Group II  |
| 8,000                                  | Group III                                        | Group III | Group III | Group III | Group III | Group III | Group III | Group III | Group II  |
| 5,000                                  | Group III                                        | Group III | Group III | Group III | Group III | Group III | Group III | Group III | Group III |

For SI: 1 square foot = 0.0929 m<sup>2</sup>, 1 foot = 304.8 mm.

a. Aircraft hangars with a door height greater than 28 feet shall be provided with fire suppression for a Group I hangar regardless of maximum fire area.

b. Groups shall be as classified in accordance with NFPA 409.

c. Membrane structures complying with Section 3102 of the *International Building Code* shall be classified as a Group IV hangar.

**C** This table is a correlation of the NFPA 409 construction and area limits with the IBC construction type requirements. It combines several tables in NFPA 409 into a single table that allows determination of the group type for aircraft hangars based on construction type and area before proceeding to the standard for the suppression requirements. Note a indicates that, regardless of size or construction type, any hangar with a door opening greater than 28 feet high (8534 mm) is required to have a fire suppression system as required for a Group I hangar. Note c provides a Group IV designation for any hangar located in a membrane structure.

**914.8.3.1 Hazardous operations.** Any Group III aircraft hangar in accordance with Table 914.8.3 that contains hazardous operations including, but not limited to, the following shall be provided with a Group I or II fire suppression system in accordance with NFPA 409 as applicable:

1. Doping.
2. Hot work including, but not limited to, welding, torch cutting and torch soldering.
3. Fuel transfer.
4. Fuel tank repair or maintenance not including defueled tanks in accordance with NFPA 409, inerted tanks or tanks that have never been fueled.
5. Spray finishing operations.
6. Total fuel capacity of all aircraft within the unsprinklered single *fire area* in excess of 1,600 gallons (6057 L).

7. Total fuel capacity of all aircraft within the maximum single *fire area* in excess of 7,500 gallons (28 390 L) for a hangar equipped throughout with an *automatic sprinkler system* installed in accordance with Section 903.3.1.1.

**C** Any of the operations listed in this section where they occur in a Group III hangar will require some level of fire suppression under NFPA 409. The hazardous operations on the list are straightforward. Additional information may be found as follows: Item 1 is discussed further in Section 914.8.4, Item 2 is regulated by Chapter 35, Items 3 and 4 are regulated by Chapter 57, Item 5 is regulated by Chapter 24.

**914.8.3.2 Separation of maximum single fire areas.** Maximum single *fire areas* established in accordance with hangar classification and construction type in Table 914.8.3 shall be separated by 2-hour *fire walls* constructed in accordance with Section 706 of the *International Building Code*. In determining the maximum single *fire area* as set forth in Table 914.8.3, ancillary uses that are separated from aircraft servicing areas by not less than a 1-hour *fire barrier* constructed in accordance with Section 707 of the *International Building Code* shall not be included in the area.

**C** Table 914.8.3 places a maximum size limit on hangars based on type of construction. For a hangar structure to exceed these sizes requires the construction of fire walls to establish fire areas that stay within the limits. This section is more stringent than the definition of “Fire area,” which allows fire barriers to establish a fire area. For hangars, the fire areas must be created by the exterior walls of a building or a combination of exterior walls and fire walls. Note that there is some allowance for the use of fire barriers to create fire areas where separating ancillary uses such as business offices, maintenance shops and storage areas. The fire suppression requirements contained within NFPA 409 are primarily focused on the protection of aircraft within the servicing and storage area. The fire protection requirements in the ancillary areas are not as extensive as those required for the aircraft servicing and storage areas, and less restrictive separations are considered appropriate. This allowance is consistent with NFPA 409 requirements.

**914.8.4 Finishing.** The process of “doping,” involving the use of a volatile flammable solvent, or of painting shall be carried on in a separate detached building equipped with automatic fire-extinguishing equipment in accordance with Section 903.

**C** Doping is a type of lacquer used to protect, waterproof and make taut the cloth surfaces of airplane wings. It is used on lighter-than-air, ultralight and some light aircraft. It is essentially painting on fabric. Doping is not used on metallic surfaces; however, the use of flammable paints is also addressed in this section. When flammable finishes are applied, the process must occur in a separate building not attached to the hangar. Because the code text refers to Section 903, the intent is for an automatic sprinkler system to be installed, unless otherwise approved (see commentary, IBC Section 412.3.5).

**914.8.5 Residential aircraft hangar smoke alarms.** Smoke alarms shall be provided within residential aircraft hangars in accordance with Section 907.2.21.

**C** A residential aircraft hangar, as defined in IBC Section 202, is “... an accessory building less than 2,000 square feet (186 m<sup>2</sup>) and 20 feet (6096 mm) in building height constructed on a one- or two-family property where aircraft are stored...” The requirements for the residential hangar are very similar to that for a residential garage. The hangar can be either detached or attached. If attached, it must be separated from the dwelling by construction with not less than a 1-hour fire-resistance rating. Section 907.2.21 requires a minimum of one smoke alarm in a residential aircraft hangar. See the commentary to Section 907.2.21 for additional information.

**914.8.6 Aircraft paint hangar fire suppression.** Aircraft paint hangars shall be provided with fire suppression as required by NFPA 409.

**C** To minimize the fire hazards associated with aircraft paint hangars, all such buildings are required to be protected with a fire suppression system. This requirement is applicable regardless of the size of the hangar in terms of height or area or the types and quantities of aircraft that are being cleaned or painted. The fire suppression system must be designed and installed in accordance with NFPA 409.

This section assumes the primary function of the hangar is as a paint hangar. Minor painting may constitute finishing in accordance with Section 914.8.4. While not referenced in the code, NFPA 410 provides additional guidance on aircraft maintenance, including the painting of aircraft.

**914.9 Application of flammable finishes.** An *automatic sprinkler system* or fire-extinguishing system shall be provided in all spray rooms and spray booths, and shall be installed in accordance with Chapter 9.

**C** Chapter 24 provides detailed information about spaces and rooms involved in the application of flammable finishes. This section references Chapter 9 for the appropriate type of fire-extinguishing system.

**914.10 Drying rooms.** Drying rooms designed for high-hazard materials and processes, including special occupancies as provided for in Chapter 4 of the *International Building Code*, shall be protected by an *approved automatic fire-extinguishing system* complying with the provisions of Chapter 9.

**C** This section reiterates the requirements from IBC Section 417.4. See the commentary to IBC Section 417.4 for additional information.

**914.11 Ambulatory care facilities.** Occupancies classified as ambulatory care facilities shall comply with Sections 914.11.1 through 914.11.3.

**C** This section simply introduces the fire protection requirements for ambulatory care facilities. See also the commentary to IBC Section 422.

**914.11.1 Automatic sprinkler systems.** An *automatic sprinkler system* shall be provided for ambulatory care facilities in accordance with Section 903.2.2.1.

**C** See the commentary to Section 903.2.2.1.

**914.11.2 Manual fire alarm systems.** A manual *fire alarm system* shall be provided for ambulatory care facilities in accordance with Section 907.2.2.

**C** See the commentary to Section 907.2.2.

**914.11.3 Fire alarm systems.** An *automatic smoke detection system* shall be provided for ambulatory care facilities in accordance with Section 907.2.2.1.

**C** See the commentary to Section 907.2.2.1.

## SECTION 915—CARBON MONOXIDE (CO) DETECTION

**915.1 General.** Carbon monoxide (CO) detection shall be installed in new buildings in accordance with Section 915.1.1. Carbon monoxide detection shall be installed in existing buildings in accordance with Section 1103.9.

**Exception:** Carbon monoxide detection is not required in Group S, Group F and Group U occupancies that are not normally occupied.

**C** CO is a colorless, tasteless, odorless gas that interrupts the attachment of oxygen molecules to hemoglobin in blood cells and can cause headaches, confusion and dizziness. At higher concentrations, CO can cause loss of consciousness and eventual death. Exposures above 100 parts/million are dangerous to human health. It is not a toxic or highly toxic gas as defined in Chapter 2 but is classified as a flammable gas.

These provisions detail what locations and conditions in new buildings require CO detection.

Retroactive requirements that apply to existing buildings are found in Chapter 11. The exception recognizes that any of the three specific occupancy types listed could be in buildings that are normally unoccupied; thus, in these situations, CO detection would be unnecessary.

**915.1.1 Where required.** Carbon monoxide detection shall be installed in the locations specified in Section 915.2 where any of the following conditions exist.

1. In buildings that contain a *CO source*.
2. In buildings that contain or are supplied by a CO-producing forced-air furnace.
3. In buildings with attached private garages.
4. In buildings that have a CO-producing vehicle that is used within the building.

**C** CO detection is to be provided to protect the occupants of buildings with any of the four conditions listed in this section and in the locations listed in Section 915.2. Condition 1 is a carbon monoxide source, which is defined in Chapter 2 as a piece of commonly used equipment or permanently installed appliance, fireplace or process that produces or emits carbon monoxide gas. Condition 2 addresses forced-air furnaces that physically move air. This potential source of CO is more indirect than having the fuel-burning appliance or fuel-burning fireplace within the space itself but still creates a potential hazard. Condition 3 includes buildings with attached private garages. A private garage, which is a term defined in Chapter 2, is often a source of CO generation. Condition 4 includes buildings that have any CO producing vehicles in use inside of the building.

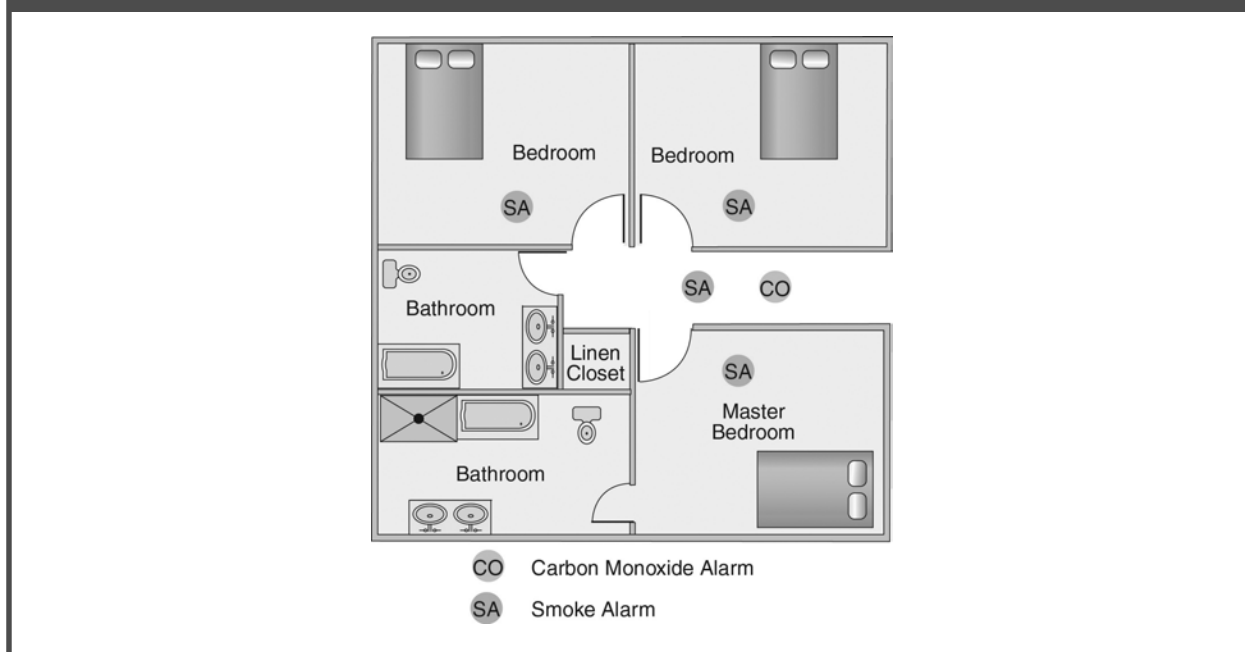
**915.2 Locations.** Carbon monoxide detection shall be installed in the locations specified in Sections 915.2.1 through 915.2.3.

**C** Sections 915.2.1 through 915.2.6 provide more detail on where detection is specifically required to be installed. This is a clarification from previous editions. NFPA 72 does not provide enough detailed information as to where detection should be located, so specific direction is provided in the code. In some cases, the locations differ from that provided in NFPA 72.

**915.2.1 Dwelling units.** Carbon monoxide detection shall be installed in *dwelling units* outside of each separate sleeping area in the immediate vicinity of the bedrooms. Where a *CO source* is located within a bedroom or its attached bathroom, carbon monoxide detection shall be installed within the bedroom.

**C** The language is similar to that required for smoke alarms, but the detection is only required outside of each sleeping area. If the CO source is within the bedroom or the attached bathroom, detection is required within the bedroom itself. Commentary Figure 915.2.1 shows the layout of carbon monoxide alarms versus smoke alarms. Note that if they are combined (smoke and CO alarms), the smoke alarm placement would be more restrictive.

Commentary Figure 915.2.1—Carbon Monoxide Alarm Placement Versus Smoke Detection Placement



**915.2.2 Sleeping units.** Carbon monoxide detection shall be installed in *sleeping units*.

**Exception:** Carbon monoxide detection shall be allowed to be installed outside of each separate sleeping area in the immediate vicinity of the *sleeping unit* where the *sleeping unit* or its attached bathroom does not contain a *CO source* and is not served by a CO-producing forced-air furnace.

**C** The intent of this section is the same as Section 915.2.1 except that it is applicable to sleeping units. Sleeping units can be as simple as a hotel room or a suite-type layout. This allows the detection to be either in the sleeping area itself, as with a typical hotel room, or outside the sleeping area as is the case for dwelling units.

**915.2.3 Group E occupancies.** A carbon monoxide system that uses carbon monoxide detectors shall be installed in Group E occupancies. Alarm signals from carbon monoxide detectors shall be automatically transmitted to an on-site location that is staffed by school personnel.

**Exception:** Carbon monoxide alarm signals shall not be required to be automatically transmitted to an on-site location that is staffed by school personnel in Group E occupancies with an *occupant load* of 30 or less.

**C** This section requires that a carbon monoxide system that has carbon monoxide detectors be installed in all Group E occupancies and that alarm signals from the carbon monoxide detectors be automatically transmitted to a location on-site where school personnel can react to the situation, such as the school office. The exception provides relief to small schools where it is very easy to communicate a hazardous situation throughout the building.

**915.2.4 CO-producing forced-air furnace.** Carbon monoxide detection complying with Item 2 of Section 915.1.1 shall be installed in all enclosed rooms and spaces served by a fuel-burning, forced-air furnace.

**Exceptions:**

1. Where a carbon monoxide detector is provided in the first room or space served by each main duct leaving the furnace, and the carbon monoxide alarm signals are automatically transmitted to an *approved* location.
2. *Dwelling units* that comply with Section 915.2.1.

**C** This section addresses forced-air furnaces that physically move air. This potential CO source is more indirect than having the fuel-burning appliance or fuel-burning fireplace within the space itself, but it still creates a potential hazard.

Exception 1 addresses the fact that if detection is provided closer to the source, then further detection within enclosed rooms and spaces is not necessary—provided that a CO alarm signal is transmitted to an approved location from which emergency actions can be initiated and occupants in the portions of the building farther from the initial detection can be notified. CO generation will continue until action is taken. Exception 2 provides a direct reference to the specific requirements for CO detection within dwelling units in Section 915.2.1.

**915.2.5 Private garages.** Carbon monoxide detection complying with Item 3 of Section 915.1.1 shall be installed within enclosed occupiable rooms or spaces that are contiguous to the attached private garage.

**Exceptions:**

1. In buildings without communicating openings between the private garage and the building.

2. In rooms or spaces located more than one story above or below a private garage.
3. Where the private garage connects to the building through an *open-ended* corridor.
4. An open parking garage complying with Section 406.5 of the *International Building Code* or an enclosed parking garage complying with Section 406.6 of the *International Building Code* shall not be considered a private garage.
5. *Dwelling units* that comply with Section 915.2.1.

**C** This section addresses attached private garages. Private garages, a term defined in Chapter 2, are often sources of CO generation. When attached to a building containing enclosed occupiable rooms or spaces, CO detection is required. There are four exceptions to the requirement for CO detection. Exception 1 applies if there are no communicating openings between the private garage and the building. In this situation, the likelihood of CO entering the building is greatly reduced. An interior door between the private garage and a contiguous enclosed occupiable room or space, even if it is self-closing, would not allow this exception to apply. Exception 2 addresses buildings where the enclosed occupiable rooms or spaces are more than one story above or below the private garage. The CO would need to travel through an entire story before reaching the areas of concern. Exception 3 does not require CO alarms to be provided where the private garage is attached to the building by an open-ended corridor (a term defined in Chapter 2 and commonly called a “breezeway”). This allows the CO concentration to dissipate before becoming a hazard to building occupants. Exception 4 provides clarification that open parking garages and enclosed parking garages complying with the IBC are not private garages; thus, they are not subject to the requirements of this section. Exception 5 provides a direct reference to the specific requirements for CO detection within dwelling units in Section 915.2.1.

**915.2.6 All other occupancies.** For locations other than those specified in Sections 915.2.1 through 915.2.5, carbon monoxide detectors shall be installed on the ceiling of enclosed rooms or spaces containing CO-producing devices or served by a *CO source* forced-air furnace.

**Exception:** Where environmental conditions prohibit the installation of carbon monoxide detector in an enclosed room or space, carbon monoxide detectors shall be installed in an *approved* enclosed location contiguous with the room or space that contains a *CO source*.

**C** This section addresses additional locations that also contain CO producing devices or are served by a CO source forced-air furnace and are not specifically listed in the preceding sections. There are locations other than the specified dwelling units, sleeping units, Group E occupancies and private garages that may also contain CO-producing sources. Therefore, based on the same hazard level, they are also required to have carbon monoxide detectors installed. The exception recognizes that there may be locations where the installation of carbon monoxide detectors does not conform to the specific equipment listing requirements. In these situations, it is required that they be installed in all of the contiguous enclosed locations approved by the fire code official.

**915.3 Carbon monoxide detection.** Carbon monoxide detection required by Sections 915.1 through 915.2.3 shall be provided by carbon monoxide alarms complying with Section 915.4 or carbon monoxide detection systems complying with Section 915.5.

**C** This section provides the option of providing the CO detection through single- or multiple-station CO alarms or through a CO detection system. This is a similar concept to smoke alarms and smoke detection systems. Each type of CO detector is listed to a different UL standard, depending on whether it is to be used as a stand-alone or interconnected detector (UL 2034) or a detector that is part of a CO detection system (UL 2075). Section 915.4 addresses CO alarms and Section 915.5 addresses CO detection systems.

**915.3.1 Alarm limitations.** Carbon monoxide alarms shall only be installed in *dwelling units* and in sleeping units. They shall not be installed in locations where the code requires carbon monoxide detectors to be used.

**C** This section provides the limits on the locations where carbon monoxide alarms can be installed and prohibits the substitution of carbon monoxide alarms in place of carbon monoxide detectors when the code requires that a detector be installed.

**915.3.2 Fire alarm system required.** New buildings that are required by Section 907.2 to have a *fire alarm system* and by Section 915.2 to have carbon monoxide detectors shall be connected to the *fire alarm system* in accordance with NFPA 72.

**C** This section clarifies that if a new building is already required by Section 907.2 to have a fire alarm system installed and by Section 915.2 to have carbon monoxide detectors installed, then the carbon monoxide detectors are to be connected as a part of a combined system in accordance with NFPA 72. This ensures that a combined means of alarm, identification and occupant notification is provided for the separate hazards that could occur within new buildings.

**915.3.3 Fire alarm systems not required.** In new buildings that are not required by Section 907.2 to have a *fire alarm system*, carbon monoxide detection shall be provided by one of the following:

1. Carbon monoxide detectors connected to an *approved* carbon monoxide detection system in accordance with NFPA 72.
2. Carbon monoxide detectors connected to an *approved* combination system in accordance with NFPA 72.
3. Carbon monoxide detectors connected to an *approved fire alarm system* in accordance with NFPA 72.

4. Where *approved* by the *fire code official*, carbon monoxide alarms maintained in accordance with the manufacturer's instructions.

**C** This section lists four options for providing carbon monoxide detection in new buildings that are not required by Section 907.2 to have a fire alarm system. The fire code official must determine which of the four options is required depending on the situation. Option 1 is a carbon monoxide detection system without a fire alarm system installed according to the requirements of NFPA 72. Option 2 is a combination carbon monoxide detection system installed according to the requirements of NFPA 72. Option 3 is carbon monoxide detectors installed in the locations connected to a fire alarm system and installed according to the requirements of NFPA 72. Option 4 is the installation of carbon monoxide alarms in the locations required by this section and maintained according to the manufacturer's instructions. If Option 4 is used in accordance with the approval of the fire code official, Section 915.4 provides the requirements. Regardless of which of the four options is complied with, it is first required to be acceptable to the fire code official in each situation.

**915.3.4 Installation.** Carbon monoxide detection shall be installed in accordance with NFPA 72 and the manufacturer's instructions.

**C** This section provides the installation reference standard for carbon monoxide detection. Such detection is required to be installed according to the manufacturer's instructions and in compliance with the code and NFPA 72. NFPA 72 identifies the minimum performance, location, mounting, testing and maintenance requirements for carbon monoxide detection systems. The manufacturer's instructions will identify the limitations on the use of equipment, including environmental conditions such as humidity, temperature and airflow.

**915.4 Carbon monoxide alarms.** Carbon monoxide alarms shall comply with Sections 915.4.1 through 915.4.4.

**C** If the option of using CO alarms is chosen, compliance is required with Sections 915.4.1 through 915.4.4. CO alarms can be stand-alone or interconnected (single- or multiple-station). Section 915 does not address whether such alarms are required to be interconnected.

CO alarms are designed to initiate an audible alarm when the level of CO is still below that which can cause a loss of the ability to react to the dangers of CO exposure. UL specifies that CO alarms activate at a level where the CO concentration over a given period of time can achieve 10 percent carboxyhemoglobin (COHb) in the body. Ten percent COHb will not cause physiological injury but is a level at which increases in the CO concentration will begin to affect the human body.

**915.4.1 Power source.** Carbon monoxide alarms shall receive their primary power from the building wiring where such wiring is served from a commercial source, and when primary power is interrupted, shall receive power from a battery. Wiring shall be permanent and without a disconnecting switch other than that required for overcurrent protection.

**Exception:** Where installed in buildings without commercial power, battery-powered carbon monoxide alarms shall be an acceptable alternative.

**C** This section is very similar to that required for smoke alarms (i.e., Section 907.2.11.6). The power supply must be provided by the building with battery backup. As with smoke alarms, the exception allows the use of battery power if commercial power is not available.

**915.4.2 Listings.** Carbon monoxide alarms shall be *listed* in accordance with UL 2034.

**C** Single- or multiple-station CO alarms are required to be listed in accordance with UL 2034. This standard is specific to CO alarms that provide both detection and notification.

**915.4.3 Combination alarms.** Combination carbon monoxide/smoke alarms shall be an acceptable alternative to carbon monoxide alarms. Combination carbon monoxide/smoke alarms shall be *listed* in accordance with UL 217 and UL 2034.

**C** Since smoke alarms are required in many occupancies, often a single combination alarm is desired. In fact, in some jurisdictions the CO alarms are required to be combined with smoke alarms. To meet this requirement, the combination CO and smoke alarm must be listed to both UL 217 and UL 2034. See Commentary Figure 915.4.3 for an example of a combination CO and smoke alarm.

Commentary Figure 915.4.3—Combination Carbon Monoxide and Smoke Alarm



**915.4.4 Interconnection.** Where more than one carbon monoxide alarm is required to be installed, carbon monoxide alarms shall be interconnected in such a manner that the actuation of one alarm will activate all of the alarms. Physical interconnection of carbon monoxide alarms shall not be required where listed wireless alarms are installed and all alarms sound upon activation of one alarm.

**C** The installation of carbon monoxide alarms in areas remote from the sleeping area will be of minimal value if the alarm is not heard by the occupants. Interconnection of multiple carbon monoxide alarms within an individual dwelling unit or sleeping unit is required in order to alert a sleeping occupant of a remote fire within the unit before the combustion products reach the smoke alarm in the sleeping area, thus providing additional time for evacuation.

The term “interconnection” refers to either hard-wired systems or listed wireless systems. UL has listed carbon monoxide alarms that use this technology.

**915.5 Carbon monoxide detection systems.** Carbon monoxide detection systems shall be an acceptable alternative to carbon monoxide alarms and shall comply with Sections 915.5.1 through 915.5.3.

**C** If a CO detection system is preferred over CO alarms, compliance with Sections 915.5.1 through 915.5.3 is required.

**915.5.1 General.** Carbon monoxide detectors shall be *listed* in accordance with UL 2075.

**C** This section indicates that carbon monoxide detectors are required to comply with UL 2075.

**915.5.2 Locations.** Carbon monoxide detectors shall be installed in the locations specified in Section 915.2. These locations supersede the locations specified in NFPA 72.

**C** This section is provided to clarify that the locations required for CO detectors in Section 915.2 differ from NFPA 72 and that those locations will supersede the standard. This simply reaffirms the provisions of Section 102.7.1 regarding conflicts between the code and a referenced standard. This section is provided to clarify that the locations required for CO detectors in Section 915.2 differ from NFPA 72 and that those locations will supersede the standard. This simply reaffirms the provisions of Section 102.7.1 regarding conflicts between the code and a referenced standard.

**915.5.3 Combination detectors.** Combination carbon monoxide/smoke detectors shall be an acceptable alternative to carbon monoxide detectors, provided that they are *listed* in accordance with UL 268 and UL 2075.

**C** Similar to CO alarms, there is often a desire or a requirement to combine smoke detection and CO detection systems. If the detectors themselves are combined, they need to be listed in accordance with both UL 268 and UL 2075. NFPA 72 requires the CO alarm to be capable of transmitting a distinct audible signal that is different from the smoke alarm signal.

**915.5.4 Occupant notification.** Activation of a carbon monoxide detector shall annunciate at the control unit and shall initiate audible and visible alarm notification throughout the building.

**Exception:** Occupant notification is permitted to be limited to the area where the carbon monoxide alarm signal originated and other signaling zones in accordance with the fire safety plan, provided that the alarm signal from an activated carbon monoxide detector is automatically transmitted to an *approved* on-site location or off-premises location.

**C** Carbon monoxide detection activation is required to both annunciate at the control unit and notify the occupants with audible and visual devices throughout the building in an alarm condition.

The exception recognizes that there are situations where an alternative limited area notification can be approved instead of a full-occupant notification throughout the building. This is subject to approval by the fire code official and the conditions listed. This includes a notification limited to the area where the signal originated or other affected signaling zones that have been specifically identified in the fire safety plan. In addition, there is an automatic detector alarm signal transmission notification at an approved on-site or off-premises location.

**915.5.5 Duct detection.** Carbon monoxide detectors placed in environmental air ducts or plenums shall not be used as a substitute for the required protection in Section 915.

**C** This section is intended to prevent a potential life safety issue by prohibiting duct-mounted carbon monoxide (CO) detectors from being installed in lieu of “open area” CO detectors or alarms. This requirement is necessary for the following reasons: (1) duct-mounted CO detectors are commercially available, but there are not currently any listed to or in compliance with UL 2075 or UL 2034; (2) duct-mounted CO detectors are for use in specific applications where CO detection in the duct is needed, which can create confusion for designers and fire code officials; (3) when the HVAC system is not moving significant quantities of air, duct-mounted CO detectors cannot accurately detect potentially hazardous levels of CO in the rooms served by the HVAC system.

**915.6 Maintenance.** Carbon monoxide alarms and carbon monoxide detection systems shall be maintained in accordance with NFPA 72. Carbon monoxide alarms and carbon monoxide detectors that become inoperable or begin producing end-of-life signals shall be replaced.

**C** This section is simply a reference to the maintenance requirements of NFPA 72. In addition, when detectors become inoperable or start producing end-of-life signals, they are required to be replaced.

**915.6.1 Enclosed parking garages.** Carbon monoxide and nitrogen dioxide detectors installed in enclosed parking garages in accordance with Section 404.1 of the *International Mechanical Code* shall be maintained in accordance with the manufacturer’s instructions and their listing. Detectors that become inoperable or begin producing end-of-life signals shall be replaced.

**C** This section is specific to carbon monoxide and nitrogen dioxide detectors. NFPA 72 does not address the maintenance of this type of detector. Thus, the required maintenance is to be provided by the manufacturer in their written instructions as well as the individual product listing. Also, similar to the previous section, when these detectors become inoperable or start producing end-of-life signals, they are required to be replaced.

## SECTION 916—GAS DETECTION SYSTEMS

**916.1 Gas detection systems.** *Gas detection systems* required by this code shall comply with Sections 916.2 through 916.11.

**C** Gas detection systems are required for many different applications in the code. This section includes basic requirements for all gas detection systems and covers construction documents, equipment, power connections, emergency and standby power, sensor locations, gas sampling, system activation, signage, fire alarm system connections, maintenance, testing and sensor calibration. These are important safety requirements that are applicable to all gas detection systems, including those that are installed in a small operation up to those in large industrial facilities. Gas detection system equipment that can comply with these requirements is commercially available.

**916.2 Permits.** Permits shall be required as set forth in Section 105.6.11.

**C** A construction permit is required for the installation of gas detection systems. The requirements are located in Section 105.6.11.

**916.2.1 Construction documents.** Documentation of the *gas detection system* design and equipment to be used that demonstrates compliance with the requirements of this code shall be provided with the application for permit.

**C** The permit application must include the design drawings and equipment data sheets in order to document that the gas detection system conforms to the code requirements.

**916.3 Equipment.** *Gas detection system* equipment shall be designed for use with the gases being detected and shall be installed in accordance with manufacturer’s instructions.

**C** The equipment to be used in the gas detection system is to be specific to the type of gas that is being used and detected. The system installation is to conform to the manufacturer’s specific equipment instructions so that it will operate as required.

**916.4 Power connections.** *Gas detection systems* shall be permanently connected to the building electrical power supply or shall be permitted to be cord connected to an unswitched receptacle using an *approved* restraining means that secures the plug to the receptacle.

**C** The electric power supply for the gas detection system is to be installed permanently so that it will be able to operate continuously as required. As an alternative, subject to the approval of the fire code official, the power supply can be connected to an outlet that does not have a separate ON-OFF switch on the condition that it has an acceptable fixed restraint.

**916.5 Emergency and standby power.** Standby or emergency power shall be provided or the *gas detection system* shall initiate a trouble signal at an *approved* location if the power supply is interrupted.

**C** The gas detection system is to have a standby or emergency electric power supply. However, if it is not provided and the normal power supply is interrupted, the system is required to transmit a trouble signal to an approved location.

**916.6 Sensor locations.** Sensors shall be installed in *approved* locations where leaking gases are expected to accumulate.

**C** Each individual gas detection sensor is to be located where leaks are anticipated to occur. The locations are to be acceptable to the fire code official.

**916.7 Gas sampling.** Gas sampling shall be performed continuously. Sample analysis shall be processed immediately after sampling, except as follows:

1. For HPM gases, sample analysis shall be performed at intervals not exceeding 30 minutes.
2. For toxic gases that are not HPM, sample analysis shall be performed at intervals not exceeding 5 minutes, in accordance with Section 6004.2.2.7.
3. Where a less frequent or delayed sampling interval is *approved*.

**C** The required performance of the gas detection system sampling is that it is to be constant and without delay. However, this section provides three situations where the gas sampling is allowed to be less than continuous.

**916.8 System activation.** A gas detection alarm shall be initiated where any sensor detects a concentration of gas exceeding the following thresholds:

1. For *flammable gases*, a gas concentration exceeding 25 percent of the lower flammability limit (LFL).
2. For nonflammable gases, a gas concentration exceeding one-half of the IDLH, unless a different threshold is specified by the section of this code requiring a *gas detection system*.

Upon activation of a gas detection alarm, alarm signals or other required responses shall be as specified by the section of this code requiring a *gas detection system*. Audible and visible alarm signals associated with a gas detection alarm shall be distinct from fire alarm and carbon monoxide alarm signals.

**C** The required local alarm is intended to alert occupants to a hazardous condition in the vicinity of the inside storage room or area. The alarm is not intended to be an evacuation alarm; however, it is required to be monitored to hasten emergency personnel response.

**916.9 Signage.** Signs shall be provided adjacent to *gas detection system* alarm signaling devices that advise occupants of the nature of the signals and actions to take in response to the signal.

**C** This section requires informational signs in order to communicate the type of alarm and what action is to be taken upon activation.

**916.10 Fire alarm system connections.** Gas sensors and *gas detection systems* shall not be connected to fire alarm systems unless *approved* and connected in accordance with the fire alarm equipment manufacturer's instructions.

**C** This section requires that where gas sensors and gas detection systems are approved to be connected to a fire alarm system, they are only to be connected in conformance to the equipment manufacturer's written instructions. This ensures that the fire alarm system will operate as required in response to the independent operation of the gas detection system.

**916.11 Inspection, testing and sensor calibration.** Inspection and testing of *gas detection systems* shall be conducted not less than annually. Sensor calibration shall be confirmed at the time of sensor installation and calibration shall be performed at the frequency specified by the sensor manufacturer.

**C** This section provides the maximum allowable inspection and testing frequency. The requirement for an annual test and inspection provides confirmation that the system is operating properly and will function as designed. All systems components are to be verified at the time of the inspection with the exception of the sensor calibration. The required frequency of the calibration of the sensors is to be in accordance with the manufacturer's instructions.

## SECTION 917—MASS NOTIFICATION SYSTEMS

**917.1 College and university campuses.** Prior to construction of a new building requiring a *fire alarm system* on a multiple-building college or university campus having a cumulative building *occupant load* of 1,000 or more, a mass notification risk analysis shall be conducted in accordance with NFPA 72. Where the risk analysis determines a need for mass notification, an *approved* mass notification system shall be provided in accordance with the findings of the risk analysis.

**C** This section does not specifically require that a mass notification system be installed. Instead, it requires that a risk analysis be completed in accordance with NFPA 72 prior to the construction of a new building that requires a fire alarm system on a multiple-building college or university campus and has a cumulative building occupant load of 1,000 or more. If the results of the risk analysis state that a mass notification system shall be provided, then it is to be installed in accordance with the results and the acceptance of the fire code official.

**917.2 Group E occupancies** Prior to construction of a new building containing a Group E occupancy requiring a *fire alarm system* and having an occupant load of 500 or more, a mass notification risk analysis shall be conducted in accordance with NFPA 72. Where

the risk analysis determines a need for mass notification, an *approved* mass notification system shall be provided in accordance with the findings of the risk analysis.

**C** This section requires that a mass notification risk analysis be performed prior to the construction of a new building containing Group E occupancies. If, and only if, the findings of the risk analysis conclude a mass notification system is needed, then a mass notification system shall be provided.

Every facility is unique and has specific risks. A risk analysis is a process that determines the likelihood, vulnerability and magnitude of all potential emergencies. The complexity of the risk analysis should be commensurate to the complexity of the building(s) and hazards under consideration. The risk analysis will determine if a mass notification system is needed and the type of a mass notification system that best meets the specific needs of the building.

Moreover, the key to determining if a mass notification system is required is to review the fire safety and evacuation plan in conjunction with the mass notification risk analysis. The risk analysis may identify hazards that are specific to the facility, thereby necessitating specific response plans and communication system requirements for that facility.

It should be noted that an in-building fire alarm emergency voice/alarm communication (EVAC) system is one type of a mass notification system that provides on-way voice messages to people inside buildings. Instead of horns, the code requires a fire alarm system using EVAC speakers for occupant notification to be installed in new schools with an occupant load greater than 100. If the risk analysis determines a mass notification system is needed, there may be no additional cost for the control unit because many EVAC systems are listed for mass notification in accordance with UL 2572, *Mass Notification Systems*.

Requiring a risk analysis will result in a more comprehensive emergency response plan that is customized for the specific hazards and risks associated with the building or campus. The risk analysis and emergency response plan can be as elaborate or as basic as the fire code official and building owner deem necessary. In some cases, other types of one-way communications may be needed to provide effective protection. Section 403 requires a fire safety and evacuation plan be developed for educational occupancies, and Section 404 requires that when a lockdown plan is developed, it is approved by the fire code official. These sections require that communication of the emergency to the building occupants be included in the plan. This section only emphasizes the need to document how communication with the occupants of the building and any occupants outside the building will be accomplished. It encourages the use of the EVAC system that is already required as a part of the fire alarm construction.

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